



Government of the People's Republic of Bangladesh
Ministry of Health and Family Welfare

HEALTH BULLETIN 2019



Management Information System
Directorate General of Health Services
Mohakhali, Dhaka 1212
www.dghs.gov.bd



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2019

June 2020

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We acknowledge the contributions of all other personnel who were involved directly or indirectly in the production of this bulletin by sending data or providing logistical support

ACKNOWLEDGMENTS

We would like to thank the following personnel for their contributions in publication of this bulletin:

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ACRONYMS

ACSM	Advocacy, Communication, and Social Mobilization	BHMS	Bachelor of Homeopathic Medicine and Surgery
ACR	Annual Confidential Report	BITID	Bangladesh Institute of Tropical and Infectious Disease
ACT	Artemisinin-based Combination Therapy	BMDC	Bangladesh Medical and Dental Council
ADB	Asian Development Bank	BMRC	Bangladesh Medical Research Council
ADL	Adeno-Lymphangitis	BNC	Bangladesh Nursing Council
ADP	Annual Development Program	BNHA	Bangladesh National Health Accounts
AEFI	Adverse Events Following Immunization	BIRDEM	Bangladesh Institute of Research and Rehabilitation for Diabetes, Endocrine and Metabolic Disorders
AeHIN	Asian eHealth Information Network		
AFP	Acute Flaccid Paralysis	BNNC	Bangladesh National Nutrition Council
AHI	Assistant Health Inspector	BNSB	Bangladesh National Society of Blind
a2i	Access to Information	bOPV	bivalent Oral Polio Vaccine
AIDS	Acquired Immunodeficiency Syndrome	BOR	Bed-occupancy Ratio/Rate
ALB	Albendazole	BPL	Below Poverty-line
ALS	Average Length of Stay	BRAC	Bangladesh Rural Advancement Committee
AMC	Alternative Medical Care	BRIS	Birth Registration Information System
AMR	Antimicrobial Resistance	BRN	Birth Registration Number
ANC	Antenatal Care	BSL	Biosafety Level
API	Annual Parasite Incidence/Application Programming Interfaces	BSMMU	Bangabandhu Sheikh Mujib Medical University
APIR	Annual Program Implementation Report	BST	British Summer Time
APR	Annual Program Review	BTRC	Bangladesh Telecommunication Regulatory Commission
ARC	American Red Crescent/Antimicrobial Resistance Containment	BUET	Bangladesh University of Engineering and Technology
ARI	Acute Respiratory Infection	BUMS	Bachelor of Unani Medicine and Surgery
ARSH	Adolescent Reproductive Strategy for Health		
ART	Antiretroviral Treatment/Antiretroviral Therapy	CAGR	Compound Annual Growth Rate
ASCEND	Accelerating the Sustainable Control and Elimination of Neglected Tropical Diseases	CAPABLE	Cambridge Program to Assist Bangladesh in Lifestyle and Environmental Risk Reduction
ASD	Autism Spectrum Disorder	CBE	Clinical Breast Examination
ASP	AIDS/STD Program	CBHC	Community-based Healthcare
AWD	Acute Watery Diarrhea	CC	Community Clinic/ City Corporation
		CCC	Chattogram City Corporation
BAMS	Bachelor of Ayurvedic Medicine & Surgery	CCU	Cardiac/Coronary Care Unit
BBS	Bangladesh Bureau of Statistics	CDC	Communicable Disease Control
BCC	Behavior Change Communication/ Barisal City Corporation	CDD	Control of Diarrheal Diseases
BCG	Bacillus Calmette–Guérin	CD-VAT	Customs Duty and Value Added Tax
BCPS	Bangladesh College of Physicians and Surgeons	CEmOC	Comprehensive Emergency Obstetric Care
BCS	Bangladesh Civil Service	CES	Coverage Evaluation Survey
BDHS	Bangladesh Demographic and Health Survey	CFR	Case-fatality Rate
BDT	Bangladeshi Taka	CG	Community Group
BEOC	Basic Emergency Obstetric Care	CHCP	Community Healthcare Provider
BHFS	Bangladesh Health Workforce Survey		

CHT	Chattogram Hill Tracts	DH	District Hospital
CHW	Community Health Worker	DHIS2	District Health Information System software version 2
CIDA	Canadian International Development Agency	DHMS	Diploma in Homoeopathic Medicine and Surgery
CIDD	Control of Iodine Deficiency Disorder	DHS	Demographic Health Survey
CIN	Cervical Intra-epithelial Neoplasia	DICs	Drop-in-Centers
CIPRB	Centre for Injury Prevention and Research, Bangladesh	DLI	Disbursement Link Indicator
CMBT	Center for Medical Biotechnology	DLS	Department of Livestock
CME	Centre for Medical Education	DMCH	Dhaka Medical College Hospital
CNCP	Comprehensive Newborn Care Package	DNA	Deoxyribonucleic Acid
CMSD	Central Medical Store Depot	DNCC	Dhaka North City Corporation
COPD	Chronic Obstructive Pulmonary Disease	DOTS	Directly-observed Treatment-Short Course
CPR	Contraceptive Prevalence Rate	DPA	Direct Project Aid
CR	Civil Registration	DR	Disaster Recovery
CRD	Chronic Respiratory Disease	DRPCC	District Rabies Prevention and Control Center
CRPD	Convention on the Rights of Persons with Disabilities	DSCC	Dhaka South City Corporation
CRVS	Civil Registration and Vital Statistics	DSF	Demand-side Financing
CS	Civil Surgeon	DUMS	Diploma in Unani Medicine and Surgery
CSBA	Community-based Skilled Birth Attendant		
CSG	Community Support Group	EA	Enterprise Architecture
CSO	Civil Society Organization	EBS	Event-based Surveillance
CT	Computed Tomography	ECG	Electrocardiogram
CtC	Child-to-Child	ECNEC	Executive Committee of National Economic Council
ComCC	Comilla City Corporation	eHealth	Electronic Health
CVASU	Chittagong Veterinary Animal Science University	EmONC	Emergency Obstetric and Newborn Care
CVD	Cardiovascular disease	EMR	Electronic Medical Record
cVDPV	circulating Vaccine-Derived Polio Virus	EOC	Emergency Obstetric Care
		EPI CES	Expanded Program on Immunization, Coverage Evaluation Survey
DALYs	Disability-Adjusted Life Years	ERD	Economic Relations Division
DAMS	Diploma in Ayurvedic Medicine and Surgery	ERP	Emergency Response Preparedness
DCC	District Coordination Committee	ESD	Essential Service Delivery
DDO	Drawing and Disbursing Order	ETL	Extract Transform Load
DEC	Diethyl-Carbamazine	EU	Evaluation Unit
DEPB	Develop Evidence-based Planning and Budgeting	EWARS	Early Warning Alert and Response System
Dev.	Development		
D4D	Data for Decision	FAO	Food and Agriculture Organization
DFATD	Department of Foreign Affairs, Trade and Development	FBIS	Foodborne Illness Surveillance
DF/DHF	Dengue Fever/Dengue Hemorrhagic Fever	FCPS	Fellow of the College of Physicians and Surgeons
DFID	Department for International Development	FCTC	Framework Convention on Tobacco Control
DGDA	Directorate General of Drug Administration	FD	Finance Division
DGFP	Directorate General of Family Planning	FDMNs	Forcibly Displaced Myanmar Nationals
DGHEU	Directorate General of Health Economics Unit	FETP/B	Field Epidemiology Training Program, Bangladesh
DGHS	Directorate General of Health Services	FP	Family Planning
DGME	Directorate General of Medical Education	FPI	Family Planning Inspector
DGNM	Directorate General of Nursing and Midwifery	FSW	Female Sex Worker

FWA	Family Welfare Assistant	HPV	Human Papillomavirus
FWV	Family Welfare Visitor	HR	Human Resource
FY	Fiscal Year	HRH	Human Resource for Health
FYP	Five-year Plan	HRIS	Health Resource Information System
		HRM	Human Resource Management
GAVI	Global Alliance for Vaccines and Immunization	HRMS	Human Resource Management System
GBS	Guillain-Barre Syndrome	HSM	Hardware Security Module
GCC	Gazipur City Corporation	HSS	Health System Strengthening
GDD	Global Disease Detection	HWF	Health Workforce
GDP	Gross Domestic Product		
GED	General Economic Division	IAEG-SDGs	Inter-agency and Expert Group on SDG Indicators
GFATM	Global Fund to Fight AIDS, Tuberculosis and Malaria	IBS	Irritable Bowel Syndrome
GH	General Hospital	ICD-10	International Statistical Classification of Disease
GHS	Global Health Security	icddr'b	International Centre for Diarrhoeal Disease Research, Bangladesh
GHSA	Global Health Security Agenda	ICMH	Institute of Child and Mother Health
GMP	Good Manufacturing Practice	ICT	Information and Communication Technology
GNI	Gross National Index	ICU	Intensive Care Unit
GOARN	Global Outbreak Alert and Response Network	ideSHi	Institute of developing Science and Health initiatives
GOB	Government of Bangladesh	ID-RV	Intra-dermal Route Vaccine
Govt.	Government	IEDCR	Institute of Epidemiology, Disease Control and Research
GPELF	Global Program to Eliminate Lymphatic Filariasis	IHR	International Health Regulations
GPS	Global Positioning System	IHT	Institute of Health Technology
GR	Geographical Reconnaissance	IMCI	Integrated Management of Childhood Illness
		IMPACT	Improving Public Health Management for Action
HA	Health Assistant	IMNCI	Integrated Management of Neonatal and Childhood Illness
HAV	Hepatitis A Virus	INFOSAN	International Network of Food Safety Authority
HBV	Hepatitis B Virus	IPGMR	Institute of Postgraduate Medicine and Research
HCPs	Healthcare Providers	IPH	Institute of Public Health
HCV	Hepatitis C Virus	IPHN	Institute of Public Health Nutrition
HDI	Human Development Index	IPV	Inactivated Polio Virus Vaccine
HDU	High Dependency Unit	ISDP	Integrated Service Delivery Platform
HEU	Health Economics Unit	ISP	Internet Service Provider
HI	Health Inspector	IT	Information Technology
HIE	Health Information Exchange	IV	Intravenous Fluid
HIS	Health Information System	IVM	Integrated Vector Management
HISPBd	Health Information System Program, Bangladesh		
HIV	Human Immunodeficiency Virus	JE	Japanese encephalitis
HMIS	Health Management Information System	JEE	Joint External Evaluation
HNP	Health and Nutrition Program	JHU	Johns Hopkins University
HNPSIP	Health, Nutrition and Population Strategic Investment Plan	JICA	Japan International Cooperation Agency
HPNSDP	Health, Population and Nutrition Sector Development Program	JCC	Joint Coordination Committee
HPNSP	Health, Population and Nutrition Sector Program	JPEG	Joint Photographic Experts Group
HPSP	Health and Population Sector Plan		

KA	Kala-azar	MPDSR	Maternal and Perinatal Death Surveillance and Response
KCC	Khulna City Corporation	MPH	Master in Public Health
KMC	Kangaroo Mother Care	M Phil	Master of Philosophy
KSM	Kala-azar Search Volunteer	MR	Measles-Rubella Vaccine
LAN	Local Area Networking	MRI	Magnetic Resonance Imaging
LD	Line Director	MRORI	Measles-Rubella Outbreak Response Immunization
LEEP	Loop Electrosurgical Excision Procedure	MRS	Medical Record System
LF	Lymphatic Filariasis	MSD	Measles Second Dose
LG	Local Government	MSH	Management Sciences for Health
LHB	Local Health Bulletin	MSM	Men having Sex with Men
L&HEP	Lifestyle, Health Education & Promotion	MSW	Male Sex Worker
LLIN	Long-Lasting Insecticidal Nets	MT-EPI	Medical Technologist of Expanded Program on Immunization
M&E	Monitoring & Evaluation	MTR	Mid-term Review
MARP	Most-at-risk Population	NAC	National AIDS Committee
MAT	Medical Assistance Training	NAP	National Action Plan
MATS	Medical Assistant Training School	NASP	National AIDS and STD Program
MBBS	Bachelor of Medicine, Bachelor of Surgery	NAVC+	National Vitamin A plus Campaign
MBDC	Mycobacterium Disease Control	NBPH	National Bulletin of Public Health
MBT	Medical Biotechnology	NCC	Narayanganj City Corporation
MCC	Mymensingh City Corporation	NCD	Non-communicable Disease
MCCoD	Medically Certified Cause of Death	NCT	National Competitive Tendering
MCH	Medical College Hospital	NFC	National Fistula Center
MCPS	Member of the College of Physicians & Surgeons	NFM	New Funding Model
MCV1	Measles Coverage Vaccine first dose	NFR	New Funding Request
MCWC	Maternal and Child Welfare Centers	NGOs	Non-governmental organizations
MDA	Mass Drug Administration	NHA	National Health Accounts
MDGs	Millennium Development Goals	NHSDP	NGO Health Service Delivery Project
MDR	Multi-drug Resistance	NIC	National Influenza Center
MDT	Multi-drug Therapy	NICU	Neonatal Intensive Care Unit
MDV	Mass Dog Vaccination	NID	National ID/National Immunization Day
MEHM	Minimum European Health Module	NIDCH	National Institute of Diseases of the Chest & Hospital
MICS	Multiple Indicator Cluster Surveys	NINH	National Institute of Neurology & Hospital
MIS	Management Information System	NINS	National Institute of Neurosciences & Hospital
MMDP	Morbidity Management and Disability Prevention	NIO	National Institute of Ophthalmology
MMed	Masters in Medical Education	NIPORT	National Institute of Population Research and Training
MMR	Maternal Mortality Ratio	NIPSOM	National Institute of Preventive and Social Medicine
MNCAH	Maternal, Neonatal, Child, and Adolescent Health	NISB	National Influenza Surveillance, Bangladesh
MNHI	Maternal and Newborn Health Initiative	NITOR	National Institute of Traumatology and Orthopedics Rehabilitation
MNT	Maternal and Neonatal Tetanus	NiV	Nipah Virus
MOCHTA	Ministry of Chittagong Hill Tracts Affairs	NKEP	National Kala-azar Elimination Program
MOHA	Ministry of Home Affairs	NLEP	National Leprosy Elimination Program
MOHFW	Ministry of Health and Family Welfare		
MOLGRD	Ministry of Local Government, Rural Development and Cooperatives		

nm	Nautical Mile	PH&DTL	Public Health and Drug Testing Laboratory
NMCP	National Malaria Control Program	PHEIC	Public Health Emergency of International Concern
NMR	Neonatal Mortality Rate	PKDL	Post Kala-azar Dermal Leishmaniasis
NMSS	National Micronutrients Status Survey	PLHIV	People living with HIV
NNP	National Nutrition Program	PLSM	Product Line Sales Manager/ Police Long Service Medal
NNHP	National Newborn Health Program	PMIS	Personnel Management Information System
NNS	National Nutrition Services	PMMU	Program Management & Monitoring Unit
NPAN2	Second National Plan of Action for Nutrition	PMO	Prime Minister's Office
NPELF	National Program on Elimination of Lymphatic Filariasis	PNC	Postnatal Care
NQAS	National Quality Assurance Standards	PoE	Point of Entry
NRPCC	National Rabies Prevention and Control Center	PPE	Personal Protective Equipment
NSC	National Steering Committee	PPM	Public-Private Mix
NSP	National Strategic Plan	PPR	Public Procurement-related Rules
NTBB	National Taskforce on Biotechnology of Bangladesh	PRM	Population, Refugees and Migration/ Physician Relationship Management
NTC	National Technical Committee	PRS	Population Registration System
NTCMB	National Technical Committee on Medical Biotechnology	PRSP	Poverty Reduction Strategy Paper
NTDs	Neglected Tropical Diseases	PS&	Planning, Supply and ownership management program
NTP	National TB Program	PSM	Preventive and Social Medicine
NTRL	National TB Reference Laboratory	PSTN	Public Switched Telephone Network
NTV	Nerve Tissue Vaccine	PUD	Peptic Ulcer Disease
NTWG	National Technical Working Group	PWID	People Who Inject Drugs
OCV	Oral Cholera Vaccination	RangCC	Rangpur City Corporation
OH-APP	OneHealth Assessment for Planning and Performance	RADP	Revised Annual Development Program
OMR	Optical Mark Recognition	RBC	Red Blood Cell
OP	Operational Plan	RBM	Roll Back Malaria
OPD	Outpatient Department	RCC	Rajshahi City Corporation
OpenMRS	Open Medical Record System	RCHCIB	Revitalization of Community Healthcare Initiative in Bangladesh
OPV	Oral Polio Vaccine	RDT	Rapid Diagnostic Tests
ORG	Office of Registrar General	RDU	Research and Development Unit
ORI	Outbreak Response Immunization	REP	Rabies Elimination Program
ORS	Oral Rehydration Salt	Rev.	Revenue
ORT	Oral Rehydration Therapy	RFI	Result Framework Indicators
OSD	Officer on Special Duty	RHC	Rural Health Center
OT	Operation Theater	RHIS	Routine Health Information System
PA	Project Aid	RIG	Rabies Immunoglobulin
PCR	Polymerase Chain Reaction	RKA	Relapsed Kala-azar
PCV	Packed Cell Volume	RMCS	Revised Malaria Control Strategies
PDA	Personal Digital Assistant	RPA	Reimbursable Project Aid
PEP	Post-exposure Prophylaxis	RRRC	Refugee Relief and Repatriation Commissioner
Pf	Plasmodium falciparum	RRT	Rapid Response Team
PH	Public Health	rRT-PCR	Real-time Reverse Transcription (RRT)-Polymerase Chain Reaction
PHC	Primary Healthcare/Primary Health Center	RT-PCR	Reverse Transcription Polymerase Chain Reaction
		RTRL	Regional TB Reference Laboratory

SAC	School-aged Children	UHC	Universal Health Coverage/ Upazila Health Complex
SACMO	Sub-Assistant Community Medical Officer	UHFWC	Union Health and Family Welfare Center
SAP	Strategic Action Plan	U5MR	Under-five Mortality Rate
SBA	Skilled Birth Attendant	UN	United Nation
SBCC	Social and Behavioral Change Communication	UNAIDS	Joint United Nations Program on HIV/AIDS
SBTP	Safe Blood Transfusion Program	UNDP	United Nations Development Program
SCC	Sylhet City Corporation	UNESCO	United Nations Educational, Scientific and Cultural Organization
SCAPAND	Strategic and Convergent Action Plan on Autism and Neurodevelopment Disabilities	UNFPA	United Nations Population Program
SCANU	Special Care Newborn Unit	UNGASS	United Nations General Assembly Special Session
SDGs	Sustainable Development Goals	UNICEF	United Nations Children's Fund
SEARO	Southeast Asian Regional Office	UPHCP	Urban Primary Healthcare Project
SHR	Shared Health Record	UPS	Uninterruptible Power Supply
SIP	Strategic Investment Plan	USAID	United States Agency for International Development
SKKRC	Surya Kanta Kala-azar Research Center	USC	Union Sub-center
SMEE	Subject Matter Expert Exchange	USD	United States Dollar
SMoL	Start-up Mortality List	USG	Ultrasonography
SOP	Standard Operating Procedure	USI	Universal Salt Iodization
SRH	Sexual and Reproductive Health	VA	Verbal Autopsy
SSK	Shashthyo Surokhsha Karmasuchi	VAC	Vitamin A Capsule
STD	Sexually Transmitted Disease	VAD	Vitamin A Deficiency
STG	Standard Treatment Guideline	VIA	Visual Inspection with Acetic Acid
STH	Soil-transmitted Helminthes	VL	Visceral Leishmaniasis
SVRS	Sample Vital Registration System	WPT	Water Purification Tablets
SWAp	Sector-wide Approach	WASH	Water, Sanitation and Hygiene
TAS	Transmission Assessment Survey	WAZ	Weight-for-age z-score
TAST	Technical Assistance Support Team	WB	World Bank
TB	Tuberculosis	WBC	White Blood Cell
TCV	Time, Cost, and Visit	WCBA	Women of Childbearing Age
TCV	Tissue Culture Vaccine	WHO	World Health Organization
TEMU	Transport & Equipment Maintenance Unit	WiMAX	Worldwide Inter-operability for Microwave Access
TFR	Total Fertility Rate	WOAH	World Organization for Animal Health
THE	Total Health Expenditure		
tOPV	trivalent Oral Polio Vaccine		
TOT	Training of Trainers		
TT	Tetanus Toxoid		
TTGA	Taurocholate-tellurite-gelatin Agar		
TTI	Transfusion-transmitted Infection		
TTU	Technical Training Unit		
TWG	Technical Working Group		
UAT	User Acceptance Test		
UCI	Universal Child Immunization		
UESDS	Utilization of Essential Service Delivery Survey		
UH&FPO	Upazila Health and Family Planning Officer		

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Note: Through a recent government announcement, spellings of the names of some places, like Chiitagong, Comilla, Jessore, have been changed for adherence to their Bangla pronunciations. The new spellings could not be fully used in this publication because the institutions having names of these places have not yet changed the spellings. Following the principle of spelling proper nouns, we had to retain the old spellings of the names of these places associated with some institutions. However, Chattogram for Chittagong is used while we meant the name of the administrative division



# Message



*Zahid Maleque*

**Zahid Maleque, MP**

Honorable Minister  
Ministry of Health and Family Welfare  
Government of the People's Republic of  
Bangladesh

I am extremely delighted to know that Health Bulletin 2019 is going to be published with all necessary information on the health sector of Bangladesh, innovative ideas, and contextual improvements.

This yearly publication of the Management Information System (MIS) of the Directorate General of Health Services is considered the mirror of the health scenario of our country reflecting the health situation and the impact of the decisions and policies taken in the health sector.

Under the strong and wise leadership of Honourable Prime Minister Sheikh Hasina, Bangladesh has made significant progress in the health sector as reflected through achievements of MDG targets.

The Government of Bangladesh is now considering attainment of the targets of Sustainable Development Goal (SDG) both as an opportunity and a challenge and has already initiated the new 4th sector-wide program called Health, Population and Nutrition Sector Program (HPNSP) 2017-2022 for this purpose.

Health Bulletin is a reliable source of data, which portray general health situation of the country with authenticity, thereby assisting efficiently in health-related national planning and policy-making.

I expect it to improve continually with the newer demands arising in this geographical area for maintenance of health and well-being of our population.

I appreciate the efforts of Director General of Health Services, Director MIS, MIS officials, editors, consultants, and personnel at the field level engaged in data-entry, compilation, and organization. I thank the development partners, NGOs, and other stakeholders for their valuable inputs for publishing in this document.

I expect more innovative ideas and detailed data of health research in the public and private sectors to enrich the Health Bulletin leading us to achieve our goal of health and prosperity for all.

Joy Bangla, Joy Bangabandhu.





## Message



**Md. Abdul Mannan**

Secretary  
Health Services Division  
Ministry of Health and Family Welfare  
Government of the People's Republic of  
Bangladesh

I congratulate MIS-Health for bringing out Health Bulletin 2019, the annual publication of the DGHS containing authentic updated information regarding health situation of Bangladesh. Bringing out a voluminous publication like this amid the grave situation of Covid-19 pandemic is really an appreciable task.

I expect this document will assist us in the evaluation of the effectiveness of our existing policies and for planning newer ones in the coming years.

Hard work of the designated personnel of MIS of DGHS involved in data collection, compilation, presentation, and publication deserves appreciation, especially in the nationwide lockdowns situation for coronavirus pandemic.

I would like to thank the DGHS personnel for their efforts of the overall activities and of the process of publication.

I expect that Health Bulletin will keep improving every year and continue to contribute in our goal of achieving the targets of health-related SDG by 2030.



## Message



**Md. Ali Noor**

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Government of the People's Republic of  
Bangladesh

MIS-DGHS has been showing its outstanding performance in health-related data management and publishing the Health Bulletin, the annual flagship publication of the DGHS. The Bulletin pays attention to include feedback for improvements and modifications each year as much as possible.

The pattern of organization of Health Bulletin data, which contribute in the policy-making for well-being of people to achieve the targets of health-related SDG in Bangladesh, is highly appreciable.

I would like to convey my special thanks to the personnel of MIS-DGHS for their innovative method of collecting, organizing, processing, and interpreting data in a well-coordinated way from various sources. Finally, I would like to appreciate the DGHS personnel for their uninterrupted intellectual coordination among different departments/sections relating to the overall publication process.



**Professor Dr. Abul Kalam Azad**

Director General  
Directorate General of Health Services  
Government of the People's Republic of  
Bangladesh

## Message

I am overtly pleased to know that Health Bulletin 2019 is being published by MIS of DGHS despite the unexpected COVID-19 pandemic situation. The Bulletin highlights the overall activities and achievements of our health sector during the past one year.

This annual publication of the MIS-DGHS is a pragmatic initiative to present the structured data for the use in policy-making and planning by monitoring their effectiveness. Hence, the Bulletin has always been trying to document reliable and authentic data as far as possible. Every year, we are improving the quality and presentation of data, and the software for data collection and data analysis.

The remarkable achievements of Bangladesh in accomplishing the targets of MDGs are expected to be continued in the SDG era as well. In this context, Health Bulletin 2019 will try to include data that portray a cumulative scenario of our achievements in the health sector. I extend my thanks to the Director of MIS-Health for his supervision and editorial guidance to the publication team comprising MIS personnel and external personnel who completed this valuable document. I am grateful to Honourable Minister, the Secretary of Health Services and the Secretary of Medical Education and Family Welfare Division of the Ministry of Health and Family Welfare for guiding us in every step of publishing Health Bulletin 2019.

As always, I like to invite suggestions for improvement of this publication both in terms of contents and the look from the readers, including from our colleagues, development partners, and critics.



## Message



**Professor Dr. AHM Enayet Hussain**

Director General  
Directorate General of Medical Education  
Ministry of Health and Family Welfare  
Government of the People's Republic of  
Bangladesh

Publication of Health Bulletin is one of the outstanding performances of MIS-DGHS. This publication is brought out each year with improvements and modifications in response to its readers at various levels.

The data for Health Bulletin are collected from the field level by using the own staff of health services, medical colleges, NGOs, and development partners of the health sector. The method of collecting data is unique and almost errorless. This unique pattern of data facilitates formulating policies in both medical education and service delivery. We can easily forecast what to do in the future.

I would like to convey my special thanks to the personnel of the MIS-DGHS for their method of collecting, organizing, processing, and interpreting data in a well-coordinated manner starting from the source to the MIS.

Finally, I offer my congratulations to Director, MIS, and the DGHS for intellectual coordination among relevant stakeholders associated with bringing out this valuable document.



## From Chief Editor's Desk



**Dr. Md. Habibur Rahman**

Director  
MIS and Line Director, HIS and eHealth  
Directorate General of Health Services  
Government of the People's Republic of  
Bangladesh

Despite a bit delay due to unexpected situation arising out of coronavirus pandemic, I am happy to hand over Health Bulletin 2019 to our instrumental readers. In repeated lockdowns and isolation of some key persons in the publication team, it was really difficult for me as Chief Editor to have all inputs ready on time. Finally, those who made it happen deserve special thanks.

A revolutionary step of the Government led by Honourable Prime Minister Sheikh Hasina was to take health services at the doorstep of the common people and improve access to healthcare with an aim to achieve Universal Healthcare Coverage by establishing community clinics (CCs).

Health Bulletin 2019 includes data collected ranging from these grassroots-level CCs to the tertiary-level health facilities. The purpose is to present structured and authentic primary data in a single volume so that policy-makers, planners, and development partners can readily access to the overall health scenario of the country.

I would like to thank Honourable Minister, Secretaries of the Ministry of Health and Family Welfare since it would never have been possible to publish the Bulletin without their kind and cordial cooperation.

My special thanks go to respected Professor Dr. Abul Kalam Azad, Director General of Health Services, whose guidance and supervision during his tenure as Director of MIS has established a standard for the structure and the working protocol for publishing Health Bulletin.

Finally, I express my ardent gratitude to the members of the editorial board, the former Director of MIS Dr Samir Kanti Sarkar for supervising the publication process, during his term in the office. I am also thankful to the personnel engaged in data-collection, the field-level workers as well as development partners whose assistance and feedback were essential for publishing the Bulletin.

At the end, I would like to welcome constructive criticism and suggestions to help us improve our performance and publication in the coming years.

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# Bangladesh Health Situation

## Major indicators at a glance

Over the last 49 years since Independence, Bangladesh has made a lot of strides in the Health Sector. Health infrastructure has been proliferated remarkably. Medical colleges, medical universities, private medical colleges, private clinics, private hospitals, district hospitals, rural health centers, and community clinics are being increased in numbers and improved in quality with modern facilities. Many NGOs are also playing a great role and contributing toward improving the healthcare delivery system. Much progress has been made in the pharmaceutical sector providing affordable medicines, intravenous fluids, anti-cancer drugs, and so on. There is also increased awareness in the general public on health issues. Public and private-level campaigns are ongoing to promote mental and child health, vaccination programs, mass deworming programs, use of safe water and latrines, hand-washing, etc. Family planning awareness is one of the most important successes of Bangladesh. The population growth rate has been brought down. A huge and thriving network of private physicians stretched and spread all over the country is trying to meet the needs of day-to-day medical problems. There have been significant gains in terms of polio and small pox eradication. Extensive vaccination and case isolation have resulted in diminishing the prevalence of diphtheria and tetanus. Progress in immunization (valid vaccination coverage) is excellent in Bangladesh. With improvement

of overall hygiene and sanitation standard, there is a discernible reduction in cholera, typhoid, and dysenteries. A great number of satellite clinics for tuberculosis now offer free treatment with contribution from NGOs. HIV/AIDS are at a low level of prevalence. Kala-azar and malaria has been significantly curved down. Malaria and tuberculosis-related death rates are currently very low in Bangladesh. Maternal mortality and neonatal mortality rate have also been reduced.

The number of hospitals under the DGHS is 739 contributing greatly to the healthcare in Bangladesh. The government hospitals of secondary and tertiary levels, hospitals at the upazila and union levels, registered private hospitals and clinics, and registered private diagnostic centers are dedicated to providing healthcare to the people. The numbers of hospital beds in the public and private health facilities are on the rise. Teaching and training institutions for healthcare are increasing, and the numbers of seats in these teaching/training facilities have also been increased in the recent years.

### Some basic information and data on Bangladesh

#### Administrative Units

|                   |      |
|-------------------|------|
| Division          | : 8  |
| City corporation  | : 12 |
| Metropolitan city | : 4  |



Municipality : 328  
 District : 64  
 Upazila : 492  
 Union : 4,554  
 Ward : 40,987  
 Village (approx.) : 87,320

### Demography

Sample Vital Registration System (SVRS)  
 2018

Estimated population on 1 July 2018: 164.6 million

Population density (per sq. km): 1,116;  
 the area is 1,47,570 square kilometer in  
 Bangladesh (1971)

Population growth rate: 1.37%

Sex ratio (M/F) x 100: 100.2/100.0

### Population by broad age-group (%):

| Age-group (years) | Both sexes | Male | Female |
|-------------------|------------|------|--------|
| 00-14             | 28.8       | 29.2 | 28.4   |
| 15-49             | 54.6       | 54.1 | 55.1   |
| 50-59             | 8.7        | 8.5  | 8.8    |
| 60+               | 7.9        | 8.2  | 7.7    |

Dependency ratio (percentage):  
 Total: 51; Rural: 55, Urban: 46

Child-woman ratio (per 1,000 women of 15-49 years): Total: 304; Rural: 332, Urban: 273

Crude birth rate (per 1,000 population):  
 Total: 18.3; Rural: 20.1, Urban: 16.1

Total fertility rate (per woman of 15-49 years):  
 Total: 2.05, Rural: 2.38, Urban: 1.68

### Age-specific fertility rate (per 1,000 women of reproductive age)

| Age-group (years) | Rate |
|-------------------|------|
| 15-19             | 74   |
| 20-24             | 132  |
| 25-29             | 106  |
| 30-34             | 62   |
| 35-39             | 26   |
| 40-44             | 7    |
| 45-49             | 3    |

General fertility rate: Total: 67, Rural: 77, Urban: 56

Net reproduction rate (per woman of 15-49 years): Total: 0.99, Rural: 1.15, Urban: 0.82

Crude marriage rate (per 1,000 population):  
 Total: 14.7, Rural: 17.2, Urban: 11.5

### Mean age at marriage (years)

| Area  | Male | Female |
|-------|------|--------|
| Total | 25.5 | 18.9   |
| Rural | 25.0 | 18.3   |
| Urban | 26.3 | 20.0   |

### Household characteristics and utilities

Sample Vital Registration System (SVRS)  
 2018

Household-size (no. of persons): 4.2

Male-headed households: 85.8

Female-headed households: 14.2

### Water and sanitation (% households)

Drinking-water: Access to tap and tube-well water: 98.0

Toilet facility: Sanitary: 78.1; Others: 19.9; None: 2.0

**Source of light (% household)**

| Source of light | Total |
|-----------------|-------|
| Electricity     | 90.1  |
| Solar           | 4.8   |
| Kerosene        | 5.0   |
| Others          | 0.1   |

**Information technology**

> Internet subscribers (BTRC, 31 December 2019)

- Total : 99.428 million
- Mobile Internet : 93,681 million
- WiMAX : 5 million
- ISP + PSTN : 5.742 million

> Mobile phone-users (BTRC, 31 March 2019)

- Total: 159.780 million

**Literacy**

Sample Vital Registration System (SVRS) 2018

Literacy rate of population of 7+ years (percentage): Both sexes: 73.2, Female: 71.2, Male: 75.2

**Health service and medical education**

Adult literacy rate among 15+ years (percentage): Both sexes: 73.9, Female: 71.2, Male: 76.7

- Total number of government facilities under the DGHS: 2,258
- Total number of primary-level facilities (except community clinics): 2,003
- Total number of secondary and tertiary-level facilities: 255

- Total number of registered private hospitals and clinics: 5,321
- Total number of registered private diagnostic centers: 9,529
- No. of hospital beds under the DGHS: 54,660
- No. of hospital beds in private hospitals registered by the DGHS: 91,537
- Total number of facilities run by DGHS, including registered private hospitals: 7,579

**Teaching/training institutions for healthcare**

| Teaching/<br>training<br>institutions                              | Total | Govt. | Private | Autono-<br>mous |
|--------------------------------------------------------------------|-------|-------|---------|-----------------|
| Postgraduate<br>medical<br>teaching<br>institutions                | 39    | 22    | 10      | 7               |
| Medical<br>colleges                                                | 106   | 36    | 70      |                 |
| Armed Forces<br>and Army<br>Medical<br>Colleges*                   | 6     | 6     |         |                 |
| Dental<br>colleges and<br>dental units<br>in medical<br>colleges   | 35    | 9     | 26      |                 |
| Nursing<br>institutions<br>offering Basic<br>BSc Nursing<br>Course | 60    | 15    | 45      |                 |

|                                                             |     |    |     |
|-------------------------------------------------------------|-----|----|-----|
| Nursing institutions offering Post-basic BSc Nursing Course | 41  | 4  | 37  |
| Nursing institutions offering Specialized Diploma           | 4   | 1  | 3   |
| Institutions offering Diploma in Nursing and Midwifery      | 183 | 43 | 140 |
| Unani and Ayurvedic Medical College                         | 4   | 2  | 2   |
| Homeopathic Medical College                                 | 2   | 1  | 1   |
| Unani and Ayurvedic Diploma College                         | 23  | 1  | 22  |
| Homeopathic Diploma College                                 | 61  | 0  | 61  |
| Medical Assistants Training Schools                         | 209 | 9  | 200 |
| Institutes of Health Technology (IHT)                       | 108 | 11 | 97  |

\*No. of Armed Forces Medical College - 1; No. of Army Medical Colleges - 5

### Available no. of seats in teaching/training institutions

| Name of degree                                                                                        | Total  | Govt. | Private | Foreign |
|-------------------------------------------------------------------------------------------------------|--------|-------|---------|---------|
| Postgraduate medical degree (CME Survey, 2017) (MD, MS, Diploma, M Phil, MPH, MMed, FCPS, MCPS, etc.) | 2,486  | 1,048 | 1,112   | 112     |
| Bachelor of Medicine & Surgery (MBBS)                                                                 | 12,082 | 4,068 | 6,231   | 1,783   |
| MBBS offered by Armed Forces and Army Medical Colleges                                                | 375    |       |         |         |
| Bachelor of Dental Surgery (BDS)                                                                      | 2,050  | 532   | 1,405   | 113     |
| Bachelor of Unani Medicine and Surgery (BUMS); Bachelor of Ayurvedic Medicine and Surgery (BAMS)      | 200    | 100   | 100     |         |

|                                                                                                |        |       |        |
|------------------------------------------------------------------------------------------------|--------|-------|--------|
| Bachelor of Homeopathic Medicine and Surgery (BHMS)                                            | 150    | 50    | 100    |
| Diploma in Unani Medicine and Surgery (DUMS); Diploma in Ayurvedic Medicine and Surgery (DAMS) | 1150   | 50    | 1100   |
| Diploma in Homeopathic Medicine and Surgery (DHMS)                                             | 3050   | 0     | 3050   |
| Medical Assistants Training                                                                    | 14,358 | 818   | 13,540 |
| BSc and Diploma courses in health technology                                                   | 11,625 | 2,685 | 8,940  |

## Health Status

### Family Planning

- > Contraceptive prevalence rate (%): 63.1 (SVRS 2018)
- > Unmet need for family planning (%): 12.0 (BDHS 2014)

## Maternal and child health, obstetric care

- Antenatal care coverage (at least 4 visits (%): 36.9 (MICS 2019)
- Antenatal care coverage (at least one visit by skilled health professionals) [%]: 75.2 (MICS 2019)
- Births attended by skilled health personnel (%): 59.0 (MICS 2019)
- C-section rate 36.0% (MICS 2019)
- Home delivery rate: 46.1% (MICS 2019)
- Infant mortality rate (per 1,000 livebirths): 22 (SVRS 2018)
- Institutional delivery rate (%): 53.8%  
Public facility: 16.0, Private facility: 37.8, Other facilities: 0.2 (MICS 2019)
- Maternal mortality ratio (per 100,000 livebirths): 169 (SVRS 2018)
- Neonatal mortality rate/1,000 livebirths: 16 (SVRS 2018)
- Under-5 mortality rate (per 1,000 livebirths): 29 (SVRS 2018)

## Immunization status: Valid full vaccination coverage 2019 (EPI-CES 2019)

✓ ≤12months old children

Valid full vaccination coverage by the age of 12 months was 83.9%

- BCG-99.7
- Penta1-99.7; Penta2-99.2; Penta3- 93.3
- OPV1-99.7; OPV2-99.2; OPV3-93.3
- PCV1-99.7; PCV2-99.2; PCV3-92.5
- MR1-88.6
- FVC-83.9

Valid tetanus toxoid coverage (%) among women of childbearing age: TT1-98.2%; TT2-96.8%; TT3-89.2%; TT4-73.2%; TT5-52.3% (EPI CES 2016)

Vitamin A coverage: Infant (6-11-months) 86.3%; Children (12-59-months) 90.7% (EPI CES 2016)

Life-expectancy at birth (years): Both sexes: 72.0; Female: 73.5; Male: 70.6 (SVRS 2017)

## Communicable diseases

### Malaria

- Population at risk: 18.1 million
- Endemic districts: 13 (71 upazilas)
- Total cases: 10,523 in 2018
- Total number of deaths: 7 in 2018
- *P. falciparum* cases 8848 (84.1%)
- *P. vivax* 1675 (15.9%)
- Malaria-positive cases/1,000 population (in endemic areas): 0.58 (in 2018)
- Malarial mortality/100,000 population (in endemic areas): 0.0004 (in 2018)

### HIV/AIDS situation (2019)

- > HIV prevalence among key populations in 2019: less than 1%
- > New HIV infections reported in 2019: 919
- > A total of 7,374 HIV-infected people have been detected in the country in 2019
- > A total of 1,242 have died of the disease within 2019
- > Number of ART recipients up to August 2018: 3,265

## Tuberculosis

WHO estimated the incidence and mortality for all ages and all forms of TB in 2018 in Bangladesh to be 221 (95% CI: 161-291) and 29 (95% CI: 18-42) per 100,000.

### WHO estimates of TB burden, 2017

| Estimates of TB burden 2017 | Number (thousand) | Rate (per 100,000 population) |
|-----------------------------|-------------------|-------------------------------|
| Mortality (excludes HIV+TB) | 47 (30-67)        | 29 (18-42)                    |
| Mortality (HIV+TB only)     | 0.19 (0.094-0.32) | 0.12 (0.06-0.2)               |
| Incidence (includes HIV+TB) | 357 (260-469)     | 221 (161-291)                 |
| Incidence (HIV+TB only)     | 0.73 (0.36-1.2)   | 0.45 (0.23-0.76)              |
| Incidence (MDR/RR-TB)       | (3.2-9.6)         | 3.7 (2-5.9)                   |

Source: Global TB Report 2019

The estimated prevalence of bacteriologically-confirmed pulmonary TB cases and smear-positive bacteriologically-confirmed TB cases among the population aged  $\geq 15$  years were 287 per 100,000 population (95% CI: 244-330) and 113 per 100,000 population (95% CI: 87-139) respectively. Total TB case notification was 268,596, and TB treatment coverage (notified/estimated incidence) was 75%.

### Kala-azar: 2017 (Source: DHIS2)

Around 32 million people, residing in 100 upazilas under 26 districts where kala-azar is endemic, are at risk in Bangladesh:

Total number of kala-azar cases (reported):  
380 cases

New kala-azar (NKA): 210 cases

Relapse kala-azar (RKA): 61 cases

Kala-azar treatment failure (KATF): 3 cases

Post-kala-azar Dermal Leishmaniasis (PKDL):  
102 cases

Death due to kala-azar: 6 cases

### **Non-communicable diseases**

- > Reduction in the prevalence of current tobacco-use in persons aged over 15 years: 35.3% (2017)
- > Reduction in the prevalence of insufficient physical activity: 15.1%(STEPS 2018)
- > Relative reduction in the prevalence of raised blood pressure: 25.4% (STEPS 2018)

**Prevalence of major NCDs as percentage (Source: Global Burden of Disease Study 2016)**

| Type of disease        | Prevalence of major NCDs (%) |
|------------------------|------------------------------|
| Ischemic heart disease | 3.4                          |
| Stroke                 | 0.9                          |
| CVD                    | 0.4                          |
| Diabetes               | 6.4                          |
| COPD                   | 2.97                         |

Death rate due to major NCDs (CVDs, cancer, diabetes, and chronic respiratory diseases among population aged 30 to 70 years) per 100,000 population: 398.1-503.7

- CVD-standardized deaths/100,000 population: 171.68

- CRD-standardized deaths/100,000 population: 53.5
- COPD-standardized deaths/100,000 population: 41.87
- Cancer-standardized deaths/100,000 population: 54.98
- Diabetes-standardized death/100,000 population: 19.43

### **Cancer**

Five most frequent cancers and others in male and female

| Type of cancer      | Incidence |
|---------------------|-----------|
| Breast              | 8.50      |
| Esophagus           | 13.90     |
| Cervix uteri        | 5.40      |
| Lung                | 8.20      |
| Lip and oral cavity | 8.90      |
| Others              | 55.20     |

### **✓ Mental health situation in Bangladesh as per STEPS 2018 survey**

- Any mental disorder among adults: 16.8% (Range 15.5-18.2)
- Any mental disorder among children and adolescents: 13.6% (Range 11.5-15.7)

### **Health Workforce**

Personnel currently working under DGHS of MOHFW (No. of sanctioned posts are given in parentheses)

- > No. of personnel: 78,619 (105,254) [excluding nursing posts]
- > No. of doctors: 25,594 (27,002)
- No. of registered physicians:
- > MBBS: 101,538; BDS: 9875 (according to BMDC)

|                                                                                                                                                                                                                                                                           |                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• No. of medical technologists:</li> </ul>                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>&gt; Population per registered physician: 1,487</li> </ul>                                                     |
| <ul style="list-style-type: none"> <li>&gt; Total: 5,208 (8,146); Dental: 476 (627); MT (EPI): 483 (498); Lab: 1,581 (2,389), Pharmacy: 1,441 (2,915); Physiotherapy: 126 (337); Radiography: 584 (795); Radiotherapy: 36 (86); Sanitary inspection: 478 (498)</li> </ul> | <ul style="list-style-type: none"> <li>&gt; No. of registered physicians per 10,000 population: 6.73</li> </ul>                                       |
| <ul style="list-style-type: none"> <li>&gt; No. of Sub-Assistant Community Medical Officers (SACMOs): 3,709 (5,337)</li> </ul>                                                                                                                                            | <ul style="list-style-type: none"> <li>&gt; No. of doctors working under the DGHS per 10,000 population: 1.55</li> </ul>                              |
| <ul style="list-style-type: none"> <li>&gt; No. of Community Healthcare Providers (CHCPs) for community clinics: 13,907</li> </ul>                                                                                                                                        | <ul style="list-style-type: none"> <li>&gt; No. of medical technologists working under the DGHS per 10,000 population: 0.60</li> </ul>                |
| <ul style="list-style-type: none"> <li>• No. of Domiciliary Staff:</li> </ul>                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>&gt; No. of community and domiciliary health workers working under the DGHS per 10,000 people: 2.26</li> </ul> |
| <ul style="list-style-type: none"> <li>&gt; Health Inspectors (HI) 1,057 (1,410); Assistant Health Inspectors (AHI): 3,753 (4,220); Health Assistants (HA) 15,020 (20,908)</li> </ul>                                                                                     | <ul style="list-style-type: none"> <li>&gt; No. of beds in the DGHS-run public hospitals per 10,000 population: 3.30</li> </ul>                       |
| <hr/> <b>Population-Health Workforce Ratio</b> <hr/>                                                                                                                                                                                                                      |                                                                                                                                                       |
| <b>DGHS 2019</b> <hr/>                                                                                                                                                                                                                                                    |                                                                                                                                                       |
| (According to SVRS 2018 total population, 165.6 million was considered the denominator in applicable cases):                                                                                                                                                              |                                                                                                                                                       |
|                                                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>&gt; No. of beds in private hospitals (registered by DGHS) per 10,000 population: 5.53</li> </ul>              |



# Healthcare Network of the Public Sector in Bangladesh

## Facilities expanded down to rural communities

The healthcare network of the public sector in Bangladesh led by the Ministry of Health and Family Welfare (MOHFW) is now expanded down to rural communities. Healthcare facilities in the private sector and NGOs are also expanding rapidly. The threads in this complicated web comprise implementing agencies, regulatory bodies, and care facilities starting from community clinics to the tertiary-level hospitals. The Ministry is responsible for developing national policies, planning, and decision-making. The directives from the Ministry are implemented by various authorities and healthcare delivery systems. Health education and research continue to produce skilled manpower and innovative healthcare delivery systems. The Ministry and its relevant regulatory bodies control the private-sector and NGO facilities through specific rules and regulations.

A new administrative structure in the MOHFW has been built up by dividing it into two divisions in March 2017. The new divisions are: (I) Health Services Division and (II) Medical Education and Family Welfare Division. The Health Services Division acts in various ways, such as making policy regarding health-related matters, management, and maintenance of nursing care, health financing,

The Directorate General of Health Services (DGHS) is the largest implementing agency under the MOHFW, with more than a hundred thousand officers and staff members...

and so on. The Medical Education and Family Welfare Division acts in making policies regarding medical education, family planning, matters relating to medical colleges and medical universities, registration of birth and deaths, and so on.

The Directorate General of Health Services (DGHS) is the largest implementing agency under the MOHFW, with more than a hundred thousand officers and staff members. The Ministry seeks technical assistance from the DGHS when new programs and interventions are required to be improved.

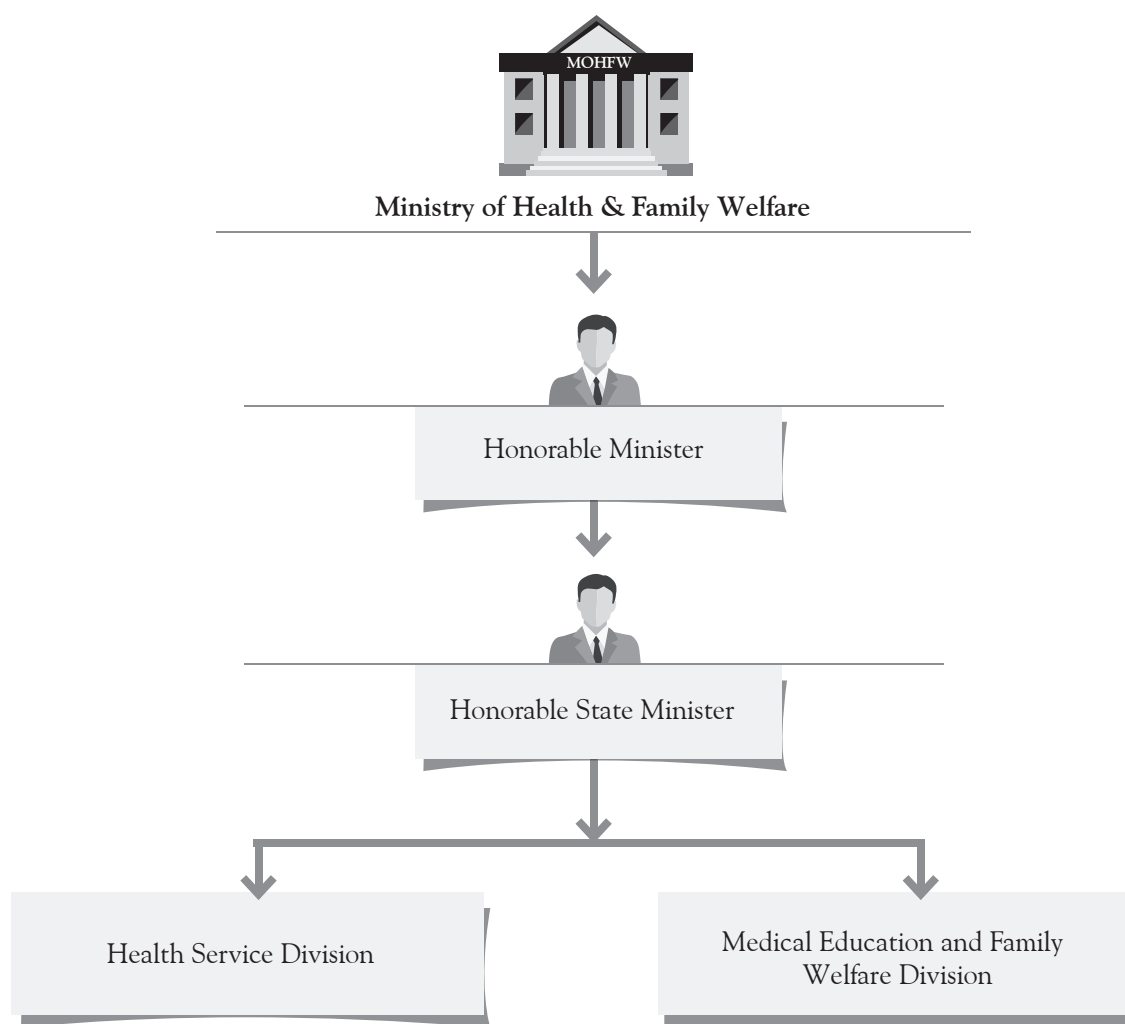
There are six tiers of healthcare infrastructure under the DGHS: national, divisional, district, upazila (sub-district), union, and ward levels. At the national level, there are institutions both for specialized tertiary care to patients and for postgraduate medical education and training.

In the administrative setup of the DGHS, The Director General is assisted by two Additional Director Generals—one for Administration and another for Planning and Development. There are several Directors and Deputy Directors under both of them.

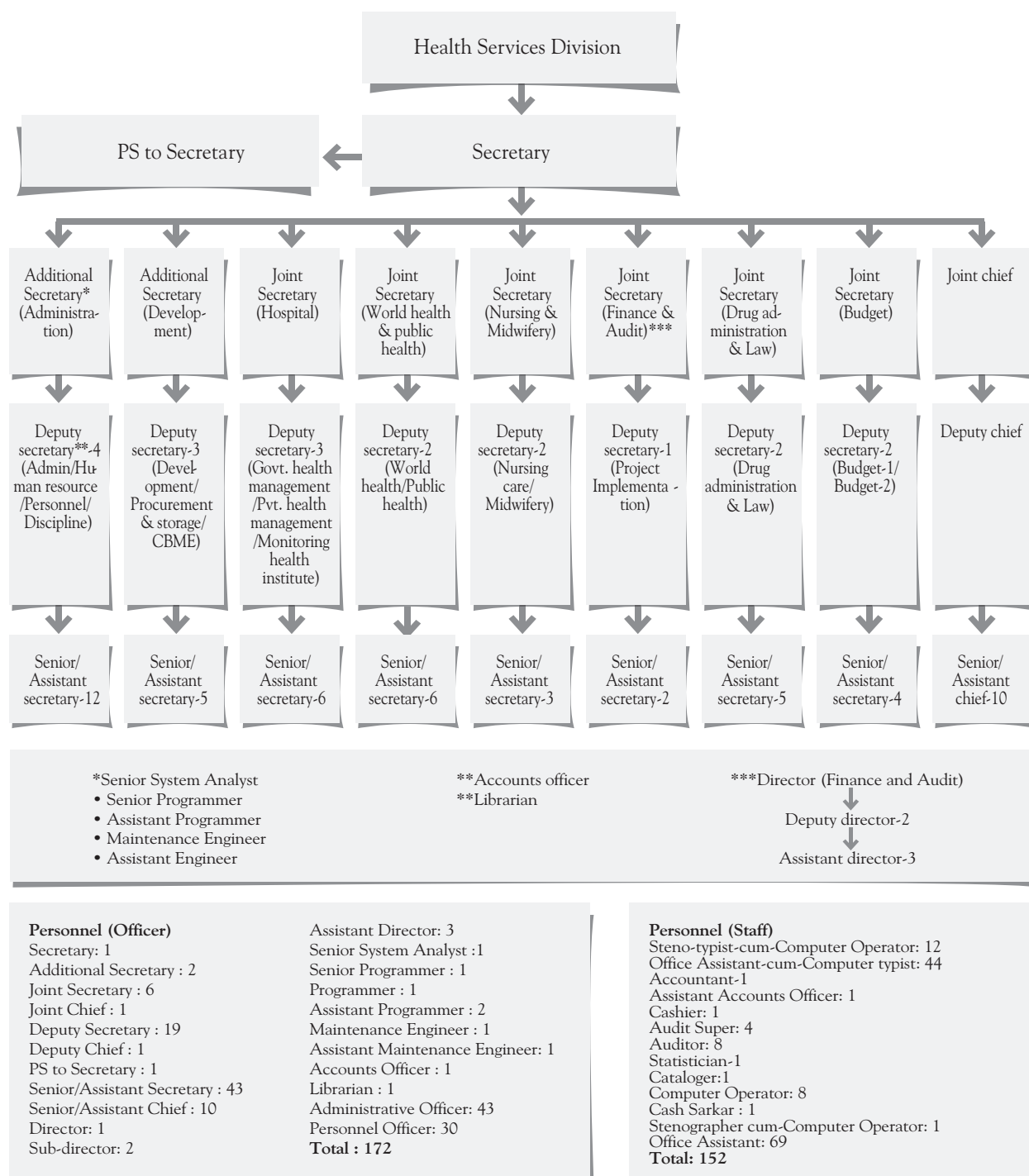
In each administrative division of the country, a Divisional Director monitors the healthcare activities. They are assisted by Deputy Directors and Assistant Directors. There is one infectious disease hospital and one or more medical

college(s) with own hospital at the divisional headquarters.

At the district level, the Civil Surgeon (CS) is the chief healthcare manager responsible for delivering secondary and primary-care services. In each district, there is a district hospital; superintendents look after the hospital management. In some districts, Civil Surgeons look after the district hospitals. Some of the district headquarters have medical colleges with attached hospitals, medical assistant's training schools, and nursing training institutes.



**Figure 2.1.** Organogram of the MOHFW

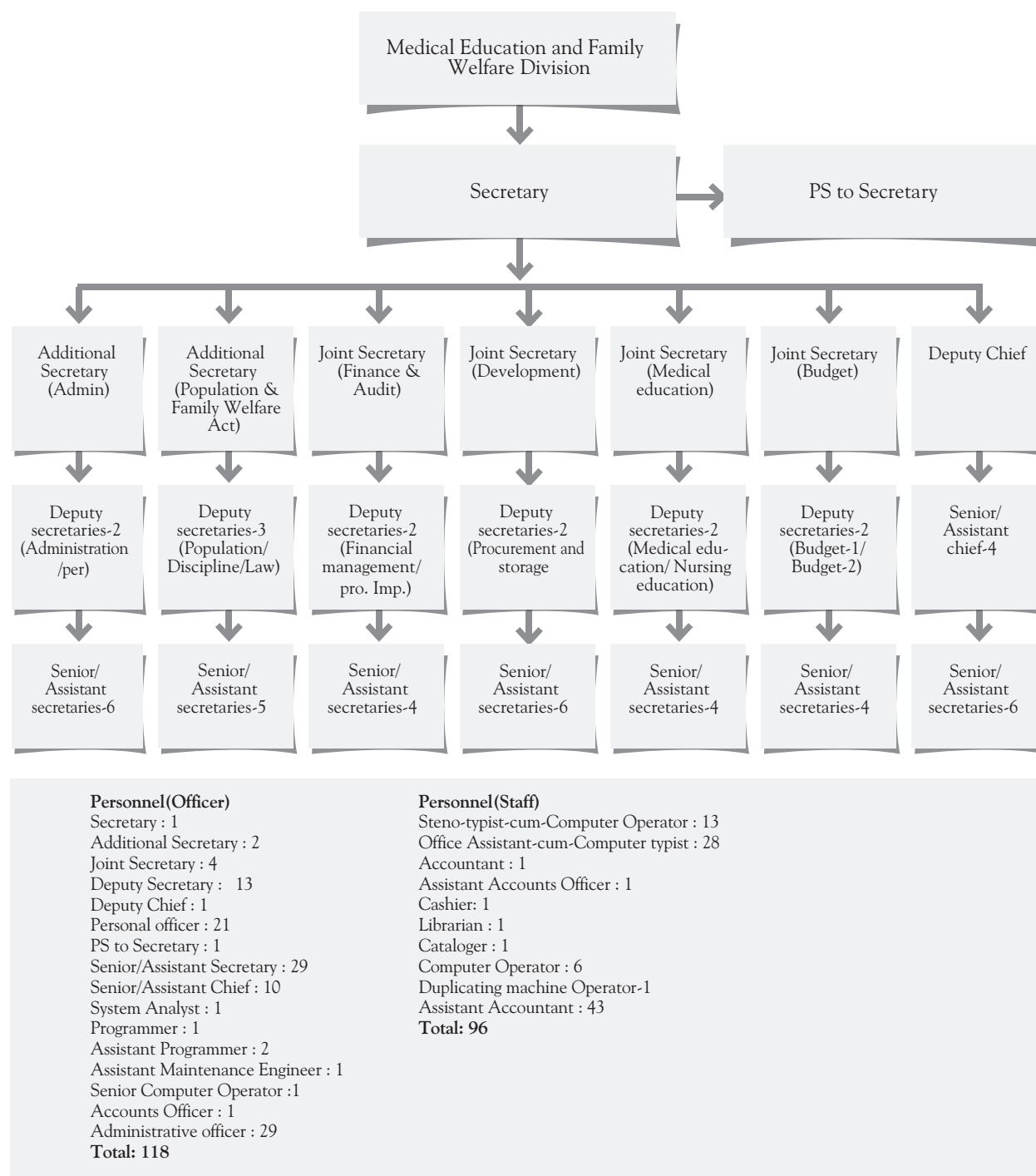


Note: (\*) Under Additional Secretary

(\*\*) Under Deputy Secretaries

(\*\*\*) Under Joint Secretary (Finance and Audit)

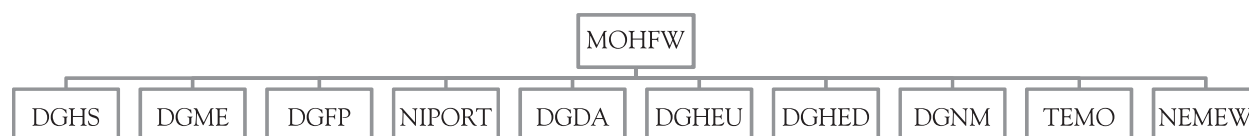
**Figure 2.2.** Hierarchy of personnel in the Ministry of Health and Family Welfare (under Health Services Division) (ref. MOHFW website, last update April 2017)



**Figure 2.3.** Hierarchy of personnel in the Ministry of Health and Family Welfare (under Medical Education and Family Welfare Division) (ref. MOHFW website)

## Implementing Authorities

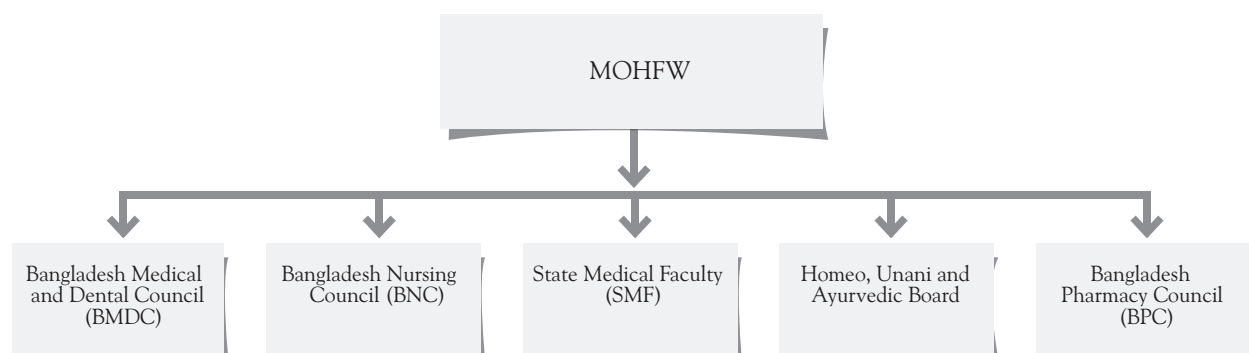
There are ten implementing authorities under the MOHFW as follows:



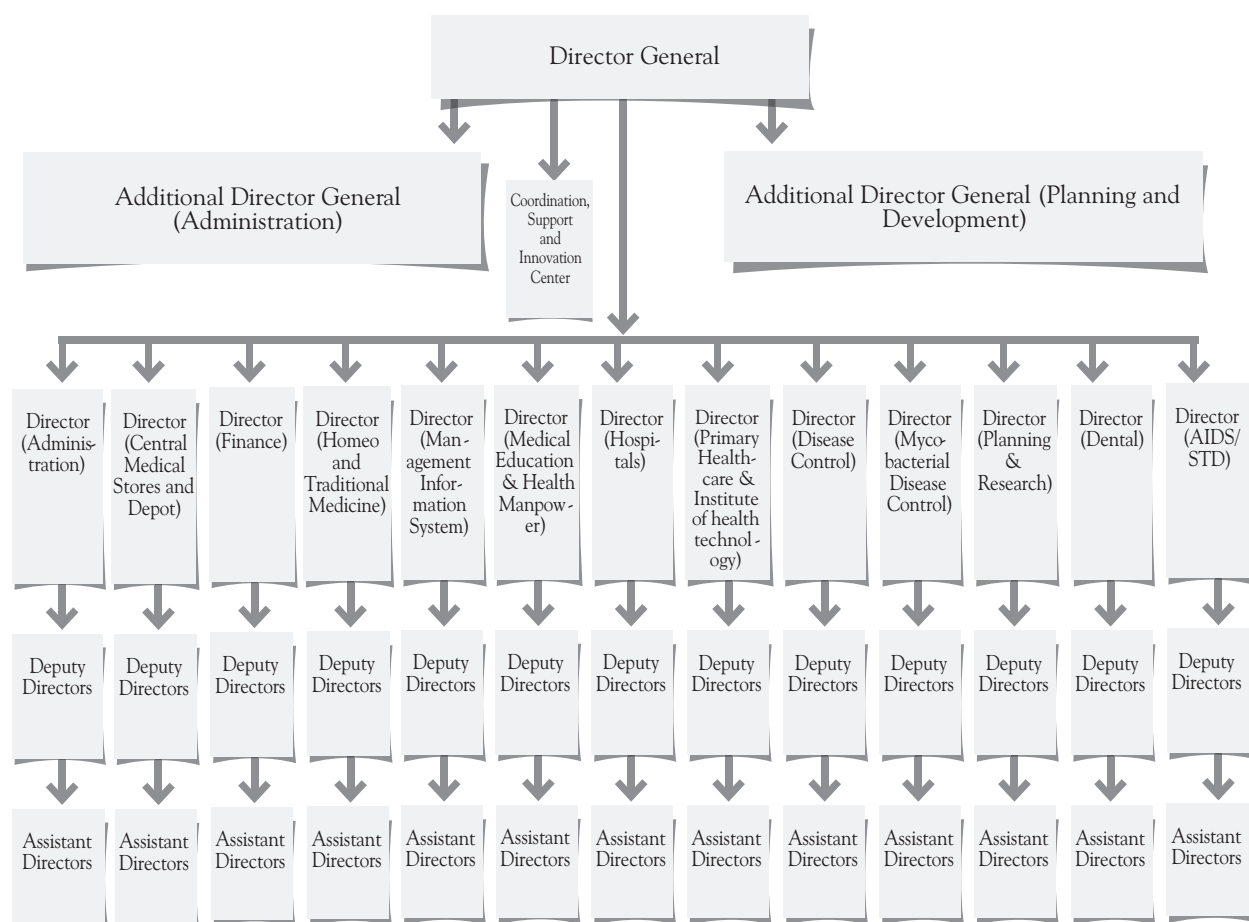
**Figure 2.4.** Implementing authorities under the Ministry of Health and Family Welfare

## Regulatory Bodies

There are five regulatory bodies under the MOHFW as follows:



**Figure 2.5.** Regulatory bodies under the MOHFW



**Figure 2.6.** Hierarchy of personnel in the Directorate General of Health Services (ref. HRIS, DGHS)

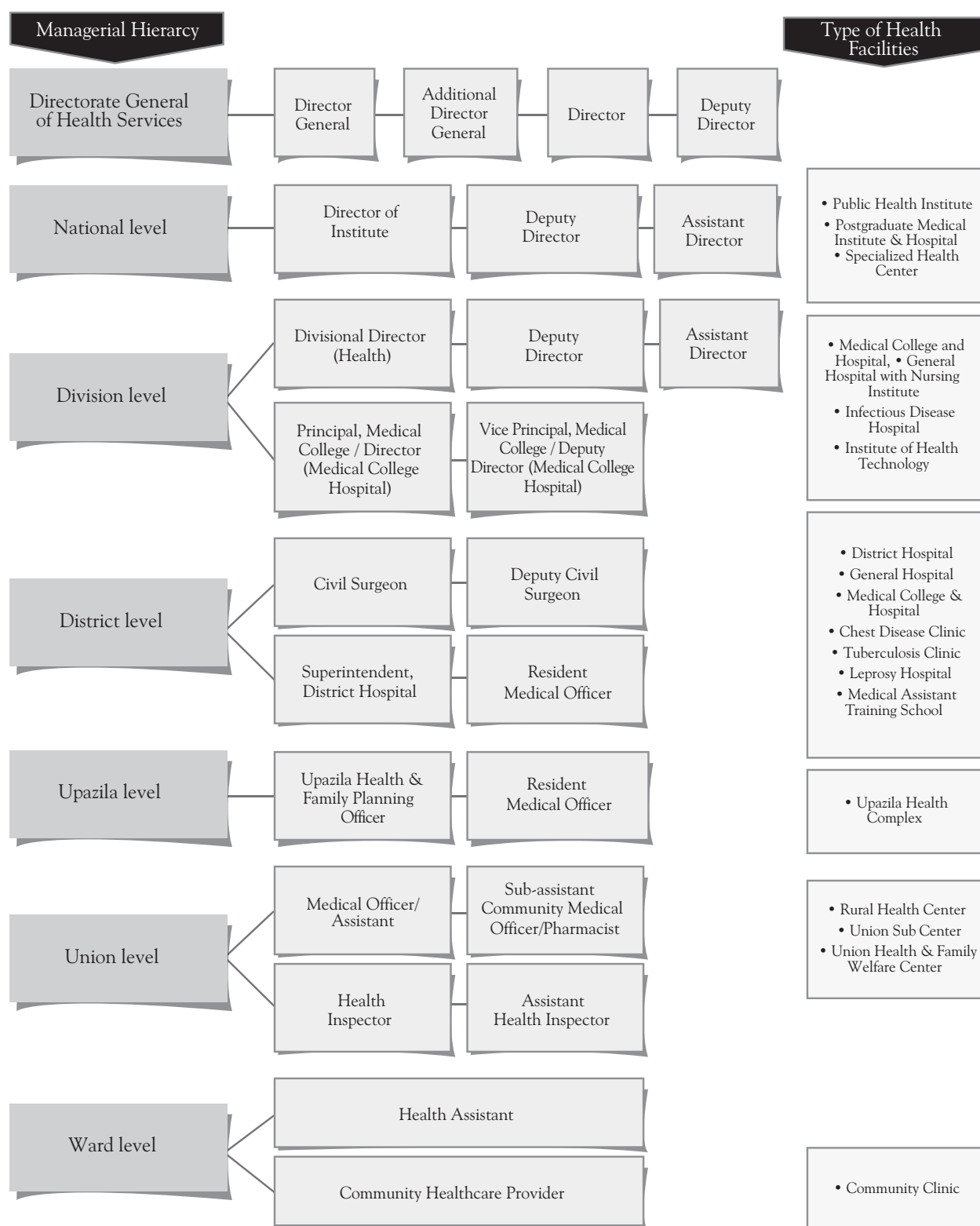


Figure 2.7. Types of facilities from national to the ward level, with managerial hierarchy



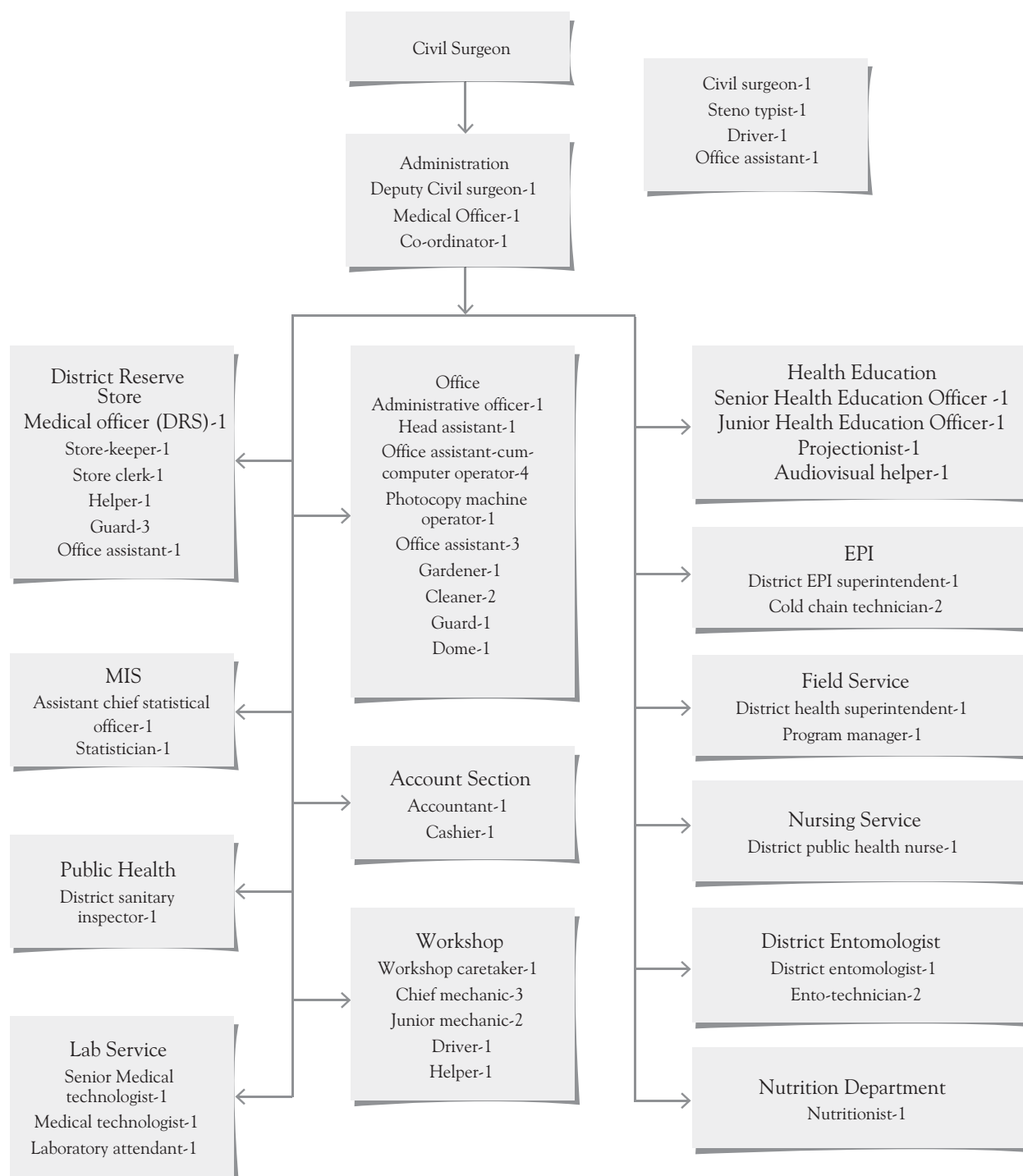


Figure 2.8. Organogram of Civil Surgeon Office (ref. MIS, DGHS)

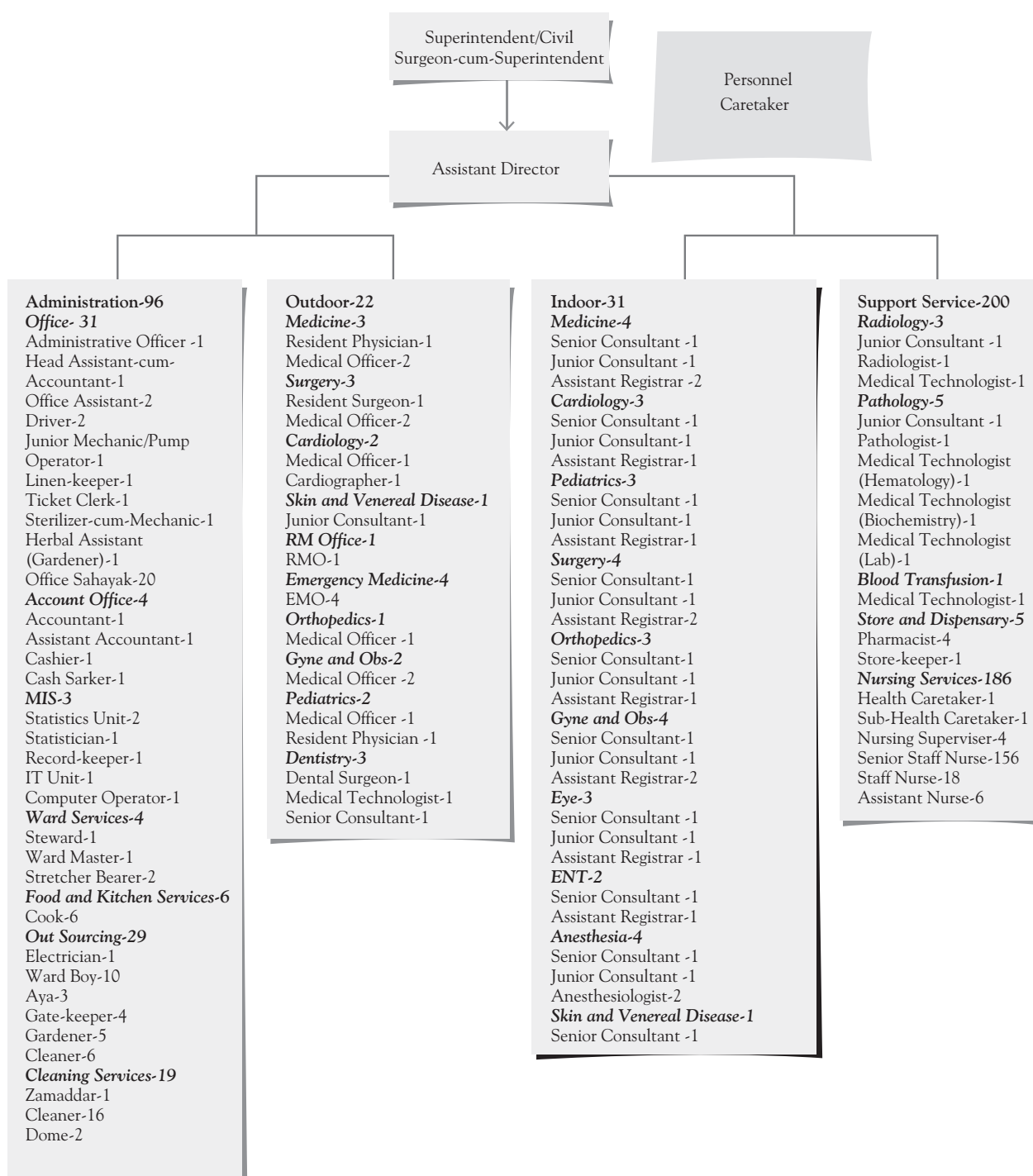


Figure 2.9. Placement of personnel in 250-bed hospitals

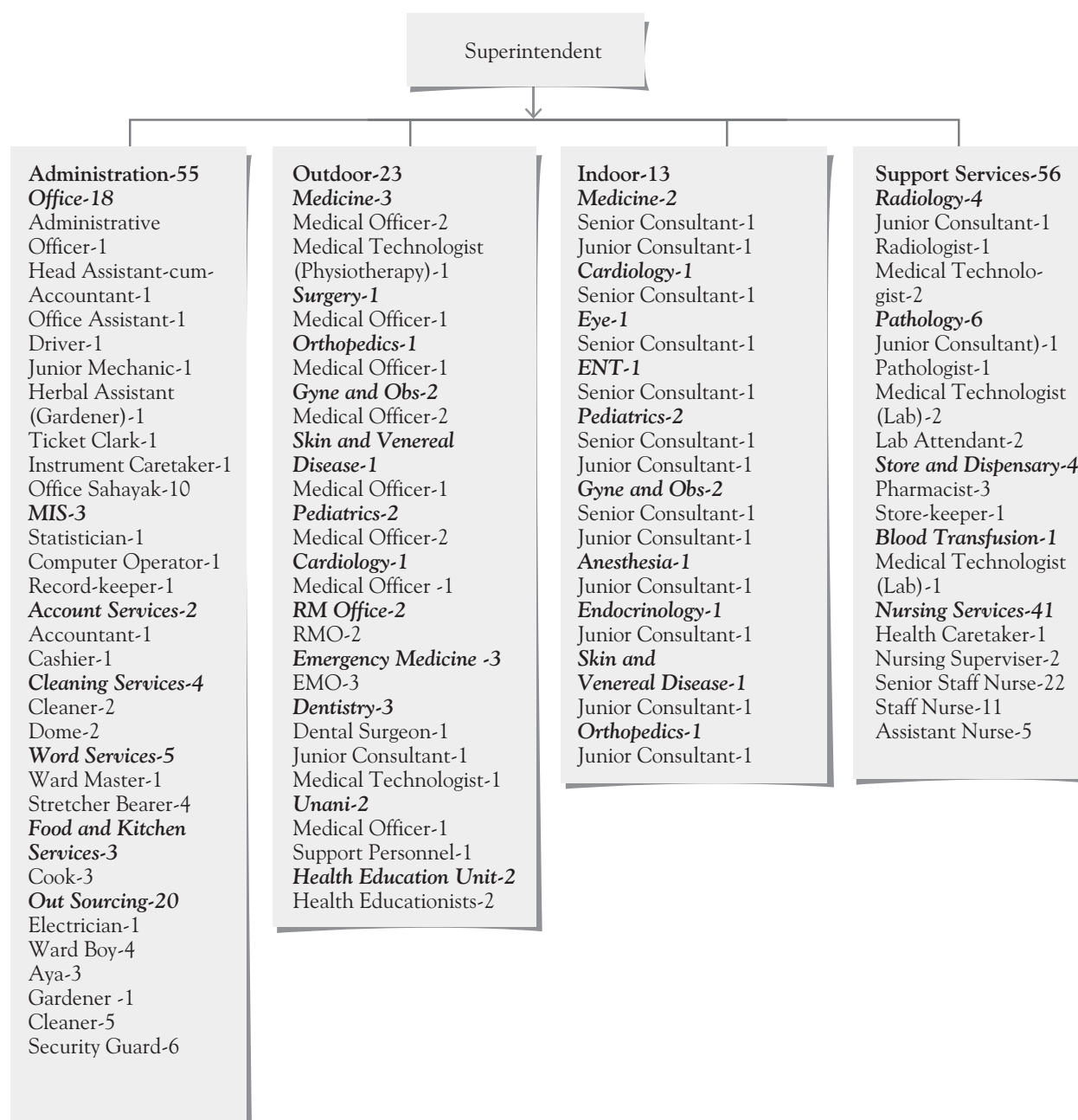


Figure 2.10. Placement of personnel in the 100-bed hospitals (ref. MIS, DGHS)

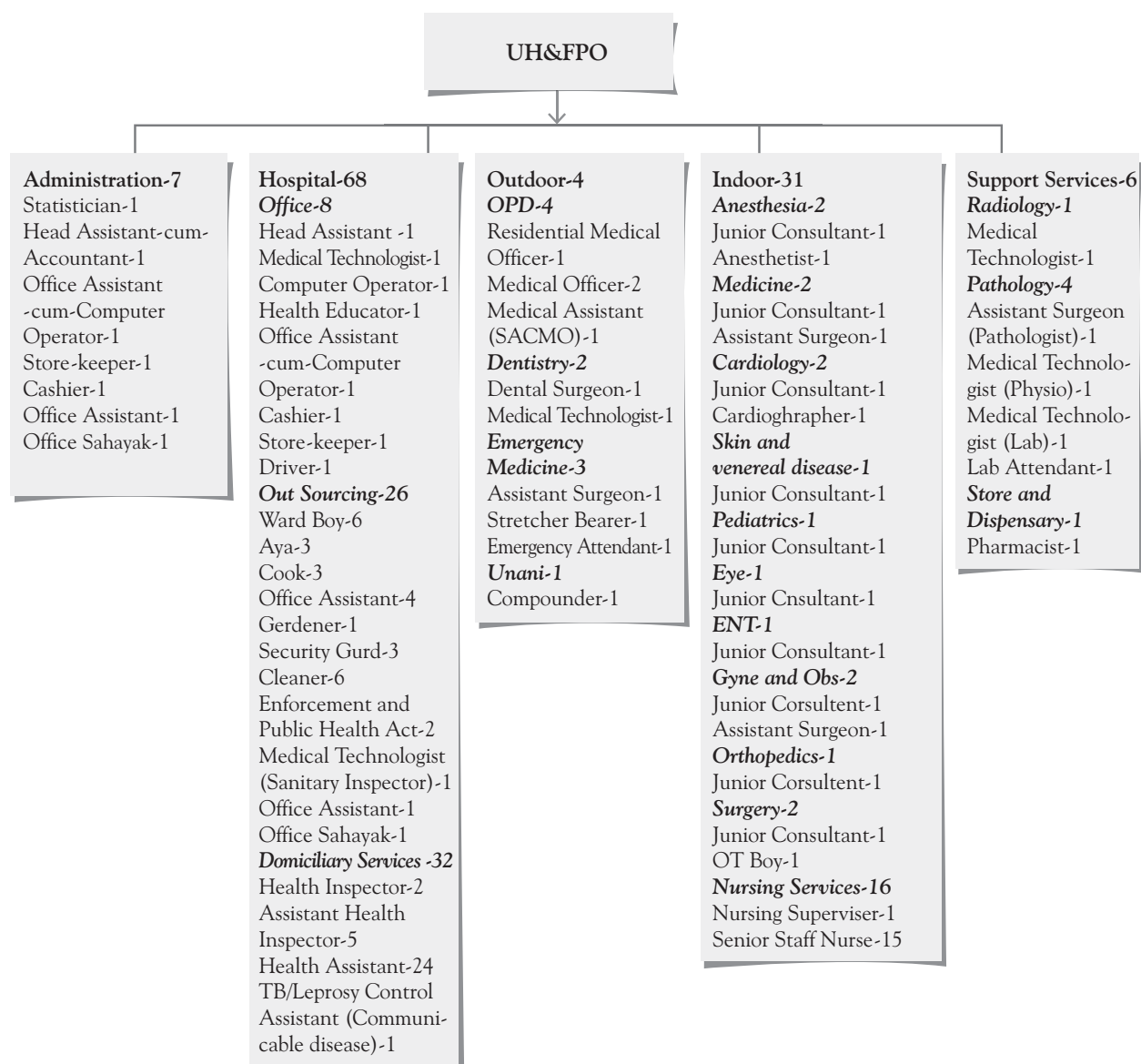
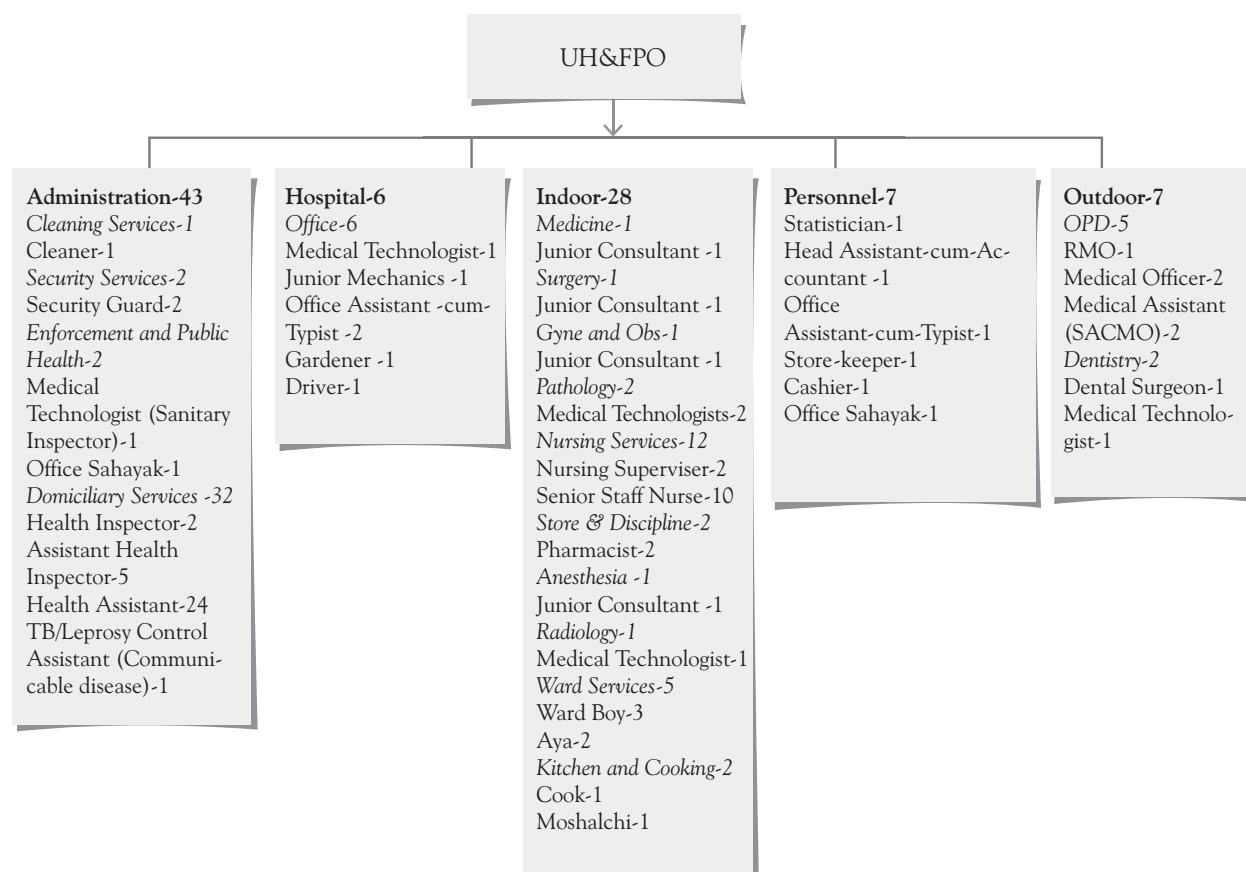
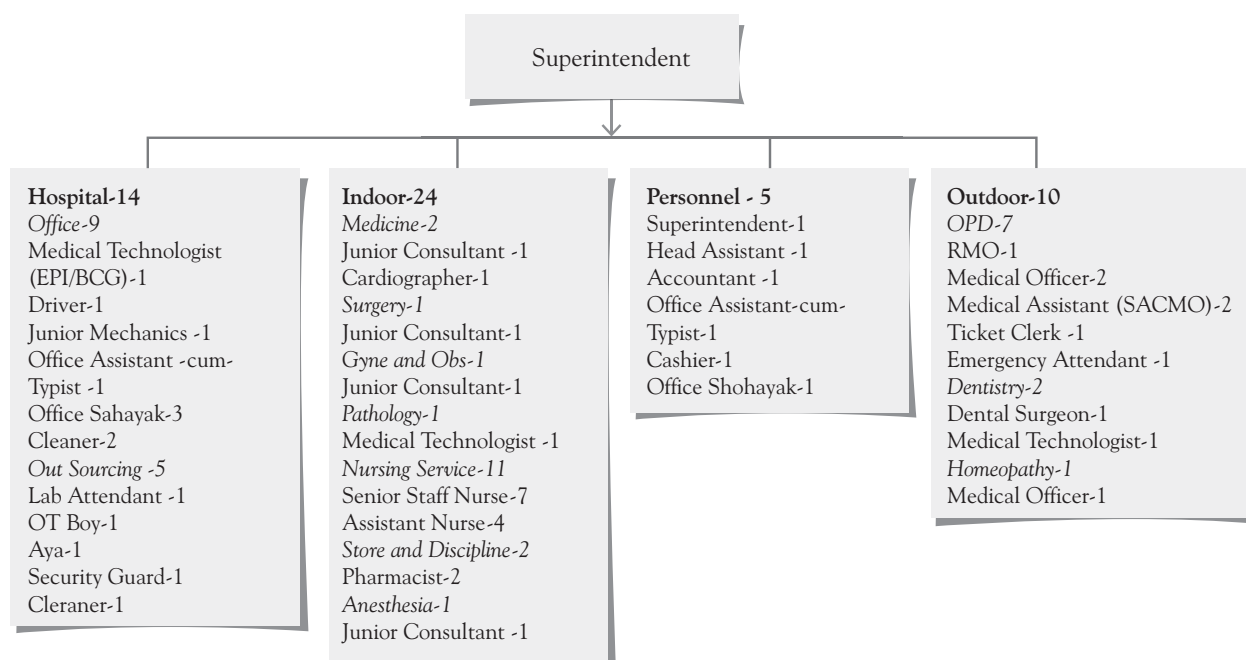


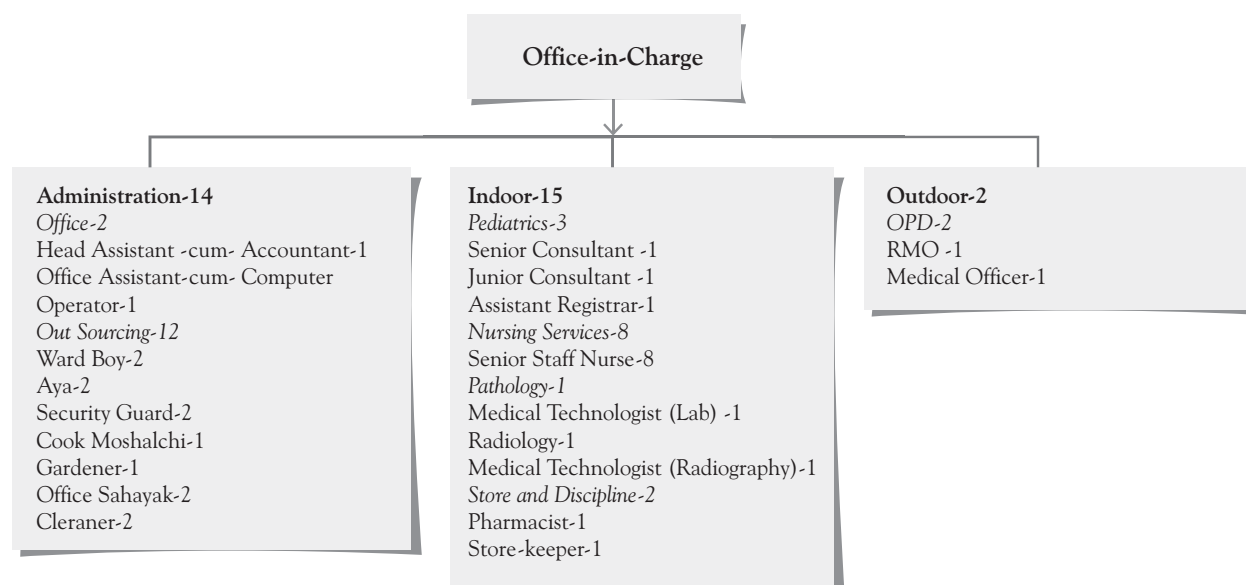
Figure 2.11. Placement of personnel in the 50-bed hospitals (ref. MIS, DGHS)



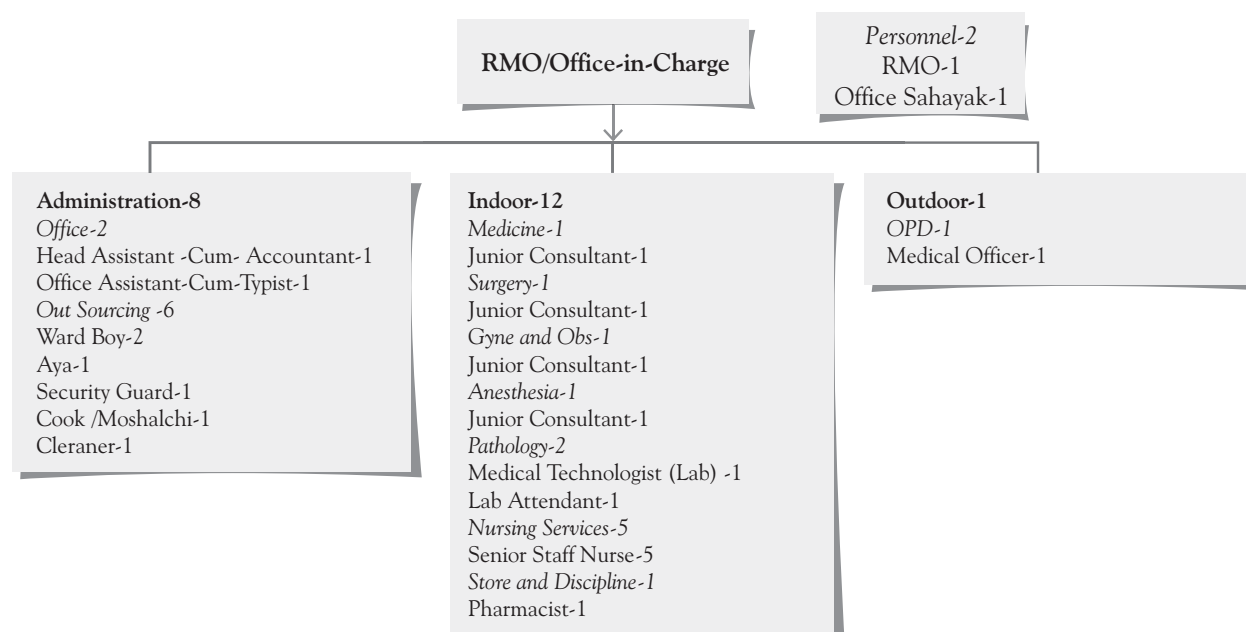
**Figure 2.12.** Placement personnel in the 31-bed hospitals (ref. MIS,DGHS)



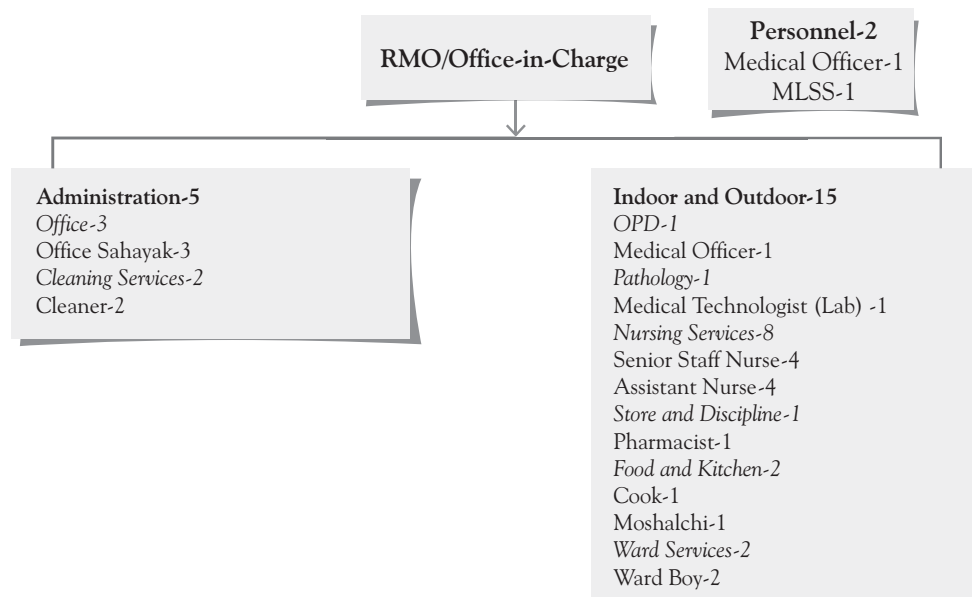
**Figure 2.13.** Placement of personnel in the 30-bed hospitals (ref. MIS, DGHS)



**Figure 2.14.** Placement of personnel in the 25-bed hospitals (ref. MIS, DGHS)



**Figure 2.15.** Placement of personnel in the 20-bed hospitals (ref. MIS, DGHS)



**Figure 2.16.** Placement of personnel in the 10-bed hospitals (ref. MIS, DGHS)

# Health Rights, Acts, and Policies

## Compliant with the Bangladesh Constitution

Since the emergence of Bangladesh as an independent country, utmost priority has been given to ensuring human rights and dignity of the population. The Constitution of the People's Republic of Bangladesh guarantees that "Health is the basic right of every citizen of the Republic. Article 15 of the Constitution says, "It shall be a fundamental responsibility of the state to attain, through planned economic growth, a constant increase of productive forces and a steady improvement in the material and cultural standard of living of the people."

Accordingly, the country's first Five-year Plan (1FYP) was chalked out for the period 1973-1978, which was started by the very first President of the country Bangabandhu Sheikh Mujibur Rahman. In that plan, health was a very important sector. Bangladesh is currently implementing its Seventh Five-year Plan (7FYP) for the period 2016-2020. To ensure its progress, the Government of Bangladesh adopted a number of policies, acts, rules, etc. Recently, a number of acts (relating to communicable diseases, mental health, organ transplantation, community clinic, etc.) have either been passed or amended to strengthen the capacity and quality of the health services of the country. Since the late 1990s, the Government of Bangladesh and its Development Partners (DP) have pursued a sector-wide approach (SWAp) in the health, nutrition and

Recently, a number of acts (relating to communicable diseases, mental health, organ transplantation, community clinic, etc.) have either been passed or amended to strengthen the capacity and quality of the health services of the country...

population (HNP) sector. The Ministry of Health and Family Welfare (MOHFW) has implemented three consecutive sector programs in between July 1998 and December 2016. The MOHFW carefully designed the 4th sector-wide approach (SWAp), namely the 4th Health, Population and Nutrition Sector Program (4th HPNSP) to initiate its journey towards achieving the health-related SDG. The articulation and design of the 4th HPNSP have been linked to the 7th Five-year Plan (FYP) of the Government. Other guiding principles are drawn from the national policies on health, nutrition and population (HNP), various HNP sector-related strategies approved by the Government, and the experiences of implementing three successive sector programs in Bangladesh. The 4th HPNSP started at a time of transition from the Millennium Development Goals (MDGs) to the newly-set Sustainable Development Goals (SDGs) to be achieved by 2030. The MDGs had a major focus on specific goals relating



to health sector while the SDGs encompass much broader agenda for change. SDGs have been taken into consideration in the 4th sector program so that the global development agenda can be illustrated in the national plan.

About 919,000 forcibly displaced people have come to Bangladesh from Myanmar since August 2017. They are placing an immense strain on an already resource-constrained social service delivery system, including health services. Thus, a fourth component, besides the original three components, has been added for extra financing to enhance the capacity of the MOHFW to respond to the crisis. This supports in extending HNP services to the Forcibly Displaced Myanmar Nationals (FDMNs) in the Cox's Bazar district.

## Health in the Constitution of Bangladesh

Bangladesh began its journey as an independent nation in 1971, and the first Constitution was established in 1972. Since the first Constitution, health has historically been a state issue. Under the current Constitution, medical care is included as one of the basic necessities of the citizenry, and it is the fundamental responsibility of the state to ensure health services.

## Important Health Sector Policies/ Strategies

- ◆ National Health Policy 2011
- ◆ Health, Nutrition and Population Sector of the 7th Five-year Plan
- ◆ National Nutrition Policy 2015
- ◆ 4th Health, Population and Nutrition Sector Program (HPNSP) 2017-2022

- ◆ Healthcare Financing Strategy 2012-2032: Expanding Social Protection for Health towards Universal Coverage
- ◆ Bangladesh Health Workforce Strategy 2016–2021.

## National Health Policy 2011

### Goal and objectives

To overcome the existing challenges and to create a regulatory framework, the National Health Policy 2011 was implemented. It aims at the following principles:

First: Make necessary basic medical utilities reach people of all upazilas and develop the health and nutrition status of the people

Second: Develop a proper system to ensure easy and sustained availability of health services

Third: Ensure optimum quality, acceptance, and availability of primary healthcare and governmental medical services

Fourth: Reduce the intensity of malnutrition among people and implement effective and integrated programs for improving nutrition status

Fifth: Undertake programs for reducing the rates of childhood and maternal mortality

Sixth: Adopt satisfactory measures for ensuring improved maternal and child health and install facilities for safe and hygienic child delivery

Seventh: Improve overall reproductive health resources and services

Eighth: Ensure the presence of full-time doctors, nurses, and other officers/staff,

**Table 3.1.** Health-related statements written in the Constitution of the People's Republic of Bangladesh

|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Provision of basic necessities                | <p>15. It shall be a fundamental responsibility of the State to attain, through planned economic growth, a constant increase of productive forces and a steady improvement in the material and cultural standard of living of the people, with a view to securing its citizens –</p> <p>(a) the provision of the basic necessities of life, including food, clothing shelter, education and medical care;</p> <p>(b) the right to work, that is the right to guaranteed employment at a reasonable wage having regard to the quantity and quality of work;</p> <p>(c) the right to reasonable rest, recreation and leisure; and the right to social security, that is to say, to public assistance in cases of undeserved want arising from unemployment, illness or disablement, or suffered by widows or orphans or in old age, or in other such cases.</p> |
| Rural development and agricultural revolution | <p>16. The State shall adopt effective measures to bring about a radical transformation in the rural areas through the promotion of an agricultural revolution, the provision of rural electrification, the development of cottage and other industries, and the improvement of education, communications and public health, in those areas, so as progressively to remove the disparity in the standards of living between the urban and the rural areas.</p>                                                                                                                                                                                                                                                                                                                                                                                                |
| Public health and morality                    | <p>18. (1) The State shall regard the raising of the level of nutrition and the improvement of public health as among its primary duties, and in particular shall adopt effective measures to prevent the consumption, except for medical purposes or for such other purposes as may be prescribed by law, of alcoholic and other intoxicating drinks and of drugs which are injurious to health.</p> <p>(2) The State shall adopt effective measures to prevent prostitution and gambling.</p>                                                                                                                                                                                                                                                                                                                                                               |

provide and maintain necessary equipment and supplies at each of the upazila health complexes and union health and family welfare centers (UHFWCs)

Ninth: Devise necessary ways and means to make optimum usage of available opportunities in government hospitals and the health service system

Tenth: Formulate specific policies for medical colleges and private clinics and to introduce laws and regulations

Eleventh: Strengthen and expedite the family planning program with the objective of attaining the target of replacement-level of fertility

Twelfth: Explore ways to make the family planning program more acceptable, easily available, and effective among the extremely poor and low-income communities

Thirteenth: Arrange special health services for the mentally-retarded, the physically-disabled and elderly population

Fourteenth: Determine ways to make family planning and health management more accountable and cost-effective by equipping with more skilled manpower

Fifteenth: Introduce systems for treatment of all types of complicated diseases in the country and minimize the need for foreign travel for medical treatment abroad.

## 7<sup>th</sup> Five-year Plan: 2016–2020

### Goals, strategies, and targets of the 7th Five-year Plan (FYP) in the health, nutrition and population sector

Bangladesh will strive to attain a number of targets on the road towards Universal Health Coverage, consistent with the proposed major targets under the proposed Sustainable Development Goals (SDG) of the UN, promoting and sustaining health and nutrition, along with containing population growth as priorities in the human development strategy. Vision 2021 aims at establishing a middle-income Bangladesh, with drastic reduction in poverty and developing conditions that allow individuals to reach and maintain the highest attainable level of health. To realize that vision, the Government has set certain objectives and targets towards achieving Universal Health Coverage (UHC). This includes the following:

- Ensure access to and utilization of HNP services for every citizen of the country, with particular emphasis on the elderly, women, children, poor disadvantaged people, and those living in difficult areas
- Reduce total fertility rate
- Ensure adolescent and reproductive healthcare
- Strengthen community support and involvement to obtain better results in the implementation of programs
- Improve nutritional status of children and women
- Take effective measures to promote alternative medicines and improve the quality of care
- Meet challenges of emerging, re-emerging, and non-communicable diseases, health hazards due to climate change, and emergency response to catastrophes
- Enhance national capacity for the pre-service education (SBA/nursing, paramedics, midwifery), provide in-service training and better management of human resources
- Improve the quality of hospitals and maternity services and to make these accessible, especially to the women, children, and the poor.

Bangladesh has already made great strides towards achieving these objectives. Past and ongoing programs have been promoted, and they are still being promoted for ensuring good health and nutrition. It also concentrates on reducing population growth.

To prosecute this progress and to address areas of insufficiency, some of the specific targets for the HNP sector under the 7th FYP are shown in Table 3.2.

These targets can be viewed as ambitious; however, these are certainly achievable with timely actions. Bangladesh is now trying to implement these associated policies and institutional reforms. Bangladesh is currently implementing the 4th SWAP in the HNP sector which aligns with the 7th FYP targets for achieving UHC and SDGs by 2030.

#### **4<sup>th</sup> Health, Population and Nutrition Sector Program 2017-2022**

The MOHFW is currently implementing its fourth sector program titled the ‘4th Health, Population and Nutrition Sector Program (4th HPNSP)’ covering a 5.5-year period between January 2017 and June 2022, at an estimated cost of US\$ 14.7 billion.

The 4<sup>th</sup> HPNSP is guided by Bangladesh’s Vision 2021 (transforming the country from a developing country into a middle-income country), which acknowledges that improved health is a necessary and critical condition for the achievement of this vision.

#### **Milestones for Vision 2021 in HNP Sector**

- 85% of the population has standard nutritional food.
- Poor people ensure a minimum of 2100 kilo calories of food. All kinds of contagious diseases are eliminated. Longevity increases to 72 years.
- Infant mortality comes down to 15 from 54 per thousand at present.
- Use of birth control methods increases to 80%.

#### **4<sup>th</sup> HPNSP: Strategy and Objectives**

##### ***Main objectives***

The 4th HPNSP is built upon the existing achievements to improve equity, quality, and efficiency with a view to moving gradually towards UHC and achieving health-related Sustainable Development Goal (SDG) by pursuing the following strategic objectives (SOs) during the next five-and-a half years, starting from January 2017:

**SO 1:** Strengthen governance and stewardship of the public and private health sectors

**SO 2:** Undertake institutional development for improved performance at all levels of the system

**SO 3:** Provide sustainable financing for equitable access to healthcare for the population and accelerated progress towards Universal Health Coverage

**SO 4:** Strengthen the capacity of the MOHFW’s core health systems (financial management, procurement, infrastructure development)

**SO 5:** Establish a high-quality health workforce available to all public and private health service providers

**SO 6:** Improve health measurement and accountability mechanisms and build a robust evidence-base for decision-making

**SO 7:** Improve equitable access to and utilization of quality health, nutrition and family planning services

**SO 8:** Promote healthy lifestyle choices and a healthy environment

**Table 3.2.** Targets of the 7th FYP in Health, Population and Nutrition Sector Program

| Sl. No.                                       | Indicator                                                                                                   | Base year's information<br>(source with year) | 7th FYP target<br>(FY 2020) |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------|
| Impact/Outcome                                |                                                                                                             |                                               |                             |
| 1.                                            | Life-expectancy at birth                                                                                    | 72.22 years (World Bank 2015)                 | 72                          |
| 2.                                            | Total fertility rate (children per woman)                                                                   | 2.13 (World Bank 2015)                        | 2.0                         |
| 3.                                            | Under-5 mortality rate (per 1,000 livebirths)                                                               | 46 (BDHS 2014)                                | 37                          |
| 4.                                            | Infant mortality rate (per 1,000 livebirths)                                                                | 38 (BDHS 2014)                                | 20                          |
| 5.                                            | Maternal mortality ratio (per 100,000 livebirths)                                                           | 176 (World Bank 2015)                         | 105                         |
| 6.                                            | Proportion of underweight among under-five children (%)                                                     | 32.6 (BDHS 2014)                              | 20                          |
| 7.                                            | Proportion of stunting among under-five children (%)                                                        | 36.1 9 (BDHS 2014)                            | 25                          |
| Output                                        |                                                                                                             |                                               |                             |
| 8.                                            | Proportion of births attended medically-trained care providers (%)                                          | 42.1 (BDHS 2014)                              | 65                          |
| 9.                                            | Contraceptive prevalence rate (%)                                                                           | 62.4 (BDHS 2014)                              | 75                          |
| 10.                                           | Proportion of children fully-vaccinated by 12 months (%)                                                    | 78 (BDHS 2014)                                | 75                          |
| 11.                                           | Proportion of births in health facilities by wealth quintile (ratio of the lowest and the highest quintile) | 15: 69.5 (BDHS 2014)                          | 1: 3.5                      |
| 12.                                           | TB case detection rate (%)                                                                                  | 53 (GTR 2014)                                 | 75                          |
| Source: Ministry of Health and Family Welfare |                                                                                                             |                                               |                             |

### Vision, Mission, and Goal of the Program

The PIP is based on the vision, mission, goal, development objectives, strategic objectives, and resource envelope as mentioned in the Strategic Investment Plan (SIP) of MOHFW developed in mid-2016.

### Vision 2021

To see the people healthier, happier and economically productive to make Bangladesh a middle-income country by 2021

### Mission

To create conditions whereby the people of Bangladesh have the opportunity to reach and maintain the highest attainable level of health

### Components

The eight strategic objectives of the 4th HPNSP have been categorized under three main components. An additional (fourth) component is added with additional financing to provide health services delivery to the

FDMNs and Host Communities in Cox's Bazar District under Chattogram division. The 4th component will also be implemented by the MOHFW. The MOHFW will continue to deliver humanitarian aid directly and through the agencies of the United Nations (UN) and local/international NGOs for supply of food, HNP services, sanitation, water and other essential services to these FDMNs.

The components of 4th HPNSP are:

1. Governance and Stewardship
2. HNP System's Strengthening
3. Provision of Quality HNP Services
4. Development of HNP Services for FDMNs in Cox's Bazar district.

Like the previous three sector programs, it is expected that a significant proportion of development partners' support will be channeled through on-budget financing. Since October 2017, the World Bank is implementing its Health Sector Support Project (HSSP), having three components through which the Bank is providing support to the 4th HPNSP. The original HSSP is consistent with the government program and policies and will play an important role in advancing key result areas with the use of Disbursement-linked Indicators (DLIs). The original HSSP will use a set of 16 DLIs in responding to the key HNP challenges. Results Framework (RFW) goal-level indicators for the 4th HPNSP are showed in the Annex to Chapter 3.

In addition, there are process/output indicators at the OP level, covering all the OPs for monitoring OP-wise progress of work on a six-monthly/annual basis. A number of tools will also be used for documenting and

tracking outputs and indicators, for regular monitoring of the program, as illustrated in Annex to Chapter 3.

For implementation of the 4th HPNSP, there are now 29 OPs that are distributed across the DGHS (14), DGFP (7), MOHFW (5), and other agencies (one for each of the DGDA, DGNM, and NIPORT), each of which reflects priority areas of the 4th HPNSP. Each OP is led by a Line Director (LD) and has a functional administrative structure, including staff, budget, and infrastructure.

## Sustainable Development Goals

### *Health and Nutrition Is Essential for the Success of All SDGs*

The SDGs have taken into consideration in the 4th sector plan so that the global development agenda can be illustrated in the national plan. The goal of the 4th sector plan is to achieve sustainable improvement in health, nutrition, and reproductive health, including family planning for all, particularly of the vulnerable groups –women, children, the elderly and the poor. Conditions will be created in the HNP sector whereby the people of Bangladesh have the opportunity to reach and maintain the highest attainable level of health.

Overall, the SDGs comprise 17 goals, 169 targets, including 230 indicators.

### **Goal 3 –Ensure healthy lives and promote well-being for all at all ages**

Number of targets in Goal 3=13

Number of SDG health indicators: 25

Goal 3 of the SDGs has been particularly focused on health,, with the statement “to



ensure healthy lives and promote well-being for all at all ages.” However, health sector has the responsibility and stake in all the other 16 goals to apply the principles of SDGs in its own settings as well as to complement achievement of the related goals by other sectors.

To achieve targets of the health-related goals, the DGHS has taken many initiatives to achieve the following:

For strengthening health information system by deployment of DHIS2 software, the entire health sector has been digitalized. The DGHS introduced telemedicine service operates a nationwide call center ‘Shasthya Batayon’ for providing health services. The DGHS also holds routine weekly video-conferences to review health system performance and progress of health-related SDG.

To improve maternal health, many government hospitals provide emergency obstetric care services round the clock, electronically register and track pregnant women to ensure effective maternal health services. Provision of antenatal care (ANC)

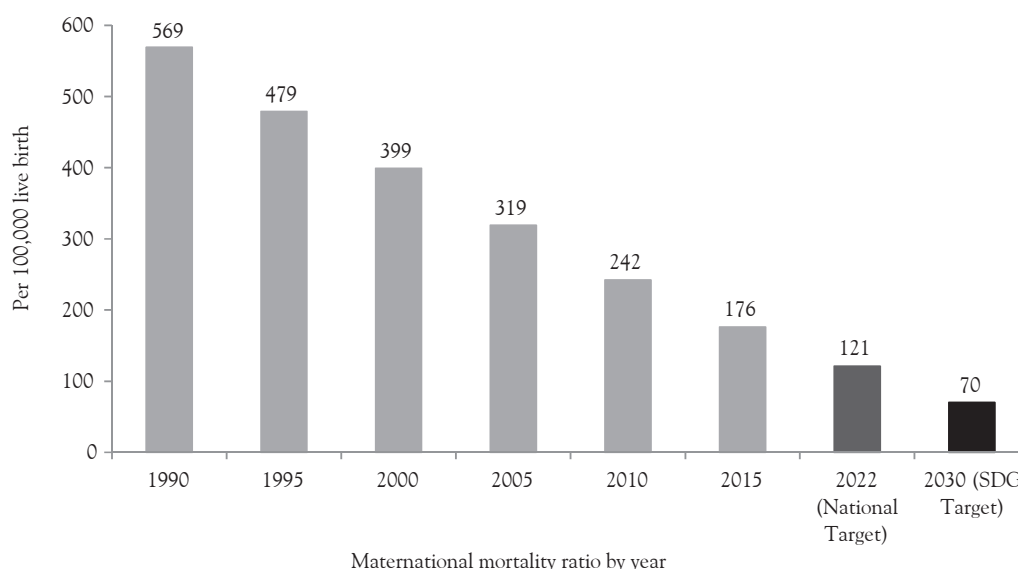
services for all pregnant women, TT vaccination, training of community skilled birth attendant (CSBAs) are ensured.

Comprehensive Emergency Obstetric and Newborn Care (CEmONC) has been expanded in more upazila health complexes. Piloting of demand-side financing through providing maternal vouchers is in place. Demand for utilization of ANC, post-natal care (PNC) and institutional deliveries is being created.

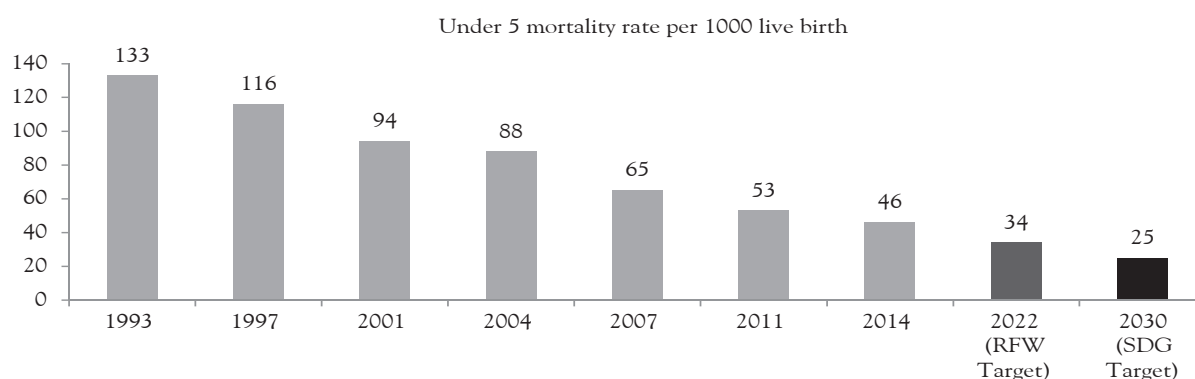
The MOHFW strengthened national nutrition and immunization program to enhance child vaccination coverage. The 4th HPNSP strongly addresses the NCD burden, prevention program through raising awareness and mass mobilization, screening, diagnosis, and treatment.

### Maternal and child mortality (SDG 3.1, 3.2)

Figure 3.1 and 3.2 show the maternal mortality and child mortality status respectively. Figure 3.1 shows the trend of reduction in maternal



**Figure 3.1.** Maternal mortality status



**Figure 3.2.** Under-5 child mortality status (Source: BDHS)

mortality with 5 years interval since 1995, and Figure 3.2 shows the trend of reduction in child mortality from 1993 to 2014 with irregular intervals. Both figures present data against the targets to be achieved by 2022 and 2030.

### Communicable Diseases (SDG 3.3)

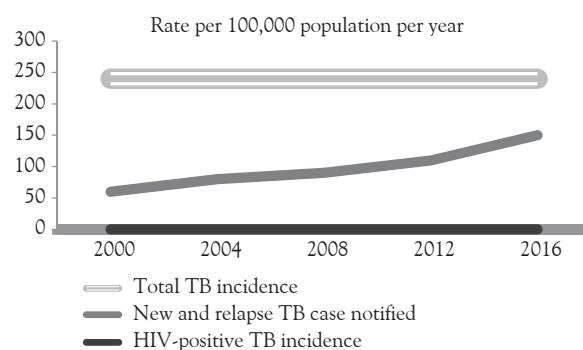
Figure 3.3. shows the status of malaria prevalence in Bangladesh (source: World Malaria Report 2018). Table 3.4 presents data on the rate of hospitalization and deaths due to TB.

## Goal 2 End hunger, achieve food security and improved nutrition, and promote sustainable agriculture

**2.2** By 2030, [Goal 2 of SDG aims to] end all forms of malnutrition, including achieving, by 2025, the internationally-agreed targets on stunting and wasting in children below 5 years of age, and address the nutritional needs of adolescent girls, pregnant and lactating women, and older persons.

The SDGs seek to achieve what the MDGs did not achieve. On 25 September 2015,

the UN General Assembly adopted the 2030 Agenda for Sustainable Development to redirect humanity towards a sustainable path that was developed following the United Nations Conference on Sustainable Development in Rio de Janeiro, Brazil, in June 2012. A key feature of the 2030 Agenda for Sustainable Development is its universality and indivisibility. It addresses all countries from the Global South and the Global North as target countries. All countries subscribing to the 2030 Agenda are to align their own development efforts with the aim of promoting prosperity while protecting the planet in order to achieve sustainable development.



**Figure 3.3.** TB prevalence and status (source WHO)



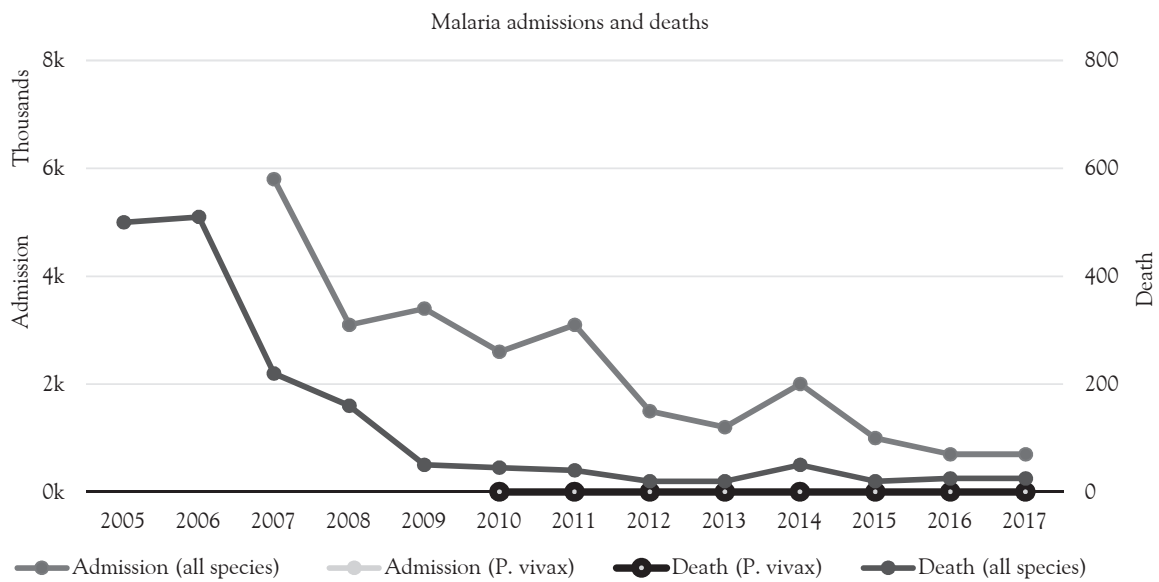


Figure 3.4. Hospitalization and deaths due to TB



Figure 3.5. Seventeen Sustainable Development Goals to be achieved by 2030

The DGHS is prepared for monitoring health indicators to achieve the SDG targets.

According to General Economic Division (GED) mapping, the DGHS is contributing in measuring the progress of health-related SDG, reporting and accountability of health

The DGHS is prepared for monitoring health indicators to achieve the SDG targets...

for country SDG progress, along with monitoring and accountability of also global

**Table 3.3. SDG 2 (Nutrition) and SDG 3 (Health) indicators (where data are available) according to WHO thematic areas**

| Goal 2. End hunger, achieve food security and improved nutrition, and promote sustainable agriculture                                                                                                         |                     |                                          |                                                                       |                     |                                            |                         |                      |                                           |                 |                     |                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------|-----------------------------------------------------------------------|---------------------|--------------------------------------------|-------------------------|----------------------|-------------------------------------------|-----------------|---------------------|---------------------|
| SDG Indicator                                                                                                                                                                                                 | Tier Classification | Data source                              |                                                                       | Reporting frequency | Baseline data                              |                         |                      | Current status (year and source)          |                 | Target 2025         | Target 2030         |
|                                                                                                                                                                                                               |                     | Institution (Ministry/ Division/ Agency) | Activity (Survey/ Census/ Administrative/ MIS/ AI/Big Bland Data/IoT) |                     | Data                                       | Year                    | Source               | Data                                      | Source and year |                     |                     |
| 1                                                                                                                                                                                                             | 2                   | 3                                        | 4                                                                     | 5                   | 6                                          | 7                       | 8                    | 9                                         | 10              | 12                  | 13                  |
| 2.2.1 Prevalence of stunting [(height-for-age <-2 standard deviation from the median of the World Health Organization (WHO) Child Growth Standards)] among children under 5 years of age                      | Tier I              | BBS                                      | MICS                                                                  | Triennial           | 36.1%                                      | 2014                    | BDHS                 | 28%                                       | MICS 2019 BBS   | 16%                 | 12%                 |
| 2.2.2 Prevalence of malnutrition [(weight-for-height > +2 or <-2 standard deviation from the median of the WHO Child Growth Standards)] among children under 5 years of age, by type (wasting and overweight) | Tier I              | BBS                                      | MICS                                                                  | Triennial           | (a) Wasting: 14.3%<br>(b) Overweight: 1.6% | (a) 2014<br>(b) 2012-13 | (a) BDHS<br>(b) MICS | (a) Wasting: 9.8%<br>(b) Overweight: 2.4% | MICS 2019 BBS   | (a) 10%<br>(b) 1.0% | (a) <5%<br>(b) 1.0% |

**Table 3.3 Contd.**

Table Continued...

**Goal 3. Ensure healthy lives and promote well-being for all at all ages**

| SDG Indicator                                                                                        | Tier Classification | Data source                                              |                                                                  | Reporting frequency | Baseline data                                           |      |        | Current status (Year and source)           |                 | Target 2025 | Target 2030 |
|------------------------------------------------------------------------------------------------------|---------------------|----------------------------------------------------------|------------------------------------------------------------------|---------------------|---------------------------------------------------------|------|--------|--------------------------------------------|-----------------|-------------|-------------|
|                                                                                                      |                     | Institution (Ministry/ Division/ Agency)                 | Activity (Survey/ Census/ Administrative/ MIS/ AI/Big Data/ IoT) |                     | Data                                                    | Year | Source | Data                                       | Source and year |             |             |
| 1                                                                                                    | 2                   | 3                                                        | 4                                                                | 5                   | 6                                                       | 7    | 8      | 9                                          | 10              | 12          | 13          |
| 3.1.1: Maternal mortality ratio                                                                      | Tier I              | BBS                                                      | SVRS                                                             | Annual              | 181                                                     | 2015 | SVRS   | 169                                        | SVRS 2018 BBS   | 85          | 70          |
| 3.1.2: Proportion of births attended by skilled health personnel                                     | Tier I              | BBS                                                      | MICS                                                             | Annual              | 42.1%                                                   | 2014 | BDHS   | 59%                                        | MICS 2019 BBS   | 72%         | 80%         |
| 3.2.1: Under-five mortality rate                                                                     | Tier I              | BBS                                                      | CMNS/ MICS                                                       | Annual              | 36                                                      | 2015 | SVRS   | 29                                         | SVRS 2018 BBS   | 30          | 25          |
| 3.2.2: Neonatal mortality rate                                                                       | Tier I              | BBS                                                      | SVRS                                                             | Annual              | 20                                                      | 2015 | SVRS   | 16                                         | SVRS 2018 BBS   | 16          | 12          |
| 3.3.1: Number of new HIV infections per 1,000 uninfected population, by sex, age and key populations | Tier I              | (a) DGHS (NASP), MOHFW<br>(b) IEDCR, MOHFW<br>(c) UNAIDS |                                                                  | Annual              | 0.04 (Women 15-49 years: <0.1%, Men 15-49 years: <0.1%) | 2016 | UNAIDS | <0.01 (All ages) 0.02 (adults 15-49 years) | UNAIDS, 2017    | 0.02        | 0.01        |
| 3.3.2 Tuberculosis incidence per 100,000 population                                                  | Tier I              | BBS                                                      | HMSS                                                             | Annual              | 287                                                     | 2016 | NTP    | 221 (Global Tuberculosis Report)           | WHO, 2017       | 200         | 150         |

**Table 3.3 Contd.**

Table Continued...

| SDG Indicator                                                                                              | Tier Classification | Data source                              |                                                                  | Reporting frequency | Baseline data |       |        | Current status (Year and source)                                                                                                                                                                                            |                                                                                                           | Target 2025 | Target 2030 |
|------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------|------------------------------------------------------------------|---------------------|---------------|-------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------|-------------|
|                                                                                                            |                     | Institution (Ministry/ Division/ Agency) | Activity (Survey/ Census/ Administrative/ MIS/ AI/Big Data/ IoT) |                     | Data          | Year  | Source | Data                                                                                                                                                                                                                        | Source and year                                                                                           |             |             |
| 1                                                                                                          | 2                   | 3                                        | 4                                                                | 5                   | 6             | 7     | 8      | 9                                                                                                                                                                                                                           | 10                                                                                                        | 12          | 13          |
| 3.4.1 Mortality rate attributed to cardiovascular disease, cancer, diabetes or chronic respiratory disease | Tier I              | BBS                                      | SVRS                                                             | Annual              | 18%           | 2016  | WHO    | 21.6% (30-70 years)<br>Cardiovascular: 12.1%<br>Cancer: 3.8%<br>Diabetes: 1.2%<br>Respiratory disease: 2.1% (SVRS 2018)<br>Cardiovascular: 10.5%<br>Cancer: 4.9%<br>Diabetes: 0.9%<br>Respiratory disease: 2.8% (SVRS 2017) | (Health SDG Profile: Bangladesh, WHO, 2019)<br>2018: Health SDG Profile, Bangladesh, WHO<br>SVRS 2018 BBS | 10%         | 6%          |
| 3.6.1 Death rate due to road-traffic injuries (per 100,000 population)                                     | Tier I              | BBS                                      | SVRS                                                             | Annual              | 2.49%         | W2015 | BP     | 1.65                                                                                                                                                                                                                        | BP, 2018                                                                                                  | 1.5         | 1.2         |
|                                                                                                            |                     |                                          |                                                                  |                     |               |       |        | 1.25                                                                                                                                                                                                                        | BP, 2017                                                                                                  |             |             |

Table 3.3 Contd.

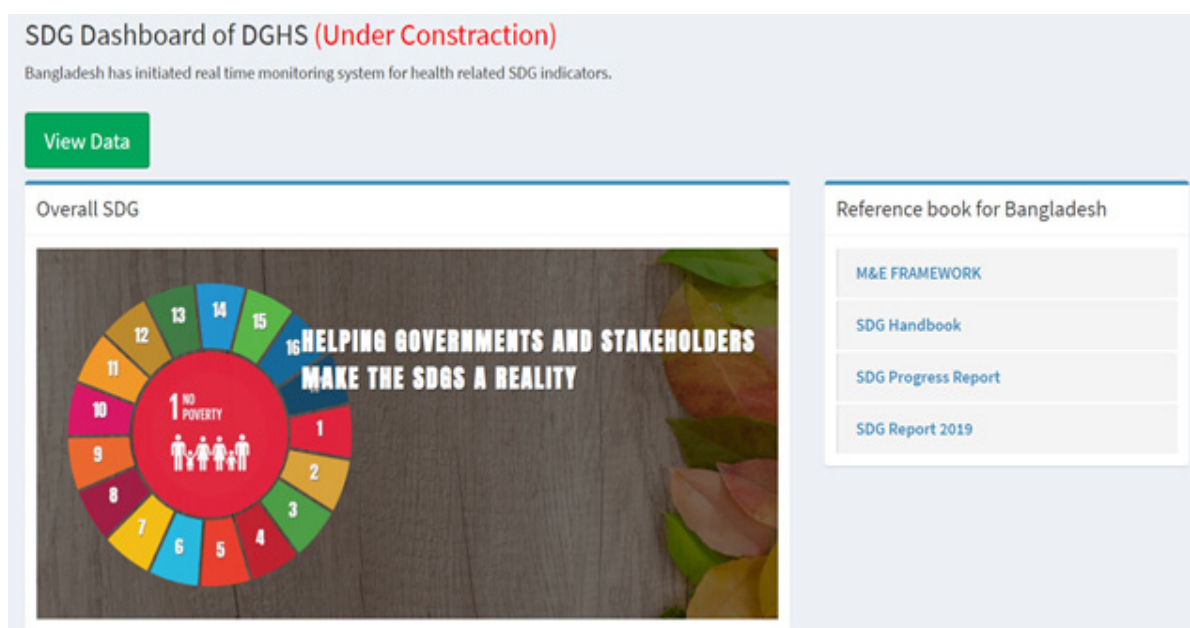
Table Continued...

| SDG Indicator                                                                                                                                                      | Tier Classification | Data source                              |                                                                  | Reporting frequency | Baseline data |      |           | Current status (Year and source) |                                           | Target 2025 | Target 2030 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------|------------------------------------------------------------------|---------------------|---------------|------|-----------|----------------------------------|-------------------------------------------|-------------|-------------|
|                                                                                                                                                                    |                     | Institution (Ministry/ Division/ Agency) | Activity (Survey/ Census/ Administrative/ MIS/ AI/Big Data/ IoT) |                     | Data          | Year | Source    | Data                             | Source and year                           |             |             |
| 1                                                                                                                                                                  | 2                   | 3                                        | 4                                                                | 5                   | 6             | 7    | 8         | 9                                | 10                                        | 12          | 13          |
| 3.9.2: Mortality rate attributed to unsafe water, unsafe sanitation and lack of hygiene (exposure to unsafe water, sanitation and hygiene for all (WASH) services) | Tier I              | (a) DGHS, MoHFW<br>(b) WHO               | -                                                                | Annual              | 5.96          | 2016 | WHO       | 11.9                             | 2018: Health SDG Profile, Bangladesh, WHO | 5.0         | 4.5         |
| 3.9.3: Mortality rate attributed to unintentional poisoning                                                                                                        | Tier I              | (a) DGHS, MoHFW<br>(b) WHO               | -                                                                | Annual              | 5.72          | 2016 | WHO       | 0.3 (per 100,000 population)     | 2018: Health SDG Profile, Bangladesh, WHO | 4.5         | 4.0         |
| 3.a.1 Age-standardized prevalence of current tobacco-use among persons aged 15 years and older                                                                     | Tier I              | BBS                                      | GATS                                                             | Triennial           | 43.3%         | 2009 | GATS, WHO | 35.3%                            | GATS 2017 BBS                             | 30%         | 25%         |

Table 3.3 Contd.

Table Continued...

| SDG Indicator                                                                            | Tier Classification | Data source                              |                                                                  | Reporting frequency | Baseline data |      |        | Current status (Year and source) |                                                                                | Target 2025 | Target 2030 |
|------------------------------------------------------------------------------------------|---------------------|------------------------------------------|------------------------------------------------------------------|---------------------|---------------|------|--------|----------------------------------|--------------------------------------------------------------------------------|-------------|-------------|
|                                                                                          |                     | Institution (Ministry/ Division/ Agency) | Activity (Survey/ Census/ Administrative/ MIS/ AI/Big Data/ IoT) |                     | Data          | Year | Source | Data                             | Source and year                                                                |             |             |
| 1                                                                                        | 2                   | 3                                        | 4                                                                | 5                   | 6             | 7    | 8      | 9                                | 10                                                                             | 12          | 13          |
| 3.d.1: International Health Regulations (IHR) capacity and health emergency preparedness | Tier I              | (a) DGHS, MOHFW (NHC-MC&CR)<br>(b) WHO   | -                                                                | Annual              | 87.5          | 2016 | WHO    | 58                               | Health SDG Profile: Bangladesh, WHO, 2019<br>World Health Statistics WHO, 2018 | 95          | 100         |
|                                                                                          |                     |                                          |                                                                  |                     |               |      |        | 78                               |                                                                                |             |             |



**Figure 3.6.** Screenshot of SDG dashboard at the DGHS

SDG reporting. Overall support is given for strengthening health system with strong evidence base to fulfill Vision 2030. The DGHS also measures timely the accurate indicators of health-related SDG and visualizes health-related SDG information at the DGHS dashboard on the website.

### Health-related SDG monitoring Activities: Progress at the DGHS

- Health-related SDG indicator tracker, online real-time SDG health indicators developed at dashboard on the DGHS website
- Contributing to information on health-related SDG indicators sent to national SDG reporting as well as global reporting
- Coordination with Prime Minister's office to provide information on health-related SDG indicators for SDG Tracker-Bangladesh Mirror (16 health indicator values)

### SDG health tracker, online real-time SDG health indicators on the dashboard/HMIS

The SDG dashboards for health indicators tracking are on the DGHS website and represent the available data on SDG achievement according to indicators, with emphasis on the country's most acute challenges.

### A Few Challenges to Monitoring SDG health indicators

- Coordination among BBS and other government offices, especially for health, DGHS, DGFP, and other health organizations
- Private sector for addressing the methodological gap at this moment in real-time data
- Harmonized and aligned support for strengthening analytical capacity of SDG health indicators

- Completeness and data quality of routine HMIS to measure health-related SDG indicators.

### **Next Initiatives to Measure SDG Health Indicators at the DGHS**

- Core group with focal points at the DGHS
- Better reporting (sub)national and global progress on SDGs
- Strengthen statistical analytical capacity to measure indicators
- Address health data gaps according to General Economic Division (GED)
- Improve facility and community data systems

- Enhance use of data and accountability
- Technical Working Groups and regional networks: leveraging existing collaborations and initiatives where possible.

Bangladesh achieved most of the MDGs, including health-related ones due to strong commitment of the Government. The Government is also determined to achieve the SDGs through all out efforts. Health-related SDG remains in the core of SDGs as they relate to health, well-being and quality of life of every citizen throughout the course of life. As per global commitment, the Government will be able to achieve all targets of the SDGs, including health-related ones.



# Maternal, Newborn, Child and Adolescent Health

## Institutional deliveries markedly increased

The Ministry of Health and Family Welfare of the People's Republic of Bangladesh, in collaboration with UNICEF, has been running the facility-based Emergency Obstetric Care (EOC) Program, in all districts of Bangladesh to improve the maternal and newborn child health situation. All the government medical college hospitals, district hospitals, upazila health complexes, and maternal and child welfare centers (MCWCs) provide obstetric care services. A number of private clinics or hospitals and health-related NGOs are also serving as partners in this program. Obstetric care is classified into two categories in this program; they are: Comprehensive Emergency Obstetric Care (CEmOC) and Basic Emergency Obstetric Care (BEOC). Currently, all medical college hospitals, 59 district hospitals, 3 general hospitals, 132 upazila health complexes, and 71 MCWCs provide CEmOC, and rest of the upazila health complexes provide BEOC. The list also includes NGOs and private care providers from a number of districts. Under a program, jointly operated by the Management Information System (MIS) of the DGHS and UNICEF, data are collected from the EOC facilities.

### Demand-side Financing (DSF): Maternal Health Voucher Scheme

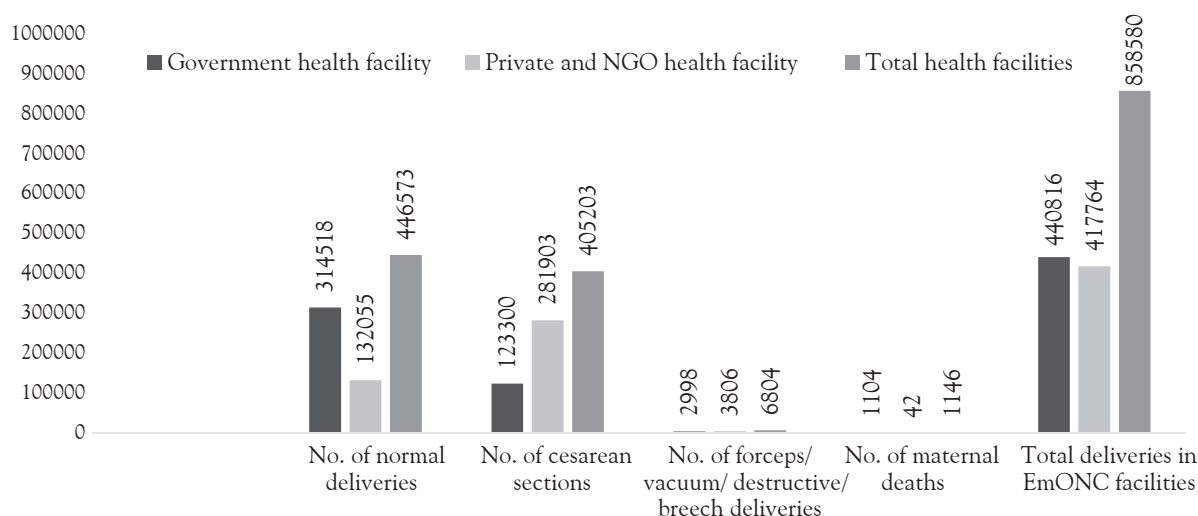
The Bangladesh Ministry of Health and Family Welfare, in collaboration with WHO,

**Table 4.1.1.** Data collected from emergency obstetric care facilities, which have been analyzed on services in 2018

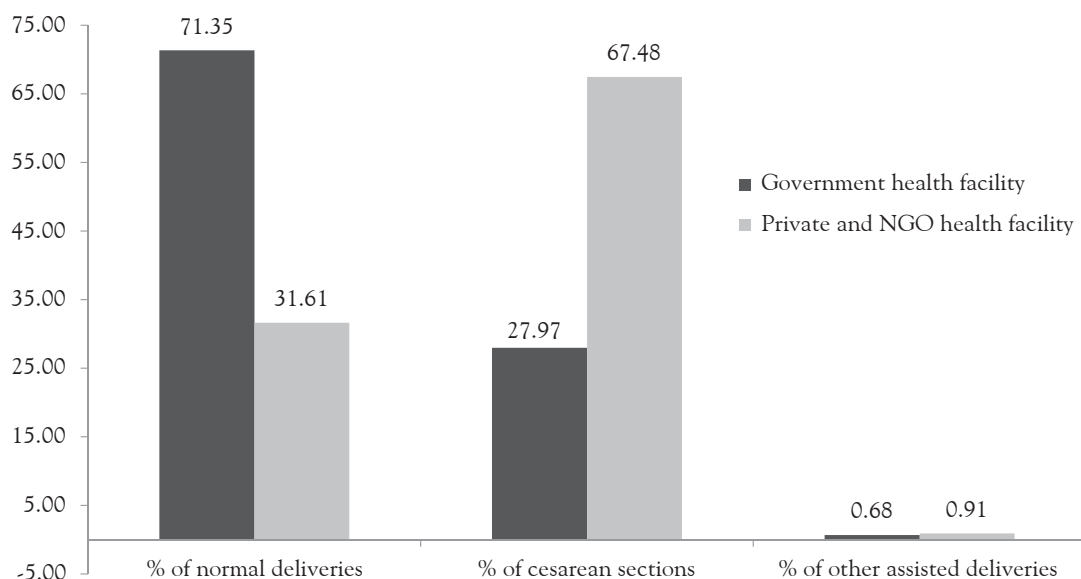
| Type of hospital/<br>facility      | Number     | Percentage<br>(%) |
|------------------------------------|------------|-------------------|
| Govt. medical college hospitals    | 19         | 2.8               |
| District and general hospitals     | 61         | 9.0               |
| Upazila health complexes           | 419        | 61.6              |
| Maternal and child welfare centers | 68         | 10.0              |
| Other government hospitals         | 5          | 0.7               |
| Private medical college hospitals  | 1          | 0.1               |
| NGO hospitals                      | 46         | 6.8               |
| Private clinics                    | 61         | 9.0               |
| <b>Total</b>                       | <b>680</b> | <b>100.0</b>      |

introduced in 2007 an innovative Maternal Health Voucher Scheme as a demand-side financing (DSF) initiative to improve access to and use of quality maternal health services for the poor pregnant women. The salient features of the scheme are as follows:

- ◆ The number of DSF upazilas was increased in phases. Currently, this program is being implemented in 55 upazilas of 41 districts.



**Figure 4.1.1.** Number of different obstetric care encounters in the government, private and NGO healthcare facilities in Bangladesh in 2018



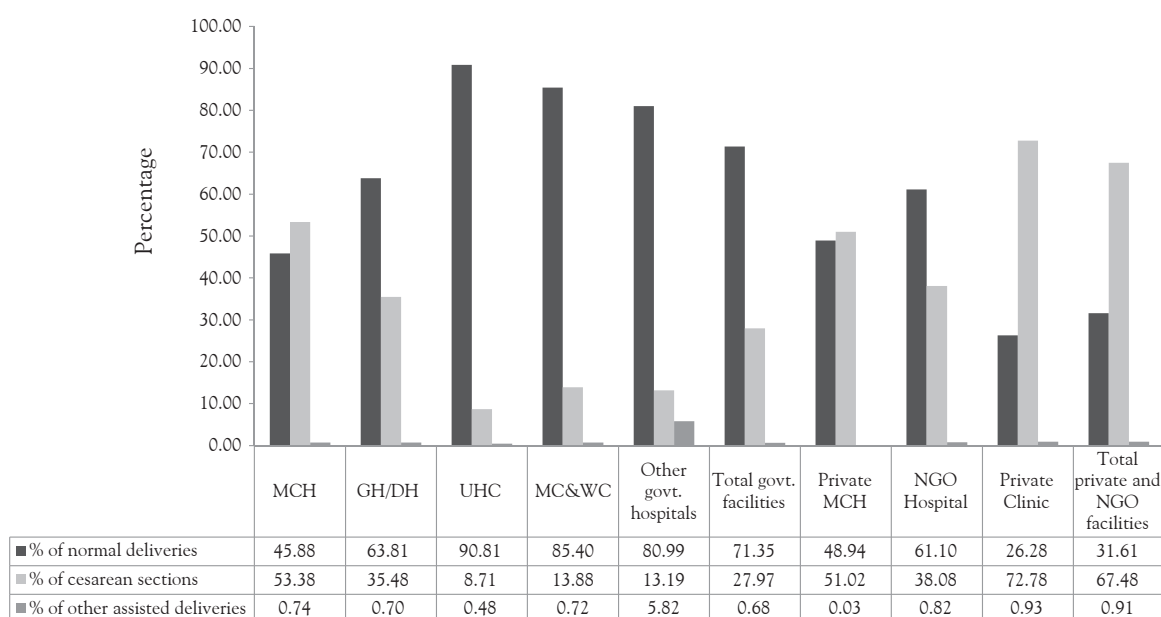
**Figure 4.1.2.** Percentage of deliveries by type between government and non-government (private and NGO) health facilities in 2018

- ◆ Poor women are defined by specific selective criteria (approximately 35% of all the pregnant woman of an upazila), and pregnant woman are selected by local government representatives, health managers, and other stakeholders.
- ◆ Registration is done by field workers. The total number of cumulative beneficiaries reached 15,46,121 (from July 2006 to June 2019).
- ◆ All the voucher-holders received health services free of charge (ANC, PNC, safe

**Table 4.1.2.** Obstetric care services provided by the government and non-government emergency obstetric care facilities in 2018

| Type of delivery/facility               | Government facility |        |        |         |                       | Non-government facility |             |              |                | Grand total of facilities |
|-----------------------------------------|---------------------|--------|--------|---------|-----------------------|-------------------------|-------------|--------------|----------------|---------------------------|
|                                         | MCH                 | GH/DH  | UHC    | MC & WC | Other govt. hospitals | Total                   | Private MCH | NGO hospital | Private clinic | Total                     |
| No. of normal deliveries                | 53782               | 72710  | 154018 | 30570   | 3438                  | 314518                  | 1530        | 37823        | 92702          | 132055                    |
| % of normal deliveries                  | 45.88               | 63.81  | 90.81  | 85.40   | 80.99                 | 71.35                   | 48.94       | 61.10        | 26.28          | 31.61                     |
| No. of forceps/vacuum/breech deliveries | 859                 | 765    | 816    | 177     | 247                   | 2864                    | 1           | 83           | 1981           | 2065                      |
| % of forceps/vacuum/breech deliveries   | 0.73                | 0.67   | 0.48   | 0.49    | 5.82                  | 0.65                    | 0.03        | 0.13         | 0.56           | 0.49                      |
| No. of destructive/invasive operations  | 12                  | 38     | 3      | 81      | 0                     | 134                     | 0           | 427          | 1314           | 1741                      |
| % of destructive/invasive operations    | 0.01                | 0.03   | 0.00   | 0.23    | 0.00                  | 0.03                    | 0           | 0.69         | 0.37           | 0.42                      |
| No. of cesarean sections                | 62576               | 40428  | 14768  | 4968    | 560                   | 123300                  | 1595        | 23572        | 256736         | 281903                    |
| % of cesarean sections                  | 53.38               | 35.48  | 8.71   | 13.88   | 13.19                 | 27.97                   | 51.02       | 38.08        | 72.78          | 67.48                     |
| Total deliveries in facilities          | 117229              | 113941 | 169605 | 35796   | 4245                  | 440816                  | 3126        | 61905        | 352733         | 417764                    |

Note: MCH- Medical College Hospital; GH/DH-General Hospital/ District Hospital; UHC-Upazila Health Complex



**Figure 4.1.3.** Percentage of normal, assisted and cesarean deliveries (obstetric care service) provided by government and non-government emergency healthcare facilities in 2018

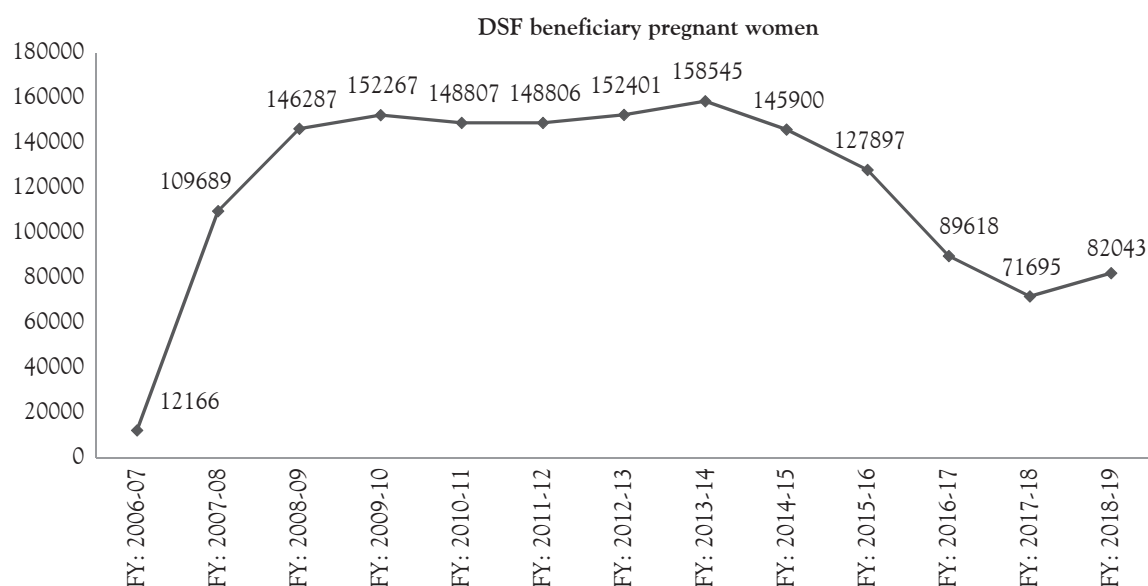
delivery and treatment of complications, including cesarean section, transportation cost, and laboratory test).

- ◆ If the voucher-holder mother has her delivery in health facility or by SBA at home, she gets cash incentive for nutritious food.
- ◆ Both public and NGO as well as private facilities participate in the DSF scheme.
- ◆ Under the current sector program, there is a target to scale up the program every year.

### Maternal and Newborn Health Initiative

The Maternal and Newborn Health Initiative (MNHI) is being implemented by the Director of Primary Healthcare of the DGHS in 11 districts of Bangladesh, with the assistance of UNFPA, UNICEF, and WHO. Some of them are funded by DFATD Canada. The following are the salient features of the initiative:

- The districts are Thakurgaon, Jamalpur, Narail, Maulvibazar, Panchagarh, Sirajganj, Patuakhali, Barguna, Rangamati, Sunamganj, and Bagerhat.
- The program focuses on saving maternal and newborn lives through creating need-based demand and priority-based actions.
- The broad principle of this program is to find the bottlenecks through data analysis.
- The health managers develop evidence-based planning and budgeting (DEPB) for every upazila and hospital. Around 5 districts have been covered under DEPB by UNICEF.
- The civil surgeon and deputy directors of family planning of the respective districts serve as the local focal points for the program.
- UNICEF has designed a comprehensive model to improve health of neonates, mothers, and young children in hard-to-



**Figure 4.1.4.** Number of DSF beneficiary pregnant women by fiscal year (total 15,46,121)

reach areas of Bandarban, Cox's Bazar, and Netrakona district. It has introduced a default-tracking system to track every mother and child.

- Under health system strengthening program, 14 consultants are working to improve the information system by using data for planning.
- Innovative dashboard has been created for all health managers.
- As a part of improving quality, Maternal and Perinatal Death Surveillance and Response (MPDSR) program has been introduced in 39 districts. This has shown a good impact in reducing maternal and neonatal deaths.

### Community-based skilled birth attendants and midwives

Activities of the community-based skilled birth attendants and midwives are highlighted in the sections that follow:

- > The MOHFW undertook a short-term measure to tackle shortage of skilled manpower by providing training courses.
- > The Directorate General of Health Services has also been implementing community-based skilled birth attendant (CSBA) training program since 2003 to train and educate the family welfare assistants/female health assistants on midwifery skills.
- > The CSBAs are trained to conduct normal safe deliveries at home and to identify the risks and complicated cases so that they can motivate the women and their family members to refer to the nearby health facilities where comprehensive EOC services are available. The CSBA training program is now organized in 465 upazilas of 64 districts.
- > The total number of CSBAs in Bangladesh (up to December, 2017) was 11,509. In November 2018; this number has increased to 11,555 in 2018.

- > Training is given to female CHCPs and female HAs.
- > Training was imparted to 54 participants from Sher-e-Bangla MCH, Adhunik Sadar Hospital of Brahmanbaria, Maulvibazar, and Civil Surgeon Office of Feni during 2017-2018 fiscal year.
- > Training was provided to 80 participants in 2018-2019 fiscal year.

**Table 4.1.3.** Levels of training institutions in the whole country

| Training institutions at the divisional level                                                                         | Training institutions at the district level                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Medical college hospital</li> <li>• Nearest family welfare center</li> </ul> | <ul style="list-style-type: none"> <li>• Sadar hospital</li> <li>• Nursing institution</li> <li>• Family welfare center</li> </ul> |

### Obstetric Fistula Program

In Bangladesh, obstetric fistula and other maternal morbidities affect thousands of women. Currently, it is estimated that approximately 50,000 women in the country are living with obstetric fistula. Measures taken to address this health problem are highlighted below:

- UNFPA has been assisting the Government of Bangladesh in strengthening the quality of service

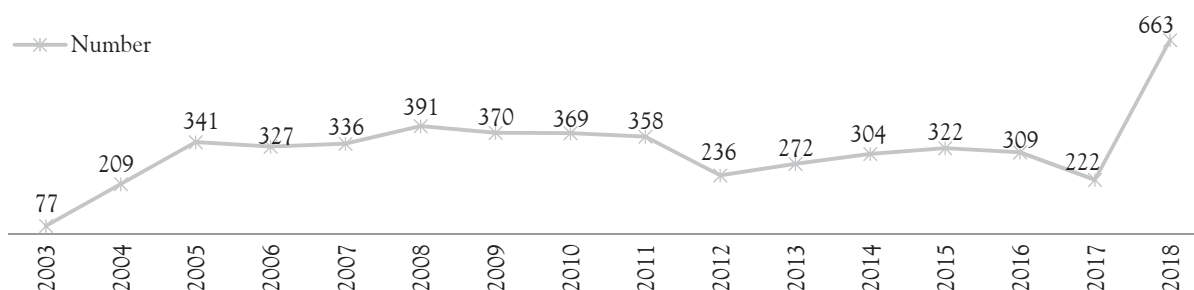
delivery and capacity-development of service/care providers at 10 government medical college hospitals.

- Bangladesh endorses the UN declaration to end obstetric fistula within a generation. In the new HPNSP (2017-2022), the Government has made lots of references for the care and prevention of fistula.
- UNFPA and other partners have been providing both technical and financial support to the Government to make Bangladesh fistula-free.

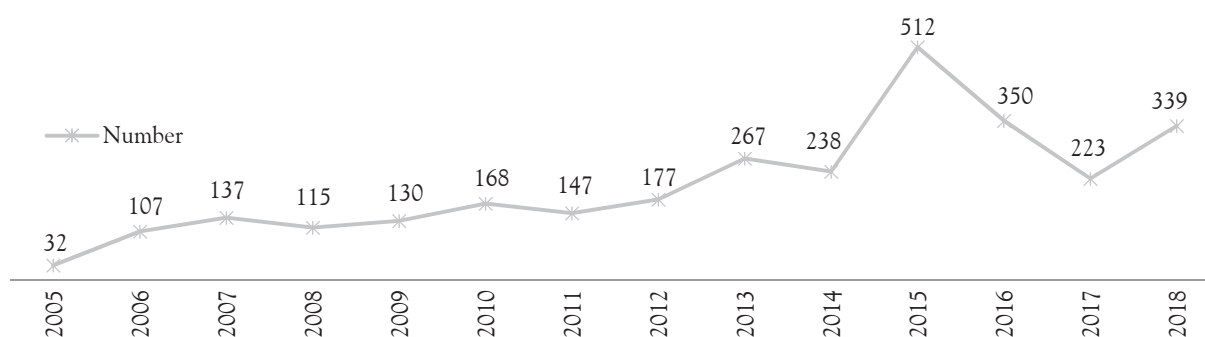
### Cervical and Breast Cancer Screening Program in Bangladesh

Cervical and breast cancers are the most common cancers among Bangladeshi women, and these contribute to a significant disease burden of Bangladesh. Routine screening and early detection of cervical and breast cancer can reduce the death rates and can improve life-expectancy of cancer patients. Activities during the reporting period are as follows:

- The Government of Bangladesh has developed a cervical and breast cancer screening program in all districts of Bangladesh with the technical assistance of Bangabandhu Sheikh Mujib Medical University (BSMMU) and UNFPA.



**Figure 4.1.5.** Number of fistula repair surgeries (N=5,106) at the government medical college hospitals during 2003-2018



**Figure 4.1.6.** Number of fistula repair surgeries (N=2,942) at private hospitals during 2005-2018

- Since 2005, a total of 431 centers (Table 4.1.4) have been established throughout the country, and about 2,036 service providers (doctors=386, nurses/FWVs/paramedics =1,650) in 64 districts were trained on cervical and breast cancer screening, based on visual inspection with acetic acid (VIA) and clinical breast examination (CBE).
- Fourteen government medical college hospitals, along with BSMMU, have been developed as referral centers through development of colposcopy clinics with facilities for colposcopy, histopathology, and management of precancerous condition of the cervix. A total number of 3,70,446 VIA tests were performed since January 2005 to June 2019 at different service centers. Of the tested women, 17,036 (4.6%) were found VIA+ve. (Figure 4.1.7).

The coverage of the screening tests has been increasing every year. All the VIA+ve cases were referred to the colposcopy clinic at BSMMU and also to different medical college hospitals.

Figure 4.1.9 shows that a total of 22,24,729 CBE tests were performed at service centers from

**Table 4.1.4.** Total number of VIA and CBE centers established at various institutions (2005-2019)

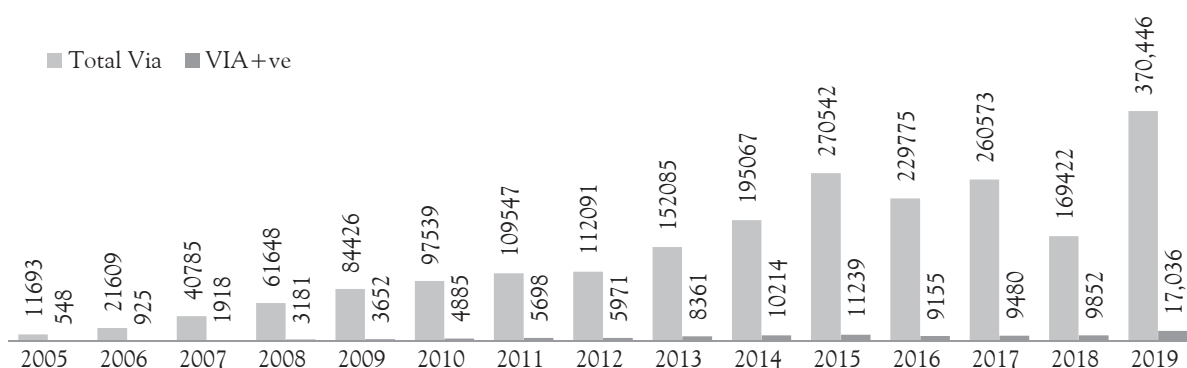
| Name of location                              | Number of centers |
|-----------------------------------------------|-------------------|
| District hospitals                            | 57                |
| BSMMU, NICRH, ICMH, medical college hospitals | 15                |
| MCHTI, MFSTC, MCWC                            | 61                |
| Upazila health complexes                      | 248               |
| Union health and family welfare centers       | 30                |
| BGB, CMH/private facility/NGOs                | 20                |
| <b>Total</b>                                  | <b>431</b>        |

2007 to 2019; among them, a total of 35,634 (1.6%) women were CBE-positive. The coverage of CBE screening is increasing every year.

In 2019, a total of 384,267 screening tests were conducted. Of those, 2.75% were CBE+ve. All these CBE-positive cases were referred to the Department of Surgery at BSMMU/MCHs and DHs.

In Bangladesh, LEEP is a commonly-used outpatient treatment procedure for CIN (Cervical Intraepithelial Neoplasia) under local anesthesia and thermo-coagulation without local anesthesia (Basu et al. 2010).





**Figure 4.1.7.** Number of VIA tests performed and VIA+ve results (2005-June 2019)

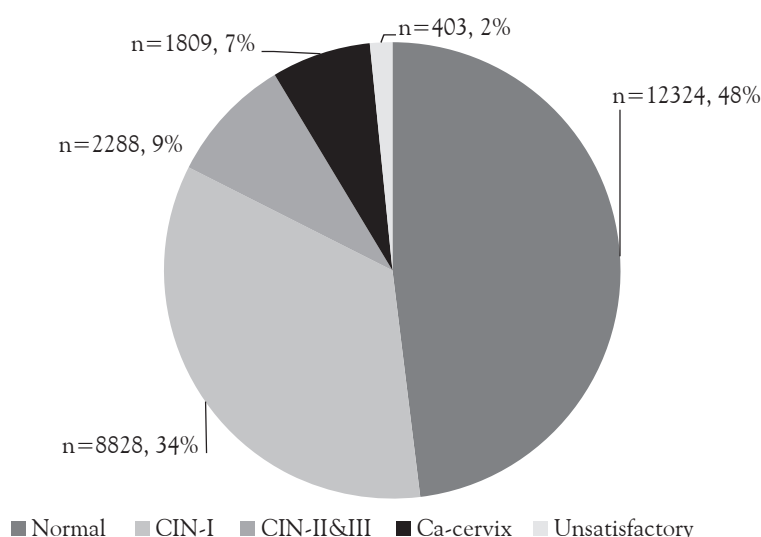
**Table 4.1.5.** Total number of colposcopy tests done in 2018 at different institutions

| Name of institution                               | Number of colposcopy tests |
|---------------------------------------------------|----------------------------|
| Bangabandhu Sheikh Mujib Medical University       | 574                        |
| Rajshahi Medical College Hospital                 | 1976                       |
| Chittagong Medical College Hospital               | 971                        |
| Khulna Medical College Hospital                   | 1113                       |
| Sher-E-Bangla Medical College Hospital, Barisal   | 570                        |
| Faridpur Medical College Hospital                 | 360                        |
| Mymensingh Medical College Hospital               | 913                        |
| Dhaka Medical College Hospital                    | 981                        |
| Dinajpur Medical College Hospital                 | 0                          |
| Comilla Medical College Hospital                  | 571                        |
| Rangpur Medical College Hospital                  | 0                          |
| Shaheed Suhrawardy Medical College Hospital       | 397                        |
| Sir Salimullah Medical College & Mitford Hospital | 86                         |
| Shaheed Ziaur Rahman Medical College Hospital     | 200                        |
| <b>Total</b>                                      | <b>1019</b>                |

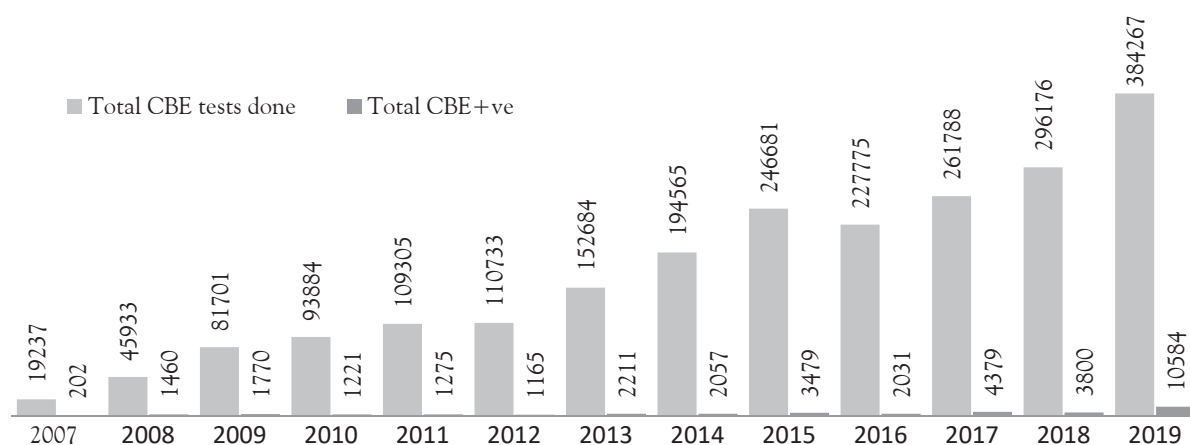
**Table 4.1.6.** Management of referred patients with cervical and breast cancer at BSMMU (January 2005 to December 2019)

| Management                                            | Number       |
|-------------------------------------------------------|--------------|
| Follow-up after 3 years                               | 11224        |
| Follow-up after 1 year                                | 3313         |
| LEEP (Loop Electrosurgical Excision Procedure) biopsy | 3450         |
| Cryotherapy                                           | 131          |
| Cold Coagulation (local ablative method)              | 2550         |
| Referred to oncology due to cancer                    | 1553         |
| Failure of treatment and follow-up                    | 2905         |
| <b>Total</b>                                          | <b>25126</b> |





**Figure 4.1.8.** Colposcopy findings of VIA+ve women attending BSMMU (since January 2005 to December 2019, N=25,652)



**Figure 4.1.9.** Number of CBE tests done, with results (2007-2019)

### Child Health Situation in Bangladesh

Bangladesh has been able to reduce the under-five mortality, well ahead of the MDG 4 target (from 144/1000 livebirths in 1990 to 40/1000 livebirths in 2019 [MICS]), reducing by 72.2% since 1990, with the annual rate of reduction being 2.4. The situation is highlighted below:

- There has been a significant shift in the epidemiological pattern of under-five mortality. Neonatal deaths still account for almost 65% of all under-five mortality (U5MR).
- Diarrheal deaths among under-five children have decreased substantially over the last few years (29.5%). However, ARI/pneumonia continues to be the single most cause (46.4%) of under-5 mortality (U5MR).
- The care-seeking from trained providers for pneumonia and diarrhea has increased remarkably over the last few years. ORS use-rate for diarrhea management has increased to more than 72%, vitamin A supplementation coverage among under-five children has increased from 85% in

2005 to 91.3% in 2016 (CES, 2016) while exclusive breastfeeding practices have stood at 62.6% (in 2019).

- Activities of EPI and introduction of new vaccines have progressed well, and scaling up of C-IMCI and IMCI and nutrition corners is taking place. About 82.3% of children aged 12-23 months received all the scheduled vaccines by 12 months of their age (CES, 2016).
- Under the Reach Every District Community Strategy, routine EPI has further been strengthened in low-performing 32 districts and 4 city corporations. NGO/private sector partnership exists for EPI program in a limited scale, especially in urban areas.

## Challenges

- Despite reduction in mortality over time, continued disparities have been observed between the richest and the poorest quintiles, between urban and rural areas, and among divisions (e.g. Khulna, Sylhet)
- Mothers' education was found to be associated with U5MR. The highest mortality rate of 82/1,000 was noted among mothers with no education.

Among the mothers with secondary-level education or higher, the rate was 39/1000.

- Although deaths due to preventable communicable diseases continued to decrease over time, the proportion of deaths due to drowning has resulted in a sharp rise from 26% in 2007 to 42% in 2011 among the age-group of 1-4 year(s). Currently, a clear multi-sector strategy to addressing child drowning is absent.
- The current child health interventions mainly focus on reduction of mortality, with limited emphasis on child development. Sustaining the already-achieved high coverage in EPI and reaching the unreached to further increase the coverage remain a challenge.

To improve newborn and child survival, the National Newborn Health Program (NNHP) was launched through the sector program; the IMCI program has also been strengthened. Activities of the program during the reporting period are as follows:

- The NNHP focused on achieving the equitable and effective coverage of high-impact newborn services and practices at scale through quality implementation of a comprehensive newborn healthcare package by the public health system.

**Table 4.1.7.** Relevant result framework indicators (RFW)

| Indicator | Unit of measurement         | Baseline data (Year and source) | Achievement     | Projected target (2022) |
|-----------|-----------------------------|---------------------------------|-----------------|-------------------------|
| U5MR      | No. among 1000 livebirths   | 46 (BDHS 2014)                  | 40 (MICS 2019)  | 34                      |
| NMR       | No. among 1000 livebirths   | 28 (BDHS 2014)                  | 26 (MICS 2019)  | 18                      |
| MMR       | No. among 100000 livebirths | 176 (UN Est. 2015)              | 169 (SVRS 2018) | 121                     |

- The activities under the IMCI program are implemented at the community and all levels of facilities in line with national policy and strategy for achieving high coverage and quality of facility- and community-based IMCI services.

## Program management

- The existing IMCI unit under Line Director MNCAH is strengthened and renamed National Newborn Health Program and IMCI. The unit has one Program Manager (PM-NNHP & IMCI), five Deputy Program Managers (DPM-Admin & Finance; DPM-Newborn Health; DPM- Training and Child Injury; DPM-Field Monitoring and Data Quality; DPM- Coordination and Logistics Management).
- Along with strong leadership and ownership of MOHFW officials at all levels, functional coordination among different departments and facilitation role from development partners are crucial to ensure successful implementation of NNHP and IMCI.
- A District MNCH Focal Person works at the district level to strengthen supervision and monitoring of all maternal, newborn and child health-related activities at the district level and below. With the support of development partners, a group of consultants worked from district and division levels to facilitate the implementation of NNHP throughout the country.

## Activities

- > Promotion of birth preparedness, newborn care preparedness, and proper care-seeking through a comprehensive SBCC approach

- > Immediate and essential newborn care (ENC), including application of 7.1% Chlorhexidine to the umbilical cord and postnatal care (PNC) in the facility and in the home by visiting community health workers for management of sick newborn and resuscitation using HBB protocol
- > Kangaroo Mother Care (KMC) for management of preterm/LBW babies
- > Judicial use of antenatal corticosteroid in threatened premature deliveries
- > Care of critically ill newborn at Special Care Newborn Unit (SCANU)
- > Strengthen effective referral systems, including reliable and affordable transportation
- > Development of NNHP management toolkit and training of health managers and MNCH focal persons on Comprehensive Newborn Care Package (CNCP) implementation
- > Capacity-building of the healthcare providers and field workers at different levels on CNCP for KMC training for doctors and nurses to manage preterm low-birthweight babies at upazila health complex and higher-level facilities
- > Training of doctors and nurses for management of sick newborn at SCANU/ NSU
- > Ensure logistics commodities and medicines at the union-, upazila- and district-level facilities under the DGHS for the management of sick newborn as per national guideline; establishment of standard KMC corners in all upazila and district-level facilities and ensuring

commodities and logistic supplies, and functional newborn resuscitation devices (Bag & Mask) in all facilities

- > Provision of a district MNCH focal person to coordinate, supervise, and monitor all maternal, newborn and child health-related activities at the district level and below, and report to Line Director MNCAH
- > Revision of National Neonatal Health Strategy and Standard Operating Procedure (SOP) for newborn health services
- > Community engagement by using community group (CG) and community support groups (CSG) of community clinics in coordination with Community-based Healthcare Service (CBHCS)
- > Strengthen societal engagement by orienting school teachers, local representatives, multi-sector officials, and NGOs through different existing platforms
- > Organization of national newborn campaign.

### Adolescent and School Health Program

Adolescent health has been highlighted frequently during the HPNSP as a priority area that needs further and focused attention. The Bangladesh national Adolescent Reproductive Strategy for Health (ARSH) was formulated in 2006, and related action plan was developed in 2013. Two major operational plans included plans to carry out selected health activities of ARSH strategy.

Adolescent pregnancy and childbearing entail a high risk of maternal death among

adolescents, and children of the young mothers have higher level of morbidity and mortality rates. There are estimated 569,000 annual births among 15-19 years old adolescent girls in Bangladesh. They face a number of important health risks arising out of early pregnancies, violence, and inadequate nutrition. Almost twice as many newborn deaths occur among them.

### Objectives of the program

- Improve knowledge of adolescents on sexual and reproductive health (SRH), nutrition, violence against adolescents, and mental health and others.
- Promotion of and ensuring access to adolescent-friendly health services at the facility and community levels
- Creation of positive changes in the behavior and attitude of the gatekeepers of the adolescents towards reproductive health.

Activities under the program during the reporting period are as follows:

- Training modules developed by the DGHS. Training provided to the service providers, school teachers, and adolescent peers in 10 districts and, adolescent-friendly health services (AFHS) established in 6 districts and 2 city corporations. A guideline on AFHS (2015) and strategy for IV/STI/AIDS/MR/PAC formulated
- Recently (2017), Bangladesh National Adolescent Health Strategy for 2017 to 2030 developed and approved. Related action plan for adolescent health also developed and soon will be approved

- Emphasis given on the following 3 thematic areas: (i) strengthen adolescent, sexual and reproductive health issues; (ii) nutrition; and (iii) violence against adolescents and mental health. All the activities and strategies planned according to these 3 thematic areas
- Training of 690 trainers (TOT) of district- and upazila-level managers of the DGHS, DGFP, and DG of Education on adolescent healthcare in Chandpur and Sunamganj districts
- Training of 1,050 secondary school and college teachers on adolescent healthcare in Chittagong, Sunamganj, and Chandpur districts
- Training of 510 secondary school students as peer educators from the adolescent groups in Maulvibazar, Sunamganj, Chandpur, and Barisal districts
- Orientation of 750 persons as gatekeepers (chairmen, members, imams, kazis, school management committee, and local population) in Maulvibazar, Sunamganj, Chandpur districts
- Orientation of 44,100 field-level workers
- Workshop for capacity-development, coordination and performance appraisal for 40 persons
- Special program undertaken in the low-performing areas of Chattogram and Sylhet division
- Operational research on adolescent health
- Developed strategy on social and behavior change communication for adolescent health
- Developed dashboard for adolescent health indicators and strengthened HMIS system for adolescent health
- Training of trainers (TOT) of secondary school teachers of Dhaka City Corporation on adolescent health, with support of UNICEF and WHO
- Training of frontline basic health workers of Tangail district on adolescent health with the support of UNICEF and Embassy of the Kingdom of The Netherlands
- Assessment of current school-based HPN services in Chattogram and Sylhet division. An action plan was developed for school-based HPN services, with the support of UNFPA.

# Vaccination Service Delivery

## Expanded Program on Immunization (EPI)

### Brief History of the Expanded Program on Immunization EPI

- EPI in Bangladesh was launched on 7 April 1979.
- Initial vaccination started against six diseases: BCG, DPT, OPV, TT, and Measles.
- The program was intensified through Universal Child Immunization (UCI) both in rural and urban areas from 1985.
- EPI service was available to all target groups by 1990.
- Polio Eradication and Elimination of Maternal and Neonatal Tetanus (MNT) activities started in 1995.
- AFP and NT Surveillance started in 1997.
- Hepatitis B vaccine was introduced in 2003.
- AD syringes was introduced in 2004.
- Hib vaccine as pentavalent form was introduced in 2009.
- MR vaccine and Measles second dose were introduced in 2012.
- PCV and IPV vaccination were introduced in March 2015.
- EPI switched from tOPV to bOPV in April 2016.
- The program switched from TT to Td in 2019.
- Inactivated Polio Vaccine (IPV) was introduced in March 2015 and shifted to fIPV (0.1 mL) since November 2017.

EPI Coverage Evaluation Survey (EPI CES) 2019 was conducted but report is not yet published completely.

### Objectives of EPI

1. At least 95% full vaccination coverage among children below one year of age at the national level and 90% full vaccination coverage in each district
2. Td5 coverage among women of childbearing age targeted to reach at least 80% at the national level and 75% at the district level
3. Maintain polio-free certification status of Bangladesh
4. Achieve 95% Measles and Rubella coverage at the national level; reach Measles and Rubella elimination status, and control CRS by 2020
5. Prevention of diseases protectable by new and under-used vaccines
6. Maintain maternal and neonatal tetanus elimination validation status



7. Ensure safe injection practices and waste disposal.

### Target Population

1. All children aged 0-11 month(s)
2. All children aged 15 months
3. All women of childbearing age (15-49 years).

### Tetanus Toxoid (TT) for women of childbearing age

Like the childhood vaccination coverage, TT vaccination coverage is assessed as crude and valid coverage. The valid TT coverage is assessed in terms of the valid doses that a woman receives. The crude TT coverage is assessed in terms of all TT doses—both valid and invalid—that a woman receives.

A TT dose administered before the recom-

mended interval is considered to be invalid. Thus, a TT3 dose given earlier than the recommended 6-month interval after a valid TT2 is enumerated as an invalid TT3 dose. The information of TT vaccination is obtained from a woman's TT card (if available). If it is not available, the information is collected from the woman's vaccination history reported by her.

- The country is maintaining the maternal and neonatal tetanus-free status since 2008. The immunization program of Bangladesh aims to immunize the women of childbearing age by administering TT before the age of 18 years.
- A period of 2 years and 7 months is required to complete all the 5 doses of TT vaccines. If a woman starts TT vaccination at the age of 15 years and maintains the exact interval, she would be able to complete all the doses before she reaches

**Table 4.2.1.** Vaccination schedule for children

| Vaccine              | No. of doses | Age of vaccination     | Doses interval | Route          |
|----------------------|--------------|------------------------|----------------|----------------|
| BCG                  | 1            | After birth            | -              | Intra-dermal   |
| Pentavalent vaccine  | 3            | W6, W10, W14           | 4 wk           | Intra-muscular |
| PCV                  | 3            | W6, W10, W14           | 4 wk           | Intra-muscular |
| bOPV                 | 3            | W6, W10, W14           | 4 wk           | Oral           |
| IPV                  | 2            | W6, W14                | 8 wk           | Intra-dermal   |
| Measles-Rubella (MR) | 2            | 9 months and 15 months | -              | Sub-cutaneous  |

**Table 4.2.2.** Td schedule for women of 15-49 years of age

| Td Doses | Starting dose/Minimum interval between doses |
|----------|----------------------------------------------|
| Td1      | 15 years of age                              |
| Td2      | At least 4 weeks after TT1                   |
| Td3      | At least 6 months after TT2                  |
| Td4      | At least 1 year after TT3                    |
| Td5      | At least 1 year after TT4                    |

the age of marriage, ensuring protection for her entire reproductive life.

- However, the coverage gradually falls for the subsequent TT doses for pregnant women and is remarkably lower for the third and fourth doses. TT coverage does not increase as expected considering the number of target women of childbearing age. This aspect needs attention to ensure effective coverage.

## Universal Routine Child Immunization

The EPI CES 2019 validated the immunization coverage rates by cross-checking EPI cards, with history taken from mothers and caregivers of children.

Bangladesh created a success story on polio eradication. The country is polio-free since 2000, with 18 exceptional cases of wild polio virus imported from neighbouring India in 2006. National Immunization Day is observed every year.

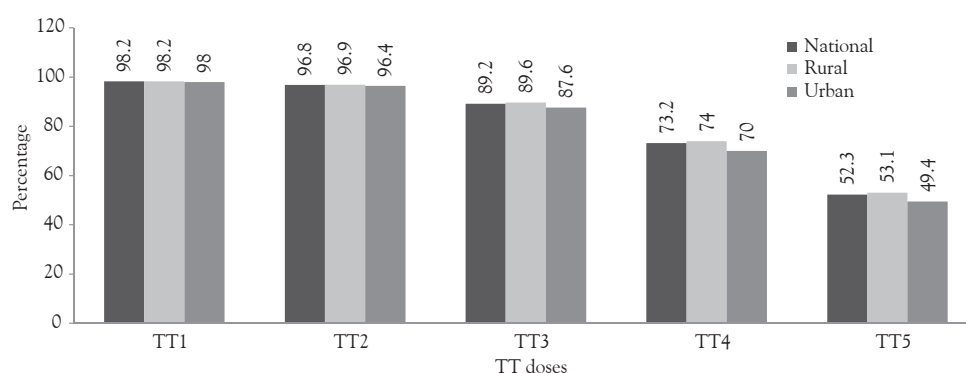
The polio eradication program in Bangladesh illustrates the Government's commitment through providing 100% cost of routine polio immunization and 95% cost of supplementary polio immunization activities. Bangladesh, despite being free from polio for a long time, could not achieve polio-free certification as one country of the WHO South-East Asia Region, viz. India, could not eradicate the last

traces of polio. However, in India, the last case of polio was detected in 2011.

After being polio-free for more than 36 months, Bangladesh, along with other 10 member countries of the South-Asia Region, obtained the polio-free certification in March 2014. As per the global polio eradication endgame strategic plan (2013-2018), Bangladesh introduced inactivated polio virus vaccine (IPV) in March 2015 for prevention of vaccine-derived polio viruses due to type 2 component of polio vaccine.

As per the country's plan, Bangladesh switches from trivalent oral polio vaccine (tOPV) to bivalent oral polio vaccine (bOPV) in April 2016 for prevention of outbreak due to type 2 component of OPV.

The country is also satisfactorily progressing toward achieving the measles elimination



**Figure 4.2.1.** Valid TT vaccination coverage at the national level, rural, and urban areas in 2016

**Table 4.2.3** Valid vaccination coverage of  $\leq 12$  months old children as found in EPI Coverage Evaluation Survey (CES), 2019

| Area     | BCG  | OPV1 | OPV2 | OPV3 | Penta1 | Penta2 | Penta3 | MR1  | PCV1 | PCV2 | PCV3 | FVC  |
|----------|------|------|------|------|--------|--------|--------|------|------|------|------|------|
| National | 99.7 | 99.7 | 99.2 | 93.3 | 99.7   | 99.2   | 93.3   | 88.6 | 99.7 | 99.2 | 92.5 | 83.9 |

**Table 4.2.4** Valid vaccination coverage of  $\leq 23$  months old children as found in EPI CES 2016

| Age-group        | BCG  | OPV1 | OPV2 | OPV3 | Penta1 | Penta2 | Penta3 | MR1  | MR2  |
|------------------|------|------|------|------|--------|--------|--------|------|------|
| $\leq 23$ months | 99.5 | 97.9 | 97.2 | 90.4 | 97.9   | 97.2   | 90.4   | 92.3 | 86.8 |



goal in the WHO's South-East Asia Region by 2020. National measles control activities have been accelerated since 2004, and all recommended strategies for measles elimination and rubella/congenital rubella syndrome control have already been implemented. Valid measles-rubella vaccination coverage was 87.5% and 92.3% among  $\leq 12$  and  $\leq 23$  months old children respectively according to EPI CES 2016. Coverage of the first dose (MCV1) of measles vaccine was estimated to increase from 64% in 2001 to 87.5% in 2016 among  $\leq 12$  months old children. Coverage of the first dose (MCV1) of Measles Vaccine was estimated to increase from 78% in 2005 to 92.3% in 2016 among  $\leq 23$  months old children.

### Valid Vaccination Coverage within the Age of 12 Months

- Improvement in valid vaccination coverage within the age of 12 months is the ultimate goal of the EPI. The program has set a target of achieving 95% vaccination coverage nationally and at least 90% in each district.
- Figure 4.2.2 indicates the increasing trend in the coverage from 2001 to 2019. A remarkable increase in vaccination coverage has occurred in the last 18 years. It increased up to 31.9 percentage points: from 52% in 2001 to 83.9% in 2019.

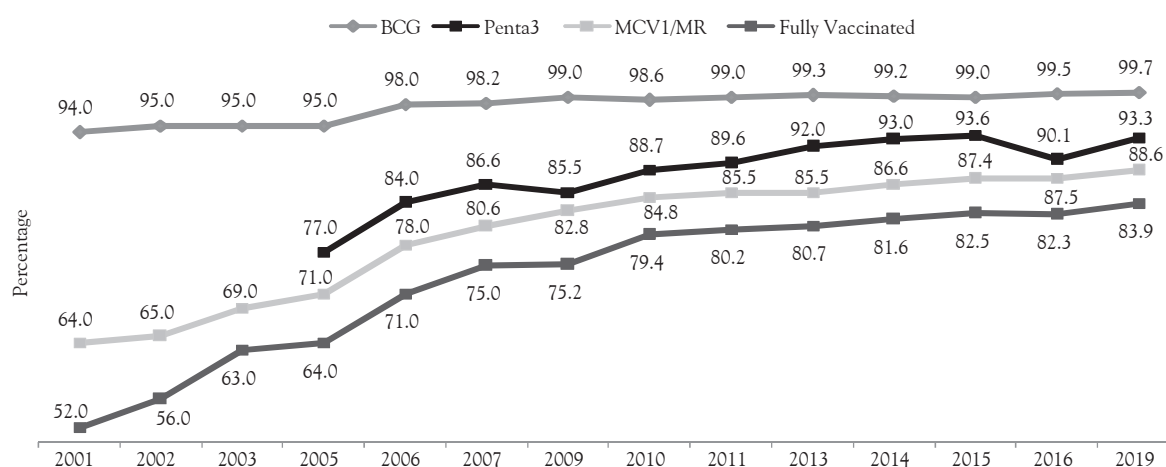
- The analysis of trend indicates that gradual improvement in BCG, Penta3, and MR coverage attributed to the continuous improvement in the full valid vaccination coverage.

**Valid Full Vaccination Coverage by the Age of 12 Months:** Overall, by the age of 12 months, 83.9% of the children country-wide received all the scheduled vaccines, following the EPI-recommended minimal age for administration and valid intervals between doses. Valid BCG coverage at 99.7% was the same as at 23 months, and the Penta1 and OPV1 coverages were also almost the same. Among all the antigens, valid MR1 coverage was the lowest with 88.6%. By area, coverage of MR1 was lower in urban areas (84.6%) than that in rural areas (89.5%). The urban-rural analysis shows that rural children were more likely to receive valid doses (85.0%) compared to their urban counterparts with 79.2% (Table 4.2.5).

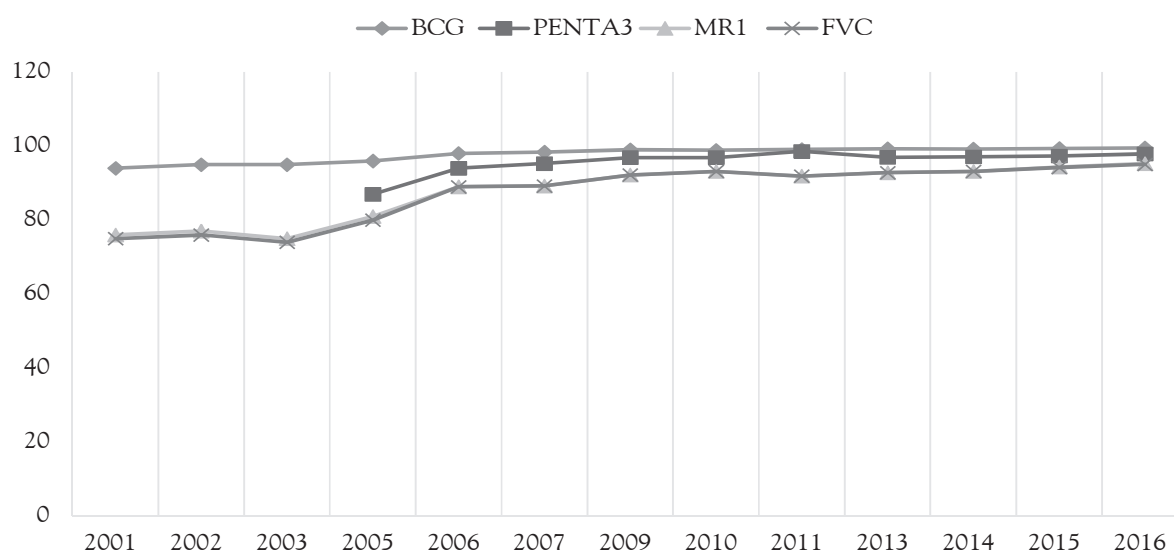
**Valid Full Vaccination Coverage by the Age of 12 Months:** Barisal division had the highest valid full vaccination coverage (91.8%) while Rajshahi division attained the second highest position with 86.0% coverage. The valid vaccination coverage was 85.8% in Sylhet, 84.8% in Khulna, 83.9% in Rangpur, 82.4% in Dhaka, 82.2% in Chattogram, and 80.4% in Mymensingh division (Figure 4.2.4). The coverage analysis and computation of valid coverage show that low drop-out rates as well as the act of administering higher valid doses contributed to the higher valid vaccination coverage.

**Table 4.2.5.** Valid vaccination coverage by age of 12 months at the national, rural and urban areas in EPI CES 2019

| Area     | BCG  | OPV1 | OPV2 | OPV3 | Penta1 | Penta2 | Penta3 | MR1  | PCV1 | PCV2 | PCV3 | FVC  |
|----------|------|------|------|------|--------|--------|--------|------|------|------|------|------|
| Urban    | 99.4 | 99.4 | 98.6 | 90.1 | 99.4   | 98.6   | 90.1   | 84.6 | 99.4 | 98.6 | 89.4 | 79.2 |
| Rural    | 99.8 | 99.8 | 99.4 | 94.0 | 99.8   | 99.4   | 94.0   | 89.5 | 99.8 | 99.4 | 93.2 | 85.0 |
| National | 99.7 | 99.7 | 99.2 | 93.3 | 99.7   | 99.2   | 93.3   | 88.6 | 99.7 | 99.2 | 92.5 | 83.9 |



**Figure 4.2.2.** Annual trend of national-level valid vaccination coverage among children of  $\leq 12$  months (2001-2019)



**Figure 4.2.3.** Annual trend of national-level valid vaccination coverage among children of  $\leq 23$  months (2001-2016) [EPI CES, 2016]

### Valid Full Vaccination Coverage by the Age of 12 Months

Among all the city corporations, Rajshahi City Corporation achieved the highest coverage (92.4%). The lowest coverage was observed in Sylhet City Corporation (SCC) with 63.8%. Valid coverages in all municipalities, on an average, were 82.9% (Figure 4.2.5)

### Valid Vaccination Coverage within the Age of 23 Months

Valid coverage was defined as vaccines received by following the EPI-recommended age and dose-interval for each antigen.

Figure 4.2.3 shows the valid vaccination coverage among children aged  $\leq 23$  months since 2005. The Figure portrays a gradual

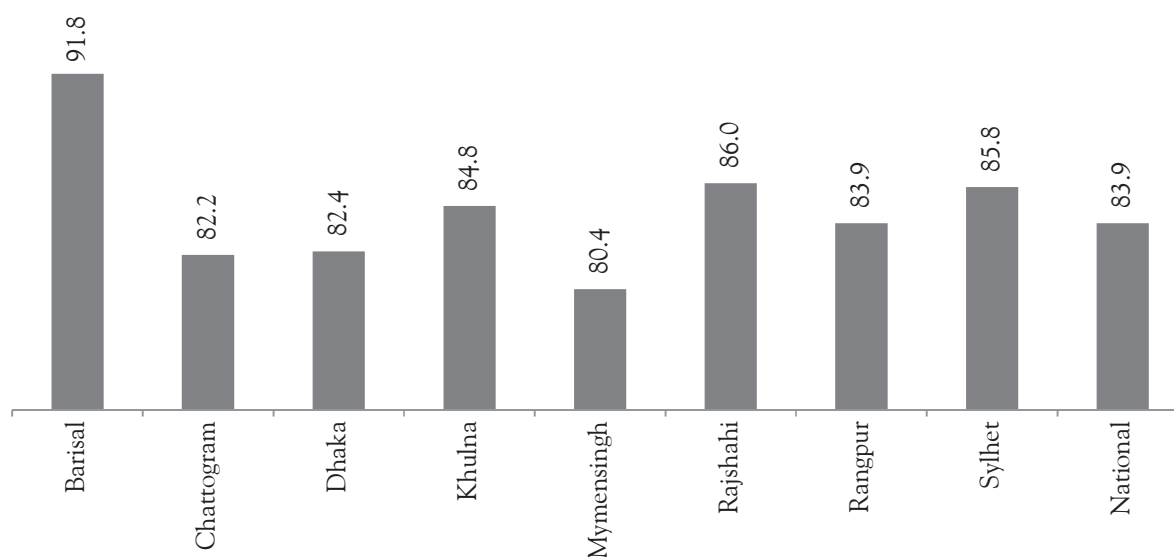
improvement in valid vaccination coverage, which increased by 17.8 percentage points: from 69% in 2005 to 86.8% in 2016 (EPI CES, 2016).

Nationally, 86.8% of children received all the scheduled doses of all antigens, with BCG coverage being at 99.5%. Penta1 coverage was 97.9%, Penta2 97.2%, and Penta3 90.4%. Moreover, MR1 coverage (92.3%) was

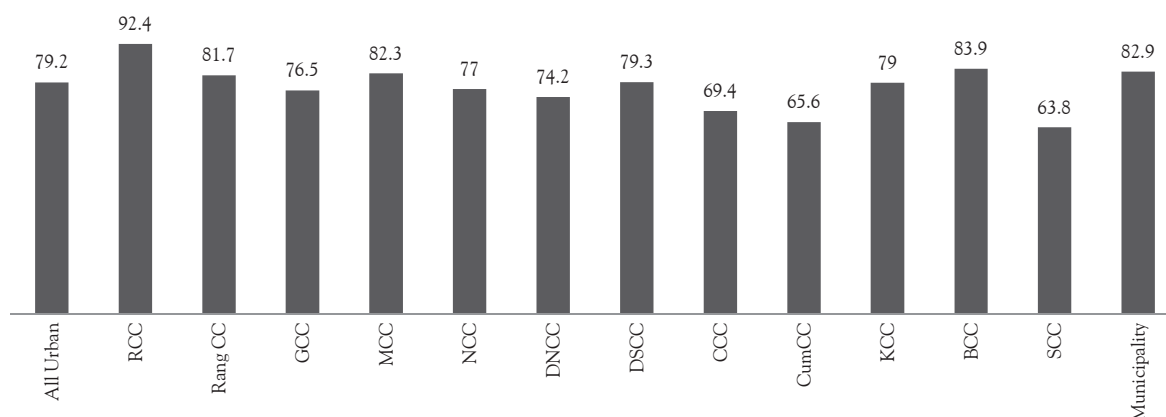
revealed to be 7.2 percentage points lower than BCG (99.5%).

### Success

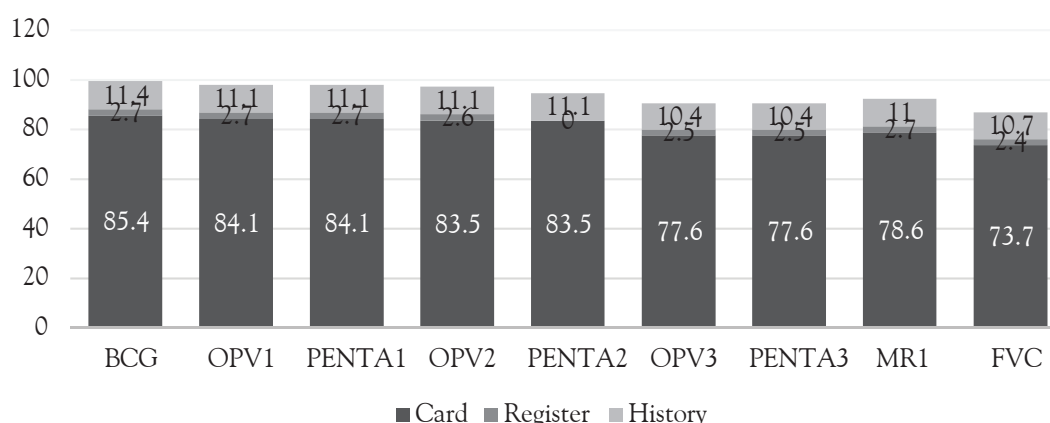
1. Maintain polio-free status:
  - > Bangladesh received polio-free certificate, with 11 other countries of South East Asia in March 2014.



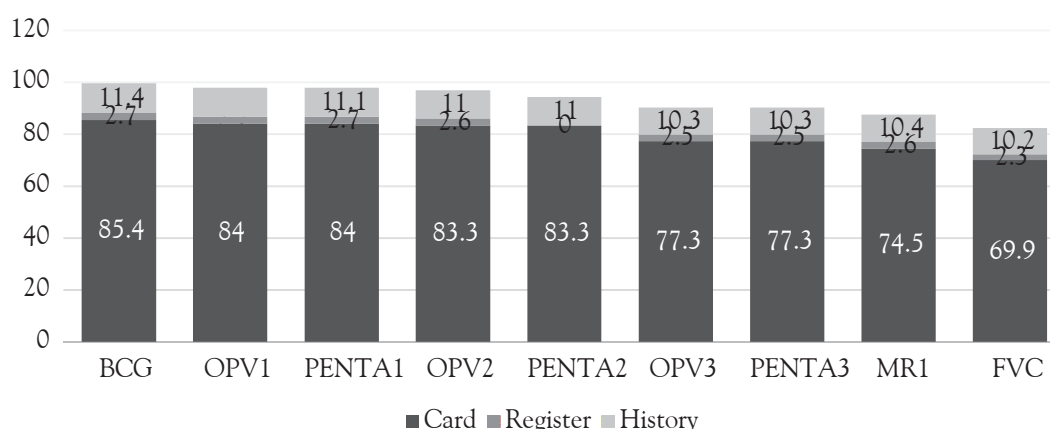
**Figure 4.2.4** Valid full vaccination coverage by the age of 12 months by division and national



**Figure 4.2.5.** National valid vaccination coverage rate among 12 months for full vaccination by city corporation (See the List of Acronyms for the abbreviations)



**Figure 4.2.6.** National-level valid vaccination coverage within the age of 23 months as revealed from cards, registers, and history in 2016 (EPI CES, 2016)



**Figure 4.2.7.** National-level valid vaccination coverage within the age of 12 months as revealed from cards, registers, and history in 2016

- > No wild polio or cVDPV (circulating Vaccine-derived Polio Virus) since 2006
- > Inactivated polio vaccine (IPV) introduced in March 2015 and shifted to fIPV (0.1 mL) from November 2017
- > The program switched from tOPV to bOPV in April 2016
- > Environmental surveillance commenced
- 2. Achieved the maternal and neonatal tetanus (MNT) elimination status in 2008
- 3. Bangladesh achieved Millennium Development Goal (MDG 4) in 2010
- 4. Achieved GAVI Alliance Award in 2009 and 2012 for outstanding performance improving the child immunization status
- 5. Achieved 99.5% coverage of BCG vaccination in 2016
- 6. Bangladesh received the polio eradication certificate from WHO on 27 March 2014
- 7. HPV demonstration Round 2 completed in 2017
- 8. Long-range Vaccine Carrier with Chilled Icepack- piloting completed in Khaliaghuri upazila of Netrakona district



**Figure 4.2.8.** Hon'ble Prime Minister Sheikh Hasina receiving Vaccine Hero Award

9. Introduction of EPI tracker in Maulvibazar and Dhaka South City Corporation area in 2019
10. Achieved goal of the control of Rubella and Congenital Rubella Syndrome (CRS) in 2018
11. Bangladesh has achieved hepatitis B control target of 2020 within July 2019

### **Vaccine Hero Award**

Hon'ble Prime Minister Sheikh Hasina received Vaccine Hero Award from Ngozi Okonjo-Iweala, Board Chair of Global Alliance for Vaccines and Immunization (GAVI), at a conference room of the UN Headquarters in New York on 23 September 2019. The award was conferred upon her in recognition of Bangladesh's outstanding success in vaccination.

# Communicable Disease Control

## Changing pattern of the burden of diseases

General objective of the Communicable Disease Control Program under the DGHS is to control/eliminate communicable diseases in Bangladesh. The specific objectives of the program are to: (i) bring malarial deaths to only 5 per year within 2019 and to zero within 2022, and achieve elimination in phases within 2025; (ii) prevent and control *Aedes* mosquito-transmitted viral diseases, including dengue, chikungunya, and Zika; (iii) achieve filaria-free status in Bangladesh by 2022; (iv) reduce the prevalence of STH among <15 years old children to 5% within 2022; (v) eliminate kala-azar by 2017 and achieve kala-azar-free Bangladesh by 2022; (vi) eliminate rabies within 2022; (vii) strengthen disease surveillance and capacity for detection, management, prevention, and control of other various communicable diseases; and (viii) prevent international spread of diseases and timely detect, assess, notify, and report events and respond to public health emergency of international concern (PHEIC).

The following sections highlight the control and management efforts of important communicable diseases, including some emerging and re-emerging diseases during the reporting period.

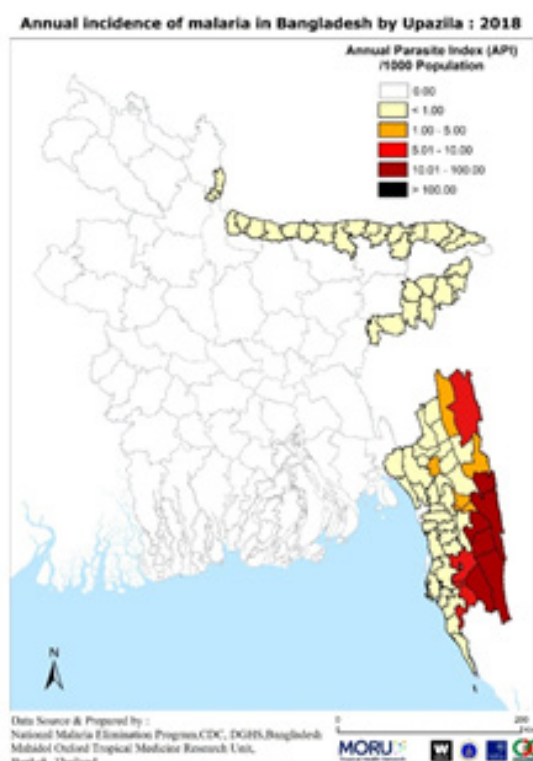
### Malaria

As in other tropical countries of the region, malaria continued to be one of the major public health problems in Bangladesh,

especially in the districts bordering with eastern states of India and part of Myanmar. Out of 64 districts in the country, malaria is endemic in 13 districts with variable transmission potentials (high, moderate, and low). In 2018, Bangladesh had an estimated population of 166 million with about 18.1 million people living in malaria-endemic districts (~10.9% of the total population). Historically, over 90% of the total cases are reported from three districts of the Chattogram Hill Tracts (CHT): Rangamati, Khagrachhari, and Bandarban, and the coastal district Cox's Bazar. The endemicity of malaria is shown in Figure 4.3.1.

Former National Malaria Control Program has been restructured in the 4th Health, Population and Nutrition Sector Program (4th HPNSP) in 2017. Now it has two sub-components: Malaria Elimination and *Aedes*-transmitted Disease Control Program. National Malaria Elimination Program is responsible for implementation of malaria control and elimination interventions under the Communicable Disease Control unit of the DGHS. It is one of the oldest programs in the health sector of the country. In the 1960s, Malaria Eradication Program was started under the Malaria Eradication Board in former East Pakistan. After the Liberation, the Eradication Program was merged with Primary Healthcare (PHC) and converted into Malaria Control Program in 1977. Bangladesh adopted the Revised Malaria





**Figure 4.3.1.** Thirteen malaria-endemic districts of Bangladesh

Control Strategies (RMCS) in the 1990s and, later on in 1998, the Roll Back Malaria (RBM) Initiative of WHO. Since 2007, activities of the National Program have been strengthened and accelerated with funding support from Round 6 and 9 as well as the New Funding Model (NFM), New Funding Request (NFR) grants of the Global Fund to Fight AIDS, Tuberculosis and Malaria (GFATM), the partnership established with BRAC-led NGO Consortium, academic/research institutions, and the private sector.

Considering the lessons learnt from successful implementation of malaria control efforts during the past decade and the recommendations of the recent program evaluation (3rd Joint Monitoring Mission), the program has adopted the phased elimination strategy. The National Strategic

### Malaria Situation in 2017

- Risk Population-18.09million
- Endemic Districts-13 (71 upazilas)
- Total Cases: 10,523 (64% reduction from 2017)
- pf-81%, pv-16%, Mixed-3%
- Death-7
- API-0.58/1,000 population
- ABER-7.19%
- Test Positivity Rate-0.81%

Plan for Malaria Elimination 2017-2021 (NSP 2017-2021) has been developed with the vision of “A Malaria-free Bangladesh by 2030”, aligning with the “Global Technical Strategy for Malaria 2016-2030.” The goal of the NSP 2017-2021 is “to ensure that Bangladesh is on track to eliminate malaria by 2030—contributing towards country development and the Sustainable Development Goals.” The following five strategic objectives have been set:

1. Reduce Annual Parasite Incidence (API) in the 13 endemic districts to less than 0.46 by 2021
2. Interrupt the transmission of malaria in 8 of the country’s 13 endemic districts by 2021
3. Ensure that the remaining 51 districts are free from malaria transmission by 2021

4. Prevent the re-emergence of malaria in the districts where transmission has been interrupted
5. Prevent the emergence of *Plasmodium falciparum* resistant to artemisinin-based combination therapies (ACT) in Bangladesh.

Over the next five years, Bangladesh aims to eliminate malaria in less-endemic areas, while accelerating control efforts in more-endemic areas. In the post-2025 period, it is expected that all areas will either be targeted for elimination of malaria, or for prevention of re-emergence so that, by 2030, Bangladesh will be malaria-free. Figure 4.3.2 illustrates the phase-wise targeted districts for elimination.

The key interventions of the programs are early case detection, effective management, prevention, and entomological surveillance, along with expanding research for innovation and improved delivery of services and strengthening the enabling environment as supporting measures.

The following groups of people are considered high-risk population for malaria infection:

- ◆ Jhum cultivators, wood-cutters, forest-goers, and seasonal agricultural laborers
- ◆ Settlers, refugees, and mobile population

- ◆ Deployed personnel of the Defense Services as well as formal and informal workers in endemic areas
- ◆ Formal and informal cross-border migrant workers
- ◆ Travelers and pilgrims from non-endemic areas
- ◆ People residing in non-endemic areas for a long time and returning home in endemic areas
- ◆ Young children, particularly under-5 children and pregnant women.

Care services for malaria are provided free of charge at the door-step of the populations at risk. Since the beginning of the accelerated efforts in program implementation in collaboration with NGO consortium, the national program has achieved a huge success in terms of reducing morbidity and mortality, and a steady decline is noted during the 2008–2013 period. However, there was a regional upsurge in 2014, and Bangladesh was not immune to that, particularly for favorable meteorological conditions during the monsoon period of that year. Due to that sudden upsurge, caseload increased by 114% compared to the previous year (from 26,891 to 57,480 cases). Malarial mortality increased 3-folds in 2014. The program took

| Districts       | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 |
|-----------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| 3 <sup>a</sup>  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| 2 <sup>b</sup>  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| 8 <sup>c</sup>  |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| 51 <sup>d</sup> |      |      |      |      |      |      |      |      |      |      |      |      |      |      |

Key:

|                                                |
|------------------------------------------------|
| Transmission reduction - API >1 per 1,000      |
| Elimination - API <1 per 1,000                 |
| Prevention of reintroduction - API 0 per 1,000 |
| Status to be determined                        |

(a). 3 CHT districts: Khagrachhari, Rangamati, Bandarban.  
 (b). Chattogram and Cox's Bazar  
 (c). Sherpur, Mymensingh, Netrakona, Kurigram, Sylhet, Habiganj, Sunamganj, Maulvibazar.  
 (d). Remaining districts

**Figure 4.3.2.** Projected number of malaria-endemic districts by phase and year



various initiatives, which resulted in the reduction of both cases and deaths since 2015. Overall, there were respectively 88% and 95% reduction in morbidity and mortality in 2018 compared to 2008. Nearly 99% case reduction has been achieved in eight elimination-targeted districts during the same period. Over 80% of the total malaria cases were diagnosed and treated in the community. Thanks go to the availability and easy access to malaria services in the community, only 2.3% of the total positive cases were severe in 2018. Table 4.3.1 summarizes year-wise epidemiological data for the period of 2000–2018.

With the exception of 2014 when the numbers of both cases and deaths increased due to a sudden upsurge, a significant progress in the decline of malaria cases and deaths has been achieved in the country over the last decade. Figure 4.3.3 shows the epidemiological trend of the malaria cases and deaths during 2007–2018.

The vast majority of malaria cases are caused by *Plasmodium falciparum* (Pf). However, a steady increase in *Plasmodium vivax* (Pv) cases was observed in the last few years. A total of 10,523 malaria cases were reported in

**Table 4.3.1.** Summary of year-wise epidemiological data (2000–2018) from the malaria-endemic districts

| Year | Positive cases |                      | <i>P. falciparum</i> |      | <i>P. vivax</i> |      | Death |                      |
|------|----------------|----------------------|----------------------|------|-----------------|------|-------|----------------------|
|      | No.            | per 1,000 population | No.                  | %    | No.             | %    | No.   | per 1,000 population |
| 2000 | 54,223         | 5.63                 | 39,272               | 72.4 | 14,951          | 27.6 | 478   | 0.049                |
| 2001 | 54,216         | 5.55                 | 39,274               | 72.4 | 14,942          | 27.6 | 490   | 0.049                |
| 2002 | 62,269         | 6.23                 | 46,418               | 74.5 | 15,851          | 25.5 | 588   | 0.058                |
| 2003 | 54,654         | 5.40                 | 41,356               | 75.7 | 13,298          | 24.3 | 577   | 0.056                |
| 2004 | 58,894         | 5.67                 | 46,402               | 78.8 | 12,492          | 21.2 | 535   | 0.052                |
| 2005 | 48,121         | 4.56                 | 37,679               | 78.3 | 10,442          | 21.7 | 501   | 0.047                |
| 2006 | 32,857         | 3.06                 | 24,828               | 75.6 | 8,029           | 24.4 | 307   | 0.029                |
| 2007 | 59,857         | 5.46                 | 46,791               | 78.2 | 13,066          | 21.8 | 228   | 0.021                |
| 2008 | 84,690         | 7.73                 | 70,281               | 83.0 | 14,409          | 17.0 | 154   | 0.014                |
| 2009 | 63,873         | 5.83                 | 57,020               | 89.3 | 6,853           | 10.7 | 47    | 0.004                |
| 2010 | 55,873         | 5.10                 | 52,049               | 93.2 | 3,824           | 6.8  | 37    | 0.003                |
| 2011 | 51,773         | 3.91                 | 49,194               | 95.0 | 2,579           | 5.0  | 36    | 0.003                |
| 2012 | 29,518         | 2.23                 | 27,819               | 94.2 | 1,699           | 5.8  | 11    | 0.001                |
| 2013 | 26,891         | 2.03                 | 25,908               | 96.3 | 983             | 5.8  | 15    | 0.001                |
| 2014 | 57,480         | 4.34                 | 54,132               | 94.2 | 3,348           | 5.8  | 45    | 0.003                |
| 2015 | 39,719         | 3.00                 | 35,708               | 89.9 | 4,011           | 10.1 | 9     | 0.0007               |
| 2016 | 27,737         | 1.58                 | 24,431               | 88.1 | 3,306           | 11.9 | 17    | 0.0009               |
| 2017 | 29,247         | 1.64                 | 23,328               | 84.8 | 4,444           | 15.2 | 13    | 0.0007               |
| 2018 | 10,523         | 0.58                 | 8,848                | 84.1 | 1,675           | 15.9 | 7     | 0.0004               |

2018, and falciparum malaria accounted for 84% of them (including mixed cases). Figure 4.3.4 illustrates the species-wise proportion of cases over the last decade.

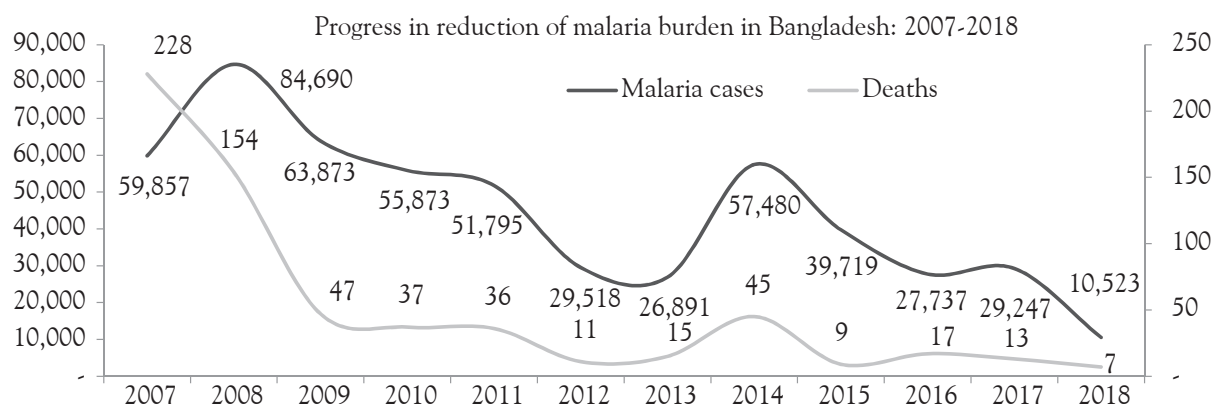
Between 2007 and 2009, the malaria case-fatality rate (CFR) dropped by 81% from 3.81 to 0.74 deaths per 1,000 cases. Since 2009, CFR has fluctuated between 0.78 and 0.23. Case-fatality rate in 2018 was 0.67. The changes in CFR over the period are presented in Figure 4.3.5.

The monsoon (May–September) is the peak period for malaria transmission. Figure

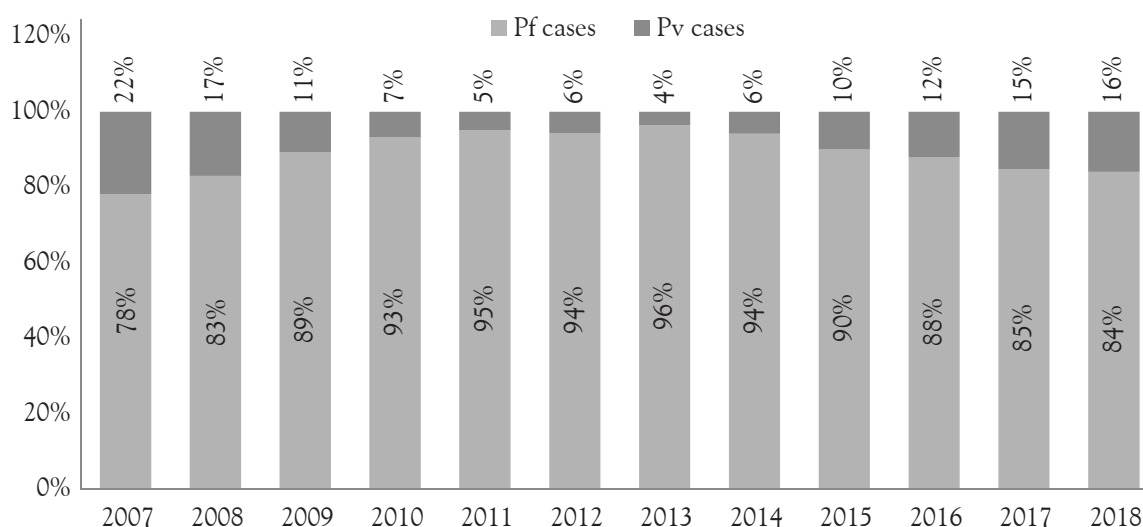
4.3.6 illustrates the seasonality of malaria transmission, which rises rapidly in June and reaches its peak between June and August.

Out of 13 malaria-endemic districts, four most-endemic districts (Bandarban, Khagrachhari, Rangamati, and Cox's Bazar) have reported nearly 98% of total cases in 2018. Two out of the total 7 death cases were among the populations from those districts.

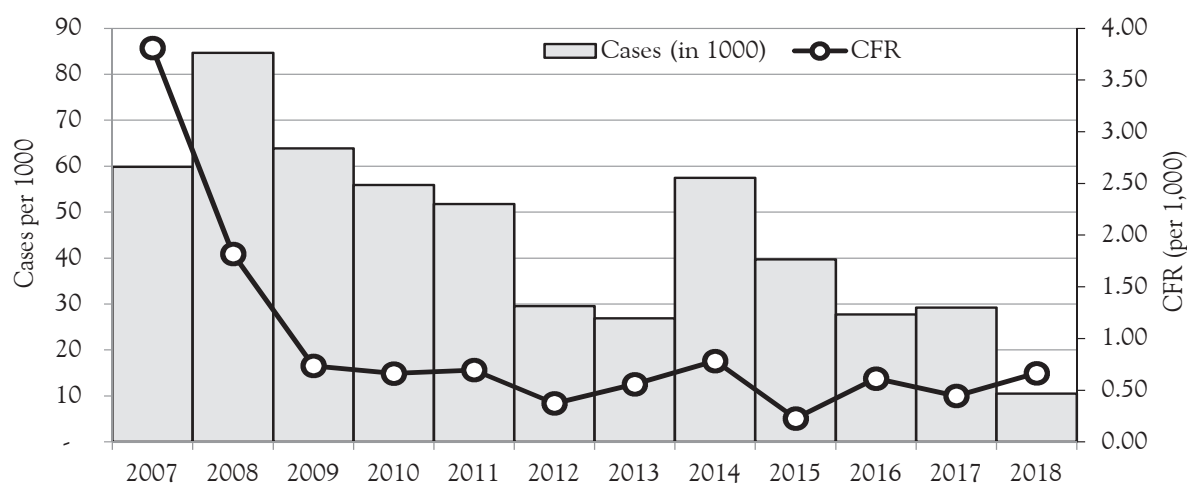
Figure 4.3.7 gives an idea about the share of total malaria burdens in endemic districts in 2018.



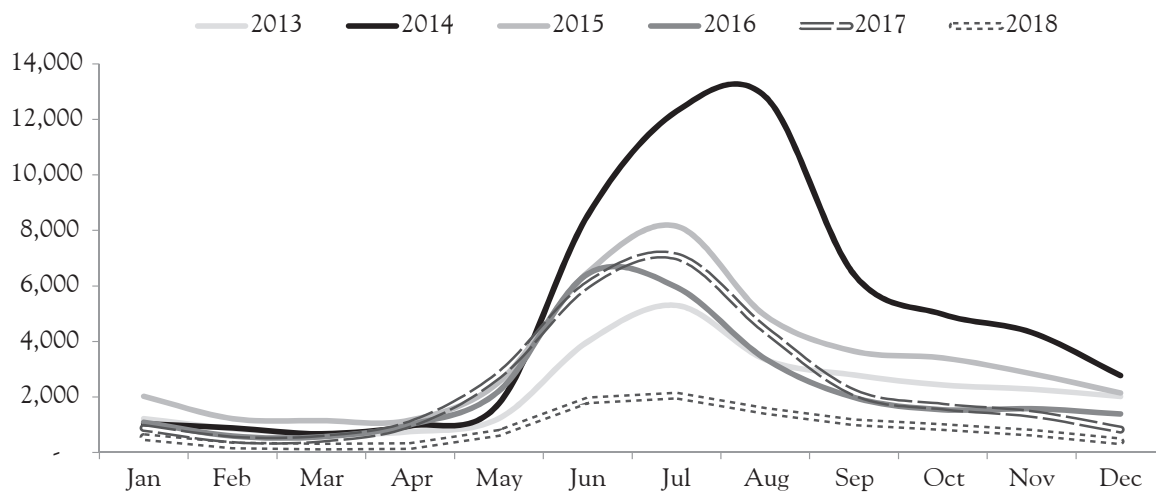
**Figure 4.3.3.** Epidemiological trend of malaria cases and deaths during 2007-2018



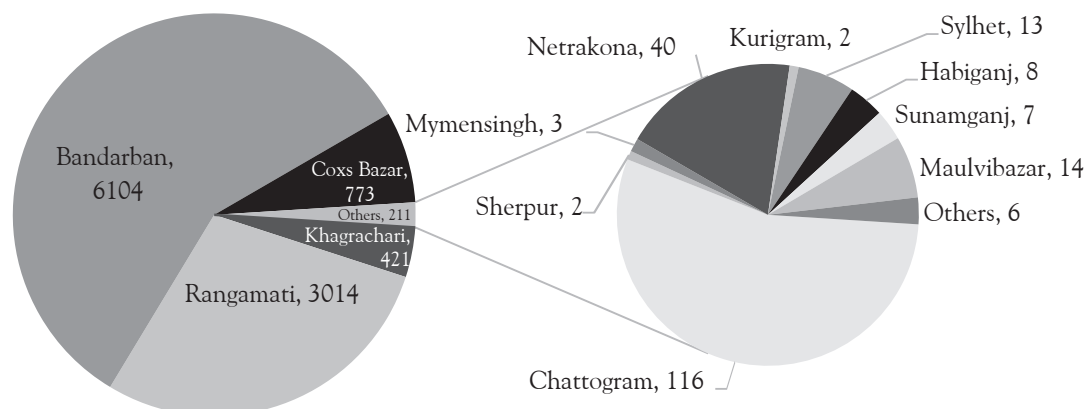
**Figure 4.3.4.** Proportion of Pf and Pv cases during 2007–2018



**Figure 4.3.5.** Changes in the case-fatality rates (CFR) during 2007-2018



**Figure 4.3.6.** Seasonality of malaria transmission (2013-2018)



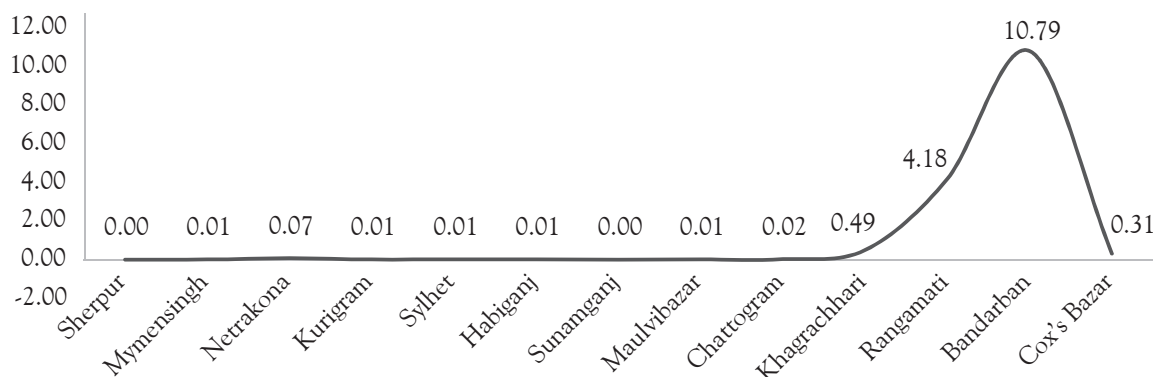
**Figure 4.3.7.** Distribution of malaria burden among 13 endemic districts in 2018

Malaria is becoming an increasingly focal disease in Bangladesh. In 2018, just two of the 13 endemic districts had an Annual Parasite Incidence greater than 1 compared to all 13 in 2008. The three districts of the Chattogram Hill Tracts (Khagrachhari, Rangamati, and Bandarban) together accounted for nearly 91% of the total confirmed malaria cases detected in the 13 endemic districts. The population of these three districts constitutes only around 12% of the total population at risk. Eight elimination-targeted districts reported only 89 cases in 2018. All those cases were investigated to identify the indigenous cases for further investigation and responses. The incidence rates in 13 malaria-endemic districts in 2018 are illustrated in Figure 4.3.8.

Distributing long-lasting insecticidal nets (LLIN) among the at-risk population is one of the key interventions in malaria prevention. About 10.69 million LLINs were distributed free of charge among the beneficiaries during the period 2008-2018. Among them, 1.56 million LLINs were distributed in 2018 only. A new initiative of distributing additional LLINs among higher-risk groups (Jhum cultivators, armed forces, pregnant women, refugees, etc.) has been started in 2017. A total of 370,362 LLINs were distributed among the population of higher-risk groups in 2018.

As part of Advocacy, Communication and Social Mobilization (ACSM) activities, various initiatives are undertaken by the program to increase awareness among the mass population. Awareness programs are broadcast through local radio in Bangla, Chakma and Marma languages. Awareness-raising messages on malaria prevention and treatment are aired (message scrolling) in local dish/video channels in three districts of CHT. Regular community drive and *uthan boithok* (court-yard meetings) with the participation of local people are conducted. Advocacy meetings are conducted at all levels. World Malaria Day is observed on 25 April each year. A malaria hotline number (01787-691370) has been introduced in 2017 for communication on matters relating to malaria.

During the ongoing catastrophic humanitarian crisis in Cox's Bazar due to influx of Rohingya refugees, the national program was highly vigilant in implementing malaria activities among Forcibly Displaced Myanmar Nationals (FDMN). The national program executed its operation in Health Camps for FDMN through mainstream health operation of the Director General of Health Services in coordination with partner NGOs and other non-governmental health agencies. Suspected malaria cases among FDMN populations



**Figure 4.3.8.** Malaria incidence in 13 endemic districts in 2018

are being diagnosed, and 14 positive cases were treated accordingly. Adequate LLINs, antimalarial and diagnostic kits are ensured for them. Regular entomological surveys are conducted in camp areas. Additional human resources were recruited through partners, and awareness messages are disseminated in their native language.

## Filariasis

Lymphatic filariasis (LF) is a parasitic infectious disease caused by three species of nematodes: *Wuchereria bancrofti*, *B. malayi*, and *B. timori*. The infection is transmitted from man to man through various species of mosquitoes, and *Culex quinquefasciatus* is the most predominant vector. The adult filarial worms live in lymphatics of man. The female worms produce tiny worms called microfilariae (Mf), and each adult worm produces millions of worms during its reproductive life-span. The microfilariae are ingested, along with blood meal, by vector mosquitoes. The transmission dynamics are such that thousands of infective mosquito-bites are necessary to produce an infection. Lymphatic filariasis (LF) infection is diagnosed at the community level through blood testing for Mf (night-time for nocturnal periodic forms) or antigen.

The adult worms in infected people produce inflammatory reactions and precipitate clinical course of the disease. It includes repeated acute adenolymphangitis (ADL) episodes, reversible chronic disease manifestations, such as lymphedema and hydrocele, which progress into irreversible stages and elephantiasis. Often, secondary bacterial infections play a major role in acute episodes and chronic disease progression. Chronic disease symptoms start appearing in adolescents, increase in frequency and severity with age, reaching

maximum levels in older adults and the elderly. The disease is diagnosed through clinical examination.

Diethylcarbamazine (DEC) and ivermectin are the well-known antifilarial drugs used for individual treatment. Either of the drugs combined with Albendazole (ALB) is widely used at the community level. Combined administration of all the three drugs is recommended recently for certain situations.

## Filariasis Elimination Program

Bangladesh is one of the major LF-endemic countries in the South-East Asia region. The parasite and vector species are *W. bancrofti* and *C. quinquefasciatus*. Epidemiological investigations showed that LF burden is very high in the northern districts of Bangladesh.

## LF elimination strategy

The MOHFW, Bangladesh, expressed its strong commitment to the World Health Assembly resolution on elimination of LF. As the first step towards launching the program, a national task force headed by the Secretary, MOHFW, was established in 2001. A technical committee formulated the goals and objectives of the program. The task force recommended converting the then National Filaria Control Program into the National Program on Elimination of Lymphatic Filariasis (NPELF) (it was later expanded to include STH control activities and re-designated as National Filariasis Elimination and STH Control Program).

To understand the distribution of LF and identify the endemic districts that required Mass Drug Administration (MDA), the program reviewed the medical and health

records, health reports from districts, filaria survey data, and research studies on LF and its vector. While these data provided important insights, Ag surveys were conducted in different districts of the country, after launching the LF elimination program, to identify the endemic districts. After review of all the data together, 30 of the 64 districts were declared non-endemic, 15 as low-endemic, and 19 districts as endemic (Table 4.3.2). Using the data, the LF map of Bangladesh was prepared (Figure 4.3.9)

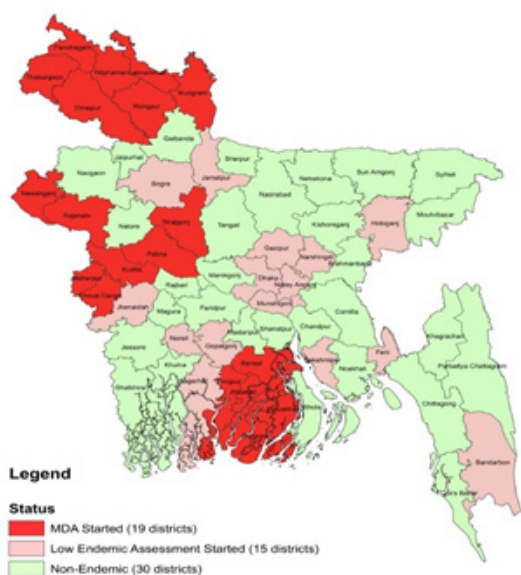
### MDA program

The first ever MDA was implemented in 2001, soon after launching the Global Program to Eliminate Lymphatic Filariasis (GPELF), in 2000. The district is designated as intervention unit (IU). In the first year of operations, the program was implemented in only one district (Panchgarh) on a pilot basis. The program was extended to all the 19 endemic districts in a phased manner. By 2008, the program was in place in all 19 districts. By the end of 2014, all districts received at least three rounds of MDA, and 10 received >5 rounds.

**Table 4.3.2.** Classification of districts into endemic and non-endemic categories

| Division   | Non-endemic districts                                                                                     | Low-endemic districts                                                   | Endemic districts                                                            |
|------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Barisal    | Bhola                                                                                                     |                                                                         | Barguna, Barisal, Jhalokathi, Patuakhali, Pirojpur                           |
| Chattogram | Brahmanbaria, Chandpur, Chattogram, Comilla, Cox's Bazar, Khagrachhari, Noakhali, Rangamati               | Bandarban, Feni, Lakshmipur                                             | -                                                                            |
| Dhaka      | Faridpur, Kishoreganj, Madaripur, Manikganj, Mymensingh, Netrakona, Rajbari, Shariatpur, Sherpur, Tangail | Dhaka, Gazipur, Gopalganj, Jamalpur, Munshiganj, Narayanganj, Narsingdi | -                                                                            |
| Khulna     | Jessore, Khulna, Magura, Satkhira                                                                         | Bagerhat, Jhenaidah, Narail                                             | Chuadanga, Kushtia, Meherpur                                                 |
| Rajshahi   | Joypurhat, Naogaon, Natore                                                                                | Bogra                                                                   | Chapainawabganj, Pabna, Rajshahi, Sirajganj                                  |
| Sylhet     | Maulvibazar, Sunamganj, Sylhet                                                                            | Habiganj                                                                |                                                                              |
| Rangpur    | Gaibandha                                                                                                 |                                                                         | Dinajpur, Kurigram, Lalmonirhat, Nilphamari, Panchagarh, Rangpur, Thakurgaon |

Note: Districts of the new division Mymensingh are shown under undivided Dhaka division



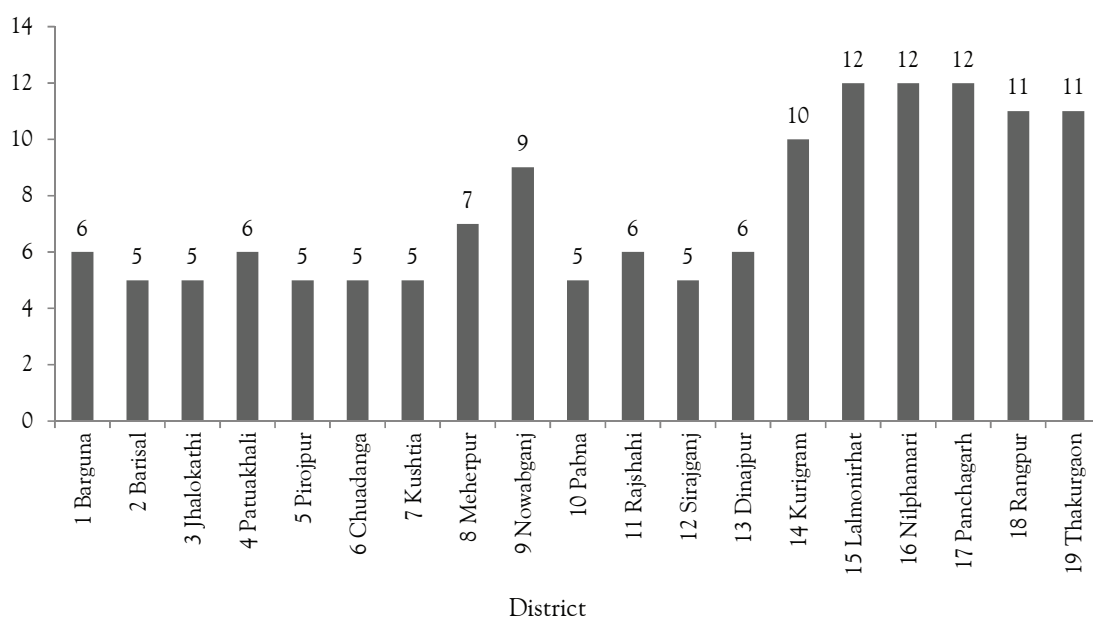
**Figure 4.3.9.** Present status of 34 LF-endemic districts

By the end of 2014, a total of 262.14 million treatments were given and 218.60 million treatments taken by a population of 36 million. On an average, a person in the targeted population received 5.74 treatments. Three to 12 MDAs were implemented in

different districts during 2001-2014, with the program treatment coverage ranging from 58% to 100%; the overall coverage was 83.39% (Figure 4.3.10.). With steady and uninterrupted progress of MDA and resulting decline in Mf prevalence, the MDA was stopped in 5 IUs in 2011. By 2014, MDA was stopped in 18 of 19 IUs, the only exception being the IU Rangpur. In 2016, IU Rangpur, a highly-endemic district in northern region, implemented its final round of MDA. The survey treatment coverage suggested that the drug distribution and people's compliance with treatment were good, and the epidemiological treatment coverage of 65% in a majority of districts surveyed were more than effective.

### Sentinel and spot-check site surveys

The effectiveness of MDA and its impact on transmission has been regularly assessed in 2 sentinel sites and 1-2 spot-check site(s) per evaluation unit (EU). In both sentinel and spot-check sites, Mf surveys were done to assess the impact of MDA. Mostly,



**Figure 4.3.10.** Number of MDAs implemented in different endemic districts, 2001-2016



500–1,000 people per sentinel or spot-check site were sampled to assess the prevalence of Mf. The Mf surveys were carried out during baseline period, after 23 rounds of MDA and, finally, after >5-6 rounds of MDA. M&E activities were made intensive in the high-endemic northern districts of Panchgarh and Thakurgaon. The Mf rate of 1.0% or less is considered the threshold infection level, at which stopping the MDA could be considered. The baseline Mf rate ranged from 1.0% to 16.0%. Wherever interim surveys showed >1.0% Mf prevalence, MDA was continued. Some districts required as many as 12 MDAs to reduce the Mf prevalence to <1.0%. By 2014, all 19 endemic districts achieved the threshold level of <1.0% Mf rate. This allowed the IUs take the next step—Transabdominal sonography (TAS) towards stopping MDA.

### Transabdominal sonography

In the IUs where the Mf prevalence was reduced to <1.0% in sentinel and spot-check sites, a stopping MDA survey or TAS 1 was conducted. The purpose of TAS 1 was to determine if the transmission levels declined below the threshold level. Seven of the 19 IUs were large with a population of 2.32 million to 3.10 million (and split into 2 EUs each), resulting in a total of 26 EUs. By 2014, all 26 EUs became eligible and underwent TAS 1. In each EU, 30 schools were sampled and 1,556 to 1,692 children tested. The critical cut-off value of Ag-positive children was 18-20. In 25 of the 26 EUs, the outcome of TAS 1 was successful—no Ag-positive child was found in 12 EUs, and 1-11 positive child(ren) were found in others, well below the critical cut-off values of 18-20. The only exception was the EU Rangpur B, carved out of Rangpur

IU where the number of Ag-positive children was 27 against the critical cut-off value of 18 (Table 4.3.3).

TAS 2 was conducted 2-3 years after completing TAS 1. By the end of 2016, TAS 2 was completed in 20 EUs. No Ag-positive child was found in 15 EUs, and 1-5 positive child(ren) found in 5 EUs against critical cut-off value of 18-20 (Table 4.3.3).

TAS 3 was conducted 2-3 years after completing TAS 2. As of 2016, TAS 3 was completed in 7 EUs. The outcome of TAS 3 was successful in all 7 EUs – 5 of 7 EUs were completely free from Ag-positive children, and only 2 children were found with Ag in the other 2 EUs (Table 4.3.3).

### Transmission assessment for LF in low-endemic districts

TAS was conducted in 2014 and 2015 in 15 low-endemic districts, which consisted of 22 EUs. In all EUs, the number of children positive for Ag was less than that of critical value. This confirms that these districts do not require MDA, and no-MDA decision for these districts was correct. Mini-TAS was conducted in 4 non-endemic districts (Joypurhat, Naogaon, Faridpur, and Manikganj) that border with the endemic districts to ascertain the status of ongoing transmission. Under mini-TAS, 480 children drawn from 30 schools each were tested. A total of 14,400 children were tested, and none was found positive.

### Morbidity management and disability prevention (MMDP)

An assessment for the number of chronic LF patients in 2013-2016 was done. In 2013-2015, a new assessment of chronic

**Table 4.3.3.** Results of TAS conducted in different EUs of Bangladesh

Summary measures and TAS 1-3 results for the 19 endemic districts (receiving MDA)

| 2011-2018             |             |                       |               |           |           |           |                            |
|-----------------------|-------------|-----------------------|---------------|-----------|-----------|-----------|----------------------------|
| District              | Sample-size | No. of schools tested | Critical      | ICT+ve in | ICT+ve in | ICT+ve in | Total ICT+ve in TSA (11-3) |
|                       |             |                       | cut-off point | TAS 1     | TAS 2     | TAS 3     |                            |
| TAS 1 started in 2011 |             |                       |               |           |           |           |                            |
| Dinajpur-A            | 1692        | 31                    | 20            | 1         | 5         | 1         | 7                          |
| Dinajpur-B            | 1692        | 30                    | 20            | 6         | 5         | 1         | 12                         |
| Meherpur              | 1556        | 30                    | 18            | 0         | 0         | 0         | 0                          |
| Rajshahi-A            | 1692        | 30                    | 20            | 1         | 0         | 0         | 1                          |
| Rajshahi-B            | 1692        | 30                    | 20            | 0         | 4         | 0         | 4                          |
| Patuakhali            | 1692        | 31                    | 20            | 0         | 0         | 0         | 0                          |
| Barguna               | 1692        | 33                    | 20            | 1         | 1         | 0         | 2                          |
| TAS 1 started in 2012 |             |                       |               |           |           |           |                            |
| Sirajganj -A          | 1692        | 30                    | 20            | 1         | 0         | 2         | 3                          |
| Sirajganj -B          | 1692        | 31                    | 20            | 0         | 0         | 0         | 0                          |
| Pabna-A               | 1692        | 31                    | 20            | 0         | 0         | 1         | 1                          |
| Pabna-B               | 1692        | 30                    | 20            | 3         | 0         | 0         | 3                          |
| Kushtia               | 1692        | 30                    | 20            | 0         | 0         | 0         | 0                          |
| Chuadanga             | 1692        | 30                    | 20            | 0         | 0         | 0         | 0                          |
| Pirojpur              | 1692        | 36                    | 20            | 2         | 0         | 1         | 3                          |
| TAS 1 started in 2013 |             |                       |               |           |           |           |                            |
| Chapainowabganj       | 1692        | 30                    | 20            | 4         | 0         | 4         | 8                          |
| Panchagarh            | 1692        | 30                    | 20            | 0         | 0         | 1         | 1                          |
| Thakurgaon            | 1692        | 30                    | 20            | 3         | 0         | 10        | 13                         |
| Barisal – A           | 1692        | 30                    | 20            | 0         | 0         | 1         | 1                          |
| Barisal – B           | 1692        | 30                    | 20            | 0         | 0         | 2         | 2                          |
| Jhalokathi            | 1556        | 30                    | 20            | 0         | 1         | 1         | 2                          |
| TAS 1 Started in 2014 |             |                       |               |           |           |           |                            |
| Nilphamari            | 1692        | 30                    | 20            | 9         | 5         | 7         | 21                         |
| Kurigram – A          | 1692        | 30                    | 20            | 2         | 0         | 0         | 2                          |
| Kurigram - B          | 1692        | 30                    | 20            | 0         | 0         | 0         | 0                          |
| Lalmonirhat           | 1692        | 30                    | 20            | 8         | 4         | 2         | 14                         |
| Rangpur – A           | 1692        | 30                    | 20            | 17        | 11        |           | 28                         |
| Rangpur – B           | 1692        | 30                    | 20            | 33        | 12        |           | 45                         |
| Total                 | 43,720      | 948                   | -             | 91        | 48        | 34        | 173                        |

patients was completed in 19 districts. Three teams—each consisting of a health assistant, a community healthcare provider, and an FWA--were deployed per community clinic area (each covering 1,200 households) to visit the households and take the census of chronic patients. Each team was allocated 400 households, and they completed the census over 10-day period, by visiting approximately 40 households per day. The estimated number of patients in this exercise was 43,708 (Table 4.3.4).

### Census of LF patients in low-endemic districts

The LF disease burden was assessed in historically low-endemic districts, which were shown to be currently non-endemic through TAS conducted in 2014 and 2015. This exercise was done as there were sporadic cases of lymphedema in these districts. The assessment of disease burden was conducted in collaboration with the

**Table 4.3.4.** Number of people identified with chronic LF disease in endemic districts in 2013-2016

| District        | Year of searching | Number of people with lymphedema |              |              | Number of hydrocele patients | Number of patients with other symptoms | Total number of patients |
|-----------------|-------------------|----------------------------------|--------------|--------------|------------------------------|----------------------------------------|--------------------------|
|                 |                   | Male                             | Female       | Total        |                              |                                        |                          |
| Dinajpur        | 2013              | 356                              | 1688         | 2044         | 1021                         | 2                                      | 3067                     |
| Kurigram        | 2013              | 162                              | 1077         | 1239         | 699                          | 20                                     | 1958                     |
| Lalmonirhat     | 2013              | 1077                             | 4306         | 5383         | 1937                         | 0                                      | 7320                     |
| Nilphamari      | 2013              | 820                              | 4655         | 5475         | 1715                         | 50                                     | 7240                     |
| Panchagarh      | 2013              | 224                              | 1117         | 1341         | 1634                         | 5                                      | 2980                     |
| Rangpur         | 2013              | 1743                             | 6789         | 8532         | 2654                         | 13                                     | 11199                    |
| Thakurgaon      | 2013              | 747                              | 2020         | 2767         | 2001                         | 50                                     | 4818                     |
| Barguna         | 2015              | 208                              | 334          | 542          | 466                          | 15                                     | 1023                     |
| Barisal         | 2016              | 124                              | 303          | 427          | 50                           | 1                                      | 478                      |
| Jhalokathi      | 2016              | 222                              | 370          | 592          | 49                           | 0                                      | 641                      |
| Patuakhali      | 2015              | 167                              | 256          | 423          | 202                          | 11                                     | 636                      |
| Pirojpur        | 2016              | 121                              | 193          | 314          | 31                           | 6                                      | 351                      |
| Chuadanga       | 2015              | 23                               | 30           | 53           | 5                            | 1                                      | 59                       |
| Kushtia         | 2015              | 74                               | 71           | 145          | 7                            | 7                                      | 159                      |
| Meherpur        | 2015              | 20                               | 10           | 30           | 0                            | 0                                      | 30                       |
| Chapainowabganj | 2014              | 141                              | 552          | 693          | 201                          | 18                                     | 912                      |
| Pabna           | 2014              | 82                               | 147          | 229          | 67                           | 20                                     | 316                      |
| Rajshahi        | 2014              | 74                               | 135          | 209          | 26                           | 12                                     | 247                      |
| Sirajganj       | 2014              | 101                              | 107          | 208          | 59                           | 7                                      | 274                      |
| <b>Total</b>    |                   | <b>6486</b>                      | <b>24160</b> | <b>30646</b> | <b>12824</b>                 | <b>238</b>                             | <b>43708</b>             |

Centre for Injury Prevention and Research, Bangladesh (CIPRB) with supervision of health inspectors of the community clinics. Close to 700 patients were identified through the activity.

### Availability of treatment

Treatment is available for chronic LF patients at the hospitals of each level –district, sub-district, union and community clinic. Each health-facility level is also equipped to train on lymphedema patient-care.

Community clinics have been chosen to play a major role in the implementation of the MMDP. Health workers at CCs attended lymphedema patients from each community in batches. They taught the patients on how to manage their lymphedema condition and improve quality of life. Each patient was given a copy of the brochure that contained the pictures and explanation of lymphedema management and a morbidity management kit that consisted of soap, gauze, towel, etc.

Hydrocele surgeries are undertaken at all sub-district and district-level health facilities. These facilities have adequate infrastructure, such as operation theaters, equipment, medicine, and commodities. In high-burden northern districts, the hospitals/health centers are instructed to give priority for hydrocele surgeries. Generally, the health workers meet with the hydrocele patients and encourage them to visit the sub-district or district-level health facilities, discuss with doctors and undergo surgery. In addition, medical camps were held in different districts, and groups of hydrocele patients were mobilized to undergo surgery. The district-level hospitals have started reporting the number of hydrocele surgeries to the national program authorities.

So far, about 2,000 affected persons have undergone surgeries.

### Goal, objectives, and strategies of the program

The goal of the program is to sustain the gains of LF elimination as a public health problem and progress towards zero incidence level and contribute to the community and country development and achieving targets of the Sustainable Development Goal 3.

The objectives of the program are the following:

- Secure the validation of elimination of LF as a public health problem by 2020
- Establish post-validation surveillance and response systems to address residual infection and eliminate transmission
- Establish strategy for enhanced post-validation morbidity management and disability prevention (MMDP) to improve quality of life of chronic LF patients
- Integrate post-validation program activities into regular healthcare system
- Promote advocacy, health education, community engagement, and partnerships to advance LF elimination.

Principal activities to achieve the goals and objectives include the following:

- Complete the transmission assessment surveys and MMDP activities in all endemic districts and prepare and submit the dossier for validation of LF elimination as a public health problem by 2020
- Conduct post-validation sentinel site and spot-check site surveys and health center-

based routine surveys and implement response measures to remove residual infection and prevent resurgence and reduce LF incidence to zero level, leading to elimination of transmission

- Implement strategy for post-validation, enhanced lymphedema management and ensure maximum hydrocele surgeries to improve the quality of life of chronic LF patients
- Strengthen healthcare system through training and laboratory networking and improvement and focus on first-line health worker to undertake post-validation surveillance, treatment, and enhanced MMDP activities
- Implement advocacy at the national and sub-national levels and engage communities for sustained surveillance activities, response measures, and vector control
- Expand partnerships for resources, innovation, and operational research to improve the efficiency of program.

The LF program made significant progress during the last 18 years. The country is on course to complete TAS 3 and MMDP in all districts, which will pave the way to prepare and submit the dossier to WHO for validation of LF elimination as a public health problem. Then, sustenance of the status of LF elimination as a public health problem remains a priority for the program in the coming years.

Recent research studies in some endemic countries showed the presence of infection above threshold level in some communities of the districts that passed TAS 3 (dossier-ready districts), indicating that TAS procedures may

not reveal several residual infection foci in a district. This may be because: often the EUs are large with more than a million population and a large number of communities, and only 30-40 schools are sampled in TAS. Although the implications of the residual infection foci are unclear for transmission occurrence, such foci constitute a threat for the programs as they could persist and remain as source of infection for decades. Such foci constitute greater risk in high vector density and highly-efficient vector situations. Therefore, post-validation surveillance activities should be implemented to detect the residual foci and treat the residual infection to sustain the status of 'Lf elimination as a public health problem'. The need for such surveillance has also been highlighted by the WHO. In parallel with post-validation surveillance, the MMDP activities should be continued and intensified to alleviate the condition of the chronic LF patients, improve quality of life, and facilitate rehabilitation.

Further, continuous and incrementally-robust post-validation surveillance and appropriate response measures provide opportunity to reduce incidence of infection to zero levels, leading to elimination of the transmission of LF.

The National strategic plan (2018-2025) focuses on surveillance and response measures to further reduce the infection and incidence, and enhance services to patients to alleviate the suffering due to chronic LF.

### Zoonotic Disease Control Program

National Rabies Elimination Program is the key focus of Zoonotic Disease Control Program that deals with elimination of rabies from Bangladesh.

Rabies, a dreadful viral zoonotic disease is one of the major public health concerns in Bangladesh. Before 2010, more than 2,000 rabies-related deaths were estimated to occur in the country. In more than 95% cases, dog-bite was found to be the single-most important cause of rabies, followed by bites of cat, jackal, etc. There is an estimated 1.6 million dog population in Bangladesh, with over 80% of them being free-roaming. There is an estimated 3-4 hundred thousand human cases with animal exposure and that of around 50 thousand cattle annually. Considering the public health impact of human rabies, Communicable Disease Control (CDC) unit of the Directorate General of Health Services, has been implementing the National Rabies Control Program since 2010. In 2017, in the 4th HPNSP, the activities have been established as a vertical program under CDC Operational Plan and shifted the target from control to elimination of rabies by 2022 in collaboration with the Department of Livestock (DLS), Local Government, non-governmental organizations (NGOs), and development partners.

### **National Rabies Elimination Program, Bangladesh**

The goal of the program is to eliminate dog-mediated rabies as a public health problem from Bangladesh by 2022. The objectives of the program are the following:

- Reduce mortality due to dog-mediated human rabies to less than 1 per 1 million population by 2022
- Reduce mortality due to dog-mediated animal rabies to less than 10 per million animals by 2022

- Ensure regular reporting and strengthen active surveillance of human and animal rabies cases.

### **Strategic activities**

- Advocacy, Communication and Social Mobilization
- Modern management for dog-bite cases
- Mass dog vaccination (MDV)
- Dog population management (DPM)
- Rabies surveillance and research.

### **Targets**

- Ensure continuous supply of human anti-rabies vaccine and rabies immunoglobulin (RIG) in all rabies prevention and control centers by 2020
- Complete three rounds of mass dog vaccination throughout the country by 2022
- Establish a referral laboratory for diagnosis of suspected rabies cases (human and animal) by 2019
- Strengthen a countrywide rabies surveillance for animal and human by 2019
- Building awareness of community people to receive PEP and to encourage them for being responsible for the dogs
- Promotion of responsible dog ownership to prevent dog-bite by 2022.

### **Important activities and achievements in 2017-2018**

#### ***Inter-sectoral coordination***

- A nationally-coordinated and supervised comprehensive program through



OneHealth approach has been working for achieving rabies elimination. The Ministry of Health and Family Welfare, along with the Ministry of Fisheries and Livestock and the Ministry of LGRD, plays the key role in eliminating rabies. Different national and international organizations, development partners, including WHO, WAP, OIE, FAO, US-based CDC are also involved in varying capacities and are working closely with Rabies Elimination Program in Bangladesh. The following committees are functioning for conducting the program:

- > National Steering Committee (NSC)
- > Joint Coordination Committee (JCC)
- > Technical Working Group (TWG)
- > District Coordination Committee (DCC)
- Review of national strategy and national MDV action plan
- Implementation of mass dog vaccination in different districts in collaboration with DLS
- Sharing of reports with stakeholders
- Facilitating animal rabies diagnosis through timely sharing of information on suspected animal that bites human cases.

### ***Animal-bite management (to save human lives)***

Three approaches have been adopted for management of animal-bites: (i) washing of wound with soap and water for 15 minutes; (ii) active immunization with intra-dermal Inj. ARV; (iii) local infiltration of RIG at the wound in category III bites.

### ***Activities and achievements***

- Functioning of 67 Rabies Prevention and Control Centers—one at the national level in Infectious Disease Hospital, Dhaka (NRPCC) and at least one at each district Sadar Hospital (DRPCCs):

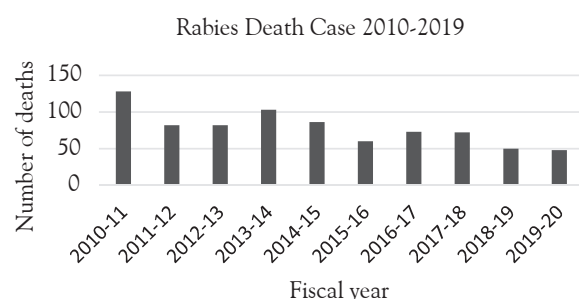
| WHO RECOMMENDED VACCINATION REGIMEN                                                                              |     |         |
|------------------------------------------------------------------------------------------------------------------|-----|---------|
| • Intra-dermal Post Exposure Prophylaxis (PEP ID) VACCINATION SCHEDULE (DELTOID) WITH Anti Rabies Vaccine (ARV): |     |         |
| VACCINE                                                                                                          | DAY | DOSAGE  |
| 1st Dose                                                                                                         | 0   | 0.1ml×2 |
| 2nd Dose                                                                                                         | 3   | Do      |
| 3rd Dose                                                                                                         | 7   | Do      |
| 4th Dose                                                                                                         | 28  | Do      |
| • RIG Dose- In case of human 20 IU/kg BW<br>- In case of equine 40 IU/kg BW                                      |     |         |

**Figure 4.3.11.** WHO-recommended vaccination regimen

- > Standard management of dog/other animal-bite cases following national guideline and WHO protocol
- > Provision of modern tissue culture vaccine (Inj. ARV) in intra-dermal route (ID-RV) in all the centers and RIG at NRPCC free of charge
- > The program distributed a total of 2,30,000 vial Inj. ARV and 25,000 vials of RIG for PEP and provided management to 270,685 affected/ animal-exposed patients
- > Provision of hands-on training and refreshers training to 800 physicians, nurses, store-keepers on animal-bite management for prevention of rabies at National Rabies Center
- > One-day training for 3,140 doctors, nurses, SACMOs, HI on animal-bite management, rabies reporting, and surveillance



- > Provision of vaccination cards, forms, and registers for documentation
- > Documentation and reporting of animal-bite cases through DHIS2 and direct reporting to CDC-DGHS
- > Monitoring of national and district rabies prevention and control centers
- > Reduction of reported rabies deaths.



**Figure 4.3.12.** Reported rabies-related deaths at NRPC (IDH) and BITID



**Figure 4.3.13.** Training on animal-bite management



**Figure 4.3.14.** Workshops for development of guideline and mass dog vaccination (MDV)

### **Mass Dog Vaccination (MDV)**

As a key strategy for elimination of human and animal rabies. MDV helps stop transmission of rabies from dog to dog/from dog to other animals and humans. The DGHS has a goal to complete at least 3 rounds of MDV throughout the country following international standard with >70% coverage by the year 2022.

By June 2019, scaling up MDV to all the district municipalities/city corporations and 40 districts and vaccination of about 13 lakh dogs have been done.

### **Surveillance and laboratory diagnosis of rabies**

- Verbal autopsy of all reported rabies cases
- Field investigations and surveillance of 5 unusual animal-bite events conducted in 2019
- Training of 2 doctors and one medical technologist on laboratory diagnosis of rabies from NIMHANS, Bengaluru, India (WHO collaborative Institute for rabies) with the technical support of SEAR, WHO
- Virology lab of IPH identified as the



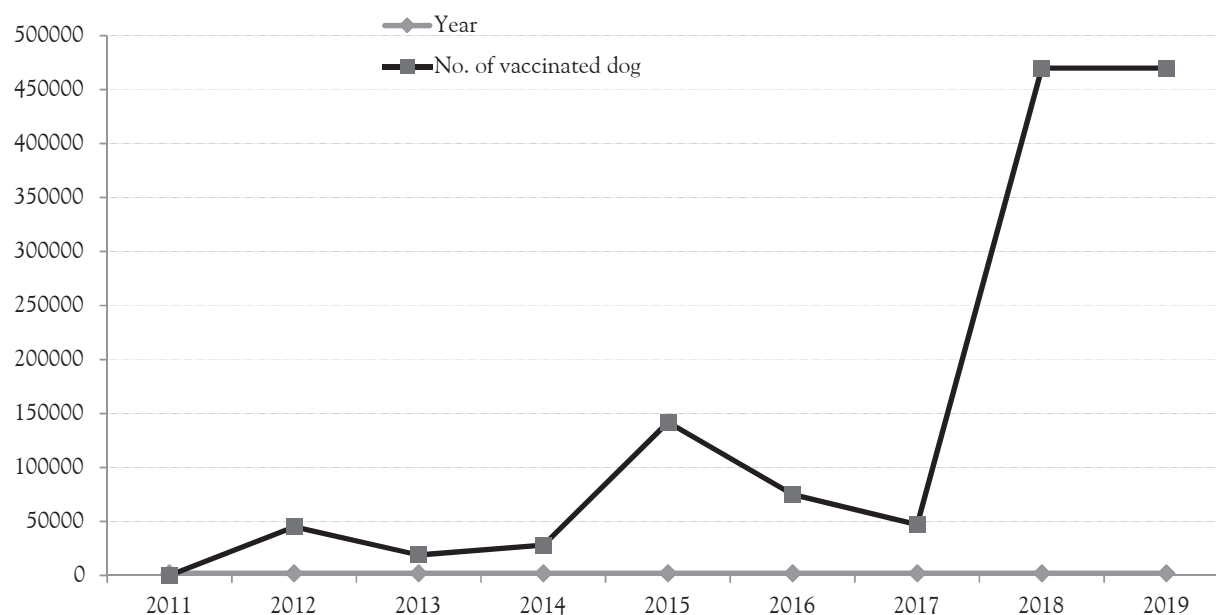


Figure 4.3.15. Number of vaccinated dogs

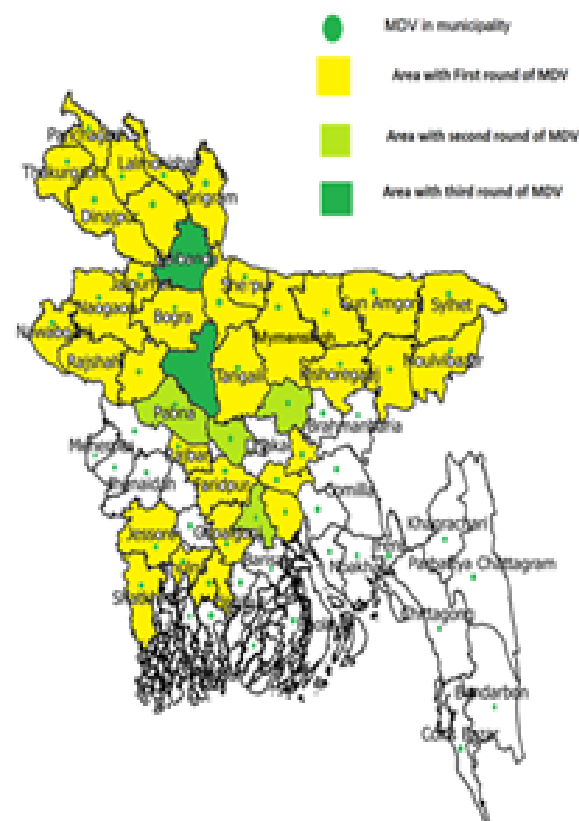


Figure 4.3.16. Locations of dog vaccination



Figure 4.3.17. MDV in field (colored dogs are vaccinated)



Figure 4.3.18. Health tips in Bangla language for healthcare after dog- or animal-bites



Figure 4.3.19. Observance World Rabies Day 2019





**Figure 4.3.20.** Awareness program arranged for school children for celebrating World Rabies Day 2019

National Reference laboratory to diagnose human rabies

- Development of draft guideline and SOP for laboratory diagnosis of human rabies
- Collection of 3 human samples (serial saliva) for laboratory testing of rabies
- Two animal samples (1 dog and 1 calf) diagnosed as confirmed animal rabies at CDIL with technical support of FAO.

### **Advocacy, Communication and Social Mobilization**

- Promotion of mass awareness for timely and appropriate management of animal-bites, including wound-washing with soap and water; community awareness for dog vaccination and promotion of responsibility in dog ownership throughout the country during MDV campaigns
- Awareness activities at different medical colleges.

### **Observance of World Rabies Day**

Since 2007, the program has been observing the World Rabies Day nationally and also at primary school levels to build awareness among caregivers as well as in the community people involving all the stakeholders. The activities include the following:

- Multisectoral involvement for observing World Rabies Day
- Programs at the national, district and sub-district levels
- ‘Little Doctor’ campaign in 1,10,000 schools all over Bangladesh for awareness-building among the students
- Stakeholder Meeting for Evaluation of National Rabies Elimination Program was participated by international experts from WHO’s HQ, UK, and experts from National level, Development Partners, and others.
- For increasing the prevention of rabies, advocacy meetings were conducted at 29 districts and upazilas with active participation of local government authority

(total participants being 2,573 from several sectors).

- Mass media have published the activities of World Rabies Day with special importance.
- Various steps taken for mass awareness-building regarding rabies control.

## **ARC, Viral Hepatitis and Diarrheal Diseases Program**

### **Antimicrobial resistance containment (ARC)**

Antimicrobial resistance (AMR) is a growing global threat to public health that is imposing serious effects on the management of infectious diseases. Irrational and inappropriate use of antimicrobials, causing AMR, is a well-understood natural consequence. An association between excessive and unnecessary use of antimicrobials and drug resistance because of selective pressure has been documented.

Following the recommendations, the health ministers of the member states of the World Health Organization for South-East Asia (WHO-SEA) region met in Jaipur, India, in September, 2011 and agreed to acknowledge the AMR as a major global public health issue, to institute a coherent, comprehensive and integrated national approach to combating AMR and 16 other activities by signing on a charter of activities namely the 'Jaipur Declaration'.

In continuation of the efforts relating to Jaipur Declaration and to review the status of implementation of AMR country plans and to align them with the AMR Global Action Plan, 2015, the designated AMR focal points of the

WHO member countries met in New Delhi, India, and made many recommendations in June 2015.

In concordance with the global and WHO activities on ARC, the Ministry of Health and Family Welfare (MOHFW) in Bangladesh has come forward and formed different committees with representation from multisectoral stakeholders, including National Steering Committee (NSC) involving several ministries, headed by Honorable Minister for the MOHFW, National Technical Committee (NTC) headed by Director General of Health Services. Subsequently, the National Strategy, for Antimicrobial Resistance Containment in Bangladesh was developed and approved by the NSC and the NTC with further recommendation of developing a National Action Plan (NAP). A multisectoral working group representing human health, animal health, and drug administration was assigned, and the NAP was developed based on the national strategy covering almost all areas recommended by the WHO and other relevant global agencies.

### ***Objectives of the National Strategy for ARC***

1. Establish multisectoral approach for planning, coordination, and implementation of antimicrobial resistance containment (ARC) activities;
2. Promote and ensure rational use of antimicrobial agents in human health, livestock and fisheries sectors;
3. Promote and strengthen infection prevention and control measures to minimize the emergence and spread of AMR;

4. Promote and strengthen biosafety and biosecurity principles and practices and containment measures;
5. Review, update, and strengthen regulatory provisions;
6. Strengthen surveillance system for AMR containment;
7. Promote operational research and education in the area of AMR;
8. Establish Advocacy, Communication and Social Mobilization (ACSM) for ARC activities.

### ***Targets of the program***

#### ***1. Ensure rational use of antimicrobials***

- a. Develop standard treatment guideline (STG) and ensure adherence to STG;
  - b. Establish/strengthen national reference laboratory and regional microbiology laboratories;
  - c. Ensure GLP at all microbiology laboratories;
  - d. Update STG annually, based on AST data;
  - e. Develop antibiotics policy;
  - f. Ensure availability of antimicrobials at all healthcare facilities;
  - g. Enforce and enhance regulatory provisions for the use antimicrobials;
  - h. Ensure good pharmacy practice (GPP).
2. Ensure good manufacturing practice (GMP) at pharmaceutical industries;
  3. Ensure infection prevention and control (IPC) activities at healthcare facilities and community level.

#### ***4. Establish surveillance system:***

- a. Web-based surveillance and laboratory networking for AMR;
- b. Surveillance for AMs use.

5. ACSM for healthcare providers (HCPs), drug-sellers, community members, animal and fisheries farm owners;
6. Ensure adequate funding for basic and operational research.

### ***Major activities in 2018-2019***

1. Advocacy, communication and social mobilization was the most focused activities in 2019. Festoons, banners, posters, leaflets, and stickers were distributed countrywide. Awareness-raising SMS was cast with the help of BTRC.
2. Rallies and seminars were held at some medical colleges, all districts and upazillas with related stakeholders to celebrate World Antibiotics Awareness Week in 2019.
3. Making of draft of laboratory SOP and training module was completed.
4. Seminar on antimicrobial resistance containment was held with stakeholders.
5. Surveillance on AMR Pattern in Action is under the consideration of Global Health Security Agenda (GHS), CDC-USA, and WHO.
6. The process of Fleming fund country grant for AMR surveillance was formalized.
7. Mobilization of multisectoral efforts for AMR containment by regular meeting of working group and National Technical Committee was done.



**Figure 4.3.21.** Seminar, workshop, advocacy meeting held at different organizations on AMR

8. Mass media to increase awareness among the people about antimicrobial resistance were involved.
9. Advocacy meetings were held at different institutions in OneHealth approach.

### Viral Hepatitis

Viral hepatitis is the seventh leading cause of death globally, accounting for 1.4 million deaths per year. In Bangladesh, the prevalence of hepatitis B and C viruses have been estimated at 5.5% and 0.6% respectively. It has been estimated that, in Bangladesh, more than 10 million are living with viral hepatitis. Liver diseases account for 8-12% admissions in medicine wards and remain the third leading cause of death in tertiary medical college hospitals of the country, which

is approximately more than 20,000 deaths per year. Hepatocellular carcinoma is the third leading cause of cancer deaths in Bangladesh, next only to cancers of lungs and stomach.

Twenty million people are suspected to die from hepatitis B globally by 2030, and the number of people living with hepatitis C is increasing, despite availability of an effective curative measure. National and global responses can no longer be delayed.

Activities at the DGHS aim at reducing 90% of viral hepatitis-related morbidity and mortality in Bangladesh by 2030.

### Specific objectives

1. Establish multi-sectorial approach for planning, coordination, and



- implementation of viral hepatitis prevention and control activities;
- 2. Promote and strengthen viral hepatitis prevention and control measures;
- 3. Promote and strengthen biosafety and biosecurity principles and practices and containment measures;
- 4. Strengthen national surveillance system for viral hepatitis containment;
- 5. Promote and ensure rational use of antiviral agents;
- 6. Review, update, and strengthen regulatory provisions;
- 7. Establish Advocacy, Communication and Social Mobilization (ACSM) for viral hepatitis prevention and control activities;
- 8. Promote operational research and education in the area of viral hepatitis prevention and control;
- 9. Adapt special care system for viral hepatitis in pregnancy and pediatric population;
- 10. Adapt special care system for viral hepatitis-related advanced liver disease patients;
- 11. Adapt special measures for people who inject drugs (PWID);
- 12. Develop training module and standard guideline for management of viral hepatitis;
- 13. Develop international cooperation and collaboration with experts, professional bodies, and development partners;

- 14. Introduction of adult vaccination program for hepatitis B and E.

### Targets of the program

- 1. Development of National Guidelines;
- 2. One of the principal concerns in tackling the issue of viral hepatitis is irrational use of antivirals. The irrational use of antivirals results in emergence of resistant viruses. Rational use of drug involves using the correct drug with proper dose and duration, keeping in mind the suitability, efficacy, safety, and cost of that drug. promotion of adherence to GPP and proper use of antivirals;
- 3. Ensure Good Manufacturing Practice (GMP) of antivirals. Quality of antivirals is also crucial for control of viral hepatitis.

### Prevention and control (IPC) measures for nosocomial viral hepatitis

The IPC measures in healthcare facilities reduce the risk of nosocomial transmission of viral hepatitis. These measures include hand hygiene, use of personal protective equipment (PPE), maintaining a clean environment, cleansing and sterilization (or high-level disinfection) of instruments, improving safety in risky areas of the health facility, safe use and disposal of sharps, appropriate waste disposal and vaccination.

### Surveillance

Surveillance for viral hepatitis will provide information on the magnitude and trends in viral hepatitis and for monitoring the effect of interventions. It is necessary to formulate policy decisions, update national formularies, inclusion of antivirals in the lists of essentials

and ensuring availability of appropriate antivirals. Surveillance will focus on the following:

- Monitoring of viral hepatitis
- Monitor use of antivirals
- Monitor manufacture of the quality-assured antivirals.

### Research and education

Researches on the use of antivirals, viral hepatitis, and IPC are essential for prevention and control of viral hepatitis. Educational works are needed to alleviate the irrational use of antivirals. Recognition of the different issues relating to viral hepatitis is limited among all levels of the population and, therefore, educating health professionals, patients, and public as a whole is the key strategy in addressing any issue relating to prevention and control of viral hepatitis.

### Advocacy, communication and Social mobilization

Use of antiviral is influenced by the interplay of knowledge, attitude, and interactions

of prescribers, patients, and promoters of manufacturers, characteristics of the health system(s) and the regulatory environment. In this context, coordinated interventions are needed to encourage rational use of antivirals.

### Major activities in 2018-2019

- Establishment of multi-sectoral approach for planning, coordination, and implementation of viral hepatitis-related activities
- Training of more than 2,000 doctors, nurses, and SACMOs on viral hepatitis from different districts
- National Plan for Viral Hepatitis has been drafted
- Awareness-building through advocacy, communication, and social mobilization
- Hepatitis B and C screening and hepatitis B vaccines were given among 6,000 healthcare providers free of charge at Dhaka Medical College and Hospital
- 20,000 injectables for hepatitis C were distributed among the patients free of charge
- Water purification tablets were supplied according to needs all over the country.



**Figure 4.3.22.** Screening and vaccination program held at Dhaka Medical College and Hospital on viral hepatitis

## Food and Waterborne Diseases

Food and waterborne diseases (F&WBD) are an important cause of morbidity and mortality worldwide, and detailed data on the costs of food and waterborne diseases in developing countries are largely missing. Access to sufficient amounts of safe food and clean drinking-water is the mainstay of sustaining life and promoting good health. An estimated 600 million (almost 1 in 10 persons) in the world fall ill after eating contaminated food, and 420,000 die annually, which results in the loss of 33 million disability-adjusted life years (DALYs). Forty percent of the foodborne disease burden is carried by children below 5 years of age with 125,000 deaths every year (WHO, 2017). Simultaneously, clean water is essential for survival and development of all children. Without clean water, children simply cannot stay alive or thrive in a healthy environment. About 663 million people are still without access to clean drinking-water, despite the MDG target for clean water being met in 2010. Eight out of 10 persons without access to clean water live in rural areas. Waterborne diseases are linked to a significant disease burden worldwide. Waterborne diarrheal diseases, for example, are responsible for 2 million deaths each year, with the majority occurring in children below 5 years of age (WHO, 2017).

In recent years, Bangladesh is in the process of rapid urbanization. About 65.7% of the population lives in rural areas but it is expected that the urban population will grow from its current (2015) level of 54 million to 81.4 million in 2029, indicating an increase of 50% in 14 years. Although growth rate in urban areas is 3.6%, slum population growth rate is 7%, which is raising concerns to health service development and public health planning.

The national goal is reduction of the burden of food and waterborne diseases to less than 50% from the baseline (2017) by 2022.

## Specific objectives

1. Reduction of hospitalization for diarrheal diseases to less than 30% from the baseline (2017) by 2022
2. Reduction of HEV-related maternal mortality by 30% of the baseline (2017) by 2022
3. Reduction of hospitalization rate for typhoid fever to less than 30% from the baseline (2017) by 2022.

## Targets of the program

### Case Management

- Ensure proper and timely case management for F&WBDs, using standard treatment guideline
- Provide specific training for proper and uniform case management, including avoidance of nosocomial infections
- Ensure sufficient pre-positioned medical supplies for case management following the list of essentials for management.

### Surveillance

- Establish and strengthen hospital-based, lab-based, and web-based surveillance through MIS, using DHIS2 for early detection of disease outbreak
- Facilitate early detection of outbreak and immediate response.

### Vaccination

- Reactive vaccination at the community level to prevent spreading of the disease or outbreak

- Provision of pre-emptive vaccination at the community level by identification of hotspots through surveillance
- Provision of vaccination in humanitarian crisis during national disasters, like flood, earthquake, etc.

### **Water, sanitation, and hygiene interventions**

- Improved access to safe water, proper sanitation, waste management, and food safety for preventing F&WBDs
- Health education at communities for adopting preventive behavior for averting contamination
- Improved communication and public information.

### **Major activities**

- National Cholera Control Plan, Bangladesh was approved.
- Round of Oral Cholera Vaccination (OCV) is ongoing among Rohingya Refugees and Bangladeshi host community at Cox's Bazar.
- Distribution of cholera saline, Water Purification Tablets (WPT) was distributed countrywide according to need.

### **National Kala-azar Elimination Program (NKEP)**

Visceral leishmaniasis (VL), commonly known as kala-azar is a neglected tropical disease transmitted by sandfly vector. It's a disease of the poor and marginalized rural people associated with malnutrition, human migration, poor housing, immunodeficiency, and lack of resources.

Around 32 million people, residing in 100 upazilas under 26 districts where kala-azar is endemic, are at risk in Bangladesh. The national Kala-azar Elimination Program (NKEP) has started since 2008 with a vision for kala-azar-free Bangladesh by 2022. Hence, the program is working with three indicators: (i) kala-azar detection rate should be close to 100%; (ii) treatment completion rate should be close to 100%; and (iii) vector control coverage rate (households coverage through IRS) should be more than 90%.

Although NKEP successfully reduced the new kala-azar and relapsed kala-azar (RKA) to less than 1 case per 10000 people at upazila level by the end of 2017 and the number of cases are declining gradually (Figure 4.3.23), a few cases are reported from non-endemic areas. However, diverse initiatives have been taken to achieve the goal. The surveillance system of NKEP is fully based on DHIS2 tracker capture and reported at the upazila level. A development partner KalaCORE through icddr'b and Accelerating the Sustainable Control and Elimination of Neglected Tropical Diseases (ASCEND) program funded by the DFID joined with NKEP. WHO is providing logistics (Liposomal amphotericin B) and technical assistance to strengthen the program.

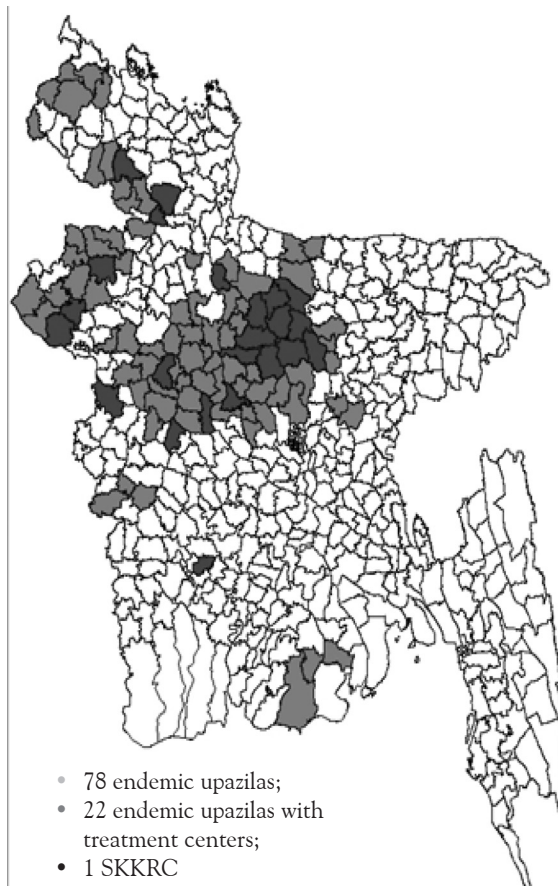
### **Program objectives**

- Reducing the incidence of new kala-azar and PKDL to less than 1 case per 10,000 population at the upazila level by the end of 2017
- Reducing the incidence of KA in the poor, vulnerable and remote endemic areas
- Reducing case-fatality rates from high to negligible level

- Strengthen diagnosis and treatment of PKDL cases to interrupt transmission of kala-azar
- Prevent and treat kala-azar-HIV-TB co-infections in the endemic areas.

### Activities completed

- Training on modern management of kala-azar for doctors and nurses provided at the central and upazila levels
- Training on supply and cold-chain management of kala-azar logistics for lab technicians and store-keeper
- Two rounds post- and pre-monsoon IRS done in selected endemic upazilas during July 2018 to June 2019
- Printed National Kala-azar Case Management Guideline published in February 2019
- IRS Monitoring Tools and Supervisory Checklist (draft developed)
- Integrated Vector Management Guideline (IVM) (draft developed)
- National Post-elimination Strategic Plan 2019-2030 updated in Bangladesh (on process)
- Based on 120 index cases kala-azar transmission control activities conducted at 12 selected UHCs for identification of any undiagnosed or hidden cases for final diagnosis and treatment
- Active case search activities through camp approach in 22 endemic upazilas
- Confirmation of kala-azar death cases through verbal autopsy.

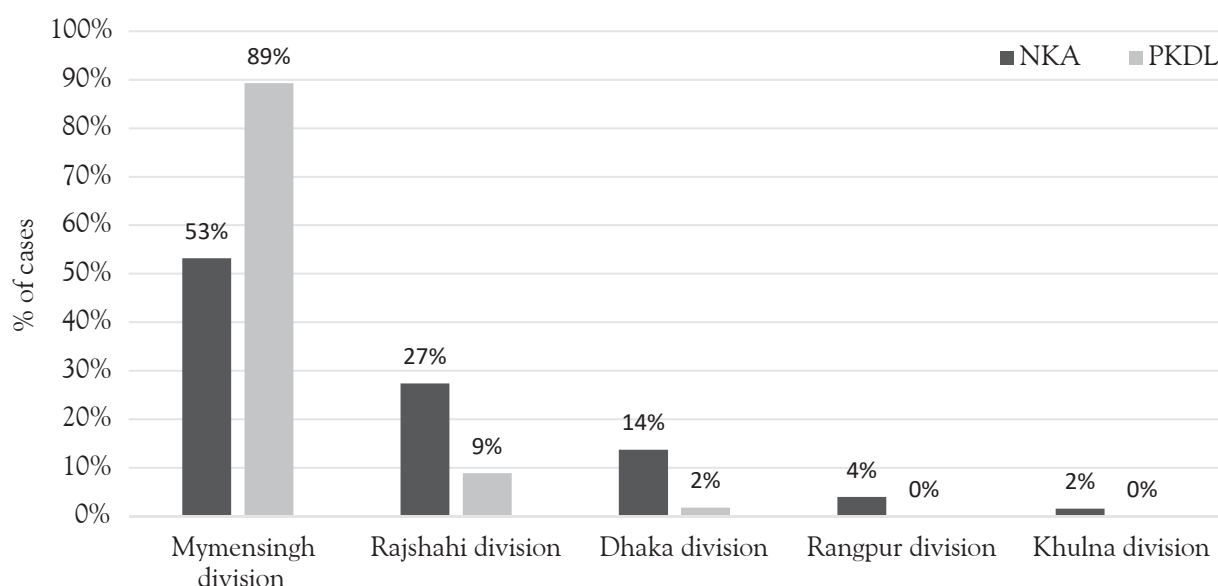


**Figure 4.3.23.** 100 endemic upazilas for kala-azar

### Challenges

- Strengthening the passive surveillance system at upazila level and ensure the real-time data in DHIS2
- Reducing the time duration between disease onset and treatment
- Increasing trend of PKDL, sub-clinical infection or asymptomatic carriers and relapse cases
- Uninterrupted logistics supply at every KA treatment centers
- Ensuring zero reporting of non-endemic upazilla.





**Figure 4.3.24.** Percentage of NKA and PKDL cases treated in different divisions from January to December 2018

### Soil-transmitted Helminthiasis

Soil-transmitted helminthiasis (STH) is a disease of public health significance in Bangladesh. Every district of the country is endemic for one or more STH species, although the level of endemicity and the specific geographic distribution of STH species are unknown. Children are at the greatest risk of infection and morbidity due to STH infection. The disease is caused by infection with one or more of these parasites: *Ascaris lumbricoides* (Round worm), *Trichuris trichiura* (Whip worm), and *Ancylostoma duodenale* or *Necator americanus* (Hookworms). In the South Asia Region, including Bangladesh, STH results in loss of over 1.6 million disability-adjusted life years annually. The greater the intensity of the STH infection, the greater the negative health impacts. In 2008, the STH Control Program of CDC, DGHS, launched the nationwide school-based deworming through biannual mass

drug administration (MDA) campaigns. Until 2018, the STH control program conducted 21 rounds of MDA among the school-aged children (SAC).

Improved water supply, sanitation, and hygiene are essential for sustainable control of STH. Under the DGHS, CDC provides STH control interventions to school-aged children. WHO, UNICEF, and CWW support the STH control program with technical assistance. Non-governmental organizations (NGOs) are working in WASH that operates under the guidance of the Department of Primary Education and Local Government. NGOs and stakeholders indirectly support the program for STH elimination by the education and nutrition sectors, e.g. in development of health education materials relevant to STH control.

In 2016, the STH control program changed its goal from controlling morbidity to elimination of STH as a public health problem by 2022.

From 2008 to 2017, the program implemented 2 rounds of MDA and, in each round, distributed about 40 million Mebendazole 500 mg. tablets to school-aged (5-11 years old) children. In 2017, the program expanded the deworming campaign to secondary school children (12-16 years old). The main interventions are deworming and health education both.

WHO defined the elimination of STH as  $\leq 1\%$  prevalence in at-risk populations with moderate to heavy intensity of infection (MHI) with one or more STH parasite(s). In the school-based surveys and also in the community-based surveys, the prevalence of MHI is around 1.69%-2.0% in Bangladesh. There is also a significant decrease in the prevalence of any STH infection over the past decade from 79.8% in 2005 to 7.85% in 2019

among the school-aged children, which may be linked to mass deworming campaigns in the schools. There is still significant regional variation in STH prevalence ranging from 1.94% to 39.8% in different risk groups at the community level of Bangladesh.

### Soil-transmitted helminthiasis (STH) status in Bangladesh

The intervention coverage and its impact on STH prevalence are monitored regularly by the STH control program through community-based and school-based surveys and supportive supervisions. In 2005, a baseline survey conducted in three districts representing three ecological zones: plains, hills and coastal agricultural areas showed a high prevalence of STH infection among school-aged children. The follow-up surveys, after implementing

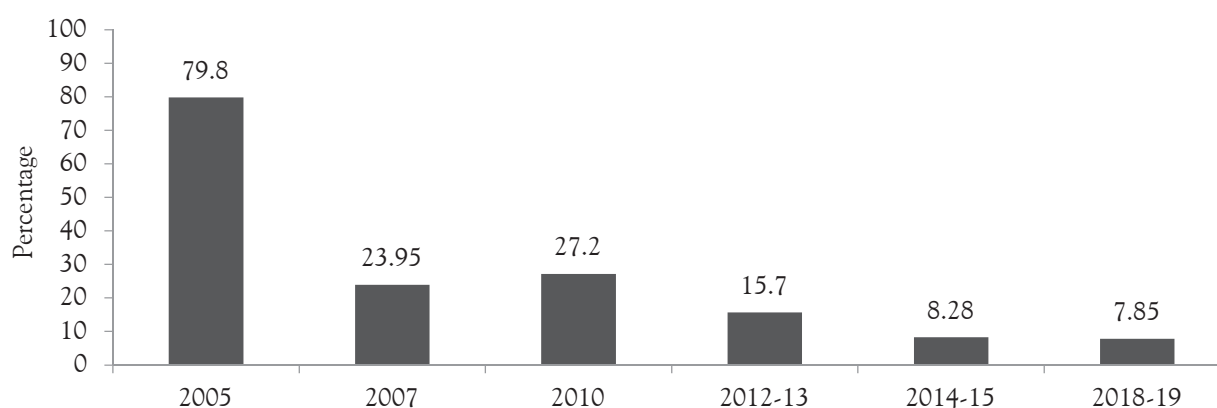


Figure 4.3.25. STH infection rate among school-aged children year-wise

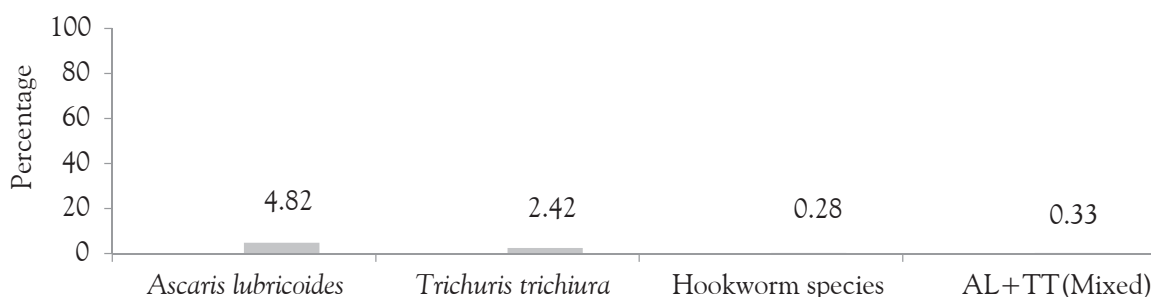
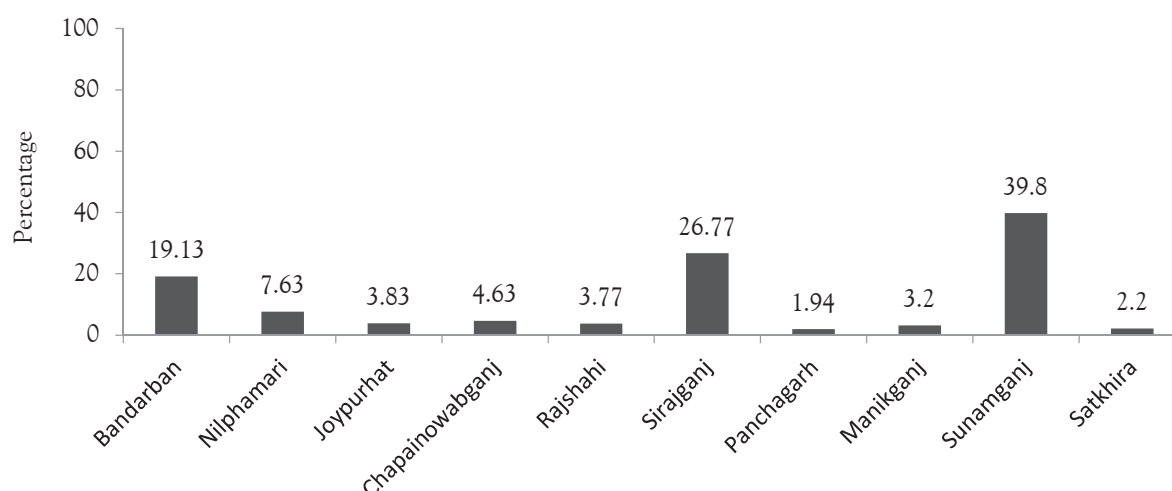


Figure 4.3.26. Infection rate of different STH species in the school-aged children





**Figure 4.3.27.** STH prevalence in the community

the mass drug administration (MDA), showed a decline in the prevalence of STH among the school-aged children (7.85 % in 2018). In 2018, this school-based STH survey was conducted by proper sampling methods in randomly-selected schools in all the 64 districts of Bangladesh (Figure 4.3.25).

From the school-based STH survey among all the species of STH, the *Ascaris lumbricoides* was more prevalent in the school-aged children. *Trichuris trichiura* was the second-most prevalent species, and hookworm infection was very low in the school-aged children (Figure 4.3.26).

In 2017, the STH control program with the technical support from CWW started to conduct community-based surveys to determine the prevalence and infection intensity of STH in all risk groups in the community in Bangladesh. This survey also aimed to estimate the association between type of household and individual risk factors for burden of STH infection. In 2017, the program conducted community-based survey in Bandarban, Nilphamari, Joypurhat,

Chapainowabganj, Rajshahi, and Sirajganj districts and, in 2018, the program conducted community-based survey in another 4 districts: Panchagarh, Manikganj, Satkhira, and Sunamganj (Figure 4.3.25).

### STH program activities and success

1. Nationwide school-based deworming through biannual mass drug administration (MDA) campaigns to school-aged children. The program distributes the deworming drug free of charge
2. Strong school-based program and Little Doctor Program (involving trained school children with trained program personnel)
3. High deworming treatment coverage (>90%)
4. Continued reductions in disease prevalence from 79.8% in 2005 to 7.85% in 2019 in the surveyed school-aged children
5. Strong government commitment for deworming activities supported through the HPNSP

6. In all risk groups, the prevalence of MHI is consistently  $\leq 2\%$  in the 10 districts surveyed
7. Among the school-aged children, the prevalence of MHI is  $\leq 1.69\%$  in 64 districts
8. Functioning STH technical committee with the Government and NGO stakeholders
9. The STH Technical Committee oversees the technical direction of the biannual deworming campaigns.

### National Tuberculosis Program

Despite remarkable progress in the health sector, Bangladesh still faces some serious health challenges, and tuberculosis (TB) is one of them. TB remains a leading cause of death among communicable diseases and the second-highest cause of disability among the productive age-group.

Tuberculosis is a major public health problem in Bangladesh for a long time. Under the Mycobacterial Disease Control (MBDC) Unit of the Directorate General of Health Services, the National Tuberculosis Control Program (NTP) is working with a mission of eliminating TB from Bangladesh. The goal of the program is to reduce morbidity, mortality, and transmission of TB until it is no longer a public health problem through achieving universal access to high-quality care for all people with TB.

The NTP adopted Directly Observed Treatment Short-course (DOTS) strategy and started its field implementation in November 1993. By 2007, the DOTS services were made available throughout the country, including the metropolitan cities. NTP also adopted the

Stop TB strategy in 2006. The Government of Bangladesh, together with its many and diverse partners from the public and private sectors, is committed to further intensifying the TB control activity in order to sustain the achieved success and to reach the TB control targets linked to the Sustainable Development Goals (SDGs) and WHO's END TB strategy.

### Government policy and strategies

Bangladesh has been implementing sector-wide approaches (SWAp) in the health sector since 1998. The first SWAp-the Health and Population Sector Program (HPSP) was implemented during 1998-2003. Within the broader context of the Bangladesh National Strategy for Economic Growth, Poverty Reduction and Social Development, the Government of Bangladesh revised its strategic approach and renamed HPSP as Health, Nutrition and Population Sector Program (HNPS) in the second SWAp. This plan was implemented in 2003-2011. The Ministry of Health and Family Welfare implemented the Health, Population and Nutrition Sector Development Program (HPNSDP) covering the period from July 2011 to December 2016 with the goal of ensuring quality and equitable healthcare for all citizens in Bangladesh by improving access to and utilization of health, population and nutrition services. To improve equitable access to and utilization of quality health, nutrition and family planning services, the current 4th Health, Population and Nutrition Sector Program (HPNSP) is being implemented from January 2017 to June 2022. In all the sector programs, tuberculosis control has been recognized as one of the priority areas.

The Government of Bangladesh is committed to providing TB diagnosis and treatment

completely free of charge to all citizens of the country. It strives to make services equally available to all people in Bangladesh, irrespective of age, sex, religion, ethnicity, social status, or race.

The Government of Bangladesh is also procuring the firstline anti-TB drugs with domestic funding from 2018 and increased the overall government budget to US\$ 51.89 million for TB control in the Operational Plan (2017-2022).

Tuberculosis was declared as a mandatory notifiable disease by the Government in 2014.

The NTP policies and strategies to date have attained international standards formulated by the World Health Organization, such as launching the DOTS strategy in 1993, and the Stop TB strategy that underpinned the Global Plan to Stop TB 2006–2015. New multi-sectoral strategic approaches and new international targets of END TB strategy have also been adopted in 2016 with a vision of a TB-free Bangladesh.

## WHO's END TB Strategy: principles, pillars and components

### Principles

1. Government stewardship and accountability, with monitoring and evaluation
2. Strong coalition with civil society organizations and communities
3. Protection and promotion of human rights, ethics, and equity
4. Adaptation of strategy and targets at the country level, with global collaboration.

### Pillars and components

#### 1. Integrated, patient-centered care and prevention

- Early diagnosis of tuberculosis, including universal drug-susceptibility testing; and systematic screening of contacts and high-risk groups
- Treatment of all people with tuberculosis, including drug-resistant tuberculosis and patient support
- Collaborative tuberculosis/HIV activities and management of co-morbidities
- Preventive treatment of persons at high risk and vaccination against tuberculosis.

#### 2. Bold policies and supportive systems

Political commitment, with adequate resources for tuberculosis care and prevention; engagement of communities, civil society organizations, and public and private care providers are ensured:

- Universal Health Coverage policy, and regulatory frameworks for case notification, vital registration, quality and rational use of medicines, and infection control
- Social protection, poverty alleviation, and actions on other determinants of tuberculosis.

#### 3. Intensified research and innovation

- Discovery, development, and rapid uptake of new tools, interventions, and strategies
- Research to optimize implementation and impact, and promote innovations.

**Table 4.3.5.** WHO's END TB Strategy: targets

| Indicator                                                    | Milestone |      | Target |        |
|--------------------------------------------------------------|-----------|------|--------|--------|
|                                                              |           |      | SDG    | End TB |
|                                                              | 2020      | 2025 | 2030   | 2035   |
| Reduction in number of TB deaths compared with 2015 (%)      | 35        | 75   | 90     | 95     |
| Reduction in number of TB incidence compared with 2015 (%)   | 20        | 50   | 80     | 90     |
| TB-affected families facing catastrophic costs due to TB (%) | Zero      | Zero | Zero   | Zero   |

National Strategic Plan for TB Control was updated in line with WHO's END TB strategy for 2018-22.

### Objectives of the National Strategic Plan 2018-2022

#### ***Pillar 1: integrated, patient-centered care and prevention***

**Objective 1:** Increase annual case detection of all forms of TB to more than 90% of all incident cases by 2022 (from baseline of 57% in 2015) with childhood TB contribution of 8% by 2022 (from baseline of 4% in 2015)

**Objective 2:** Further increase treatment success rate at levels of more than 90% in all forms of detected non-MDR TB cases and ensure quality-controlled treatment services at all implementation sites

**Objective 3:** Increase annual case detection of MDR-TB to 4,100 (from baseline of 880 in 2015), childhood MDR-TB case detection to 112 by 2022 (from baseline of 0 in 2015) and improve management of MDR-TB cases through country-wide implementation of the shorter MDR-TB treatment regimen.

#### ***Pillar 2: Bold policies and supportive systems***

**Objective 4:** Ensure that no TB-affected families face catastrophic costs due to tuberculosis by 2022

**Objective 5:** Ensure that 100% of TB service facilities receive regular supervision and monitoring with appropriate feedback, resulting in remedial actions by 2022

**Objective 6:** Ensure the long-term availability of 100% of the required funding for activities at all program levels through effective planning and partner coordination and the continuing increase of government contributions to the NTP budget.

#### ***Pillar 3: intensified research and innovation***

**Objective 7:** Ensure adequate support for operational research to foster innovation.

### **TB Epidemiology in Bangladesh**

WHO estimated the incidence and mortality for all ages and all forms of TB in 2018 in Bangladesh to be 221 (95% CI: 161-291) and 29 (95% CI: 18-42) per 100,000 (Table 4.3.6).

**Table 4.3.6.** WHO estimates of TB burden 2017

| Estimates of TB burden 2017 | Number (thousand) | Rate (per 100,000 population) |
|-----------------------------|-------------------|-------------------------------|
| Mortality (excludes HIV+TB) | 47 (30-67)        | 29 (18-42)                    |
| Mortality (HIV+TB only)     | 0.19 (0.094-0.32) | 0.12 (0.06-0.2)               |
| Incidence (includes HIV+TB) | 357 (260-469)     | 221 (161-291)                 |
| Incidence (HIV+TB only)     | 0.73 (0.36-1.2)   | 0.45 (0.23-0.76)              |
| Incidence (MDR/RR-TB)       | (3.2-9.6)         | 3.7 (2-5.9)                   |

Source: Global TB Report 2019

### National Tuberculosis Prevalence Survey in Bangladesh, 2015-2016

The recent survey of 98,710 participants, conducted between 2015 and 2016, was designed in accordance with the WHO standards.

The estimated prevalence of bacteriologically-confirmed pulmonary TB cases and smear-positive bacteriologically-confirmed TB cases among the people aged  $\geq 15$  years were 287 per 100,000 population (95% CI: 244-330) and 113 per 100,000 population (95% CI: 87-139) respectively. The prevalence was higher among males compared to females and was higher in urban compared to rural clusters. Prevalence increased with age and was much higher among  $\geq 65$  years age-group. Overall estimated prevalence (all ages, all forms of TB) was 260 per 100,000 population (95% CI 220-301).

### Major activities of the NTP

To achieve the objectives, the following main activities were conducted under the NTP:

- Developed policies, strategies and guidelines for TB control
- NTP ensured planning and budgeting for TB control activities through government's

Operational Plan and developed human resources capacity for TB control, including training

- Promoted early detection of smear-positive patients at all levels of the health services
- Implemented quality assurance system for smear microscopy
- Diagnosed smear-negative, extra-pulmonary and childhood TB
- Ensured Directly Observed Treatment Short-course (DOTS) for patients in the community
- Participation and involvement of government and non-government healthcare providers
- Maintained a high treatment success rate ( $>90\%$ ) among newly-diagnosed smear-positive patients
- Ensured uninterrupted supply of drugs, laboratory equipment, consumables, and other logistics
- Implemented standardized recording and reporting system, including electronic reporting

- Involved academic medical institutions and hospitals, private practitioners and professional associations in TB control activities
- Special services, like those in prisons, defense, industries, and other corporate sectors included in the NTP activities
- Strengthened cooperation and collaboration between the Government of Bangladesh and non-government organizations (NGOs) already involved in the control of tuberculosis

**Table 4.3.7, TB case notification and treatment success in 2018**

| Parameter                                                                   | Value   |
|-----------------------------------------------------------------------------|---------|
| Total TB case notified                                                      | 268,596 |
| TB treatment coverage (notified/estimated incidence)                        | 75%     |
| Notified TB/HIV co-infection                                                | 67      |
| Laboratory-confirmed MDR/RR-TB cases                                        | 1228    |
| MDR/RR-TB patients started treatment                                        | 1147    |
| Laboratory-confirmed XDR-TB cases                                           | 6       |
| XDR-TB patients started on treatment                                        | 6       |
| Treatment success of new and relapsed cases registered in 2017              | 94%     |
| Treatment success in HIV-positive TB cases registered in 2017               | 67%     |
| Treatment success in MDR/RR-TB cases starting second-line treatment in 2016 | 78%     |

Source: Draft Global TB Report 2018

- Conducted regular supervision, monitoring and evaluation of the NTP, thus, measuring program performance and impact
- Ensured programmatic management of drug-resistant TB
- Established linkage for management of TB-HIV co-infection
- Maintained liaison with development partners and established inter-sectoral and inter-ministerial collaboration
- Carried out operational research relating to TB control.

## Achievements of TB control in Bangladesh

Since the introduction of DOTS in Bangladesh in 1993, remarkable progress in TB control has been achieved in terms of DOTS coverage, diagnosis, and treatment of TB cases.

### DOTS coverage

Bangladesh adopted the internationally-recommended DOTS strategy in 1993, and DOTS services were made available to all upazilas by June 1998; by 2007, NTP reached 100% DOTS coverage. Yearly, more than 2 million people with symptoms of TB are being tested, and approximately 1 million lives were saved in the last decade.

### TB case notification

After introduction of the DOTS strategy in 1993, the overall progress in case finding was slow and steady until 2001 to reach case notification rate for new smear-positive cases of 59/100,000 population. From 2001

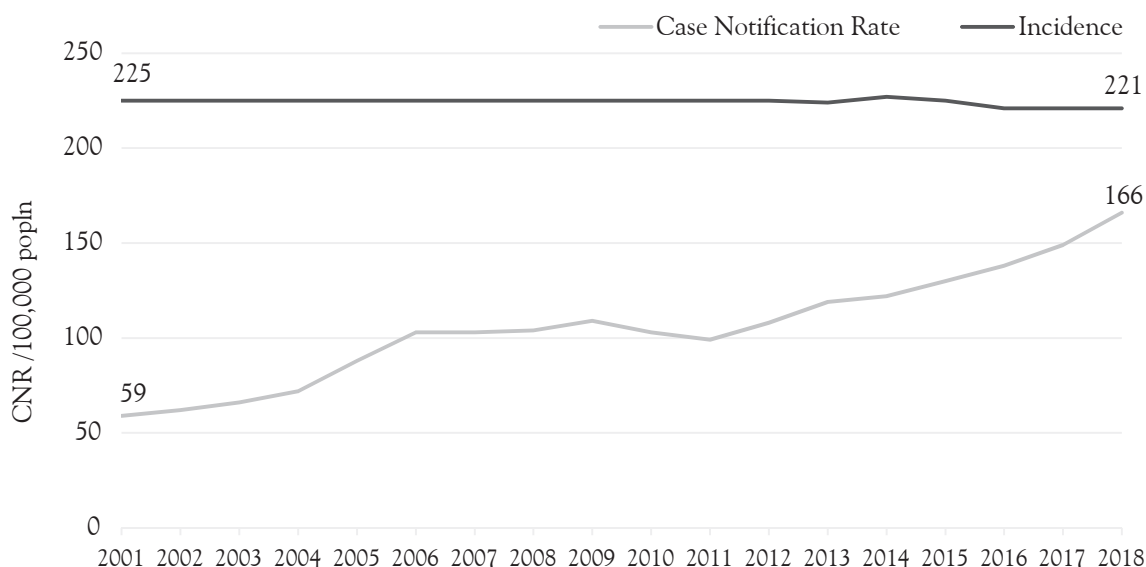


onward, case notification accelerated to reach 103/100,000 in 2006. As a result of additional effort addressing detection of smear-negative and childhood TB cases with special attention to hard-to-reach areas through social support for ultra-poor group, case notification increased to 166/100 000 population in 2017. As a result, the gap between incidence and notification is narrowing (Figure 4.3.28). In 2017, a total number of 244,201 TB cases were notified to

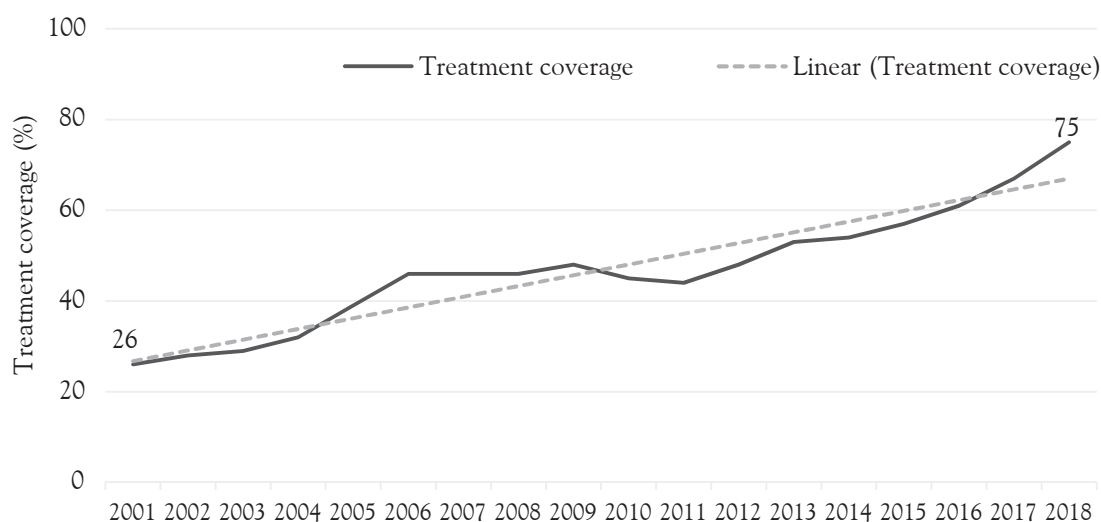
NTP and were brought under treatment. Up to September 2018, a total number of 199,625 TB cases were notified to NTP and brought under treatment. So, 73% cases were detected, and 27 % cases were missing.

### TB treatment coverage

Bangladesh is steadily progressing toward Universal Health Coverage; 75% of the



**Figure 4.3.28.** Trends of incidence and notification rate in Bangladesh



**Figure 4.3.29.** Trend of TB treatment coverage in Bangladesh



estimated TB cases were detected and brought under treatment in 2017 (Figure 4.3.29).

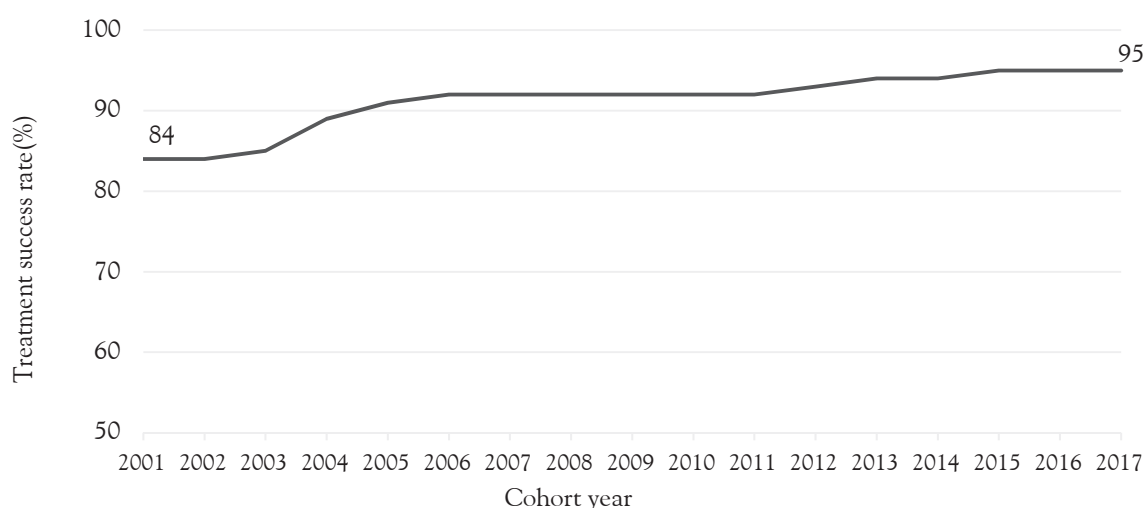
### TB treatment outcomes

Treatment success rates with DOTS have been consistently high from the beginning and crossed the global target of 85% in 2003. After strengthening DOTS and ACSM activities, the unfavorable outcomes have been remarkably reduced. As a result, the treatment

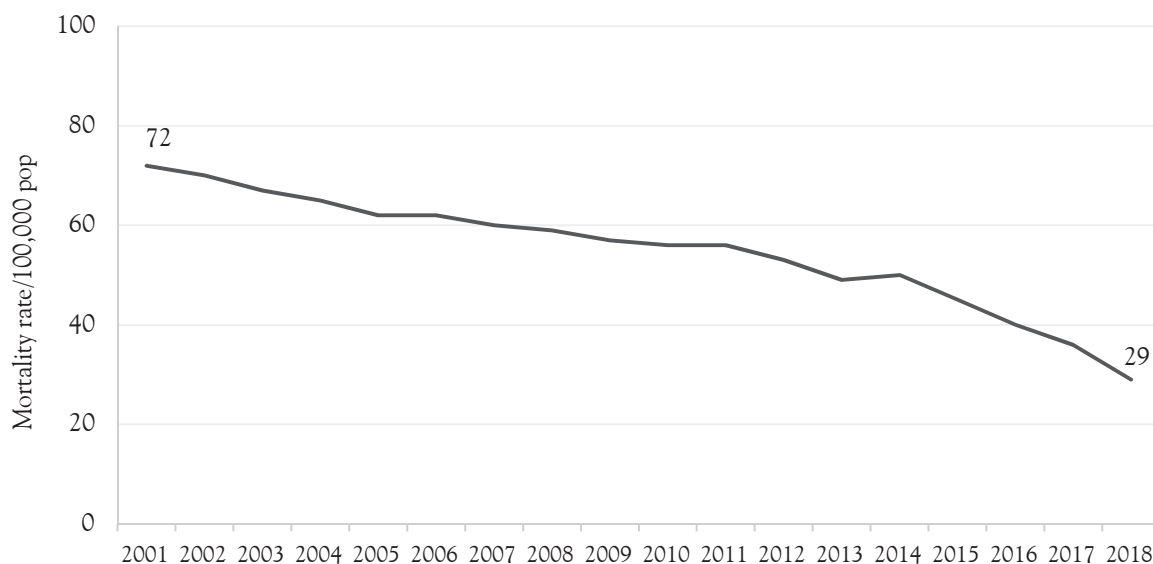
success rate has improved, and NTP has been contributing over 91% treatment success rates since 2005 (Figure 4.3.30).

### Mortality due to TB

TB-related deaths in Bangladesh has a declining trend. From 2001, there was 50% reduction in TB mortality. The SDG target is 90% reduction in the number of TB-related deaths by 20,130 compared to



**Figure 4.3.30.** Trends of treatment success rates (Bacteriologically-confirmed TB)



**Figure 4.3.31.** Trend of TB mortality in Bangladesh

that in 2015. In 2015, the absolute number of TB-related deaths was 73,000, which reduced to 59,000 in 2017; it means 19% reduction in TB-related deaths in two years (Figure 4.3.31).

### Drug-resistant TB

NTP started programmatic management of drug-resistant TB with 20-month regimen in August 2008 at NIDCH, Dhaka and, by the end of 2013, this service has been made available in CDHs Chattogram, Khulna, Sylhet, and Pabna. The CDH, Rajshahi and 3 Damien Foundation Hospitals have been managing drug-resistant TB since May 2005 with 9-month regimen under operational research.

Research on shorter treatment regimen for MDR-TB conducted in Bangladesh, popularly known as ‘Bangladesh Regimen’ has significantly reduced the MDR-TB treatment duration from 20 months to 9 months with a better treatment success. Bangladesh has started implementing this shorter treatment regimen from 1 April 2017 under programmatic condition and scaled up all over the country by 2017. By the end of 2018, a total number of 7,387 MDR-TB patients were

treated with second-line anti-TB drugs (Figure 4.3.32).

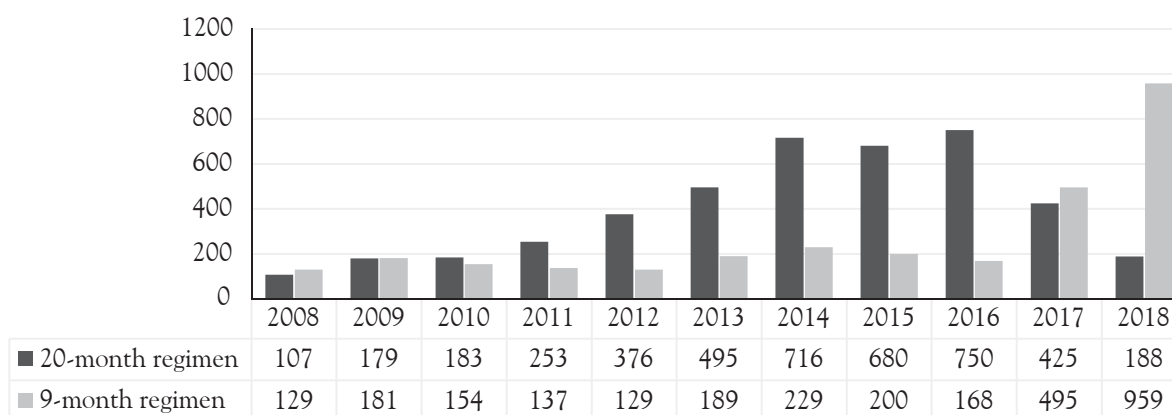
Bangladesh is maintaining a high treatment success rate in MDR-TB over the years. Strict the adoption of DOTS, quality-assured second-line drugs free of charge, regular monitoring, management of adverse events, social support and community involvement are the key components of programmatic management of drug-resistant TB in Bangladesh.

NTP in Bangladesh introduced new anti-TB drugs, i.e. Bedaquiline and Delamanid in 2016 and successfully treated XDR, pre-XDR, and complicated MDR-TB cases.

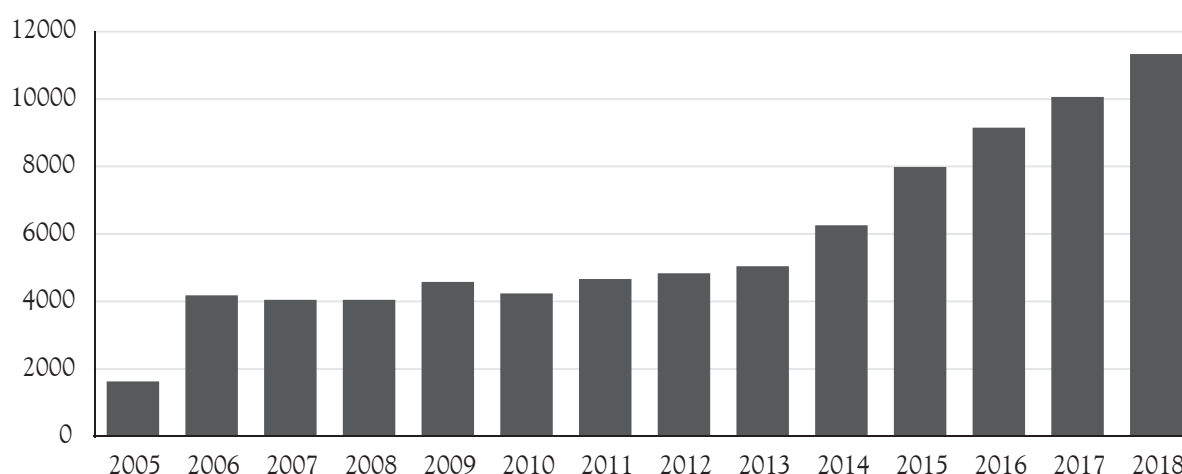
In total, 32 XDR-TB cases were diagnosed and given treatment till December 2018.

### Childhood TB

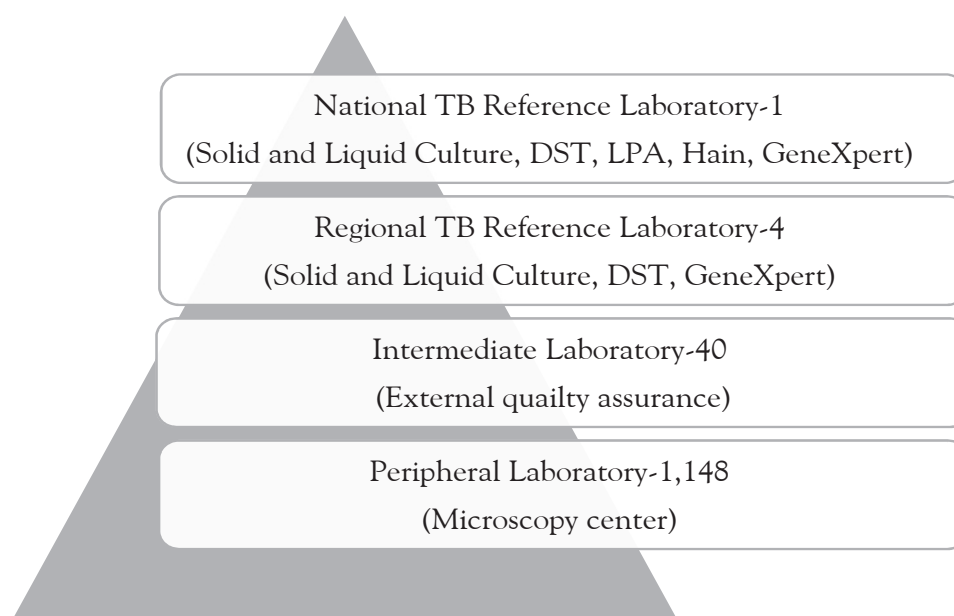
In 2017, children below 15 years of age constituted 4.37% of new TB cases. This is an increase from 3.3% in 2014 and from 2.8% in 2013. Therefore, progress is being made in identifying new pediatric cases; 10,189 childhood TB cases were identified in 2017, which increased to double in comparison with 2013 (Figure 4.3.33).



**Figure 4.3.32.** Number of MDR-TB cases given second-line treatment under different regimens



**Figure 4.3.33.** Childhood TB case notification in Bangladesh



**Figure 4.3.34.** Laboratory network of NTP

Preventive therapy for under-5 children is being successfully implemented among the children with close contact of a TB case.

Capacity-building of the pediatricians, medical doctors, and field-level staff has been completed, which helped in the significant increase of childhood TB case detection.

New child-friendly pediatric formulation of anti-TB drug has been introduced in Bangladesh from April 2017.

NTP is currently running 219 GeneXpert machines in 193 sites all over the country.

The laboratory network has rapidly expanded since 2012 in terms of the number of

diagnostic sites for GeneXpert, culture, and DST, and has greatly improved patient's access to diagnosis (Figure 4.3.34).

Bangladesh has introduced new diagnostic algorithm for testing all presumptive TB cases by GeneXpert, a more sensitive tool for diagnosis of TB cases from January 2018.

### Public-Private Mix (PPM)

The multi-sectoral response, including public-private mix (PPM), is one of the strengths of NTP to combat TB in Bangladesh. PPM is a cross-cutting strategy within the National Strategic Plan that contributes to the national TB control goals and objectives.

Currently, NTP is working with 28 NGOs with a vision of WHO's END TB, which was initiated back in 1994. The partners include: BRAC, Damien Foundation, LEpra, HEED, RDRS, TLMI, LAMB Hospital, PIME Sisters, NATAB, icddr, and BGMEA. The Government's strength includes: physical facilities from community level to the tertiary level, and TB is integrated with general health services whereas the NGOs' strength includes: extensive network from central to the grassroots level.

Existing PPM interventions in Bangladesh include the following:

- Engaging public and private medical college hospitals and selected high-volume hospitals in TB control activities
- Engagement of graduate medical practitioners
- Using social enterprise model where graduate medical care providers are linked to a designated TB screening center
- Using individual private medical care provider engagement model
- Engagement of the professional bodies
- Engagement of informal healthcare providers
- Promotion and integration of TB services in workplaces.

### Response to TB/HIV co-morbid cases

Bangladesh is maintaining a low prevalence and incidence of HIV (below 0.1%) through the national response to HIV for the last two decades. In most of the key populations, the prevalence is still less than 1% (MSM, transgender, PWID, FSW, etc.), and it is less than 0.01% in the total population.

Under TB/HIV collaborative activities, all HIV cases are being tested for TB, and around 2% of TB cases are being tested for HIV. In 2017, a total of 3995 TB cases were tested for HIV; 67 TB/HIV co-infected cases were identified; among them, 86% received ART. HIV +ve patients receive isoniazid preventive therapy to prevent TB.

NTP has recently published the 2nd edition of the guidelines for TB/HIV collaborative activities where the full package of WHO's TB-HIV Strategy has been adopted.

A total of 919 HIV-infected people have been detected in the country this year, the highest in the last 20 years, said a government survey. Conducted by the National AIDS/STD Program of the Ministry of Health and Family Welfare, the survey reported: 170 of the affected people had already succumbed to death for the virus. Since 1989, a total of 7,374 HIV-infected people have been detected in the country. Of them, 1,242 died of the disease, the report said (Source: National STD/AIDS Program, 2019).

## Monitoring and evaluation

NTP conducts routine quarterly monitoring missions in every district, including all stakeholders to review TB control activity. Routine supervision and monitoring are being conducted jointly by MOHFW, NTP, and partner NGOs; regular feedback is provided for better program implementation. NTP is collecting quarterly TB reports through DHIS2, an integrated government electronic reporting system.

Besides paper-based recording and reporting, an electronic individual case-based recording and reporting tool (eTB Manager) has been introduced in Bangladesh. The eTB Manager is functional in 246 upazilas.

## Political commitment

All the policies and strategies taken by the Government showed high political commitment toward TB control program. It is also reflected in the current Operational Plan by significantly increasing domestic funding.

Honorable Prime Minister of Bangladesh

participated in the Global Fund 5th Replenishment Conference held in Montreal, Canada, on 16-17 September 2016. In that conference, the Global Fund Highlighted Bangladesh TB Control Program as one of the successful ones for utilizing Global Fund investments and producing results.

Honorable Minister for the Ministry of Health and Family Welfare is Chair of the Bangladesh Country Coordinating Meeting (BCCM). He attended the Ministerial Meeting Towards Ending TB in the South-East Asia on 15-16 March 2017 held in New Delhi and signed the 'Delhi Call for Action' to end TB in the WHO South-East Asia Region by 2030. The Minister also attended the first WHO global ministerial conference held in Moscow, Russian Federation, on 16 -17 November 2017.

In October 2017, Honorable Minister for the Ministry of Health and Family Welfare declared "Zero TB Cities Initiatives" and calls on all stakeholders in urban areas to focus on local government participation in a comprehensive set of activities to treat, diagnose, and prevent TB.



"I firmly believe that tuberculosis is preventable and treatable. It needs commitment, determination and solidarity... let's make a pledge to work together towards that end."

**Figure 4.3.35.** Sheikh Hasina, Hon'ble Prime Minister of Bangladesh speaking in the Global Fund Replenishment Conference in Canada

## Respect for equity, ethics, and human rights

TB program was specially focused on and designed for the poor, marginalized and vulnerable population groups of the following areas:

- Urban slums
- Prisons-working in 56 prisons with laboratory and DOTS facility
- Garments industries
- Tea gardens-sputum campaign
- Stranded Bihari camp
- Refugee camp
- Migrant population

Recently, more than 8 million refugees have infiltrated into the southern part of Bangladesh from Myanmar. These Displaced Population of Rakhaine State (DPRS) is in a huge threat for all kinds of communicable diseases. The Government of Bangladesh has ensured integrated health services for these DPRS. With all kinds of health services, specially-designed TB program is also working in the area where the DPRS are living. Besides the regular health centers and services, NTP provided additional human resources to provide support to the camps. Fifteen additional microscopy laboratories have been established in that area, and one GeneXpert has been placed. NTP will provide an additional portable X-ray machine for diagnosis of TB cases besides the regular X-ray facility in the upazila health complexes. All the diagnosed TB patients are receiving TB treatment under supervised DOTS.

## Program supported by development partners

Funding sources for the Bangladesh TB Control Program are: Bangladesh Government

through MOHFW budget, the Global Fund, and USAID. Apart from this, WHO also extends financial and technical support in some areas as a technical partner. Some major highlights of the roles of development partners in TB control are as follows:

### USAID

‘Challenge TB Bangladesh’ is part of a 5-year cooperative agreement funded by USAID. It supports the National TB Program of Bangladesh to achieve the goals of its National Strategic Plan for TB. MSH is the implementing partner while KNCV and IRD provide technical guidance to the project. CTB provides technical assistance to the National Tuberculosis Control program with a country-wide coverage.

The project serves to achieve the following objectives:

- ◆ Improved access to quality patient-centered care for TB, MDR-TB, Co-morbid conditions (e.g. TB/HIV, TB/Diabetes, etc.)
- ◆ Prevention of transmission and disease progression
- ◆ Strengthened TB platforms.

### World Health Organization

WHO is providing support to increase efforts for detection of TB cases; diagnostics, and laboratory strengthening; maintain high cure rates; improve the quality of TB control services and strengthen major critical components of the service delivery system; address the issue of drug resistance; set up norms and standards; assist in taking evidence-based policy decisions; mobilizing partnerships for TB control and supporting research, monitoring, and development.



**icddr,b**

International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b), with unique proximity to the health challenges of the developing world—both urban and rural—provides cutting-edge research that is relevant, rigorously tested, and scalable in resource-limited settings. Tuberculosis remained a focused area of research in icddr,b, which respects and values all national guidelines and policies and maintains a very highly-esteemed partnership with NTP. At present, icddr,b is the only Sub-recipient (SR) of Global Fund and is implementing their TB Control Program in direct partnership with NTP.

**International Organization for Migration Bangladesh (IOM Bangladesh)**

International Organization for Migration Bangladesh (IOM Bangladesh) is affiliated with the National TB Control Program in Dhaka and Sylhet. In addition, IOM Bangladesh has recently started TB laboratory

enabled to process sputum sample for Acid fast Bacilli smear and *Mycobacterium tuberculosis* (MTB) culture, along with Drug Susceptibility Test.

In the last 12 years, IOM Bangladesh conducted health assessment for more than 225,000 people, identified over 259 TB cases, and cured 47 TB patients successfully.

**Program implemented by other partners (NGOs, private sector)**

Bangladesh represents a unique example of close collaboration between NTP and NGOs, and in TB Control Program, there is a clear geographic assignment from central level to avoid overlap. BRAC, Damien Foundation, HEED Bangladesh, LAMB Hospital, NATAB, NGO Health Service Delivery Project (NHSDP), PIME Sisters, Rangpur-Dinajpur Rural Service, Salvation Army, and The Leprosy Mission Bangladesh are implementing their respective TB programs in partnership with NTP.



# Non-communicable Disease Control

## Multi-sectoral approaches emphasized

Bangladesh is facing epidemiological transition in disease burden. Like many countries in the world, the burden of non-communicable diseases (NCDs) in Bangladesh is an emerging public health challenge. Non-communicable Disease Control (NCDC) is the Operational Plan (OP), taking the stewardship role on behalf of the Bangladesh Government. To overcome the increasing non-communicable disease burden, total allocation for this OP has been increased more than three folds in the 4th HPNSP than that in the previous one. Bangladesh is now ready to achieve the 2025 NCD targets and 2030 SDG targets. Overall mortality from cardiovascular diseases, cancers, diabetes, or chronic respiratory diseases has relatively reduced to 25% (WHO) from 2014 to 2017. Relative reduction in the prevalence of raised blood pressure is 25% from 2010 to 2018.

The Government approved the ‘Strategic Plan for Surveillance and Prevention of NCDs in Bangladesh’ and, very recently, it has endorsed ‘Multi-sectoral Action Plan for Prevention and Control of NCDs 2018-2025’, with national NCD targets. A National Multi-sectoral NCDC Coordination Committee (NMNCC) is already approved. NMNCC Secretariat has been established and equipped with necessary logistics.

The Government has started dialogue with stakeholders, including WHO on measures to improve diets, including awareness

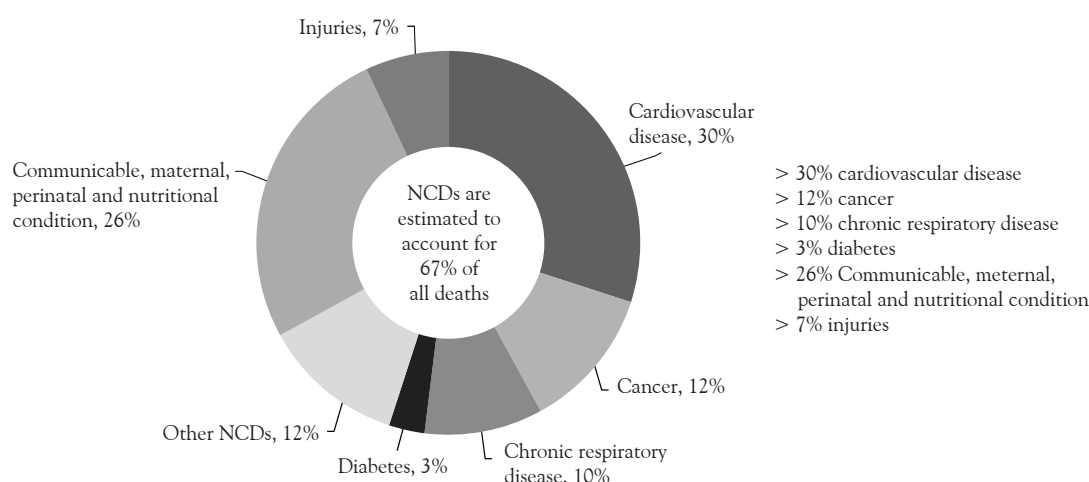
campaigns on dietary salt reduction. The NCDC program is piloting community-based NCD management model (preferably for hypertension and diabetes) in some selected sub-districts following WHO’s Package of Essential Non-communicable Disease Intervention (PEN) protocol. The cancer registry protocol and formats are on trial in eight government medical college hospitals under the guidance of NICRH and NCDC.

### Goal of NCDC Program

Reduce mortality and morbidity due to NCDs in Bangladesh through control of risk factors and improving health service delivery.

### Objectives of NCDC Program

- Formulate, promote, and implement effective, integrated, sustainable, and evidence-based public policies for non-communicable diseases, their risk factors, and determinants
- Foster, support, and promote social and economic conditions that address the determinants of chronic non-communicable diseases and empower people to increase control over their health and to adopt healthy behaviors
- Strengthen the capacity and competencies of the health systems for the integrated early detection, management, and control of the risk factors of non-communicable diseases.



**Figure 4.4.1.** Major NCDs (CVD, cancer, diabetes, COPD)

### Non-communicable disease control targets

Bangladesh is committed toward achieving the 2025 NCD targets and 2030 SDG targets. Potential indicators for the 2025 targets are listed in Table 4.4.1. Through implementation of the multi-sectoral NCD Control and Prevention of NCDs (2018–2025), Bangladesh aims to achieve the proposed 2025 targets.

The STEPS survey for risk factors of non-communicable diseases (NCDs) in Bangladesh was carried out from January to June 2018. Bangladesh followed all three steps (Step—1, Step—2, and Step—3) in this survey. A total of 8,185 adults participated in the survey. The overall response rate was 82.67%.

The trend of raised blood pressure shows an increment from 2010 (17.9%) to 2018 (25.4%), and it is consistent in both men and women.

Prevalence of diabetes was 6.4% among the sample population. Again, 6.4% of the respondents were pre-diabetic. 45.2 % of those having raised blood glucose were taking medication.

According to the report of Global Burden of Disease Study 2016, age-standardized death rate due to NCDs among all people in Bangladesh was 369.78 (range 335.79-406.84) per 100,000 population.

Death rates due to major NCDs (CVDs, cancer, diabetes, and chronic respiratory diseases among population aged 30 to 70 years ranged from 398.1 to 503.7 per 100,000 population.

Majority of the respondents never had screening for cervical cancer. Only 7.5% women aged between 30 and 49 years had cervical cancer screening test in their lifetime.

### Mental Health

Mental disorders account for a large burden of disease and disability globally. According to a WHO-supported national survey on mental health in Bangladesh, 16.8% of the adult population and 13.6% of children in the country are suffering from mental disorders. Bangladesh passed a new Mental Health Act in 2018, which replaced the out-dated 105 years

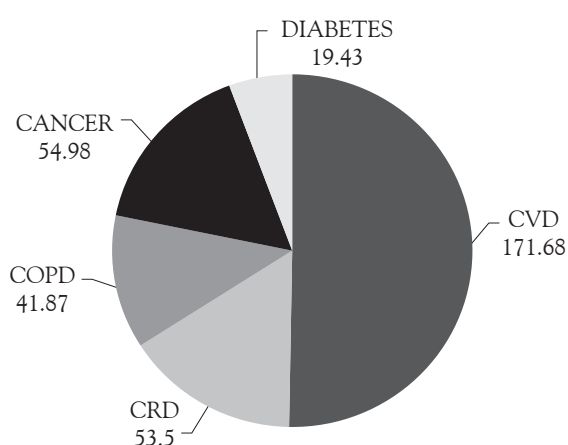
**Table 4.4.1.** NCDC baseline and proposed targets of Bangladesh

| Parameter/Indicator                                                                                                                               | Baseline                                                                                                                                                                    | 2019                                                                                                                                                                     | 2025 targets                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Overall mortality from cardiovascular diseases, cancers, diabetes or chronic respiratory diseases                                                 | 59%                                                                                                                                                                         | 67%                                                                                                                                                                      | 25% relative reduction                                      |
| Reduction in the harmful use of alcohol                                                                                                           | 0.9% (STEPS 2010)                                                                                                                                                           | 1% (STEPS 2018)                                                                                                                                                          | 10% relative reduction                                      |
| Reduction in the prevalence of current tobacco-use in persons aged over 15 years                                                                  | 43.3% (2009)                                                                                                                                                                | 35.3% (2018)                                                                                                                                                             | 30% relative reduction                                      |
| Reduction in the prevalence of insufficient physical activity                                                                                     | 27% (STEPS 2010)                                                                                                                                                            | 29.1% (27-31.3) (STEPS 2018)                                                                                                                                             | 10% relative reduction                                      |
| Reduction in mean population intake of salt/sodium                                                                                                | 10.6 g/day (95% CI 9.6-11.6)                                                                                                                                                | 16.5 g/day (95% CI 16.4-16.7)                                                                                                                                            | 30% relative reduction                                      |
| Relative reduction in the prevalence of raised blood pressure                                                                                     | 17.9% (STEPS 2010)                                                                                                                                                          | 25.2% (23.4-27)                                                                                                                                                          | 25% relative reduction                                      |
| Halt the rise in obesity and diabetes                                                                                                             | Overweight: 14.3% $\leq$ BMI 25-29.9 kg/M <sup>2</sup> , obese: 3.6% $\leq$ BMI 30 kg/M <sup>2</sup> , (STEPS 2010), Diabetes: 6.3 % of the adults (IDF), 5.5% (STEPS 2006) | Overweight: 20.3% $\leq$ BMI 25, 29.9 kg/M <sup>2</sup> , obese: 5.5% $\leq$ BMI 30 kg/M <sup>2</sup> (STEPS 2018) Diabetes: 8.4% of the adults (IDF), 5.5% (STEPS 2018) | 0%                                                          |
| Reduction in the proportion of households using solid fuels (wood, crop residue, dried dung, coal, and charcoal) as the primary source of cooking | 41.4% straw and leaves, 4.8% brans, 42 % wood, bamboo, 10.3% natural gas (BBS 2005), COPD 4.3% (2007, Bangladesh Lung Foundation), 2.97 % (WHO survey)                      |                                                                                                                                                                          | 50% reduction in the number of households using solid fuels |

**Table 4.4.1 Contd.**

Table Continued...

| Parameter/Indicator                                                                                                                                                      | Baseline                                            | 2019 | 2025 targets |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|------|--------------|
| Increase the number of eligible people receiving drug therapy and counseling (including glycemic control) to prevent heart attacks and strokes                           |                                                     |      | 50%          |
| Improve the availability of affordable basic technologies and essential medicines, including generics required to treat major NCDs in both public and private facilities | NCD corner piloting in every upazila health complex |      | 80%          |



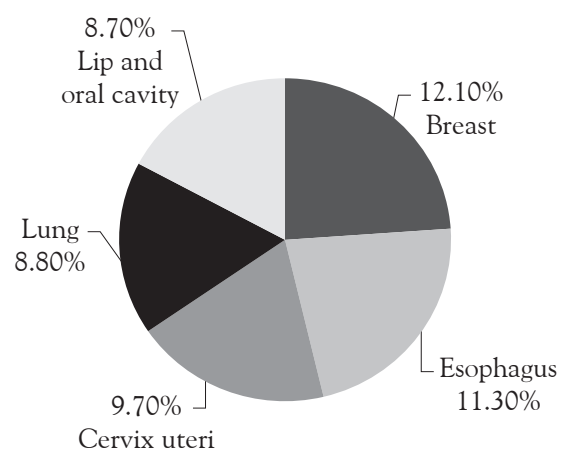
**Figure 4.4.2** Death rate due to major NCDs per 100,000 population of 30-70 years

old Lunacy Act 1912. Mental Health Policy 2019 has been developed. National Mental Health Strategy Action Plan 2020-2030 has been developed. Besides these, the National Strategic Plan for Neurodevelopmental Disorders 2016-2021 has been developed, and it is being implemented by different ministries. Widespread social stigma, limited human resource capacity, and lack of formal initiatives are hindrances to addressing mental health problems. Identification of signs of the most

**Table 4.4.2.** Prevalence of major NCDs as percentage in 2016

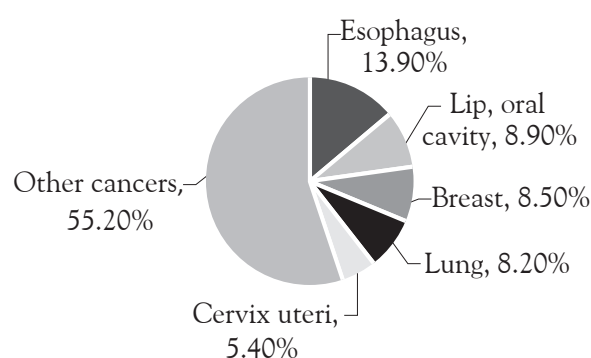
| Type of disease        | Prevalence of major NCDs (%) |
|------------------------|------------------------------|
| Ischemic heart disease | 3.4                          |
| Stroke                 | 0.9                          |
| CVD                    | 0.4                          |
| Diabetes               | 6.4                          |
| COPD                   | 2.97                         |

Source: Global Burden of Disease Study 2016

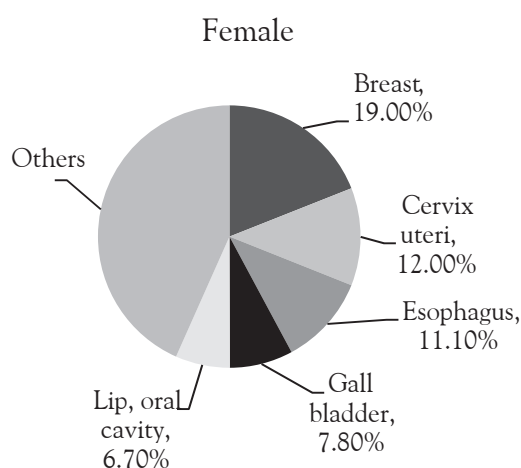


**Figure 4.4.3.** Five most frequent cancers in 2016 (percentage)

commonly-prioritized mental health conditions (autism and neurodevelopmental disorders, epilepsy, and common mental health disorders, including depression, psychosis, anxiety and substance-abuse) and their referral to UHC and DH is essential for management. Community and union-level facilities will also participate in the support to the rehabilitation of mental health patients, including fight against stigma. The DH is the main facility for organizing these services and will host the Community Mental Health Team in charge of providing specialized support to UHC and others.



**Figure 4.4.4.** Percentage of new cases of cancer during 2018 in both sexes of all ages (GLOBOCAN 2018)

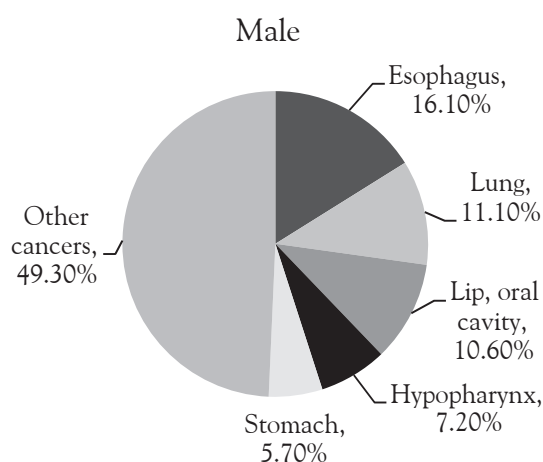


**Table 4.4.3.** Current situation of mental health disease prevalence

| Common mental disorder              | Percentage (Range) |
|-------------------------------------|--------------------|
| <b>Adults</b>                       |                    |
| Any mental disorder among adults    | 16.8 (15.5-18.2)   |
| Anxiety disorder                    | 4.5 (3.80-5.3)     |
| Depression                          | 6.7% (5.8-7.6)     |
| Schizophrenia                       | 1% (0.7-1.3)       |
| Substance-abuse                     | 0.2                |
| <b>Children and adolescents</b>     |                    |
| Any mental disorder among children  | 13.6 (11.5-15.7)   |
| Neurodevelopmental disorders (3.4%) | 5.9 (4.5-7.2)      |
| Autism (0.84%)                      | —                  |

### Shishu Bikash Kendra (SBK) or Child Development Center (CDC)

Child Development Center (CDC) or commonly known in Bangla as Shishu Bikash Kendra (SBK) are multidisciplinary specialized service centers for healthcare of children. These are located in different medical college hospitals in Bangladesh catering to the needs



**Figure 4.4.5.** Percentage of new cases of cancer among males and females of all ages during 2018 (GLOBOCAN 2018)

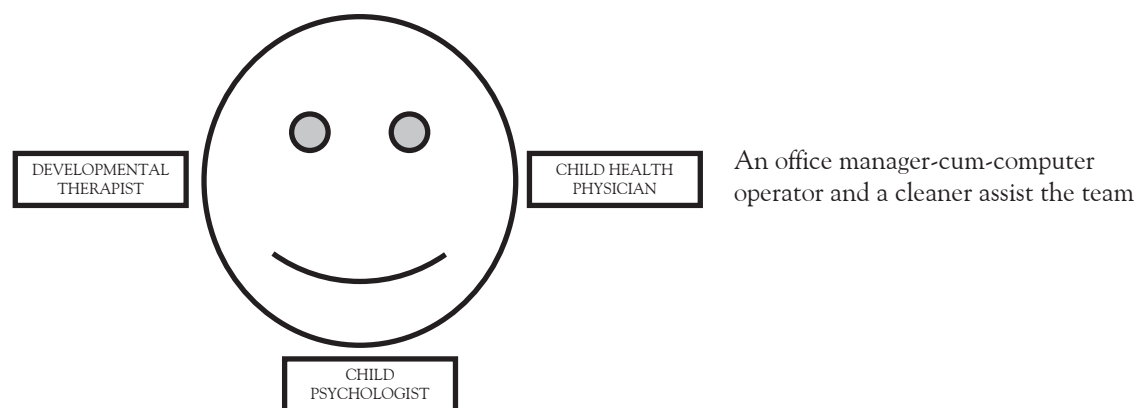
for diagnostic and therapeutic services for children with autism spectrum disorder (ASD) and other neurodevelopmental disorders (NDDs). There are 15 CDCs in the country at present in medical college hospitals; 25 more CDCs are planned to be established in the next 4 years in medical college hospitals and district hospitals. The centers emphasize on the development and psychological assessments of the children with ASD, NDDs, and the follow-up visits for ensuring functional improvement in motor, vision, hearing, speech communication skills, cognitive and behavioral domains. Each CDC is run by a multidisciplinary team comprising a child-physician, a child psychologist, and a development therapist. The team is dedicated to ensuring that every child who comes to the center is undergone a comprehensive assessment, diagnostic work-up, and selection of best-suited intervention technique is done. The teams work closely with the parents and family of the child to optimize his/her development potential, improve the quality of life, and prevent disability.

The journey began with the establishment of 5 SBKs in August 2009 in Dhaka Medical College Hospital, Shaheed Suhrawardy Medical College Hospital, Sir Salimullah

Medical College & Mitford Hospital, Sher-E-Bangla Medical College Hospital in Barisal, and MAG Osmani Medical College Hospital in Sylhet. Following successes, 5 more centers were established in 2010 that included medical college hospitals in Chattogram, Khulna, Rajshahi, Mymensingh, and Rangpur. In February 2014, additional 5 centers started providing services in Dinajpur, Faridpur, Comilla, Cox's Bazar, and Shaheed Ziaur Rahman Medical College Hospital in Bogra.

Under the 4th Health, Population and Nutrition Sector Program (4th HPNSP), Hospital Services Management Operational Plan (OP) will establish 25 more SBKs in the next three years. There are several other CDCs working in autonomous and private hospitals. They include Dhaka Shishu (Children) Hospital, BIRDEM, Shishu Shastho Foundation Hospital, ICMH, Matuail, CMH, Dhaka, and Apollo Hospitals, Dhaka.

Since inception of the services in 2009, a total of 124,649 children (till June 2018) were served in the 15 CDCs; of them, 15,839 were served from July 2017 to June 2018. In addition to diagnosis and intervention



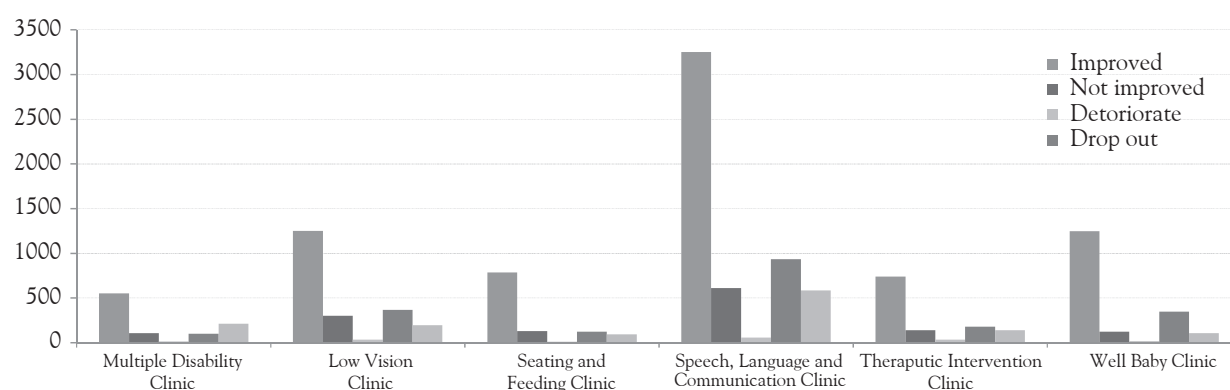
**Figure 4.4.6.** Multidisciplinary team at Shishu Bikash Kendra (SBK)

for ASD and NDDs, the CDCs are running weekly clinics for epilepsy, mental health, and Well Baby Clinic. In total, 294,432 follow-up visits were recorded during the same period. Rapid Neurodevelopmental Assessment (RNDA), Autism Diagnostic Observation Schedule (ADOS), Independent Behavior Assessment Scale (IBAS), Bayley Scales for Infant Development (BSID), Wechsler Intelligence Scale for Children (WISC IV), DSM-V, and ICD-10 are used, among other scales and methods, for assessment of children. Behavioral interventions, including motor skills, speech language communication, and sensory integration techniques are used. The professionals in each CDC have been working incessantly to serve the children regarding their development and well-being. However, workload (a CDC serves, on average, 250 children every month), high rate of attrition among employees, shortage of psychologist, and shortfall in the effective referral mechanism are the major challenges faced by the CDCs. Our CDCs provided technical assistance to various child development programs in Bhutan, Guatemala, Haiti, and Zambia in recent years. The CDCs are continuously engaged in research activities and have conducted three

major epidemiological surveys on autism and NDDs.

## Disability

Disability refers to a situation where any person who is affected with long-term or permanent physical, psychological, intellectual, developmental and sensory-related damages or obstacles which, in interaction with attitudinal and environmental barriers of the person, restricts the person's equal and effective participation in the society. The World Report on Disability 2011 estimates over a billion people to live with one or other forms of disability globally. The report states that people with disabilities generally have poorer health, higher rates of poverty than those living without disabilities, and that they face significant barriers and lack of access to services. It emphasized the need for strategy to overcoming these barriers to healthcare, rehabilitation, education, employment, and other support services. Disability is a great challenge for our country. Hon'ble Prime Minister of Bangladesh is very much concerned regarding this. Disability-inclusive primary healthcare is an important component of NCDC Operational Plan.



**Figure 4.4.7.** Number of clinics run by Child Development Centers, with number of children treated and the accompanying results



**Table 4.4.4.** Total attendance of patients in Shishu Bikash Kendras (July 2017- June 2018)

| Name of SBK<br>in medical<br>college | Number of new patients |      |     |     |       | Follow-up visits |     |     |     |     |      |      |      |      |      |     |             | Grand<br>Total |
|--------------------------------------|------------------------|------|-----|-----|-------|------------------|-----|-----|-----|-----|------|------|------|------|------|-----|-------------|----------------|
|                                      | OPD                    | PA   | WBC | IPD | Total | GDA              | TIA | MDC | S+F | LV  | SLC  | DT   | EPI. | PA   | OPD  | MH  | Total<br>FU |                |
| Dhaka                                | 605                    | 224  | 68  | 23  | 920   | 125              | 64  | 69  | 24  | 33  | 159  | 221  | 375  | 195  | 20   | 19  | 1304        | 2224           |
| Suhrawardy                           | 902                    | 219  | 63  | 54  | 1238  | 144              | 95  | 45  | 48  | 57  | 95   | 272  | 322  | 110  | 370  | 135 | 1693        | 2931           |
| Salimullah                           | 600                    | 189  | 0   | 59  | 848   | 78               | 36  | 44  | 20  | 18  | 63   | 241  | 402  | 117  | 78   | 1   | 1098        | 1946           |
| Sylhet                               | 927                    | 152  | 0   | 12  | 1091  | 239              | 4   | 3   | 0   | 10  | 25   | 583  | 16   | 46   | 1207 | 0   | 2133        | 3224           |
| Barisal                              | 575                    | 147  | 11  | 71  | 804   | 110              | 38  | 84  | 21  | 39  | 57   | 370  | 91   | 267  | 370  | 17  | 1464        | 2268           |
| Chittagong                           | 863                    | 234  | 0   | 0   | 1097  | 94               | 18  | 0   | 7   | 79  | 124  | 259  | 2339 | 185  | 723  | 0   | 3828        | 4925           |
| Rangpur                              | 971                    | 63   | 16  | 0   | 1050  | 24               | 4   | 0   | 2   | 2   | 118  | 341  | 19   | 89   | 162  | 0   | 761         | 1811           |
| Rajshahi                             | 1219                   | 173  | 9   | 15  | 1416  | 52               | 55  | 107 | 37  | 49  | 55   | 1104 | 575  | 107  | 1036 | 10  | 3187        | 4603           |
| Khulna                               | 862                    | 274  | 6   | 25  | 1167  | 140              | 13  | 99  | 10  | 18  | 53   | 65   | 135  | 306  | 133  | 54  | 1026        | 2193           |
| Mymensingh                           | 1013                   | 220  | 15  | 99  | 1347  | 58               | 44  | 24  | 92  | 415 | 33   | 917  | 258  | 137  | 1485 | 25  | 3488        | 4835           |
| Dinajpur                             | 379                    | 83   | 12  | 10  | 484   | 104              | 28  | 72  | 14  | 49  | 53   | 128  | 86   | 43   | 126  | 10  | 713         | 1197           |
| Faridpur                             | 621                    | 151  | 2   | 24  | 798   | 109              | 0   | 34  | 5   | 15  | 89   | 177  | 132  | 131  | 35   | 4   | 731         | 1529           |
| Cox's Bazar                          | 443                    | 54   | 61  | 7   | 565   | 81               | 28  | 78  | 2   | 29  | 46   | 205  | 96   | 72   | 189  | 8   | 834         | 1399           |
| Comilla                              | 1240                   | 187  | 10  | 0   | 1437  | 111              | 37  | 46  | 14  | 59  | 71   | 1894 | 180  | 119  | 1006 | 4   | 3541        | 4978           |
| Bogra                                | 1328                   | 240  | 7   | 2   | 1577  | 81               | 47  | 60  | 10  | 40  | 75   | 657  | 280  | 265  | 657  | 0   | 2172        | 3749           |
| Total                                | 12548                  | 2610 | 280 | 401 | 15839 | 1550             | 511 | 765 | 306 | 912 | 1116 | 7434 | 5306 | 2189 | 7597 | 287 | 27973       | 43812          |

## Current Status

A rapid assessment of disability found a prevalence rate of 8.9% in Bangladesh. The Household Income and Expenditure Survey (HIES) of Bangladesh (2010) found a prevalence rate of 9.1%. Taking into account the population estimate of Bangladesh (over 166 million), the national HIES (2010) estimate of 9.1%, and the WHO's estimates of 15%, Bangladesh has an estimated 15.1 million to 24.9 million people who are affected with one or other forms of disability.

Effort of the DGHS to demystify disability for primary healthcare (PHC) workforce, particularly medical doctors, is a way forward to strengthen the healthcare systems and structures by making these more responsive, equipped, and ready to cater to the people with disabilities.

Common disabilities in Bangladesh are: NDDs, ASD, cerebral palsy (CP), down syndrome, intellectual disability, mental disability, physical disability, visual disability, hearing and speech disability, deafness-blindness disability, multiple disabilities, and others.

## Ongoing activities to reduce disability

- Development of guidelines defining disability for health workforce in Bangladesh (by 6 workshops in 2017-2018)
- Development of sign language training manual
- Conducting sign language training in upazilas
- Survey done: Service Availability and Readiness Assessment (SARA) survey

for NCDC and disability service delivery system in Bangladesh

- Priority-based circulation of information for management of disability done at all levels
- Shibalay, Manikganj Pilot Project is ongoing (4 sensitization workshops for doctors already conducted, and 3 fieldworkers training ongoing)
- According to the guidelines, development of module preparation activities ongoing
- 1-3 days' compulsory training given for development of at least one doctor to identify disability at upazila level.

## Injury Prevention and Control Program

Injury is one of the leading causes of disability. WHO estimates that approximately 70,000 deaths in Bangladesh occur annually due to injury (burn, drowning, acid, and accidents at workplace) and about 18,000 deaths due to road-traffic injury. Poisoning is a leading cause of morbidity and mortality across the country, with an approximate annual admission of 64,000 cases; 2% of all admissions and 9% of deaths among hospitalized cases are related to injury and poisoning. Snake-bite is a rural health problem among the poor farming community causing more than 700,000 episodes and more than 6,000 deaths.

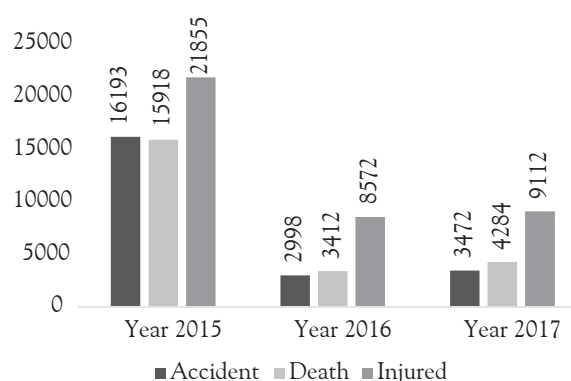
Prevention and emergency management of various injuries under the DGHS include: road-traffic injury, drowning, prevention and treatment of poisoning, burns, prevention and treatment of snake-bite, occupational injury, intentional injuries, and violence against women (GBV).

## Road-traffic injury

The National Committee to Protect Shipping, Roads and Railways (NCPSRR), a non-government organization, provided information on road-traffic accident in their yearly survey and observation report. According to the report, the number of road accidents and casualties increased in 2017 compared to 2016. The number of accidents increased by 15.82%, and death toll by 25.56% in 2017 compared to that in 2016. In 2017, total number of accidents was 3472, in which 4,284 people died, including 516 women and 539 children; 9,112 persons were injured.

### *Activities on reducing road-traffic injuries and deaths*

1. Education and publicity improved knowledge and behavior of road-users, and raised awareness about traffic rules and safety. Multi-sectoral coordination (non-health interventions and stewardship) increased.
2. Education of pedestrians regarding the use of road, footpath, and over bridge, along with imparting knowledge on traffic signals through school, community safety program, and media were prioritized. Multi-sectoral coordination (non-health interventions, and stewardship) was strengthened.
3. Recruitment, training and testing of drivers were done properly, ensuring minimum standards for, such as eligibility in terms of age and physical fitness; they were provided adequate training on safe driving practices. Multi-sectoral coordination (non-health interventions, stewardship, and regulatory actions) was emphasized.
4. Training of healthcare providers were conducted on pre-hospital care by the first responder and comprehensive training on trauma case management was also conducted for them. Cross-cutting issues linked to Essential Services Delivery (ESD), Secondary and Tertiary-level Hospital Management System (HSM) were addressed.
5. Engineering and environmental approach through multi-sectoral coordination (non-health interventions, stewardship and regulatory actions) was adopted.
6. Legislation and standards of multi-sectoral coordination (non-health interventions, stewardship and regulatory actions) were formulated.
7. Vehicle design and safety equipment through multi-sectoral coordination (non-health interventions, stewardship and regulatory actions) were ensured.



**Figure 4.4.8.** Number of accidents, injuries, and deaths in last 3 years found in NCPSRR Survey 2017

8. Emergency medical care, emergency assistance for rescue and transportation of the crash victims within golden hour were provided. Emergency medical service system—pre-hospital care and hospital care was strengthened.. Treatment of the road-crash victims in public and private hospitals was ensured.
9. Other activities included: effective coordination and collaboration among road safety council which is responsible for planning, implementation, and monitoring of road safety activities; formulating road safety policy; facilitating funding of research and dissemination of information; organizing road safety education, publicity, and training; collaboration among relevant sectors/departments/agencies for injury-related information and reporting; providing emergency medical services; effective collaboration with vehicle owners association and workers association.

## Occupational Health

Major occupations in Bangladesh are farming (two-thirds of the population are farmers producing rice, tea, mango, potato, and other crops), fishing, working in garments factories, working abroad, cleaning, raising livestock and poultry, working in forests, providing healthcare as professionals. Workers in ship-breaking industry are at increased risk of injury and death. These occupations have their own unique health-related problems. Industrial workers have different health problems as well. Occupational Safety Board of the Bangladesh University of Engineering and Technology (BUET) identified safety issues in different industries. To address the health of people in major occupations

in relation to equity, poverty, and class, the health hazards in major occupations need to be removed. A few observations suggest that important health hazards of common occupations are: injury; pesticide-related harms (self-poisoning and environmental hazards); injuries caused by animals (dog-bite and snake-bite); NCDs at early age; drowning at sea, river and pond; backpain and sciatica; and various infections (hepatitis, B, C; anthrax, typhoid, leptospirosis, malaria, melliodosis, typhus, TB, HIV infections). National health policy provided guidance for improvement of health of the farmers, industry workers, livestock and poultry workers

## Current status

No data were available during the reporting year. Sufficient budget was not allocated.

## Activities

Seminars on Safe Work Practice for healthcare professionals in selected districts were conducted in 28 batches.

## Environmental Health

Air pollution, noise pollution, soil pollution, water pollution, lead contaminations are some important and significant environmental health hazards that are related to development of NCDs. These are often neglected, although these collectively contribute to a considerable burden of NCDs. Ironically, the policy-makers and the public health specialists are not sensitized about these health problems at significant level. These are cross-cutting issues. Although the Department of Environment is taking many steps, coordination with and participation of the health sector are

mandatory. Air quality in major cities and in rural areas of Bangladesh is poor. This poor air quality is a contributor to the development of NCDs (COPD, for example) and childhood acute reparatory infection. Considering the importance of reducing air pollution, the country has developed 'Air Pollution Reduction Strategy for Bangladesh (2012)'. Lead is known to cause neurocognitive impairment. Prevention of lead contamination in food (turmeric, rice) and soil is essential.

### Arsenicosis

Contamination of groundwater with arsenic resulted in chronic arsenicosis in population in a large area of Bangladesh. Various health problems, including skin lesions, peripheral nerve problems, and even cancer can occur due to arsenic contamination of water. Collaboration between the DGHS and DHPE is ongoing to address the problem under a National Policy for Arsenic Mitigation (2004) with key interventions for providing safe water and care to the identified patients.

### *Current status of arsenicosis scenario in Bangladesh*

In a survey titled "Prevalence of Arsenicosis and the Treatment Availability" was conducted in 2017-2018 financial year funded by NCDC Operational Plan. According to the data collected from 61 of 64 districts (except Maulvibazar, Bandarban, and Sherpur), the total number of arsenicosis patients was 31,865.

The key findings are as follows:

- Heavily-contaminated districts are Bagerhat (100%), Chuadanga (100%), Lakshmipur (92%), Jessore (75%), Magura

(75%), Feni (67%), Chattogram (58%), and Satkhira (50%).

- Clinically-diagnosed patients were found in Laxmipur Noakhali, Chattogram, and Jaipurhat.
- Urinary excretion of arsenic without any clinical manifestation was seen in Chandpur (2 psn), Netrakona (1 psn), Jessore (1 psn), Narayanganj (1 psn), Sylhet (1 psn), and Maulvibazar (1 psn).

Apart from this study, provisional reports were collected from 42 districts, Information was gathered from the arsenicosis patients registrars. It revealed a total of 18,862 patients in October 2017.

Arsenic mitigation activities during the reporting period included: workshop on updating arsenicosis detection, screening tools, management protocol, updating arsenic mitigation policy, training on arsenicosis screening, management and treatment protocol for doctors at selected district-level facilities for capacity-building.

### Prevention of Thalassemia (Cross-cutting issue with HSM)

Thalassemia (Hb E disease) is the most common genetic disorder in Bangladeshi people, with a gene prevalence rate of around 6%. This translates that 6 out of every 100 Bangladeshis carry this mutant gene. With a population of about 160 million, around 10 million people are healthy carriers of this gene. Thalassemia is an autosomal recessive disease, which means that: if two carriers get married to each other, they will have 1 in 4 (25%) chance of having an offspring with B-Thalassemia

Major in every pregnancy. Over 5,000 affected children are born. These children require regular monthly blood transfusions and chelation therapy to remove the excess iron from their bodies in order to just stay alive. A large amount of money is required for each child per annum. Currently, majority of the patients in Bangladesh do not get adequate treatment, especially the transfusion of safe blood has been a big problem, and the majority of the thalassemics are positive for infections, such as hepatitis B, C, and HIV.

Since there is no easily-available cure for this disorder, the only hope lies in adopting a preventive programs in the line pursued by countries, like Italy, Cyprus, and Iran, which have resulted in either a complete control or a significant reduction in the birth of children with Thalassemia Major in these countries.

Considering the gravity of the issue, the Government of Bangladesh should take the lead with Bangladesh Thalassemia Prevention Program (BTPP), which will be unique and can be compared with any other preventive programs of the world.

### **Current status**

About 5,000 children are born with thalassemia every year in our country. This means more than 20 children are born every day. About 3-5% of people in our country carry thalassemia genes.

### **Activities**

The BTPP will be working on: (i) awareness about thalassemia. (ii) screening of carriers (extended family screening). (iii) pre-marital screening among the general public, (iv) genetic counseling, and (v) prenatal diagnosis.



# Disease Surveillance and Response of IEDCR

## Novel pathogens are of prime concern

The Institute of Epidemiology, Disease Control and Research (IEDCR) is the focal institute for conducting disease surveillances and outbreak investigations in Bangladesh. The IEDCR has been engaged in controlling diseases and conducting research on events of public health importance and developing public health workforce. The IEDCR comprises eight departments: Biostatistics, Epidemiology, Medical Entomology, Medical Social Science, Microbiology, Parasitology, Virology, and Zoonosis.

Conducting disease surveillances, outbreak investigation, public health research, and workforce development for emergency response are objectives of the IEDCR. In recent years, the country experienced a number of significant outbreaks of emerging and re-emerging infectious diseases, including dengue, Nipah virus infection, chikungunya, avian influenza, pandemic influenza H1N1, Japanese encephalitis, hepatitis E, cutaneous anthrax, etc. In response to these, the IEDCR has taken initiatives by developing workforce and augmenting the laboratory capacities to identify outbreaks or pathogens responsible for public health disasters. The IEDCR is also an academic institution for public health, offering courses, like MSc in Applied Epidemiology (FETPB), Cambridge Program to Assist Bangladesh in Lifestyle and Environmental Risk Reduction (CAPABLE) and Improving Public Health Management for Action (IMPACT) .

The IEDCR has five laboratories: (i) Virology, (ii) Parasitology, (iii) Microbiology, (iv) Entomology, and (v) OneHealth Laboratory. The IEDCR has the national reference laboratories for influenza, Nipah, and HIV. These laboratories play a significant role in the diagnosis of emerging infections, conduct and assist in training programs for laboratory personnel, and take part in quality assurance program. The IEDCR has over 170 staff members, including government officials, fellows, and project staff. In addition, there are officials seconded from ministries other than the Ministry of Health and Family Welfare (MOHFW) and various organizations.

### Special Achievement

Director of IEDCR Professor Dr. Meerjady Sabrina Flora, was elected Vice-chairman of International Association of National Public Health Institutes (IANPHI) in 2018.

### OneHealth Activities of IEDCR

OneHealth is a multi-sectoral and multi-disciplinary approach to dealing with the health issues arising at human-animal ecosystem interface. Strategic framework for OneHealth approach to infectious diseases in Bangladesh was developed in 2012 and later endorsed by the Ministry of Health and Family Welfare, Ministry of Fisheries and Livestock, and the Ministry of Forest and Environment. The IEDCR hosts the



OneHealth Secretariat. The objectives of OneHealth are to establish necessary institutional arrangements to enable effective collaboration between sectors.

### ***OneHealth activities in 2018***

- Coordination of outbreak investigations and responses to zoonotic diseases: OneHealth Secretariat coordinated zoonotic disease outbreak investigations and responses after being informed by different sources. OneHealth Secretariat so far coordinated activities against the anthrax outbreak at Rajbari, Pabna, and Sirajganj. The Secretariat also coordinated the mortality among crows in the Dhaka city and mortality among ducks in Borguna district, and mortality among poultry in Mymensingh district.
- Publication of OneHealth newsletter on quarterly basis: OneHealth Secretariat published a newsletter, namely Interface on quarterly basis. The first issue was published in March 2018.
- Training program titled “Familiarize OneHealth Approach for Future Veterinary Professionals”: A total of 63 intern students of Chittagong Veterinary Animal Science University (CVASU) participated in the training where they have been informed about overall activities of the IEDCR and other OneHealth-related activities.
- Training program on orientation of OneHealth approach: The OneHealth Secretariat organized a training program for the participants of different OneHealth partners for functional network and OneHealth advocacy.
- Training on scenario-based outbreak investigation using OneHealth approach: The OneHealth Secretariat organized a training program for capacity-building of Rapid Response Team (RRT), including physicians, veterinarians, and wildlife experts on disease outbreak investigation of known or unknown origin.
- Workshop on OneHealth Assessment for Planning and Performance (OH-APP): A workshop on OneHealth Assessment for Planning and Performance (OH-APP) was arranged to prioritize 10 areas for planning and performance of OneHealth.
- Observance of International OneHealth Day 2018: The OneHealth Secretariat and OneHealth Bangladesh jointly celebrated the International OneHealth Day, 2018. A day-long seminar on “Steps towards Antimicrobial Resistance through OneHealth Approach” was organized on 10 November 2018 at Sasakawa Auditorium, icddr,b, Mohakhali, Dhaka.

### **Entomological Activities of IEDCR**

The laboratory of medical entomology department is carrying out the biological efficacy tests of different types of public health insecticides supplied by different departments. The insecticides are tested against mosquito, housefly, cockroaches (American and German), and bedbugs. This department is providing regular services free of charge to many organizations.

In 2018, medical entomology department investigated the bio-efficacy of over 431 insecticides; among those the common ones were: coils, adulticides, larvicides, impregnated mosquito nets, aerosol, vaporizer, repellent gel, repellent cream, repellent lotion,

cockroach chalk, cockroach gel, cockroach powder, insect growth regulator, etc.

## Surveillance Activities of IEDCR

The IEDCR conducts sentinel sites, community-based, cellphone-based, event-based and web-based disease surveillances throughout the country.

## National Influenza Center

The IEDCR is recognized as the National Influenza Center (NIC) of Bangladesh by World Health Organization (WHO) since 2008. Like all NICs throughout the world, the IEDCR is designated by national ministries of health and recognized by WHO. The NICs form the backbone of the WHO's Global Influenza Surveillance and Response System (GISRS). The IEDCR collects specimens and performs preliminary analysis and ship the specimens and isolated viruses to WHO collaborating centers for advanced antigenic and genetic analyses. The results are used in developing the influenza vaccine each year. The IEDCR coordinates all the influenza activities in Bangladesh, including surveillances. The NIC in the IEDCR in recent years developed laboratory capacity and is regularly testing specimens from surveillance sites for influenza typing and sub-typing and sequencing for new/novel strains. The NIC regularly submits data to FluNet and FluID of WHO. The IEDCR publishes a bulletin regularly to disseminate influenza surveillance data and update with the latest global knowledge on influenza.

National Influenza Surveillance Bangladesh (NISB): Through NISB surveillance, the IEDCR collected 3,118 samples of severe acute respiratory infection (SARI) and patients with influenza-like illness (ILI),

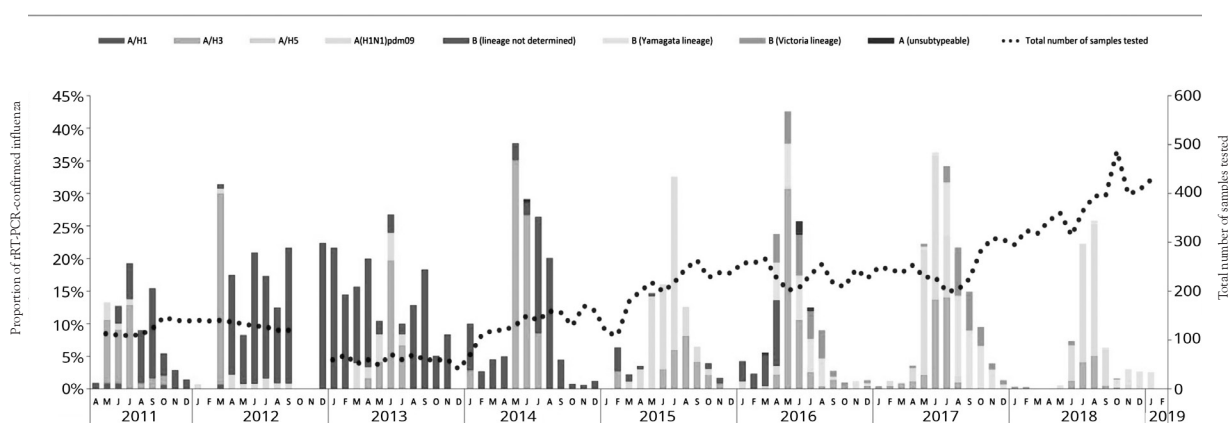
out of which 267 (8.6%) were influenza. Most of the cases were Influenza A (87%), and most common subtype was AH1pdm09 (69.3%). Males were most commonly positive for influenza (1860, 59.7%). Most of the influenza-positive cases were from 1-4 year(s) age-group (71, 9.9%). Most of the positive cases were from Habiganj district (43, 16.1%), followed by Patuakhali (38, 14.2%), and Thakurgaon (33, 14.4%) district.

## Trend of influenza in Bangladesh assessed by NISB

Hospital-based influenza surveillance (HBIS): HBIS platform of icddr,b, in collaboration with IEDCR and US-CDC, collected 2,714 samples from SARI and ILI patients in 2018, out of which 416 (15.3%) were influenza; of them, 338 (81%) were Influenza A, and 78 (19%) were Influenza B. Figure 4.5.1 shows trend of influenza in Bangladesh assessed by NISB (Source: IEDCR, Mohakhali).

Nipah virus transmission in Bangladesh: Human Nipah Virus (NiV) transmission surveillance is conducted to identify outbreaks of NiV infection in Bangladesh to characterize the conditions and risk factors for transmission of NiV. In 2017-2018 Nipah season (December to March), serum samples were collected from a total of 708 encephalitis cases. Three cases were found positive for Nipah IgM antibody by ELISA.

Surveillance of Japanese encephalitis: Surveillance of Japanese encephalitis (JE) helps in estimating the proportion of JE cases among the acute encephalitis cases to generate representative data for the policy-makers and to find out the impact of administered vaccine. In 2018, a total of 1,555 Acute Encephalitis Syndrome {AES} cases were tested for IgM



**Figure 4.5.1.** Trend of influenza in Bangladesh assessed by NISB (Source: IEDCR, Mohakhali)

antibody against JE by ELISA in the Virology Laboratory of the IEDCR; out of 1,555 cases, 103 were found JE-positive.

**Antimicrobial resistance (AMR) surveillance in Bangladesh:** The IEDCR has been conducting AMR surveillance since 2017, with technical support from US-CDC and WHO to find out the status of antimicrobial resistance in 10 selected bacteria from six types of specimen (urine, stool, wound swab, sputum, blood, and endotracheal aspirate). In 2018, three new sites: Dhaka Medical College, Khulna Medical College, and Sylhet MAG Osmani Medical College started their activities in addition to the previous five sentinel sites. In total, 8,206 samples have been tested in the sentinel sites till 31 December 2018, and positive samples have been preserved at ultra-low ( $-80^{\circ}\text{C}$ ) temperature in the freezer at the IEDCR for further analysis. Dissemination of the findings on the antimicrobial resistance was done to Subject Matter Expert Exchange (SMEE) program at Dhaka on 7 August 2018.

**Anthrax surveillance and response:** The IEDCR has started an active surveillance at nine sentinel sites of the five anthrax-endemic districts, based on previous outbreak reports



**Figure 4.5.2.** A meeting of the Subject Matter Expert Exchange (SMEE) program was held in Dhaka on 7 August 2018

since March 2018 to get early information and unreported cases from the community. OneHealth approach is being followed in conducting the surveillance activity where both human health and animal health personnel actively participate. Of the 9 sentinel sites, only one site—Gangni upzila of Mehepur district—has been reporting human anthrax cases, based on clinical presentations and meeting case definition of surveillance protocol. At the IEDCR laboratory, 97 samples were tested and 6 samples were found to be positive. All other 8 sites have been sending zero report monthly as there is no human cases of anthrax.

**Leptospirosis surveillance in Bangladesh:** The IEDCR is conducting leptospirosis surveillance in 10 sentinel sites by utilizing the existing Foodborne Illness Surveillance Platform. The existing protocol was updated in collaboration with Bacterial Special Pathogen Group of CDC-DGHS. The sentinel sites personnel were trained at the IEDCR on the updated protocol. The surveillance is ongoing at the sentinel sites. Rapid Detection Kits are being used in identifying the leptospirosis cases within the surveillance case definitions, enabling public health officials to classify and count cases ... diagnosis or determining how to meet an individual patient's health needs.... Clinical presentation includes history of fever within the 5-14 days' prescribed case definition. The IEDCR has also developed capacity to conduct PCR for leptospirosis. The surveillance will help validate the Rapid Detection Kit and epidemiological characteristics of leptospirosis in Bangladesh.

**Hospital-based dengue surveillance:** Hospital-based Dengue Surveillance is going on in four sentinel sites: Dhaka Medical College Hospital, Uttara Adhunik Medical College Hospital, Khulna Medical College Hospital, and Chittagong Medical College Hospital. In 2018, four hundred thirty-seven blood samples received from the sentinel sites have been tested in the IEDCR. Of these, 63 have tested positive for dengue IgM.

**Cholera surveillance:** The IEDCR has a cholera surveillance program in 22 sentinel sites in Bangladesh run in collaboration with icddr,b. Till December 2018, a total of 26,731 stool samples were collected. Samples were tested in the field by Rapid Diagnostic Test (RDT), and 2,300 of those (9.3%) were found to be positive for *Vibrio cholerae*. Culture was

done for 26,465 stool samples, and the number of culture-positive samples was 1,448 (5.5% of the total samples). In the month of December 2018 alone, 613 RDTs were done in the field, and positive results were found in 37 samples (6.0%). Culture was done for 667 samples in the laboratory, and 25 (3.7%) were found positive for *Vibrio cholerae*.

**Hospital-based rotavirus surveillance:** The IEDCR, as a national organization, made a transition plan in 2017 to undertake Hospital-based Rotavirus and Intussusception Surveillance (HBRIS) as rotavirus vaccine is being introduced in the country. According to the transitional plan, the IEDCR is designated to perform all the laboratory testings since January 2018 and, as per plan, samples were transported to the IEDCR from sentinel sites. In 2018, a total of 1,447 suspected samples were collected from the sentinel sites and tested for rotavirus antigen by ELISA at the Virology Laboratory of the IEDCR; 853 (58.9%) samples were positive for rotavirus antigen. A total of 170 positive samples were also tested for genotyping by sequencing at the Molecular Laboratory at the Virology Department of the IEDCR. G1P8 was the most common (52.9%) circulating rotavirus genotype in the year 2018. A total of 40 intussusception cases were also confirmed that year

**Child Health and Mortality Prevention Surveillance (CHAMPS):** Child health and mortality prevention surveillance (CHAMPS) is a network of disease surveillance sites in developing countries with high under-five mortality rates. The IEDCR, icddr,b, and Bangabandhu Sheikh Mujib Medical University (BSMMU) are jointly implementing the surveillance in Bangladesh. The Bangladesh site has established



epidemiologic surveillance of under-5 deaths and stillbirths to find out the cause of death of the children, utilizing demographic surveillance of minimal invasive tissue sampling (MITS), verbal autopsy, and clinical findings of the deceased. The tissue samples are tested by both conventional and advanced histopathology and molecular screening for various pathogens. The field site is in Baliakandi upazila of Rajbari district, situated in the west-central part of Bangladesh, and the MITS facilities are established in Faridpur Medical College Hospital, Zahid Memorial Children Hospital, and Baliakandi Upazila Health Complex.

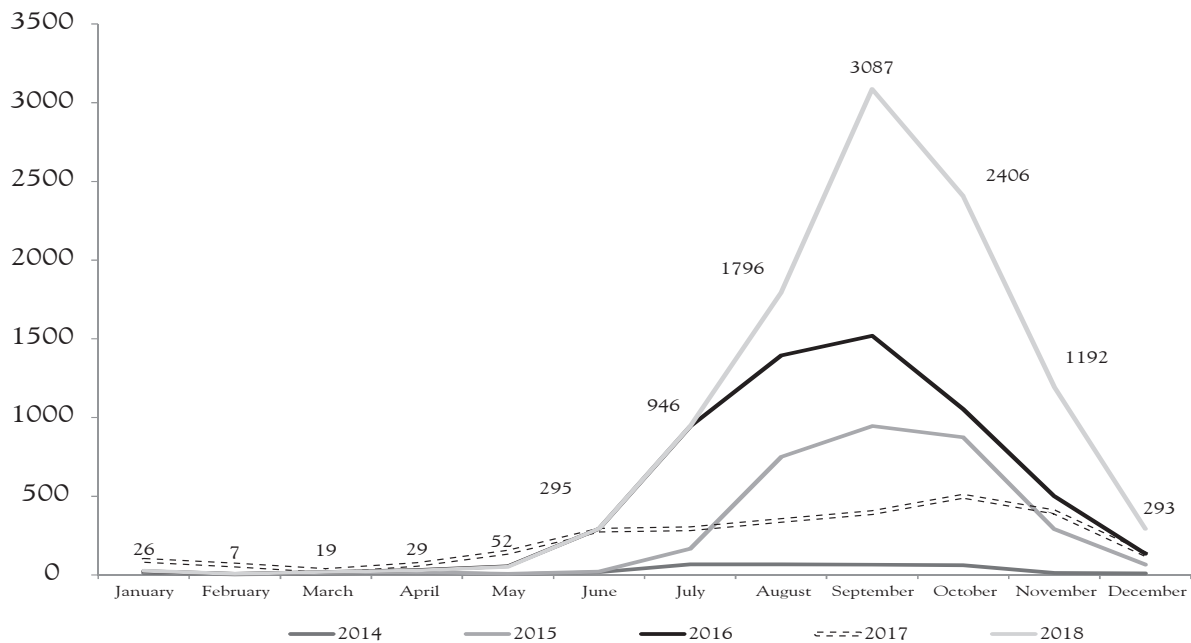
**Cell Phone-based disease surveillance system (CPBDSS):** The IEDCR conducted Cell Phone-based sexual and reproductive health surveillance of women aged between 15 and 49 years (reproductive age) in Bangladesh, using CPBDSS platform during 2017-2018. This surveillance collected information on women about their reproductive health, healthcare-seeking behavior, contraception, and challenges regarding these issues. The surveillance was funded by UNFPA. A total of 95,534 calls were made, out of which, 12,460 (13%) were complete calls. The mean age of the participants was 29.35 years. More than two-thirds of the ever-married women were married before the age of 18 years. More than half (52.2%) of the women heard of three modern contraceptive methods. Approximately 90.1% of the adolescent and youth participants (of 15-25 years) were satisfied on service received from the health center/hospital for reproductive health.

**Web-based integrated disease surveillance system:** Reporting was started from 493 upazila-level health facilities directly through web-based DHIS2 software on weekly basis

covering 18 diseases or conditions since July 2009. It is established as an integrated computer-friendly surveillance system to cater to the needs of all program activities. The diseases under this surveillance system include: acute watery diarrhea, blood dysentery, pneumonia, severe pneumonia, very severe disease, severe acute respiratory infection (SARI), acute meningitis-encephalitis syndrome, dengue fever, acute hepatitis, kala-azar, cutaneous anthrax, malaria, enteric fever, probable rabies, PTB+, PTB-, EPTB, and childhood TB.

**Web-based dengue surveillance:** The IEDCR receives reports of dengue patients through passive reporting, using the District Health Information System 2 (DHIS2) software. Thirty-two hospitals (both government and private) of Dhaka city are participating in the reporting. Figure 4.5.5 shows the number of dengue outbreaks in Bangladesh (Source: IEDCR, Mohakhali, Dhaka) [Data collected from DHIS2 but analyzed and report published by IEDCR].

**Respiratory event-based surveillance:** The IEDCR is piloting an event-based surveillance (EBS) to enhance and strengthen the existing EBS activities across the country to improve early detection and rapid reporting of emerging and re-emerging respiratory disease threats. This respiratory EBS system focused on respiratory diseases and has the potential to detect human outbreaks of MERS-CoV, avian influenza, as well as other emerging and re-emerging respiratory pathogens. The goal of this surveillance is to ensure that healthcare facilities recognize respiratory events early and report potential emerging respiratory threats promptly. In 2018, the IEDCR completed training and orientation in all districts and medical college hospitals of three divisions



**Figure 4.5.3.** Dengue reporting by month and year in Bangladesh (Source: IEDCR, Mohakhali)

(Chattogram, Sylhet, and Mymansingh), including public and private hospitals of Dhaka city. This promising platform has already identified a few signals and responded to contain any spread of those potential threats.

**Hospital-based syndromic surveillance:** PREDICT Bangladesh anticipates working with Faridpur Medical College Hospital (FMCH), using the human behavioral survey, along with biological specimen collection, focusing on diseases of unknown origins, including fever, acute encephalitis, and acute respiratory illness. A total of 162 behavioral risk questionnaire and 1,943 biological samples have been collected in 2018.

**Wildlife surveillance:** The PREDICT project, part of USAID's Emerging Pandemic Threats (EPT-2) program, is enabling global surveillance for pathogens that can spillover from animal hosts to people by building capacities to detect and discover viruses of

pandemic potential. In Bangladesh, PREDICT is implemented by EcoHealth Alliance in close collaboration with the IEDCR, DLS, icddr,b, and in cooperation with local stakeholders and communities. In 2018, the PREDICT investigated two outbreaks and collected samples from 1,935 animals (1,814 samples from wildlife and 121 samples from livestock) at high-risk human-animal interfaces for zoonotic disease transmission. The samples are being screened for novel, high-consequence viral families (coronaviruses, filoviruses, flaviviruses, paramyxoviruses, and Influenza-A viruses), using consensus PCRs. Seven known viruses were detected in 17 people. The laboratory work is carried out at the Virology Department of the IEDCR. Positive samples are being genotyped at the Virology Laboratory of icddr,b.

**Media surveillance:** Since 2009, the IEDCR has been operating media monitoring as a component of event-based surveillance to monitor print and electronic media for quick

detection of outbreaks. The institute hired a media monitoring company to search health-related news from daily national newspapers and television channels. The company scans the health-related news, video-clippings, and sends to the IEDCR every day before 11:30 am through email. The IEDCR reviews all the clippings and takes appropriate initiatives to respond to the outbreak-related events reported in media.

**Hotline:** The IEDCR receives phone calls on disease outbreaks from healthcare personnel from every upazila and district through hotlines. The IEDCR has four hotlines. The hotlines remain open round the clock. These were also used during the Hajj and outbreaks to provide health advice to general people. The hotline numbers are: +8801937000011, +8801937110011, +8801927711784, and +8801927711785.

### **National Bulletin of Public Health**

National Bulletin of Public Health (NBPH) is the first, open access, national, peer-reviewed public health bulletin of the Government of Bangladesh. It is organized and published by the IEDCR with the technical support of Bloomberg Philanthropies and US-CDC Foundation. The online version is available on the IEDCR website. NBPH is dedicated to publishing research/news/awareness-building/ promotion/information relating to public health from all disciplines and practice. NBPH accepts original surveillance reports, information on emerging diseases, research articles, review articles, medical news, short communications, correspondence, perspectives, and new drug updates, adverse drug reactions, preventive health measurements, mass awareness abstracts, and things relating to improve health literacy.

Nearly 2,000 copies of each issue are printed and distributed to upazila health complexes. It is also distributed to the government-run medical academic institutions and some private institutes/organizations. NBPH tries to publish each issue focusing on a specific theme. In the year 2018 after the introductory one, NBPH published three issues with special issues on dengue and non-communicable diseases. The issue published on dengue created a huge positive churn among local people and other international public health bulletin authorities. Under this project, ‘Scientific Communication Training’ will be held to help new writers for writing science articles from their works. NBPH will play the role of prepublication platform for them.

### **Field Activities of IEDCR**

The iEDCR’s Belabo REsearch, Action & Training Centre on HEalth at COMmunity (iBREATHE@COM) is a new platform of the IEDCR to conduct health-related research and training activities at the community level established in 2018. Belabo upazila of Narsingdi district was selected as the field site following visit to a good number of upazilas to determine the feasibility of study site. The inception program was held on 22 April 2018. Honorable Member of Parliament of the People’s Republic of Bangladesh Mr. Advocate Nurul Majid Mahmud Humayn was present as the chief guest. Upazila Chairman of Belabo, Additional Director General for Planning & Research (of DGHS), Director, IEDCR, Line Director of NCDC, and local health staff were also present in the inception program. The IEDCR conducted census of the whole upazila, facility mapping of the upazila, health facility assessment, set up surveillance field



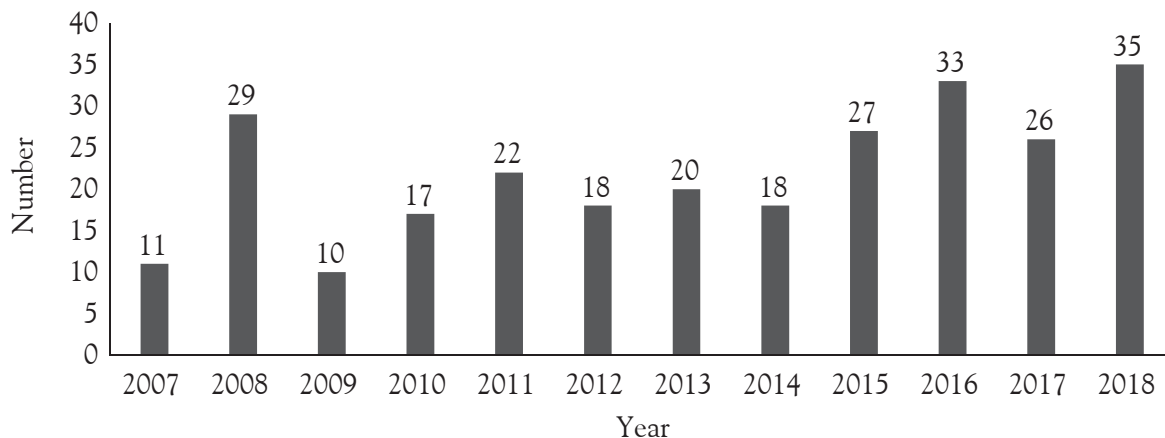
site at upazila health complex for preliminary data collection on behavioral risk factors for communicable and non-communicable diseases. The IEDCR is planning formative researches to design intervention strategies, developing protocol for intervention strategies, implementing intervention strategies and adopting nationwide strategies from the findings to prevent disease burden in the community in future. The IEDCR has the vision to uphold iBREATHE@COM platform as a center of excellence to conduct future training and research activities at the community level. Findings from the research activities will guide the policy-makers to design nationwide intervention strategies in Bangladesh.

### Outbreak Investigations and responses of IEDCR in 2018

The IEDCR is the mandated institute of MOHFW to conduct outbreak investigations and response all over the country. As National Rapid Response Team (NRRT), the IEDCR responds to any unusual health events on an emergency basis. In 2018, the IEDCR responded to 35 outbreaks throughout the country. The institute investigates and controls the outbreaks of zoonotic diseases through OneHealth approach. The IEDCR also guides and supervises District Rapid Response Team (DRRT) and Upazila Rapid Response Team (URRT) for responses to local outbreaks.



**Figure 4.5.4.** A meeting of the inception program was held at the iEDCR's Belabo REsearch, Action & Training Centre, Belabo upazila, on 22 April 2018



**Figure 4.5.5.** Number of dengue outbreaks in Bangladesh responded to by IEDCR (Source: IEDCR, Mohakhali, Dhaka)

The IEDCR performs case investigations of different patients presenting symptoms, like those of emerging and re-emerging diseases and helps physicians to confirm the diagnosis. As a focal institute, the IEDCR also performs case investigations reported through International Health Regulations (IHR 2005). In 2018, the IEDCR conducted 4 case investigations and reviewed 55 dengue death cases.

### Outbreaks investigated by IEDCR in 2018

- |                         |                                 |
|-------------------------|---------------------------------|
| • Acute febrile illness | • Encephalitis                  |
| • Avian influenza       | • Food poisoning                |
| • Atypical pneumonia    | • Guillain–Barré Syndrome (GBS) |
| • Chikungunya           | • Hepatitis: A, B, and E        |
| • Cutaneous anthrax     | • Mass psychogenic illness      |
| • Diarrheal diseases    | • Measles                       |
| • Diphtheria            | • Respiratory tract infection   |
| • Rubella               |                                 |

### Case investigations performed by IEDCR in 2018

- Atypical pneumonia
- Severe respiratory illness following the Hajj
- Suspected encephalitis

**Diphtheria outbreak among FDMNs:** In response to a diphtheria outbreak among FDMNs, an outbreak investigation team was deployed from PHEOC, IEDCR in early December 2017. The team worked together with concerned national and international agencies to control and manage the outbreak till March 2018. The IEDCR also provided laboratory support during outbreak to confirm the diagnosis of diphtheria among FDMNs.



**Figure 4.5.6.** Diphtheria outbreak investigation team visited the FDMNs area in December 2017

### Hepatitis E outbreak in Chattogram:

An outbreak investigation team was deployed from PHEOC, IEDCR in June 2018 in response to a hepatitis E outbreak in Chattogram metropolitan city. Some members from icddr,b were included in the investigation team. The team worked together with Civil Surgeon's office, Chattogram City Corporation, and WASA to control and manage the outbreak. The team identified the sources and mitigated the outbreak.



**Figure 4.5.7.** Hepatitis E outbreak investigation team working in Chattogram in June 2018 to mitigate the outbreak

**Dengue outbreak in Dhaka city:** In response to the dengue outbreak in Dhaka metropolitan city in 2018, PHEOC operated in 'alert mode'. The IEDCR provided confirmatory



**Figure 4.5.8.** Dengue outbreak investigation team working in Dhaka metropolitan city in 2018 to overcome the situation

diagnostic services, health advice by hotlines and conducted regular media briefing, and risk

communication to members of rapid response team.

# International Health Regulations, Migration Health, Emerging and Re-emerging Diseases

## Health of migrants is of prime concern

The International Health Regulations (IHR) constitute an international legal instrument that is binding on 196 countries across the globe, including the Member States of WHO. In May 2005, the 58th World Health Assembly (WHA) adopted the International Health Regulations (2005) [IHR (2005)] which subsequently were enforced on 15 June 2007, with 123 member countries signing the document. The purpose and scope of the IHR (2005) are “to prevent, protect against, control and provide a public health response to the international spread of diseases in ways that are commensurate with and restricted to public health risks, and which avoid unnecessary interference with international traffic and trade.” State parties are required by the IHR (2005) to develop minimum core public health capacities. Attaining the maximum public health core capacity as per IHR (2005) will significantly contribute to the achievements of SDG 2, 3, and 6.

### Aim of the Program

The aim of the program is to help international community prevent and respond to public health risks at scale that have the potential to cross the borders and threaten

Attaining the maximum public health core capacity as per IHR (2005) will significantly contribute to the achievements of SDG 2, 3, and 6...

people worldwide. Member States should be able to detect, assess, notify, and report events and respond to public health risks and public health emergency of international concern (PHEIC) timely.



**Figure 4.6.1.** Director General of Health Services attended the Hajj Camp of Dhaka to see the screening process of Hajjis



## Objectives of the Program

- Enactment of adequate legal framework and laws for implementation of IHR 2005
- Implementation of multi-sectoral and multi-disciplinary coordination and communication mechanism
- Establishment of cross-border surveillance of priority diseases among neighboring countries (with India, Myanmar)
- Enactment of the establishment of cross-border surveillance
- Inclusion of the component of IHR in curricula of medical and public health, and courses in related fields and in all related training
- Adoption of mechanisms and documented procedures among all relevant sectors, particularly those responsible for human and animal health to minimize the transmission of zoonotic diseases from animal to human
- Strengthening laboratory capacities, infection prevention, and control
- Plan of action for developing human resource for IHR 2005 implementation
- Plan of action for risk communication and strategy development
- Plan of action for strengthening core capacities of points of entry (PoE)
- Strengthening capacities to respond to potential hazards
- Detect, assess, and respond to food safety issues and related hazards
- Implementation of mechanism for timely detection and effective response to

potential radio-nuclear hazards/events/emergencies in collaboration with other sectors responsible for radio-nuclear managements

- Ensure timely detection and effective response to potential chemical risks and/or events in collaboration with other sectors responsible for chemical safety, industries, transportation, and safe disposal.

Achieving international public health security is one of the main challenges arising from the emerging and mosaic landscape of public health. Shared vulnerability implies shared responsibility. Strengthening disease surveillance and response systems of WHO Member Countries is central to improving public health security in each country as well as globally. Meeting the requirements of IHR (2005) compliance is a challenge that needs time, commitment, and the willingness to change. The updated IHR (2005) came into effect from 15 June 2005 with 123 Member Countries signing the document. The IHR (2005) set forth new requirements and obligations for Member States of WHO to develop capacity for detection, verification, notification, and accuracy of information.

WHO Member States, including Bangladesh, have to comply with developing core capacities for surveillance and response which covers surveillance and emergency threats from infectious diseases as well as core capacities for designated airports, ports, and ground crossings to help ensure global, regional and national health security, and the capacity to detect and combat diseases of radio-nuclear, chemical and food origin in the community.



**Figure 4.6.2.** WHO Member States, including Bangladesh attended a workshop to develop core capacities for surveillance and response

### Related activities undertaken by IHR, CDC-DGHS

1. Formation of IHR Committees (National IHR Coordination Committee, National IHR Technical Committee, National IHR Focal Point)
2. On the basis of findings of the assessment for 2009, 2011, 2013, and joint external evaluation (JEE), the National Action Plan for Health Security (NAPHS) drafted under the initiative of CDC/IEDCR (the National Technical Focal Institute)
3. Modernization of facilities made as per IHR guidelines in the Hazrat Shahjalal International Airport for health desk, screening center, health center, and screening room
4. Finalization of all hazard emergency contingency plans
5. Field simulation exercise done at Hazrat Shahjalal International Airport
6. IHR core committee meeting for chemical and radiation hazards
7. IHR core committee meeting for PoEs
8. Extended core committee meeting for risk-mapping and capacity assessment
9. Strengthening capacities at Mongla and Chittagong Seaport, and inclusion of these ports as IHR-authorized ports to issue ship sanitation certificates
10. Standard operating procedure (SOP) for public health emergencies of international concern (PHEIC)
11. Strategy and guideline for management of PHEIC at points of entry (PoE) and implementation of IHR (2005) involving 3,600 health managers (health officers of points of entry, immigration officials of airports, clinicians/pediatricians, health administrators, medical officers of government and private medical colleges, autonomous and private hospitals of Dhaka city)
12. Leaflets and booklets for Zika, Ebola, MERS-CoV, and influenza both in Bangla and English developed
13. National strategy was also finalized for Zika virus, although there was no active case of Zika identified in Bangladesh till now.

### Migration Health

Migration health addresses the physical, mental and social needs of migrants and their families, and the public health needs of communities of the countries of origin and destination, through policies and practices corresponding to the emerging challenges





**Figure 4.6.3.** Health of migrants has become a prominent concern of the Government

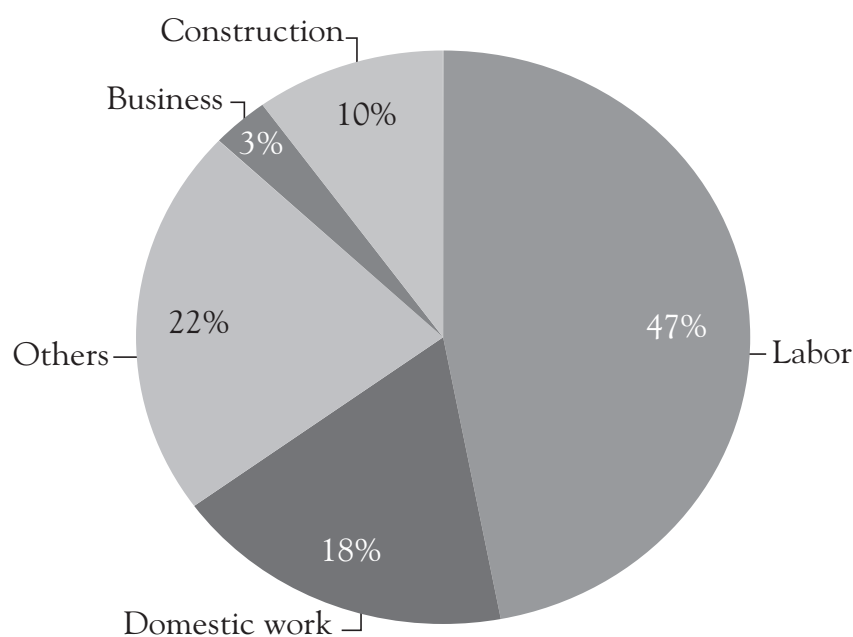
faced by mobile population today. Migrants constitute a large number of people moving out of the country and generating huge remittances. Despite the migrants' great contribution to the economy of Bangladesh, it is unfortunate that they are often exposed to occupational health hazards and have very limited access to health benefits. Health of migrants has become a more prominent concern of the Government as Prime Minister stated in the 64th World Health Assembly for the country's commitment to ensure the welfare of migrants, especially by migrant-inclusive health policies. The Director General of Health Services has developed a national strategic action plan on migration health for the period 2019 to 2022.

The Director General of Health Services has developed a national strategic action plan on migration health for the period 2019 to 2022...

### Strategy

The Government of Bangladesh is committed to improving the health status of all categories of migrants, including inbound, outbound and internal migrants throughout the migration process, and for the progressive realization and protection of their fundamental right to health. Specific strategic objectives are as follows:

### Types of employment by departing & Returnee migrants



Source: Health Vulnerabilities of Migrants from Bangladesh, Baseline assessment IOM, Dhaka, August 2015

**Figure 4.6.4.** Type of employment of migrants

- Establish a strategic monitoring and information system (MIS) for migrants' health
- Enhance policy and legal framework for migration health to provide support for the health and social protection of migrants
- Enable existing health and social welfare system to be more migrant-sensitive
- Facilitate multi-sectoral partnerships for migration health at national and regional levels.

# Health Education, Promotion, and Lifestyle Modification

## Innovative tools used for mass awareness-building

Health education/counseling is an important part of public health management. Health counseling is a profession of educating people about health. It is a principle by which individuals and groups of people at the community learn how to behave in a manner conducive to the promotion, maintenance, or restoration of health. Through effective counseling and lifestyle modification, a country can reduce both mortality and morbidity rates and health expenditure. The 4th HPNSP has an Operational Plan on Lifestyle, Health Education & Promotion (L&HEP) under the DGHS. Main goal of this Operational Plan is to influence and promote healthy lifestyle and change behavior of individuals and the community.

### General objective

Influence the healthy behavior of individuals and the community, and living conditions that influence health by improving their knowledge, attitude, practices, and skills by creating a 'health-literate society'.

### Specific objectives

- Establish legislative framework, communication strategy, implementation

Through effective counseling and lifestyle modification, a country can reduce both mortality and morbidity rates and health expenditure...

strategy, and inter-sectoral collaboration for healthy lifestyle and healthy environment

- Improve the activities for implementation of lifestyle and health counseling and promotion at the individual and community level
- Identify different target audiences and address communicable, non-communicable diseases, MNCAH issues, and emerging and re-emerging health problems
- Use and establish linkage with print, electronic, and social media and effective community engagement for social and behavioral change communication (SBCC).

## Activities in 2018-2019

### *Health education/counseling training for manpower development*

During the reporting period, 1,150 officers and employees of the national, divisional and district level (Sr. Health Education Officer, Jr. Health Education Officer, SACMO, Health Educator, SI, HI, AHI, CHCP) were provided training on health education/counseling, environmental health, and occupational health. To strengthen school health education, along with staff/employees of health departments, 2,565 primary school teachers and 2,100 high school teachers, were trained on healthy lifestyle and other health-related issues.

Advocacy meeting/workshop/seminar were held at different districts and upazilla levels to raise public awareness on the following issues (the figures indicate the numbers of participants):

- ✓ Healthy lifestyle and environment-150
- ✓ Prevention and control of non-communicable diseases-94
- ✓ Cervical and breast cancer-72
- ✓ Awareness on arsenic-free water- 40
- ✓ Personal hygiene and sanitation-40
- ✓ Prevention of Nipah virus-10
- ✓ Food safety and safe food-71
- ✓ Adolescent health-37
- ✓ Student orientation-40
- ✓ Prevention of smoking, tobacco and tobacco-related products-12

### **Health education/counseling: promotion through television and electronic media**

- Documentaries on development and success of the health sector under the present government were prepared and disseminated through different TV channels and different district and upazilas.
- TV talk-shows in Bangladesh Television were arranged in celebration of the World Health Day.
- TV spot-shows on healthy lifestyle, dengue, chikungunya, cancer, cervical and breast cancer, environmental hazards, diabetes, heart disease, fast food and overweight, were telecast through different channels.
- Four episodes/dramas (*Jibon Dhara*) on healthy lifestyle and occupational health were produced and telecast through 4 popular TV channels; a drama (*Porir Banshi*) on the reduction of NCD risk behaviors were produced and showed in different districts and upazilas by video-shows. In addition, a drama (*Ekhoni Somoy*) was produced that focused on fast food, overweight, and ARI. This was telecast through different TV channels.
- TV scrolls on prevention and control of cervical and breast cancer, diarrhea, dengue, and chikungunya were prepared and shown. TV scrolls on the National Health Service Week, flood, safe food in Ramadan, *Bishaw Ijtema* were prepared and shown in different TV channels.
- Two mega billboards with light-emitting diode (LED) at Kurmitola General Hospital and the Institute of Mental Health premises were installed.

## Promoting health education/ counseling through print media

- > Activities were performed on the occasion of the World Health Day celebration on 7 April (production and distribution of different types of SBCC materials, such as posters, leaflets, stickers, folders, theme songs, road decoration, banners, festoons, newspaper supplements, etc.).
- > About 800 advertisements/features on dengue and chikungunya, flood, heart disease, stroke, cancer, diabetes, thundering, high blood pressure, drowning, Nipah virus, cervical and breast cancers, and other communicable and non-communicable diseases were prepared and published in different national-level daily newspapers.
- > Newspaper supplements on the occasion of different Days and Weeks. (Safe Motherhood Days, World No Tobacco Day, Mental Health Day, National Health Service Week) were published.

## Print materials on health education

- Printed around 3 lakh leaflets on different health issues
- About 3,000 flip-charts printed on adolescent health and control and prevention of non-communicable diseases and distributed in different districts and upazilas
- Around 4,50,000 posters printed at BHE printing press and distributed all over the country
- 20 buses branded with health messages regarding occupational health risk in industrial areas
- Establishment of 740 digital banners (wall writing) in and outside 64 upazilla health complexes
- Installed 160 signage on non-communicable diseases at different community clinics
- Printed 10,000 booklets on malaria and tuberculosis and distributed in 3 hill districts
- Printed 8,000 posters on cervical and breast cancers and distributed in different districts
- Installation of 104 roadside sign boards on malaria and tuberculosis in 3 hill districts.

## Other health education activities

- ◆ Awareness-raising activities on personal hygiene among the school children
- ◆ Organizing folk songs and mobile film shows in different districts and upazilas
- ◆ Observance of National Children's Day and the birth anniversary of the Father of the Nation
- ◆ Printed and distributed long banners, festoons, and stickers on the National Mourning Day on 15 August.

## Challenge Faced

The major challenge faced in implementing activities under this OP included: frequent changes of Line Director. This drawing and disbursing order (DDO) of senior health education officer has been cancelled in the ibas2 system. For this reason, many problems were created for the utilization of fund. No



manpower was deployed at the field level, especially at the upazilla level to implement and monitor the programs.

To provide healthcare services to the doorsteps of people, the present Government has undertaken various public welfare health programs. The role of health education/counseling is immense in the successful implementation of these programs. Through this program, rural people are becoming more

aware of the prevention of infectious diseases, nutrition, pregnancy and postnatal care, child health, and so on. These are helping to reduce maternal and infant mortality by improving overall health standards. The activities adopted in the Lifestyle, Health Education and Promotion Operational Plan play a significant role in achieving the SDG 3 directly and indirectly, along with other goals of the health sector.



# Health Interventions for FDMNs

## Coordination Center at Cox's Bazar

Settlement for the Forcibly Displaced Myanmar Nationals (FDMNs) in Cox's Bazar on the border with Myanmar is now the world's largest refugee camp having a total of 914,998 FDMNs. Between August 2017 and 15 March 2019, Bangladesh has received 706,364 FDMNs since the conflict started in Rakhine State of Myanmar (Source: DGHS dashboard, January 27, 2020). It has been the fastest and largest movement of refugees in decades.

Since the arrival of FDMNs in Bangladesh, the Government, through the Directorate General of Health Services (DGHS) of the Ministry of Health and Family Welfare has been in the frontline to deliver emergency responses as well as health support to them. The FDMNs have been placed in 68 settlements across Ukhiya upazila and Teknaf upazila in Cox's Bazar district as well as Naikkhongchhari upazila in Bandarban district. Yet, in order to accelerate the emergency response, the Government is bringing most of the FDMNs population together to a 3,000-acre area located in Ukhiya.

Since the influx of refugees, the collaborative partnership between the government and private sectors, civil society and international development agency has been able to provide static health centers in 216 fixed sites with 293 doctors, 186 nurses, 1,259 community health workers, 224 SACMOs, 45 ambulances, and 231 vaccination teams (Source: DGHS

Since the arrival of FDMNs in Bangladesh, the Government, through the Directorate General of Health Services (DGHS) of the Ministry of Health and Family Welfare has been in the frontline to deliver emergency responses as well as health support to them...

dashboard). To provide up-to-date health information on the FDMNs, the DGHS has created a public dashboard, namely "Forcibly Displaced Myanmar National (FDMN) Health Situation and Intervention in Bangladesh", which is being updated on a daily basis.

Given the size of population influx and the fact that the FDMNs camps are located in remote areas, the speed of service delivery, and the availability of amenities to provide healthcare services, the achievements have been significant.

### Health Sector Coordination

With the aim of better coordination in the field of health, the Government of Bangladesh established the DGHS coordination center on 15 November 2017. From 25 August 2019, this center started its activity with the new name "Ministry of Health & Family Welfare (MOHFW) Coordination Center" under World Bank support. The

MOHFW Coordination Center participates regularly at Health Sector Committee, Strategic Advisory Group and Emergency Preparedness meetings and provides necessary directions and recommendations. Moreover, other coordinators participate in other sub-committee meetings on regular basis, which are usually held weekly or fortnightly and provide their valuable suggestions for strengthening health interventions for the FDMNs. Besides these, the MOHFW Coordination Center regularly arranges coordination meetings with partners at the office of Coordination Center and/or Civil Surgeon's office. Another aim of the MOHFW Coordination Center is to support the Civil Surgeon and Deputy Director of Family Planning of Cox's Bazar on different health and family planning issues. Not only that, two immunization and outbreak officers of the Coordination Center also support the UH&FPOs of Ukhiya and Teknaf on different health issues.

### Healthcare Service Delivery

To ensure the quality of medical service and to continue the coordination between the partners on different health interventions for the FDMNs, the Coordination Center has been started in August 2019. Family planning service is also included as part of monitoring. World Bank has provided financial support to the Bangladesh Government to establish and maintain this project. Under this project, there are four UN agencies that are deployed to ensure the necessary and quality medical service for the FDMNs. The agencies are WHO, UNICEF, UNFPA, and IOM.

WHO is responsible to establish the Coordination Center and to recruit all the staff for the center. A total of 22 members

comprise this team. Among them, 17 are deployed in Cox's Bazar to provide technical support to the Civil Surgeon's office of Cox's Bazar, Upazila Health Complexes of Ukhiya and Teknaf, and 5 members are in Dhaka to provide support to the Director General of Health Services (DGHS), Director General of Family Planning (DGFP), and to the Ministry office. Other three UN agencies: UNICEF, UNFPA, and IOM are responsible for providing medical health services by establishing different health facilities in camps and adjacent areas.

Under this project, 45 health facilities are established and maintained for the FDMNs. As per necessity of health services to the FDMNs, support has been provided to 11 primary health centers, 13 health posts, 10 community clinics, 5 family welfare centers, 4 union sub-centers, and 2 upazila health complexes. A team, including medical doctors, medical assistants, nurses, midwives, pharmacy assistants, medical technologist of lab discipline, etc. were deployed in these facilities.

The MOHFW Coordination Center coordinates the total process, including recruitment, quality of service, and reporting of these 45 health facilities according to the contract with Government and UN agencies. Besides these 45 health facilities,

As per necessity of health services to the FDMNs, support has been provided to 11 primary health centers, 13 health posts, 10 community clinics, 5 family welfare centers, 4 union sub-centers, and 2 upazila health complexes...

the Coordination Center is providing any other relevant and necessary support to the Government which is required for health interventions for the FDMNs.

**Table 4.8.1.** Number of health facilities in the FDMN camps

| Type of health facility                                           | Number of facilities |
|-------------------------------------------------------------------|----------------------|
| Health post                                                       | 140                  |
| PHC                                                               | 36                   |
| Community clinic (MOHFW)                                          | 33                   |
| Diarrhea Treatment Center                                         | 2                    |
| Field hospital                                                    | 4                    |
| Health and family welfare center (MOHFW)                          | 6                    |
| Labor room or specialized sexual and reproductive health facility | 9                    |
| Other specialized facilities                                      | 23                   |
| Secondary health facility                                         | 12                   |
| Sub-center {MOHFW}                                                | 6                    |
| Tertiary health facility                                          | 1                    |
| Upazila health complex (MOHFW)                                    | 2                    |
| Women-friendly spaces                                             | 19                   |
| <b>Total</b>                                                      | <b>293</b>           |

### Essential Service Package

Under the guidance of Civil Surgeon's office in Cox's Bazar, WHO, along with the health-sector partners developed an ESP for FDMN camps in 2017 to ensure the quality health service among the FDMNs. In December 2018, it has been reviewed and updated by MOHFW Coordination Center and endorsed by Civil Surgeon, Cox's Bazar and Refugee Relief and Repatriation Commissioner (RRRC).

### Referral SOP

Referral SOP has been drafted with a view to providing possible health services inside the FDMN camps but whenever there is a need for referral services, it should follow referral pathway. A referral pathway and referral form also have been developed. These documents have been finalized and approved by Civil Surgeon, Cox's Bazar and other relevant authorities.

### Health Interventions

Vaccination is ongoing through special immunization activities, along with routine immunization.

Routine EPI has also been initiated, targeting children <2 years of age with BCG, Pentavalent, OPV, PCV and MR. Routine EPI has started from 3 June 2018 for FDMNs camps situated at Ukhiya upazila. In total, 56 teams are working to operate 56 outreach centers in Ukhiya, among which DGHS Coordination Center contributed 8 teams.

### Measles-Rubella Situation Update

A total of 3,274 cases have been reported from 2019 till date from Early Warning Alert and Response System" (EWARS). Among these cases, 2,195 (67%) were reported via Measles Case Report Form.

### Measles-Rubella Outbreak Response Immunization (MR ORI): 12 January to 12 February 2020

In response to the above MR situation, an outbreak response immunization has been already started in all 34 camps, including two registered camps. Immunization is

**Table 4.8.2.** Health posts and primary health center sites for MOHFW Coordination Center, Cox's Bazar

| SI No | Camp No./Name          | Functioning with WB funding | Population | Partner             | Status                          |
|-------|------------------------|-----------------------------|------------|---------------------|---------------------------------|
| 1     | Camp-1E                |                             | 40424      |                     |                                 |
| 2     | Camp-1W, Lambashia     | HP                          | 40913      | HAEFA<br>AFAD (PHC) | Functioning<br>MOU signed       |
| 3     | Camp- 2E               |                             | 29823      | AFAD<br>SRC (PHC)   | 1st Phase<br>Under construction |
| 4     | Camp-2W                |                             | 25696      |                     |                                 |
| 5     | Camp-3                 |                             | 39630      | AFAD (PHC)          | MOU signed                      |
| 6     | Camp-4                 |                             | 32112      | AFAD                | Land not allocated yet          |
| 7     | Camp-4 Ext             |                             | 5837       |                     |                                 |
| 8     | Camp-5, Modhurchora    | HP                          | 25093      | AFAD                | Functioning                     |
| 9     | Camp-6                 |                             | 24954      | AFAD                | MOU signed                      |
| 10    | Camp-7, TV Center      | HP                          | 39312      | UNHCR               | Functioning                     |
| 11    | Camp-8 E               |                             | 31185      |                     |                                 |
| 12    | Camp-8 W               |                             | 32351      |                     |                                 |
| 13    | Camp-9, Balukhali 1    | HP                          | 36597      | HAEFA               | Functioning                     |
| 14    | Camp-10                |                             | 32969      | IOM                 | Functioning                     |
| 15    | Camp-11, Moynarghona   | PHC                         | 31316      | SRC<br>UNHCR/GK     | Functioning<br>Functioning      |
| 16    | Camp-12                |                             | 23615      |                     |                                 |
| 17    | Camp-13, Tajnimarkhola | PHC                         | 41678      | SRC                 | Functioning                     |
| 18    | Camp-14, Hakimpara     | HP                          | 31849      | UNHCR               | Functioning                     |
| 19    | Camp-15, Jamtoli       | PHC                         | 49426      | SRC                 | Functioning                     |
| 20    | Camp-16                |                             | 21614      | SRC                 | Proposed                        |
| 21    | Camp-17                |                             | 16801      | AFAD                | Under construction              |
| 22    | Camp-18                | HP                          | 26635      | IOM                 | Functioning                     |
| 23    | Camp-19                | PHC                         | 20804      | IOM/RTMI            | Functioning                     |
| 24    | Camp-20                |                             | 7311       |                     |                                 |

**Table 4.8.2 Contd.**

Table Continued...

| SI No | Camp No./Name                   | Functioning with WB funding | Population | Partner       | Status                    |
|-------|---------------------------------|-----------------------------|------------|---------------|---------------------------|
| 25    | Camp-20 Extension               | PHC                         | 4347       | AFAD          | Functioning               |
|       | Teknaf                          |                             |            |               |                           |
| 26    | Camp-21, Chakmarkul             | HP                          | 12253      | UNHCR         | Functioning               |
| 27    | Camp-22, Unchiprang             | HP                          | 22194      | AFAD          | Functioning               |
| 28    | Camp-23, Shamlapur              | HP                          | 10963      | IOM           | Functioning               |
| 29    | Camp-24, Leda                   |                             | 33544      |               |                           |
| 30    | Camp-25, Ali Khali              |                             | 9511       |               |                           |
| 31    | Camp-26, Shalbon, Noyapara Ext. | HP                          | 41037      | UNHCR<br>ICRC | Functioning<br>MoU signed |
| 32    | Camp-27 Jadimora (Damdamia)     |                             | 14269      | ICRC          | Under construction        |

**Table 4.8.3.** Special immunization activities, along with routine immunization, conducted by round

| Special immunization activities | 1st Round | 2nd Round | 3rd Round                                    | 4th Round | 5th Round |
|---------------------------------|-----------|-----------|----------------------------------------------|-----------|-----------|
| MR vaccine                      | 135,519   | 354,982   | Outbreak response immunization (ORI) ongoing |           |           |
| Oral polio vaccine              | 72,334    | 236,696   |                                              |           |           |
| Vitamin A                       | 68,941    | 156,505   |                                              |           |           |
| Oral cholera vaccine            | 700,487   | 199,472   | 879,273                                      | 364,686   | 162,871   |
| Td                              | 165,927   | 223,065   | 259,016                                      |           |           |
| Penta, PCV, bOPV                | 149,962   | 169,617   | 172,432                                      |           |           |

provided through fixed sites within the specific camps, targeting a total of 290,116 population aged from 6 months to less than 10 years. Total number of children vaccinated so far is 160,042 (till 26 January 2020)

### **Diarrheal Situation Update**

Since September 2019, there have been 239 cases of AWD (Acute Watery Diarrhea) that have tested positive by Rapid Diagnostic Tests (RDT) for cholera, or culture, or links to

these cases (data as of 31 December 2019). A campaign for the first dose of the 5th Round of oral cholera vaccine (OCV) was conducted from 8 to 14 December 2019 in the FDMN camps in both Ukhiya and Teknaf. A total of 162,871 children were vaccinated. The date of the OCV campaign for the 2nd dose (6th Round) is not yet finalized; the target population is around 138,726 aged 1 to <5 years.

### ***Response to Increased Diarrheal Patients***

The MOHFW Coordination Center sent a team for investigation of cholera outbreak at scale. Later, based on the findings, a presentation was made in front of a joint high-level delegation from CDC, IEDCR, DGHS, and icddr,b at Civil Surgeon's office. Besides, a meeting with district public health engineering has been held to strengthen the WASH response. In addition to that, training of 100 health personnel was conducted with support from WHO; immunization and outbreak officers from MOHFW Coordination Center Participated.

### ***Family Planning, Maternal-Newborn Healthcare Services***

After the initial influx of the FDMNs, emergency control room has been opened at the District Family Planning Office to provide family planning, maternal and newborn healthcare (FP&MNH) services. Seven emergency medical teams (4 in Ukhiya and 3 in Teknaf) have been formed to provide departmental services. Provision of services is also in progress through 6 UH&FWCs, 2 Rural Dispensaries, and 1 registered family planning departments,

and 16 NGOs under Ukhiya and Teknaf upazilas after the influx. Each month in FDMN camps, a total of 10 satellite clinic services are operating: 7 in Ukhiya upazila and 3 in Teknaf upazila. Through audiovisual van, a monthly televised exhibition in Rohingya language on family planning, maternal-child health, and reproductive health is underway in Ukhiya and Teknaf upazila. Given below is a summary of the services provided since September 2017 to 27 January 2020.

A total of 303,686 pregnant women have received IFA. EPI and other activities have been continued in coordination with the health department. Emergency maternity care activities are being provided at MC& WCs of Cox's Bazar and Ukhiya and Teknaf upazila sadar clinics and UH&FWCs. A total of 3,932 pregnancy-related complications have been treated so far.

### **Health Awareness**

Health counseling sessions for different groups (male group, female group, adolescent girls group, pregnant and lactating mothers group, mixed group, children group, school students) of the FDMNs were performed on regular basis by the field staff of the DGHS..

### **Major Achievements**

Host communities were being deprived of usual health services after sudden influx of the Myanmar Nationals in Cox's Bazar. Establishment of the Coordination Center reduced the extra burden of Civil Surgeon's office and UH&FPOs so they can focus on host community.



**Table 4.8.4.** EmOC and family planning services given to the FDMNs during September, 2017 to 27 January, 2020

| Services             |                         | Adolescents (10-19 years) | Adult ( $\geq 20$ years) |
|----------------------|-------------------------|---------------------------|--------------------------|
| Delivery             | Assisted                | 31                        | 224                      |
|                      | C-section               | 93                        | 187                      |
|                      | Normal Vaginal Delivery | 2362                      | 13660                    |
| ANC                  |                         | 69375                     | 369662                   |
| PNC                  |                         | 2486                      | 21874                    |
| FP methods           | Condoms                 | 1153                      | 21063                    |
|                      | Implants                | 110                       | 2536                     |
|                      | Injectable-continuing   | 1625                      | 38656                    |
|                      | Injectable-new          | 1613                      | 19440                    |
|                      | IUD                     | 34                        | 1288                     |
|                      | OCP-continuing          | 2625                      | 66814                    |
|                      | OCP-new                 | 2790                      | 30460                    |
| Menstrual regulation |                         | 321                       | 2228                     |
| Post-abortion care   |                         | 284                       | 1155                     |

1. MOHFW Coordination Center, Cox's Bazar, coordinates the health issues of the FDMN community.
2. MOHFW Coordination Center in Cox's Bazar has collaborated with all the healthcare providers, including NGOs, which, in turn, helps avoid the duplication of health services, and thus ensures equitable distribution and quality of health services to the FDMN community.
3. Different partners are committed to supporting the MOHFW Coordination Center for construction of 17 health posts and 10 PHCs. Several MoU for supporting these facilities have been signed already.
4. Six primary health centers (PHCs) at Camp 7, 11, 13, 15, 20E, and 10 health posts at different camps have been functional and are providing limited curative care with preventive and promotive care services by the MOHFW Coordination Center while partners are providing nutrition, psycho-social support and sexual and reproductive health services.
5. Twelve medical teams of the MOHFW Coordination Center in Cox's Bazar have provided services to 341,261 patients in the FDMN camps.
6. Disease data are reviewed regularly to identify the likelihood of developing outbreak/epidemic and provide early warning and alerts.

7. The MOHFW Coordination Center in Cox's Bazar coordinated and supported the overall activities to contain several outbreaks, including diphtheria, measles, chicken pox with the support from WHO and other partner organizations.
8. The Coordination Center actively supports the Civil Surgeon's office to implement cholera vaccination campaign for hosts and FDMNs.
9. The Center supported the World Bank for detailed planning of human resources, logistics, construction, and renovation of Cox's Bazar Sadar Hospital, UHCs, and other facilities of Cox's Bazar district.
10. The Coordination Center provided technical support to medical waste

Distribution of medical consultation by disease/ailment from the Bangladesh Government and other partners in the FDMN camps during 24 October, 2017 to 28 January 2020 is presented in the Annex...

management team of WHO to accomplish their work (starting from 3 November to 21 November 2019).

Distribution of medical consultation by disease/ailment from the Bangladesh Government and other partners in the FDMN camps during 24 October, 2017 to 28 January 2020 is presented in the Annex to Subchapter 4.8.

# Nutrition and Food Safety

## Collaboration among IPHN, IPH, and DGHS

As a basic human right, nutrition should be equally ensured for every citizen of a nation. Nutrition is an essential component of the physical growth and mental development. In the absence of nutrition, the fetus does not increase properly, and a baby is often born with low birthweight. The Government of the People's Republic of Bangladesh is committed to improving the nutritional status of the people. The Constitution of Bangladesh cites nutrition in Article 18 (1), describing the principles of State governance as: "...the state shall regard raising the level of nutrition and improvement of public health as among its primary duties..." Moreover, the Bangladesh National Nutrition Policy 2015 and National Plan of Action for Nutrition (NPAN-2) were developed to address the challenges relating to nutrition. The policy takes into consideration both of the global policies: as formulated in the Second International Conference of Nutrition (ICN2) and relevant national policies and strategies in areas, such as health, food, agriculture, environment, and education reflecting its multi-sectoral approach to ensuring improvement of nutritional status.

### Current Nutritional Status

Along with undernutrition, obesity and nutrition-related non-communicable diseases are major public health problems in Bangladesh. Though it has not reached an expected level, different interventions and programs relating

to the field of nutrition have contributed to significant improvement in Bangladesh's nutrition situation over the last few decades. Childhood malnutrition has been decreasing slowly in Bangladesh, and the result is quite noticeable: 31% under-five children are stunted while 8% are wasted. About 34% of children aged 6-23 months are fed with appropriate practices (BDHS 2017-18).

NNS OP 2017-22 focused mainstreaming of nutrition services through DGHS and DGFP during the implementing period 2017-22. This new OP will focus more on system strengthening and multi-sectoral coordination to ensure universal and equitable coverage of both specific and sensitive nutrition interventions. Nutritional status indicators are listed in Table 4.9.1.

### Summary of Maternal and Child Health and Nutritional Status

#### *Maternal Health*

- ◆ A medically-trained care provider attends 53% of deliveries. In a decade, deliveries by Skilled Birth Attendants (SBAs) increased from 21% to 53%, due to sharp increase in facility delivery.
- ◆ Half of the deliveries occur in a facility, mostly in private facilities (32%). The public and NGO health sectors account for 14% and 4% of deliveries respectively.

**Table 4.9.1.** Health indicators relating to nutrition

| Indicator                                                                                                                            | 2007 | 2011 | 2014 | 2017-18 | Remarks                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------|------|------|------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maternal and newborn care                                                                                                            |      |      |      |         |                                                                                                                                                                       |
| ANC from medically-trained provider (percentage of women receiving at least one ANC from a medically-trained provider)               | 53%  | 55%  | 64%  | 82%     |                                                                                                                                                                       |
| ANC-4+ visits (percentage of women receiving 4 or more ANC)                                                                          | 22%  | 26%  | 31%  | 47%     |                                                                                                                                                                       |
| Medically-trained care provider by place of delivery (percentage of deliveries in the last three years)                              | 21%  | 32%  | 42%  | 53%     | Home: 50%, Public facility: 14%, Private facility: 32%, NGO: 4%                                                                                                       |
| Facility deliveries by the poorest wealth quintiles (percentage of facility deliveries in the last three years)                      | 6%   | 10%  | 15%  | 26%     |                                                                                                                                                                       |
| Facility deliveries by the richest wealth quintiles (percentage of facility deliveries in the last three years)                      | 49%  | 60%  | 70%  | 78%     |                                                                                                                                                                       |
| Cesarean section (percentage of deliveries in the last three years)                                                                  | 9%   | 17%  | 23%  | 33%     | Facility deliveries by CS, National facility : 67%, Public facility : 36%, Private facility: 84%, NGO: 40%                                                            |
| Postnatal care for mothers (percentage of mothers receiving care from medically-trained care providers within 2 days after delivery) | 20%  | 27%  | 34%  | 52%     | Place of delivery (percentage of mothers receiving PNC from medically-trained care provider within 2 days of delivery), Facility deliveries: 97%, Home deliveries: 7% |
| Trends in child nutritional status                                                                                                   |      |      |      |         |                                                                                                                                                                       |
| Stunting                                                                                                                             | 43%  | 41%  | 36%  | 31%     |                                                                                                                                                                       |

**Table 4.9.1 Contd.**

Table Continued...

| Indicator                                                                                                          | 2007 | 2011 | 2014 | 2017-18 | Remarks                                      |
|--------------------------------------------------------------------------------------------------------------------|------|------|------|---------|----------------------------------------------|
| Underweight                                                                                                        | 41%  | 36%  | 33%  | 22%     |                                              |
| Wasting                                                                                                            | 17%  | 16%  | 14%  | 8%      |                                              |
| Trends in infant and young child-feeding (IYCF) (percentage of 6–23 months old children fed with 3 IYCF practices) |      | 21%  | 23%  | 34%     | 11% points increase in the last three years  |
| Childhood mortality                                                                                                |      |      |      |         |                                              |
| Under-five (<5 years) mortality per 1,000 livebirths                                                               | 65   | 53   | 46   | 45      |                                              |
| Infant mortality per 1,000 livebirths                                                                              | 52   | 43   | 38   | 38      |                                              |
| Neonatal mortality per 1,000 livebirths                                                                            | 37   | 32   | 28   | 30      | Increased and holding 67% of total <5 deaths |
| Childhood vaccination                                                                                              |      |      |      |         |                                              |
| Percentage of 12–23 months old children vaccinated by 12 months of age (All vaccines)                              | 76%  | 83%  | 78%  | 86%     |                                              |
| Percentage of 12–23 months old children vaccinated by 12 months of age (measles)                                   | 77%  | 84%  | 80%  | 88%     |                                              |
| Treatment for diarrhea (5% under-five children had diarrhea in the last two weeks before the survey)               |      |      |      |         |                                              |
| ORT (ORS or homemade solutions)                                                                                    | 81%  | 81%  | 84%  | 85%     |                                              |
| ORT and zinc                                                                                                       | 20%  | 34%  | 38%  | 44%     | Probably highest in the globe                |
| Vitamin A supplementation (percentage of 6–59 months old children receiving supplement in the last 6 months)       | 84%  | 60%  | 62%  | 79%     |                                              |

ANC=Antenatal Care; CS= Cesarean section; PNC=Postnatal Care; ORT=Oral Rehydration Therapy, ORS=Oral Rehydration Salt

Source: IPHN, Mohakhali, Dhaka

- ◆ Equity in the use of facility for delivery among the poor and the rich is still high but declining over time.
- ◆ 8 out of 10 pregnant women are getting at least one ANC from a medically-trained care provider. Sharp increase is observed in ANC visits from 64% to 84% between BDHS 2011 and 2017.
- ◆ Receiving ANC4+ visits increased from 31% to 47% in the last three years.
- ◆ Less than one in five (18%) pregnant women received quality ANC4+ visits at least one from medically-trained care provider + all basic components.
- ◆ One-third of the deliveries are performed by cesarean section. This implies that one million C-section deliveries are taking place every year.
- ◆ 84% of the deliveries in the private sector are performed by C-section.
- ◆ 52% of the mothers received PNC from a medically-trained care provider within 2 days after delivery, among the non-institutional deliveries, only 7% of the mothers received PNC within 2 days.

### Childhood Mortality and Healthcare-seeking

- Under-5 mortality is 45 per 1,000 livebirths.
- Neonatal mortality is 30 per 1,000 livebirths and accounts for 67% of all under-5 deaths.
- Under-5 mortality rate has been declining gradually over the last two decades. However, between 2014 and 2017 BDHS,

the decline in under-5 mortality rate has slowed down noticeably.

- Infant and neonatal mortality have not declined between 2014 BDHS and 2017-2018 BDHS.
- 86% of 12-23 months old children received all the recommended vaccinations before their first birthday.
- 85% of children with diarrhea were given oral rehydration therapy (ORT), and 44% received both ORT and zinc.

### Child Nutrition and Feeding

- There has been steady improvement in child nutritional status over the past decade.
- Stunting among children declined from 43% in 2007 to 31% in 2017.
- Wasting came down to 8% after remaining critically high for years.
- 65% of infants below the age of 6 months were exclusively breastfed in 2017, which was markedly higher than the 2014 level (55%).
- IYCF practices among 6–23 months old children increased from 23% in 2014 to 34% in 2017.
- 79% of children received vitamin A supplementation—an increase from 62% in 2014.

### Major Activities in Nutritional Progress in 2018

Nutrition is a basic human right in terms of both equity and equality. Nutrition is an



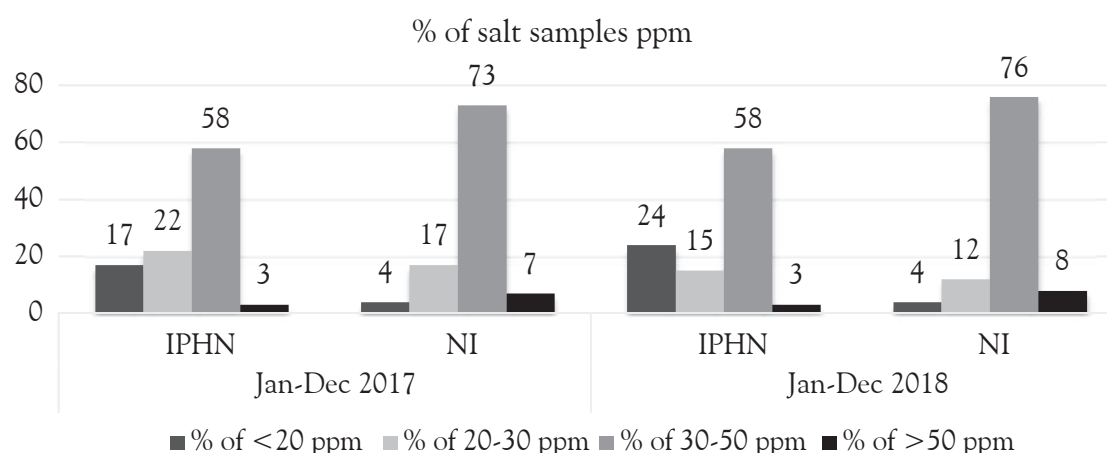
essential component of the physical growth and mental development of every person. In the absence of nutrition, the fetus does not increase properly and is born with low birthweight. Government of the People's Republic of Bangladesh is committed to improving the nutritional status of the people of the country. Moreover, the Bangladesh National Nutrition Policy 2015 and National Plan of Action for Nutrition (NPAN-2) were developed to address the challenges relating to nutrition. The policy takes into consideration both global policies, such as Second International Conference of Nutrition (ICN2) and relevant national policies and strategies in areas, such as health, food, agriculture, environment, and education reflecting its multi-sectoral approach to ensuring improvement of nutritional status.

### Nutritional activities under the Institute of Public Health Nutrition (IPHN)

The Institute of Public Health Nutrition (IPHN) under the Ministry of Health and Family Welfare (MOHFW) conducted

validation test of 888 BSCIC salt samples from January-December 2017. A total of 1,072 salt samples were tested during January-December 2018, which was supported by the Nutrition International (NI); 697 and 1,022 salt samples collected by sanitary inspectors of different sub-districts were also tested. Among the NI-collected samples, >30parts per million (ppm) of iodine per kg of salt slightly increased from 2017 (80%) to 2018 (84%) where IPHN-collected samples containing >30ppm was 61% and del to 66% in 2017 and 2018 respectively.

- > Real-time Monitoring and Response (RTMR) initiated in the last Vitamin A Plus Campaign (NVAC+) in July 2018
- > The MOHFW approved National Advocacy and Communication Strategy
- > National Dietary Guideline developed and sent to the Ministry for approval
- > Digital Archive at IPHN established
- > BCC eToolkit developed for field workers (link: [www.dghs.gov.bd](http://www.dghs.gov.bd) and <http://bdsbcc.org/>).



**Figure 4.9.1.** Percentage of salt samples tested by NI and IPHN laboratory in 2017 and 2018 for iodine contents (in ppm)

- > 395 Integrated Management of Childhood Illness (IMCI) Corners and Nutrition Corners in the district hospitals and upazila health complexes developed
- > 398 Service Availability Mapping (SAM) Unit at district hospitals and UHCs established
- > 592 Baby-friendly Hospital Initiatives (BFHI) undertaken
- > A position of Nutritionist created for each of 64 districts

### National Vitamin A Plus Campaign (NVAC+) First Round, 14 July 2018

Out of 21,055,529 children aged 6-59 months, a total of 20,805,290 (98.8%) received vitamin A capsule (including child-to-child searching).

- ◆ No significant differences were observed among coverage in administrative divisions.

- ◆ In some districts, the coverage was 100% while in the rest of the districts, the coverage was between 95% and 99%.
- ◆ Coverage in Gazipur City Corporation was 102% among infants of 6-11 months and 101% among children of 12-59 months, which indicates that the target needs to be updated; coverage in rest of the city corporation areas was similar to that in the districts/divisions (95-100%).
- ◆ Among 240 hard-to-reach unions, 49,985 children aged 6-59 months, who remained unreached by then received vitamin A capsules through child-to-child (CtC) searching; over 2.8% unreached children in 240 unions were covered through CtC (based on respective targets of the hard-to-reach areas).

Pourashava (Municipality) coverage is incorporated into respective districts; National target and coverage is the sum of values from city corporations and rural districts.

**Table 4.9.2.** National Vitamin A plus Campaign (NAVC+): First Round, 2018 coverage summary

| Area                        | Target     | Coverage   | Percentage (%) |
|-----------------------------|------------|------------|----------------|
| City corporation (CC)       | 2,059,688  | 2,038,448  | 99.0           |
| Pourashava (Municipality) 1 | 1,006,925  | 981,388    | 97.5           |
| Rural (Districts)           | 18995841   | 18,766,842 | 98.8           |
| National1 (CC + Rural) {OK} | 21,055,529 | 20,805,290 | 98.8           |

**Table 4.9.3.** National Vitamin A plus Campaign: (First Round, 2018 coverage of 46 Sub-districts (240 hard-to-reach unions)

| Hard-to-reach areas           | 6-11 months old children | 12-59 months old children | 6-59 months old children |
|-------------------------------|--------------------------|---------------------------|--------------------------|
| NVAC+ day in 46 sub-districts | 96.3%                    | 97.1%                     | 97.1%                    |
| CtC on subsequent 4 days      | 3.1%                     | 2.7%                      | 2.8%                     |
| <b>Total</b>                  | <b>99.4</b>              | <b>99.9</b>               | <b>99.8%</b>             |

### Coverage by division, district and city corporation

The NVAC+ coverage in July 2018 by division is given below. Figure 4.9.2 shows that coverage is equal, with little variance. Therefore, in case of 6-11 months old children, coverage in Dhaka division (97%) is less than in other divisions.

### Coverage by city corporation

Figure 4.9.4 shows that coverage exceeded 100% (among both age-groups) in 7 city

corporations, which indicates that uncounted floating people may have been covered, although average coverages are 97.2% and 99.5% for 6-11 months old and 12-59 months old children respectively.

### Food Safety and Quality Control

Unsafe food has been a long-standing challenge in the country and poses a serious public health risk leading to increased prevalence of foodborne diseases. The Institute of Public Health (IPH) is playing an important role in assuring safe and quality

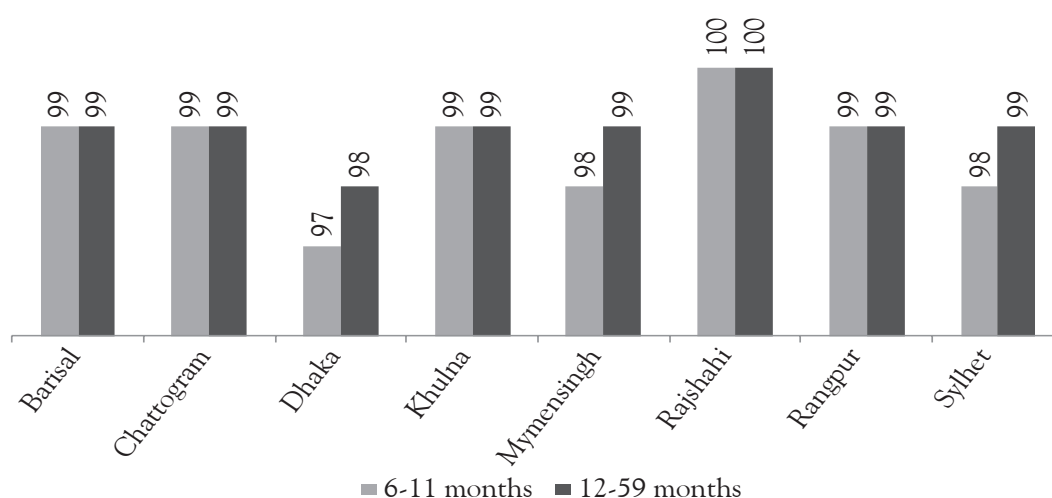


Figure 4.9.2. NVAC+ coverage during the First Round (July 2018) by division and age

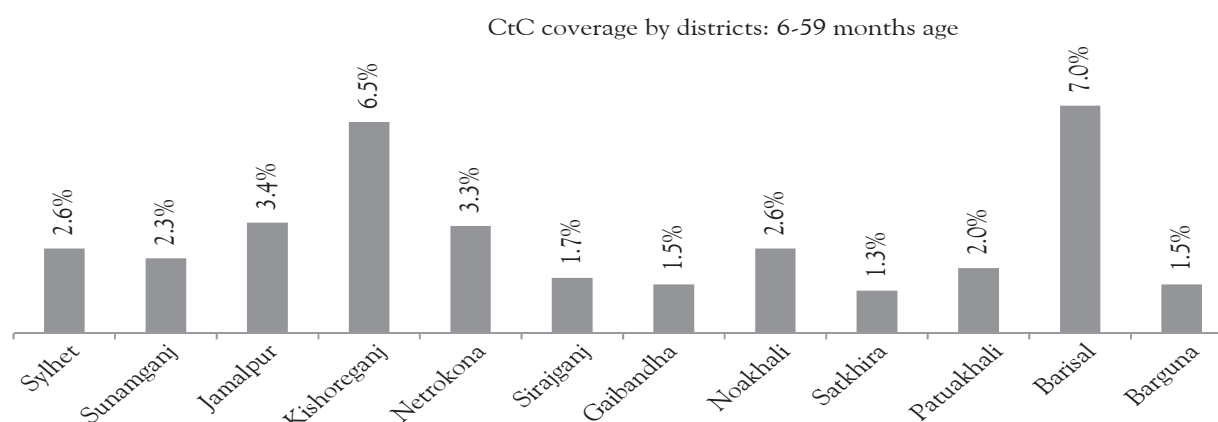
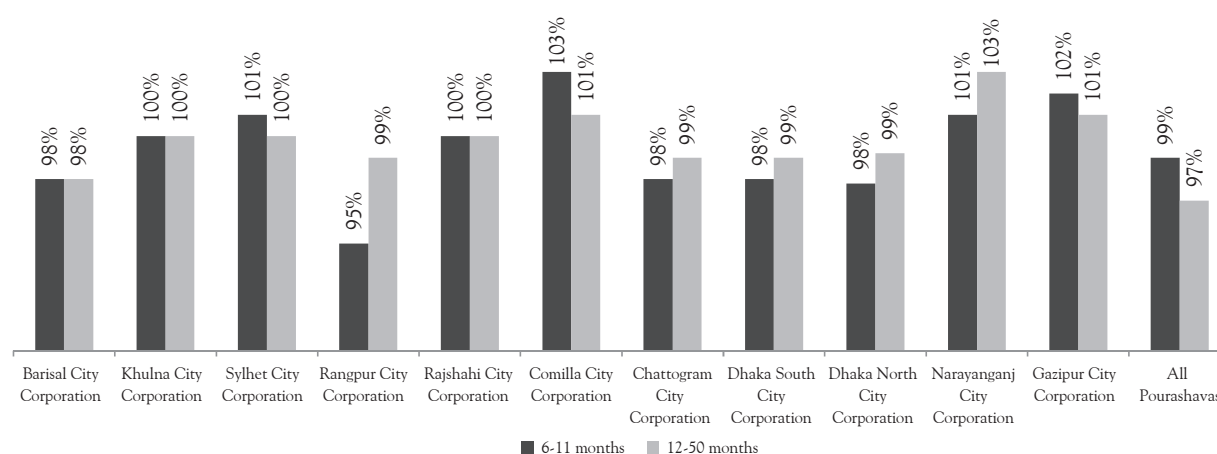


Figure 4.9.3. NVAC+ coverage during the First Round (July 2018) by district and age



**Figure 4.9.4.** NVAC+ coverage during the First Round (July 2018) by city corporation/ municipality and by age

food, providing laboratory analysis of food and water, technical expertise and coordination support through its National Food Safety Laboratory (NFSL), Public Health Laboratory (PHL), and Food Safety Unit (FSU).

### ***Laboratory analysis of food and water samples***

The NFSL and PHL conduct tests for food and water samples according to the country's laws, regulations, and standards as well as international standards. The food and water samples are sent by the sanitary inspectors working under the DGHS, municipalities, and city corporations. Food and water samples are also sent from other sources, including private organizations, international organizations, and UN bodies across the country. With the test results, the local authorities take legal actions against the offenders. It may be mentioned that some 600 well-trained and equipped sanitary inspectors under the DGHS are rendering risk-based food inspection services across the country.

The laboratories also conduct routine and regular tests for water samples sent from the

Bangabhaban, Ganabhaban, Office of the Hon'ble Prime Minister, the Parliament, and other important offices.

### ***National Food Safety Laboratory***

With financial support from the European Commission and technical support from the FAO of UN, the National Food Safety Laboratory (NFSL) was established in 2009 and started its function in 2012.

This service, research, training and reference laboratory was inaugurated in October 2012 by Hon'ble Minister for the Ministry of Health and Family Welfare. Further development of the NFSL was done with the financial support of the Government of the Netherlands and technical support from the FAO of UN during the period of 2012-2018. The Laboratory is manned by a group of scientists, chemists, and technologists trained in home and abroad. The team has validated more than 80 methods for different food items. The Laboratory has been accredited by the Bangladesh Accreditation Board (BAB) under ISO/IEC 17025:2005.

The NFSL is playing a key role in testing of pesticide residues; heavy metals; antibiotics; formaldehyde, and different additives, including coloring agents and preservatives; myco-toxins, and foodborne pathogens in different food items. The samples are being received from inspection activities, private sectors, UN and international agencies, surveillance activities, including food contaminants and foodborne disease surveillance as well as food safety emergency response. The Lab is regularly participating in proficiency tests in home and abroad.

During the reporting period, this laboratory of international standard tested a wide range of parameters in different food items as shown in Table 4.9.4.

### **Public Health Laboratory**

The Institute of Public Health (IPH) started its journey with the combined Public Health Laboratory (PHL) in 1948. Recently, with the financial support from the Dutch Government and technical support from the FAO of UN, the PHL has been refurbished and upgraded with instruments and equipment, which was inaugurated by Hon'ble Minister for the Ministry of Health and Family Welfare on 29 August 2016.

The numbers of tested food items and water samples over the last seven years (2012-2018), with satisfactory and unsatisfactory test results are shown below.[in Figure 4.9.2, Table 4.9.5 and 4.9.6] It may be mentioned that the suspected samples are being collected by Sanitary Inspectors which might to be adulterated or contaminated.

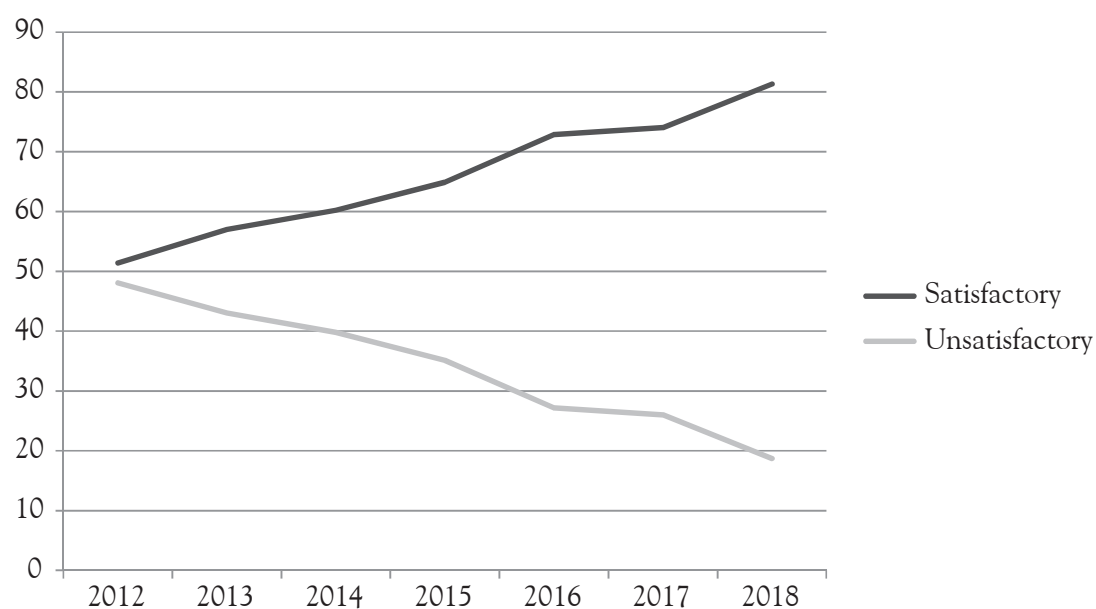
The NFSL and PHL also provided laboratory support to the IEDCR for response to the food safety emergencies and outbreak investigations during the reporting period. Furthermore, the NFSL also provided support for the research fellows and students of different institutions and universities for their theses and dissertations.

Earlier, the PHL was the Public Health and Drug Testing Laboratory (PH&DTL). Until recently, the Drug Testing Laboratory (DTL) wing was placed under the administrative and technical control of the IPH but, as per recommendations of the WHO, the Ministry of Health and Family Welfare has placed the DTL under the administrative control of the Directorate General of Drug Administration (FGDA). However, staff salaries and other recurring expenditure are still met from the existing budget of the IPH.

**Table 4.9.4.** Summary of the tested food samples by the National Food Safety Laboratory during 2013-2018

| Year                     | Total samples | Parameter tested |
|--------------------------|---------------|------------------|
| 2013                     | 214           | 1,284            |
| 2014                     | 363           | 2,541            |
| 2015                     | 1,814         | 33,875           |
| 2016                     | 1,134         | 13,706           |
| 2017                     | 835           | 9,726            |
| 2018                     | 426           | 2,211            |
| <b>Total (2013–2018)</b> | <b>4,786</b>  | <b>63,343</b>    |

Source: IPH, Mohakhali, Dhaka



**Figure 4.9.5.** Food samples tested by Public Health Laboratory over the last 7 years (2012-2018)

**Table 4.9.5.** Bacteriological test results of water samples over the last 7 years (2012 to 2018)

| Year | Total samples | Satisfactory |            | Unsatisfactory |            |
|------|---------------|--------------|------------|----------------|------------|
|      |               | No.          | Percentage | No.            | Percentage |
| 2012 | 600           | 563          | 94.0       | 37             | 6.0        |
| 2013 | 706           | 666          | 94.0       | 40             | 6.0        |
| 2014 | 595           | 545          | 92.0       | 50             | 8.0        |
| 2015 | 502           | 434          | 86.0       | 68             | 14.0       |
| 2016 | 787           | 693          | 88.0       | 94             | 12.0       |
| 2017 | 320           | 308          | 96.3       | 12             | 3.7        |
| 2018 | 523           | 393          | 75.1       | 130            | 24.9       |

**Table 4.9.6.** Chemical test results for water samples during 2012-2018

| Year | Total samples | Satisfactory |            | Unsatisfactory |            |
|------|---------------|--------------|------------|----------------|------------|
|      |               | No.          | Percentage | No.            | Percentage |
| 2012 | 350           | 347          | 99.20      | 03             | 0.80       |
| 2013 | 500           | 481          | 96.19      | 19             | 3.81       |
| 2014 | 482           | 453          | 94.52      | 29             | 5.48       |
| 2015 | 438           | 425          | 97.90      | 09             | 2.10       |
| 2016 | 644           | 602          | 95.48      | 42             | 6.52       |
| 2017 | 593           | 559          | 94.27      | 34             | 5.73       |
| 2018 | 406           | 398          | 98.00      | 08             | 2.00       |



***Technical and coordination support for food safety activities***

With support of the FAO of UN and the Dutch Government, the Ministry of Health and Family Welfare has established the Food Safety Unit in the IPH in 2012, aimed at providing technical and coordination support for food safety activities under the DGHS as well as for liaison with other agencies. The Unit is run by a group of experts in food safety and allied fields.

***Food Safety Unit***

A Food Safety Program, technically supported by the FAO of UN and financially supported

by the Dutch Government and earlier by the European Commission, is housed within the administrative set-up of the Food Safety Unit of the IPH. The Program is implementing its activities under the Operational Plan of the National Nutrition Services of the 4th Health, Population and Nutrition Sector Program. A group of experts in food safety and allied fields runs the Unit. A number of technical experts of the FAO of UN also support the Unit. The Unit has also been contributing in the development of manuals, guidelines, strategies, policies, and plans in the areas of food safety and quality control in the country. The IPH is taking necessary steps to make it a full-fledged unit to accomplish the objectives.

# Curative Care Services

## From primary care facilities to specialized hospitals

### Primary healthcare

Primary healthcare (PSC) addresses comprehensive and interrelated physical, mental and social health and well-being. It is an attitude to health and well-being centered on the essentials and circumstances of individuals, families and communities through their active participation and at an affordable cost. The salient features of primary healthcare are as follows:

- PHC is an integral part of a country's health system and, ideally, it is the main focus of the system. It is the first level of contact for individuals, families, and communities and enables healthcare to be delivered as close as possible where people live and work. PHC is, therefore, the first element of the care continuum.
- Primary healthcare is rooted in a commitment to social justice and equity and in the recognition of the fundamental right to the highest attainable standard of health as echoed in Article 25 in the Universal Declaration on Human Rights: Everyone has the right to a standard of living adequate for the health and well-being of him/herself and of his/her family, including food, clothing, housing, and medical care and necessary social services.
- Primary healthcare is well-positioned to be able to respond to rapid economic,

Primary healthcare (PSC) addresses comprehensive and interrelated physical, mental and social health and well-being...

technological and demographic changes, all of which have an impact on health and well-being. A recent analysis found that approximately half of the gains in reducing child mortality from 1990 to 2010 were due to factors outside the health sector (water and sanitation, education, economic growth).

- A primary healthcare approach draws in a wide range of stakeholders to examine and change policies to address the social, economic, environmental and commercial determinants of health and well-being. Treating people and communities as key actors in the production of their own health and well-being is critical for understanding and responding to the complexities of our changing world.
- Primary healthcare has been proven to be a highly-effective and efficient way to address the main causes and risks of poor health and well-being today as well as handling the emerging challenges that will threaten the health and well-being of tomorrow. It has also been shown to be a good value investment as

there is evidence that quality primary healthcare reduces total healthcare costs and improves efficiency by reducing hospitalizations. Stronger primary healthcare is essential to achieve the health-related Sustainable Development Goals (SDGs) and Universal Health Coverage. It will contribute to the attainment of other goals beyond the health-related goal (SDG3), including those on poverty, hunger, education,

gender equality, clean water and sanitation, work and economic growth, reducing inequality, and climate action.

### Community Clinics

Community Clinic (CC) is the brainchild of Hon'ble Prime Minister Sheikh Hasina. It is being branded in her name. It is the lowest-tier public health facility. The formation and functions of CCs are as follows:

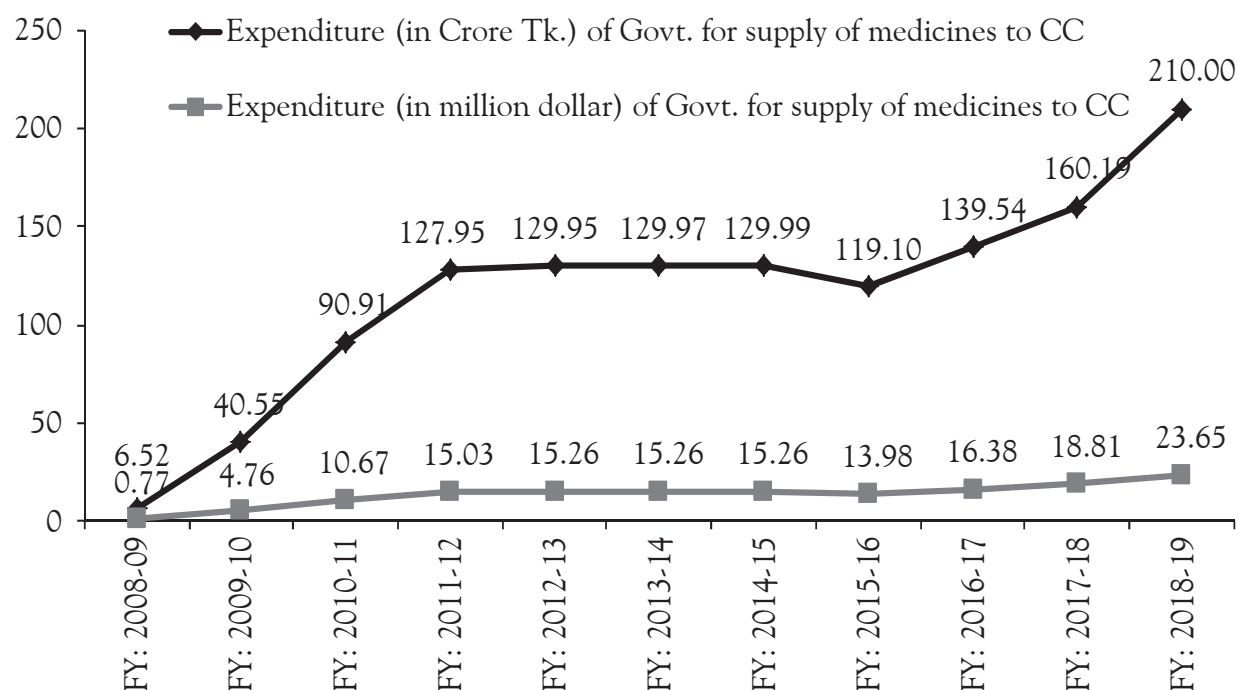
**Table 5.1.** Primary healthcare facilities run by the DGHS, 2019 (December)

| Type of facility                                                            | Type of service | Total no. of facilities | Total no. beds |
|-----------------------------------------------------------------------------|-----------------|-------------------------|----------------|
| Upazila health complex (100-bed)                                            | Hospital        | 3                       | 300            |
| Upazila health complex (50-bed)                                             | Hospital        | 345                     | 17250          |
| Upazila health complex (31-bed)                                             | Hospital        | 65                      | 2015           |
| Upazila health complex (10-bed)                                             | Hospital        | 11                      | 110            |
| <b>Subtotal of upazila health complexes</b>                                 |                 | <b>424</b>              | <b>19675</b>   |
| 50-bed hospital                                                             | Hospital        | 2                       | 100            |
| 31-bed hospital                                                             | Hospital        | 7                       | 217            |
| 30-bed hospital                                                             | Hospital        | 2                       | 60             |
| 25-bed hospital                                                             | Hospital        | 1                       | 25             |
| 20-bed hospital                                                             | Hospital        | 32                      | 640            |
| 10-bed hospital                                                             | Hospital        | 17                      | 170            |
| Upazila health office                                                       | OPD             | 60                      | 0              |
| Union Sub-center                                                            | OPD             | 1312                    | 0              |
| Union health and family welfare center (UH & FWC, Netherlands project)      | OPD             | 87                      | 0              |
| Urban dispensary                                                            | OPD             | 35                      | 0              |
| School health clinic                                                        | OPD             | 23                      | 0              |
| Tejgaon Health Complex, Dhaka                                               |                 | 1                       | 0              |
| <b>Grand total of primary-level facilities (except community clinic)</b>    |                 | <b>2003</b>             | <b>20887</b>   |
| Community clinic (functional at present)                                    | OPD             | 13907                   | 0              |
| <b>Grand total of primary-level facilities (including community clinic)</b> |                 | <b>15910</b>            | <b>20887</b>   |

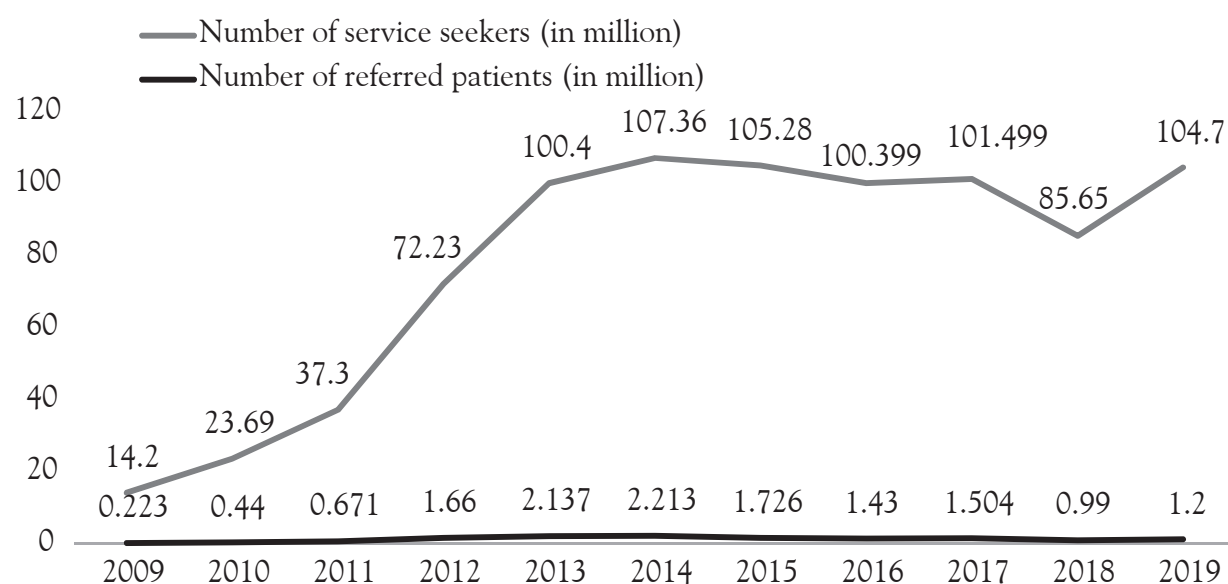
- CC is a one-stop service outlet on health, family planning, and nutrition, each covering around 6,000-12,000 rural population. It is a unique example of public-private partnership since all CCs have been constructed on land donated by the community people.
- Costs incurred for construction, medicines, service providers, logistics, and all other inputs are borne by the Government but these are managed by both community and the Government through Community Groups (CGs).
- The CG (CC management body) for each CC comprises 17 members and is headed by Elected Member of the Union Parishad; other members in the CG are from different walks of life. To support a CG in the management and community engagement, there are 3 Community Support Groups (CSGs) for each CC. In both CG and CSG, at least one-third of the members are female, including adolescents. During 1998-2001, more than 10,000 CCs were constructed, of which about 8,000 started functioning but, in 2001, shortly after commissioning, the CC activities were closed due to political change in the Government and continued as such till 2008.
- In 2009, the Government took initiative for revitalization of CCs as a topmost priority under a project titled “Revitalization of Community Healthcare Initiatives in Bangladesh” (RCHCIB), setting the implementation period from 1 July 2009 to 30 June 2015.
- Under this project, the closed-up CCs were made functional in phases after necessary repair, recruitment, and capacity-building by creating the position of a new category of service provider called Community Healthcare Provider or CHCP for each CC. Medicines, with all necessary logistics, were provided.
- As of December 2018, a total of 13,907 CCs were operational. From July 2015, all activities of CCs were being carried out under an Operational Plan of the DGHS, namely Community-based Healthcare (CBHC) under the Health, Population and Nutrition Sector Development Program (HPNSDP) 2011-2016.
- Thousands of villagers, particularly the poor and the underprivileged community mothers and children, are getting services from the nearby CCs. On an average, 9.5-10.0 million visits by service-seekers take place nationwide each month in the CCs.
- It is to be mentioned that normal child delivery is being conducted in a good number of CCs (about 20% of the functional CCs) at the demand of the community members and decision of the CG based on the availability of skilled manpower. In total, 1,935 CHCPs have been trained on child delivery and, based on the decision of the committed local management, cases can be referred easily to the nearby UHC, if necessary. From 2009 till 2019 (November), 79,054 normal child deliveries have been conducted in CCs without notable casualties to the mother and their babies.
- The use of ICT by CCs for data management and service provision is quite impressive. The Management Information System (MIS) of the DGHS

is providing all-out support to develop and maintain the ICT backbone from April 2014. In some upazilas, monitoring of CCs is being done through Skype;

telemedicine services are also being organized between CCs and UHCs, with a patient at the CC and a doctor at the other end.



**Figure 5.1.** Total expenditure of Government for supply of medicines to community clinics in crore Tk and million dollar by fiscal year



**Figure 5.2.** Total number of service-seekers (in million) at CCs and cases referred from community clinics to higher facilities for proper management

From 2011, with the deployment of CHCPs after basic training, the number of service-seekers has been increasing. From April 2009 to 2019, there were 852.70 million visits by rural people to CCs all over the country and 14.194 million emergency and complicated cases were referred to higher facilities for better management.

- Community clinic is an unprecedented instance of community participation and public-private partnership. Being inspired by community participation, some UN agencies and NGOs have started working for the community clinics. In Bangladesh, the 4th Health, Population and Nutrition Sector Program (HPNSP) is under implementation for a period from January 2017 to June 2022. The Operational Plan “Community-based Healthcare (CBHC)” will contribute substantially in achieving the targets of SDGs, as it did during the MDG era and will continue till the achievement of the SDGs by 2030.
- Due to the flagship nature of the program, community clinics deserve special mention. In addition to the community clinics, important components of primary healthcare, among others, include domiciliary healthcare, essential service delivery, along with urban primary healthcare, maternal healthcare (inclusive of some screening programs for women’s health), child healthcare, school health program, and adolescent health program.
- The MIS-DGHS has been playing the leading role in providing computers with Internet connections, which now extend down to the grassroots-level health facilities as well as health workers. The union health centers and community

clinics have laptop computers and wireless modems, and the community health workers have Android tablets.

## **Domiciliary Health Services in Rural Bangladesh**

There are domiciliary workers—one for every 5 to 6 thousand people at the ward or village level. Under the DGHS, there are 26,538 sanctioned posts of domiciliary workers, of which 20,908 are for health assistants (HA), 4,220 for assistant health inspectors (AHI), and 1,410 for health inspectors (HI). As of now, 74.72% posts were filled up.

Like the DGHS, the DGFP also has domiciliary workers to work at the ward or village level. These staff members are called family welfare assistants (FWA) and family planning inspectors (FPI).

## **Essential Service Delivery and Urban Primary Healthcare**

There was an Operational Plan, namely “Essential Service Delivery” (ESD) mainstreamed under the DGHS (HPNSDP 2011-2016) to improve service, particularly at the upazila level and below and complement urban primary healthcare. ESD will continue under the current HPNSP also. Activities are detailed below:

- The areas of services include: limited curative care, support services and coordination, medical waste management, urban health, mental health, and tribal health.
- The urban primary healthcare in Bangladesh is principally the responsibility of the Ministry of Local Government,



- Rural Development and Cooperatives (MOLGRD), carried out through the city corporations and municipalities.
- Besides, the large-scale primary healthcare activities under Urban Primary Healthcare Project (UPHCP) and Smiling Sun Franchise Program are run by NGOs in collaboration with the city corporations, municipalities with financial assistance from donors.
  - Available number of beds is one of the good proxies for measuring the strength of healthcare infrastructure in different geographic areas. These are run under the administrative control of the DGHS.

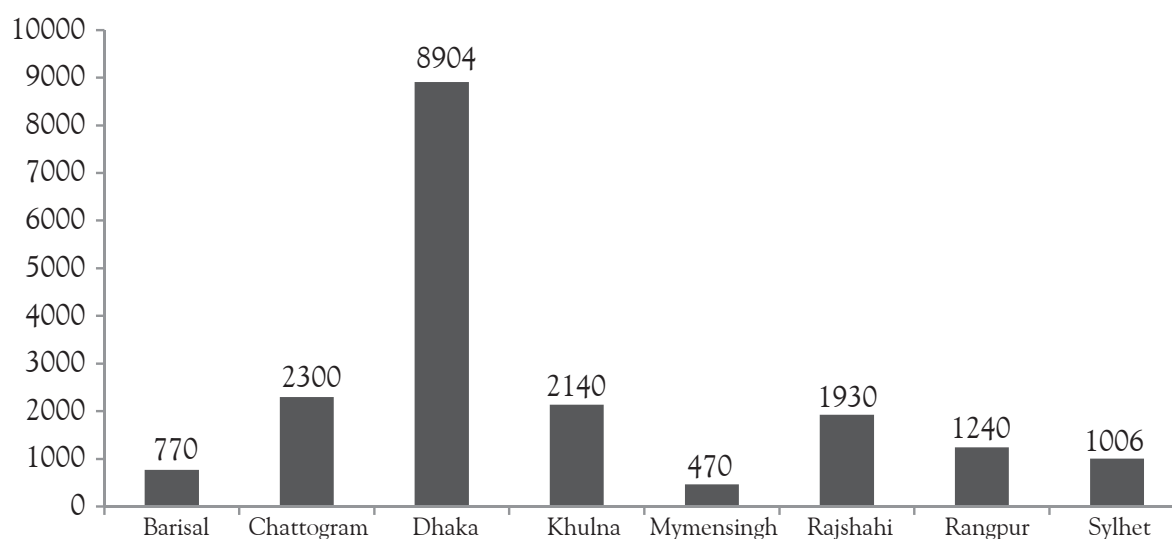
**Table 5.2.** Secondary and tertiary facilities under the DGHS, with functional beds (December 2019)

| Type                                                | Level of facilities | Name of facilities                                             | No. of facilities | No. of functional beds |
|-----------------------------------------------------|---------------------|----------------------------------------------------------------|-------------------|------------------------|
| Airport Health Office                               | Secondary           | Dhaka Airport Health Office                                    | 1                 | 0                      |
|                                                     |                     | Chittagong Airport Health Office                               | 1                 | 0                      |
| Chest disease clinic                                | Secondary           |                                                                | 42                | 0                      |
| Chest hospital                                      | Secondary           |                                                                | 14                | 816                    |
| District-level office (Civil Surgeon's Office)      | Secondary           |                                                                | 65                | 0                      |
| District-level hospital (District/General Hospital) | Secondary           |                                                                | 59                | 9950                   |
| Division-level office (Divisional Health Office)    | Tertiary            |                                                                | 8                 | 0                      |
| Dental college hospital                             | Tertiary            |                                                                | 1                 | 200                    |
| Hospital of alternative medicine                    | Tertiary            |                                                                | 2                 | 200                    |
| Infectious disease hospital                         | Secondary           |                                                                | 5                 | 180                    |
| Leprosy hospital                                    | Secondary           |                                                                | 3                 | 130                    |
| Medical college hospital                            | Tertiary            |                                                                | 18                | 15013                  |
| Port Health Office                                  | Secondary           | Chittagong Port Health Office                                  | 1                 | 0                      |
|                                                     |                     | Khulna Port Health Office                                      | 1                 | 0                      |
| Specialized Health Center                           | Tertiary            | Central Skin & Social Hygiene Center, Chittagong               | 1                 | 0                      |
|                                                     |                     | National Center for Control of Rheumatic Fever & Heart Disease | 1                 | 0                      |
|                                                     |                     | National Asthma Center                                         | 1                 | 70                     |
| Specialized hospital                                | Tertiary            |                                                                | 5                 | 1450                   |

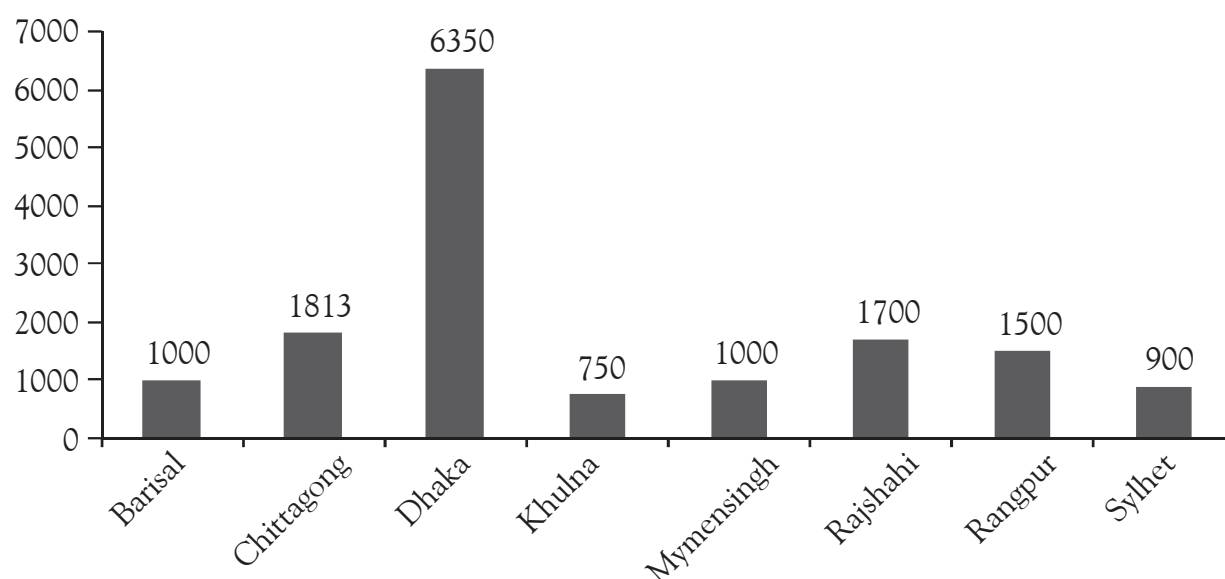
**Table 5.2 Contd.**

Table Continued...

| Type                                          | Level of facilities | Name of facilities                                          | No. of facilities | No. of functional beds |
|-----------------------------------------------|---------------------|-------------------------------------------------------------|-------------------|------------------------|
| Specialty postgraduate institute and hospital | Tertiary            |                                                             | 12                | 3784                   |
| Trauma center                                 | Secondary           |                                                             | 5                 | 100                    |
| Other hospitals                               | Tertiary            | Kurmitola 500-bed General Hospital, Dhaka                   | 1                 | 500                    |
|                                               |                     | Mugda 500 bed General Hospital, Dhaka                       | 1                 | 500                    |
|                                               | Secondary           | Narayanganj 300-bed Hospital                                | 1                 | 300                    |
|                                               |                     | Narsingdi 100-bed Hospital                                  | 1                 | 100                    |
|                                               |                     | Shaheed Ahsan Ullah Master General Hospital, Tongi, Gazipur | 1                 | 250                    |
|                                               |                     | Bangladesh Korea Moitree Hospital, Savar, Dhaka             | 1                 | 30                     |
|                                               |                     | Kuwait Bangladesh Friendship Govt. Hospital, Uttara, Dhaka  | 1                 | 100                    |
|                                               |                     | Saidpur 100-bed Hospital, Nilphamari                        | 1                 | 100                    |
|                                               |                     |                                                             |                   |                        |
| Total secondary and tertiary-level facilities |                     | 255                                                         | 33773             |                        |



**Figure 5.3.** Number of beds in all secondary and tertiary-care hospitals other than medical college hospitals by administrative division, 2018



**Figure 5.4.** Number of beds in all government medical college hospitals by administrative division of Bangladesh, 2018

- ◆ As of June 2019, the DGHS provided registration to 15,728 private hospitals, clinics, diagnostic centers, and blood banks in Bangladesh.
- ◆ The number of registered private hospitals and clinics is 5,321, and that of registered private diagnostic centers is 10,407. The total number of beds in these registered private hospitals and clinics is 91,537.
- Outpatient attendance at specialized hospitals ranged from 26,558 at the National Centre for Control of Rheumatic Fever and Heart Disease to 1,023,846 at the Kurmitola General Hospital (500 beds) (Table 5.3).
- At medical college hospitals, the outpatient attendance ranged from 219,939 at M Abdur Rahim MCH, Dinajpur to 1,311,023 at Mymensingh MCH, Mymensingh (Table 5.4).
- At the district/general hospitals, outpatient attendance ranged from 56,393 at Rangamati District Hospital, Rangamati to 464,893 at Jessore District Hospital, Jessore (Table 5.5).

### Utilization of Public Health Facilities

The information is based on hospital statistics available from the 2018 DHIS2 data, which should be interpreted with caution due to incomplete progress reporting.

### Hospital outpatient, emergency ward and inpatient attendance

Outpatient attendance at different hospitals reporting through DHIS2 in 2018 was as follows:

424 upazila health complexes reported OPD attendance data, which are presented in the Annex to Chapter 5.

Attendance at hospital emergency wards at different hospitals reporting through DHIS2 in 2018 was as follows:

- At specialized hospitals, attendance ranged from 3,866 at the National Institute of Mental Health Research, Dhaka to 116,784 at the National Institute of Cardiovascular Disease, Dhaka (Table 5.3).
- At medical college hospitals, the figures ranged from 10,018 at Khulna Medical College Hospital, Khulna to 470,780 at Dhaka Medical College Hospital, Dhaka. (Table 5.4).
- At the district/general hospitals, attendance in emergency wards ranged from 3048 at Shahid Shamsuddin District Hospital to 148,631 at Narayanganj General (Victoria) Hospital (100 beds) (Table 5.5).
- Four hundred twenty-four upazila health complexes reported emergency attendance data, which are shown in Annex to Chapter 5.

Inpatient attendance at different hospitals reporting through DHIS2 in 2018 was as follows:

- At specialized hospitals, inpatient attendance ranged from 1,548 at Pabna Mental Hospital, Pabna to 72,268 at the National Institute of Cardiovascular Diseases, Dhaka (Table 5.3).
- At medical college hospitals, the figures ranged from 31,352 at Shaheed Tajuddin Ahmad Medical College, Gazipur to 222,519 at Chittagong Medical College Hospital, Chittagong (Table 5.4).
- At the district/general hospitals, inpatient attendance ranged from 3,090 at Shahid Shamsuddin District Hospital, Sylhet to 88,411 at Pabna District Hospital, Pabna (Table 5.5).

Four hundred twenty-four upazila health complexes reported inpatient attendance data which are presented in Annex to Chapter 5.

## Hospital-reported deaths

The total number of deaths reported from different health facilities under the DGHS was 121,090.

Table 5.6 shows the number of institutions reporting deaths according to age and sex. Figure 5.6 shows the total number of deaths according to age and sex in 2018.

## Cause of Morbidity

Major causes for admission in healthcare facilities during the reporting year were: diarrhea and gastroenteritis of presumed infectious origin. The morbidity profile is visualized in the following graph (Figure 5.7):

## Cause of death

The following graph shows the causes of death among inpatients reported from medical college hospitals, district/general hospitals, and upazila health complexes in 2018 through DHIS2.

## Average Length of Stay at Hospitals

- ◆ Average length of stay at specialized hospitals ranged from 2.3 days at Dhaka Shishu Hospital to 13.9 days at the National Institute of Diseases of the Chest and Hospital (NIDCH) (Table 5.7).
- ◆ At medical college hospitals, the figures ranged from 3.3 days at Rajshahi Medical College to 6.6 days at Dhaka Medical College Hospital (Table 5.8).

**Table 5.3.** Outpatient, emergency and inpatient attendances at government specialized hospitals, 2018

| Sl. No.      | Name of specialized hospital                                                   | OPD visit      |                | Emergency visit |               | Inpatient    |               |
|--------------|--------------------------------------------------------------------------------|----------------|----------------|-----------------|---------------|--------------|---------------|
|              |                                                                                | Female         | Male           | Female          | Male          | Female       | Male          |
| 1            | Bangladesh Institute of Tropical and Infectious Diseases, Fouzdarhat           | 22074          | 18922          | 4083            | 2474          | 3763         | 1983          |
| 2            | National Institute of Cancer Research and Hospital, Dhaka                      | 117360         | 93839          | 3748            | 3021          | 3938         | 3050          |
| 3            | National Institute of Diseases of the Chest & Hospital (NIDCH)                 | 56055          | 81128          | 2776            | 7658          | 5056         | 12765         |
| 4            | Institute of Child & Mother Health (ICMH), Matuail                             | 120010         | 103516         | 41277           | 41349         | 9585         | 5351          |
| 5            | National Institute of Cardiovascular Diseases, Dhaka                           | 66323          | 104534         | 36254           | 80530         | 20059        | 52209         |
| 6            | National Institute of Kidney Diseases & Urology, Dhaka                         | 37494          | 72455          | 2722            | 4738          | 2204         | 3131          |
| 7            | National Institute of Mental Health and Research, Dhaka                        | 25025          | 29431          | 1652            | 2214          | 1869         | 2836          |
| 8            | National Institute of Neurosciences and Hospital (NINS&H)                      | 135138         | 147650         | 18271           | 17800         | 4368         | 6572          |
| 9            | National Institute of Ophthalmology (NIO), Dhaka                               | 174771         | 214161         | 2422            | 4052          | 7019         | 8463          |
| 10           | National Institute of Traumatology & Orthopedics Rehabilitation (NITOR), Dhaka | 31186          | 49831          | 0               | 0             | 513          | 1198          |
| 11           | National Institute of ENT in Dhaka                                             | 57128          | 66256          | 2276            | 4002          | 1238         | 1855          |
| 12           | Pabna Mental Hospital, Pabna                                                   | 25709          | 22403          | 0               | 0             | 369          | 1179          |
| 13           | National Centre for Control of Rheumatic Fever and Heart Disease               | 16257          | 10301          | 0               | 0             | 0            | 0             |
| 14           | Kurmitola General Hospital (500-bed)                                           | 580809         | 443037         | 32321           | 32338         | 17963        | 17533         |
| 15           | Mugda 500-bed General Hospital                                                 | 233995         | 214384         | 20580           | 20802         | 14386        | 11932         |
| <b>Total</b> |                                                                                | <b>1699334</b> | <b>1671848</b> | <b>168382</b>   | <b>220978</b> | <b>92330</b> | <b>130057</b> |

Note: Report of two facilities: Sheikh Fazilatunnessa Mujib Eye Hospital and Training Institute and Sheikh Russel Gastroenterology Institute & Hospital were not available (Source: DHIS2, MIS report, 2018)

**Table 5.4.** Outpatient, emergency and inpatient attendances at government medical college hospitals, 2018

| Sl. No.      | Name of medical hospital                                 | OPD visit      |                | Emergency visit |                | Inpatient     |               |
|--------------|----------------------------------------------------------|----------------|----------------|-----------------|----------------|---------------|---------------|
|              |                                                          | Female         | Male           | Female          | Male           | Female        | Male          |
| 1            | Sher-E-Bangla Medical College Hospital, Barisal          | 216477         | 189923         | 58271           | 58992          | 60621         | 61270         |
| 2            | Chittagong Medical College Hospital, Chittagong          | 359442         | 368348         | 119401          | 135638         | 104835        | 117684        |
| 3            | Comilla Medical College Hospital, Comilla                | 164699         | 139150         | 32850           | 28055          | 33575         | 29722         |
| 4            | Dhaka Medical College Hospital, Dhaka                    | 527694         | 465505         | 221189          | 249591         | 81573         | 95015         |
| 5            | Shaheed Suhrawardy Medical College Hospital, Dhaka       | 656817         | 628199         | 152870          | 150027         | 37432         | 35281         |
| 6            | Sir Salimullah Medical College Hospital, Mitford, Dhaka  | 417027         | 295766         | 78197           | 73726          | 40446         | 28662         |
| 7            | Faridpur Medical College Hospital, Faridpur              | 127266         | 114422         | 33863           | 29290          | 33430         | 28957         |
| 8            | Shaheed Tajuddin Ahmad Medical College Hospital, Gazipur | 127620         | 129683         | 44517           | 45855          | 17835         | 13517         |
| 9            | Khulna Medical College Hospital, Khulna                  | 183071         | 105380         | 4896            | 5122           | 35342         | 34123         |
| 10           | Mymensingh Medical College Hospital, Mymensingh          | 654059         | 656964         | 92438           | 89946          | 111057        | 100359        |
| 11           | Shahid Ziaur Rahman Medical College Hospital, Bogra      | 259014         | 208492         | 35445           | 43906          | 38929         | 45560         |
| 12           | Rajshahi Medical College Hospital, Rajshahi              | 478313         | 328228         | 4529            | 7625           | 88001         | 87867         |
| 13           | M Abdur Rahim Medical College Hospital, Dinajpur         | 110475         | 109464         | 28258           | 28326          | 27650         | 27397         |
| 14           | Rangpur Medical College Hospital, Rangpur                | 190550         | 216807         | 53398           | 60022          | 57063         | 62764         |
| 15           | MAG Osmani Medical College Hospital, Sylhet              | 512200         | 417504         | 3911            | 6514           | 74972         | 85175         |
| <b>Total</b> |                                                          | <b>4984724</b> | <b>4373835</b> | <b>964033</b>   | <b>1012635</b> | <b>842761</b> | <b>853353</b> |

Note: Report of three facilities: Shahid Syed Nazrul Islam Medical College Hospital, Kishoreganj, Sheikh Hasina Medical College Hospital, Tangail and Satkhira 250-bed Medical College Hospital, Satkhira were not available;  
Source: DHIS2, MIS report, 2018



**Table 5.5.** Outpatient, emergency and inpatient attendances at district/general hospitals, 2018

| Sl. No. | Name of district/general hospital            | OPD Visit |        | Emergency visit |       | Admission |       |
|---------|----------------------------------------------|-----------|--------|-----------------|-------|-----------|-------|
|         |                                              | Female    | Male   | Female          | Male  | Female    | Male  |
| 1.1     | Barguna District Hospital, Barguna           | 48258     | 37429  | 14222           | 15627 | 9808      | 7617  |
| 1.2     | Barisal District Hospital, Barisal           | 44891     | 43335  | 13519           | 13180 | 5112      | 4866  |
| 1.3     | Bhola District Hospital, Bhola               | 75224     | 50348  | 6626            | 8090  | 17622     | 20409 |
| 1.4     | Jhalokati District Hospital, Jhalokati       | 68475     | 47282  | 22287           | 23576 | 7681      | 5359  |
| 1.5     | Patuakhali District Hospital, Patuakhali     | 93450     | 73327  | 21509           | 19119 | 23391     | 17994 |
| 1.6     | Pirojpur District Hospital, Pirojpur         | 64266     | 49133  | 21588           | 20549 | 11092     | 8453  |
| 2.1     | Bandarban District Hospital, Bandarban       | 32358     | 26058  | 1594            | 3071  | 2816      | 2938  |
| 2.2     | B.Barua District Hospital, B.Barua           | 139123    | 221679 | 37348           | 55814 | 18095     | 16668 |
| 2.3     | Chandpur District Hospital, Chandpur         | 120040    | 95566  | 28334           | 33053 | 20346     | 16982 |
| 2.4     | Chittagong General Hospital, Chittagong      | 112929    | 118582 | 6189            | 12422 | 6587      | 4397  |
| 2.5     | Cox's Bazar District Hospital, Cox's Bazar   | 144223    | 112234 | 25741           | 24598 | 36520     | 29041 |
| 2.6     | Feni District Hospital, Feni                 | 103639    | 92964  | 37293           | 36184 | 29376     | 21249 |
| 2.7     | Khagrachhari District Hospital, Khagrachhari | 56477     | 43432  | 2382            | 3039  | 7911      | 6644  |
| 2.8     | Comilla District Hospital, Comilla           | 59866     | 43005  | 5911            | 10897 | 3934      | 3787  |
| 2.9     | Lakshmipur District Hospital, Lakshmipur     | 133532    | 98971  | 6227            | 10653 | 15197     | 13211 |
| 2.10    | Noakhali District Hospital, Noakhali         | 84854     | 66370  | 21312           | 23750 | 18050     | 20202 |
| 2.11    | Rangamati District Hospital, Rangamati       | 30433     | 25960  | 2747            | 4693  | 7555      | 5952  |
| 3.1     | Faridpur General Hospital, Faridpur          | 79126     | 75765  | 14948           | 18952 | 9359      | 10399 |

**Table 5.5 Contd.**

Table Continued...

| Sl. No. | Name of district/general hospital                  | OPD Visit |        | Emergency visit |       | Admission |       |
|---------|----------------------------------------------------|-----------|--------|-----------------|-------|-----------|-------|
|         |                                                    | Female    | Male   | Female          | Male  | Female    | Male  |
| 3.2     | Gopalganj District Hospital, Gopalganj             | 102071    | 100681 | 13175           | 13137 | 19682     | 19445 |
| 3.3     | Kishoreganj District Hospital, Kishoreganj         | 152546    | 158048 | 57093           | 62631 | 35802     | 32745 |
| 3.4     | Madaripur District Hospital, Madaripur             | 52143     | 32827  | 15861           | 16351 | 14567     | 14280 |
| 3.5     | Manikganj District Hospital, Manikganj             | 176938    | 136307 | 8915            | 11680 | 19565     | 16822 |
| 3.6     | Munshiganj District Hospital, Munshiganj           | 108997    | 98242  | 24031           | 21203 | 8812      | 6551  |
| 3.7     | Narayanganj General (Victoria) Hospital (100 beds) | 227386    | 230808 | 76103           | 72528 | 11496     | 10884 |
| 3.8     | Narayanganj 300-bed Hospital, Narayanganj*         | 154447    | 135940 | 33505           | 31920 | 5203      | 3944  |
| 3.9     | Narsingdi District Hospital, Narsingdi             | 81159     | 77429  | 51253           | 52358 | 5906      | 4271  |
| 3.10    | Narsingdi (100-bed) Zila Hospital, Narsingdi*      | 114150    | 99805  | 22967           | 22649 | 5953      | 5685  |
| 3.11    | Rajbari District Hospital, Rajbari                 | 80735     | 61381  | 24854           | 21947 | 15906     | 10185 |
| 3.12    | Shariatpur District Hospital, Shariatpur           | 67346     | 65151  | 14101           | 12816 | 10667     | 8840  |
| 3.13    | Tangail District Hospital, Tangail                 | 192094    | 139404 | 14919           | 20726 | 36174     | 44778 |
| 4.1     | Bagerhat District Hospital, Bagerhat               | 75300     | 48549  | 3128            | 4834  | 9663      | 7322  |
| 4.2     | Chuadanga District Hospital, Chuadanga             | 80779     | 108229 | 39802           | 39773 | 22206     | 18555 |
| 4.3     | Jessore District Hospital, Jessore                 | 290858    | 174035 | 10862           | 17456 | 36644     | 30090 |
| 4.4     | Jhenaidah District Hospital, Jhenaidah             | 200073    | 131976 | 5545            | 15711 | 20988     | 16241 |
| 4.5     | Khulna District Hospital, Khulna                   | 107160    | 60304  | 11110           | 14840 | 6202      | 4679  |
| 4.6     | Kushtia District Hospital, Kushtia                 | 225939    | 195495 | 12261           | 16023 | 33810     | 32874 |

Table 5.5 Contd.

Table Continued...

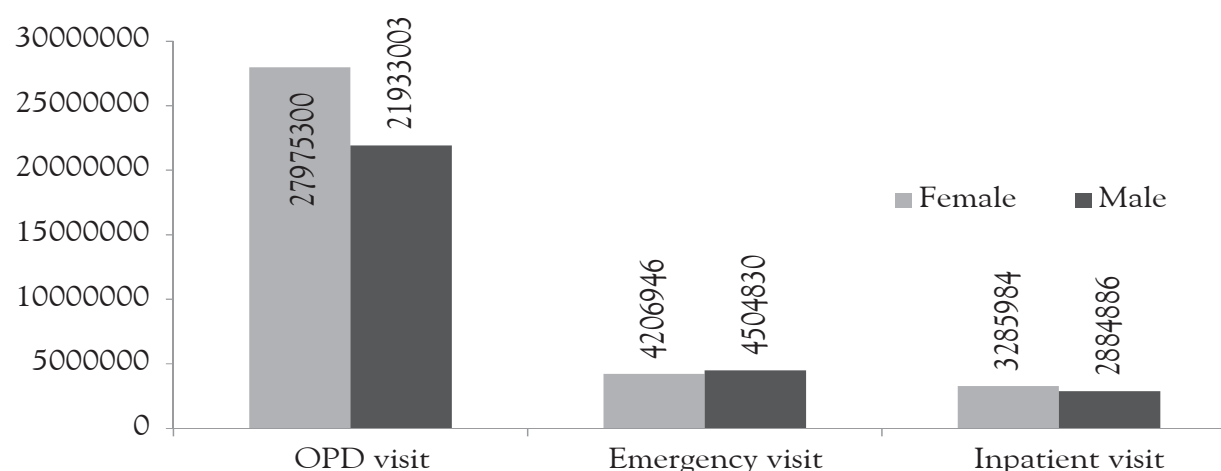
| S.N  | Name of district/general hospital          | OPD Visit |        | Emergency visit |       | Admission |       |
|------|--------------------------------------------|-----------|--------|-----------------|-------|-----------|-------|
|      |                                            | Female    | Male   | Female          | Male  | Female    | Male  |
| 4.7  | Magura District Hospital, Magura           | 101564    | 75488  | 7972            | 8485  | 20765     | 18892 |
| 4.8  | Meherpur District Hospital, Meherpur       | 106786    | 95943  | 26263           | 27022 | 19902     | 19233 |
| 4.9  | Narail District Hospital, Narail           | 56595     | 32410  | 9569            | 14114 | 11815     | 8956  |
| 4.10 | Satkhira District Hospital, Satkhira       | 77923     | 69495  | 4167            | 4470  | 13259     | 10412 |
| 5.1  | Jamalpur District Hospital, Jamalpur       | 161266    | 119182 | 34671           | 29370 | 28442     | 24724 |
| 5.2  | Netrakona District Hospital, Netrakona     | 61448     | 44522  | 47630           | 46163 | 22089     | 13341 |
| 5.3  | Sherpur District Hospital, Sherpur         | 109115    | 89826  | 13145           | 17263 | 18716     | 17625 |
| 6.1  | Mohammad Ali District Hospital, Bogra      | 146023    | 118409 | 16286           | 16336 | 9892      | 8248  |
| 6.2  | Nowabganj District Hospital, Nowabganj     | 150944    | 150271 | 13241           | 14696 | 15746     | 13338 |
| 6.3  | Joypurhat District Hospital, Joypurhat     | 108683    | 94027  | 3160            | 3120  | 17339     | 17061 |
| 6.4  | Naogaon District Hospital, Naogaon         | 209042    | 104843 | 30406           | 29586 | 15558     | 13431 |
| 6.5  | Natore District Hospital, Natore           | 116459    | 112200 | 15321           | 15942 | 16003     | 14866 |
| 6.6  | Pabna District Hospital, Pabna             | 100993    | 126734 | 20061           | 19542 | 43489     | 44922 |
| 6.7  | Sirajganj General Hospital, Sirajganj      | 128246    | 99770  | 24162           | 23092 | 19985     | 19182 |
| 7.1  | Dinajpur District Hospital, Dinajpur       | 74208     | 72869  | 7519            | 11460 | 8393      | 8158  |
| 7.2  | Gaibandha District Hospital, Gaibandha     | 95135     | 79504  | 32572           | 35614 | 15527     | 13602 |
| 7.3  | Kurigram District Hospital, Kurigram       | 102364    | 87414  | 24289           | 26502 | 17858     | 16031 |
| 7.4  | Lalmonirhat District Hospital, Lalmonirhat | 99588     | 56145  | 11670           | 11624 | 7048      | 6989  |

Table 5.5 Contd.

Table Continued...

| S.N   | Name of district/general hospital           | OPD Visit |         | Emergency visit |         | Admission |        |
|-------|---------------------------------------------|-----------|---------|-----------------|---------|-----------|--------|
|       |                                             | Female    | Male    | Female          | Male    | Female    | Male   |
| 7.5   | Nilphamari District Hospital, Nilphamari    |           |         |                 |         |           |        |
| 7.6   | Panchagarh District Hospital, Panchagarh    | 63285     | 39616   | 7537            | 7401    | 11382     | 7950   |
| 7.7   | Thakurgaon District Hospital, Thakurgaon    | 111363    | 84531   | 12152           | 17831   | 27325     | 22096  |
| 8.1   | Habiganj District Hospital, Habiganj        | 48178     | 41614   | 9979            | 8363    | 21022     | 17615  |
| 8.2   | Maulviazar District Hospital, Maulviazar    | 116846    | 110715  | 42321           | 40390   | 22136     | 18577  |
| 8.3   | Sunamganj District Hospital, Sunamganj      | 38478     | 34887   | 37529           | 35300   | 13287     | 12790  |
| 8.4   | Shahid Shamsuddin District Hospital, Sylhet | 75909     | 53816   | 1457            | 1591    | 1792      | 1298   |
| Total |                                             | 6527421   | 5438475 | 1212895         | 1295299 | 1009478   | 888405 |

Source: DHIS2, MIS report, 2018; \*These two hospitals are added to this list, albeit not characterized as other district or general hospitals

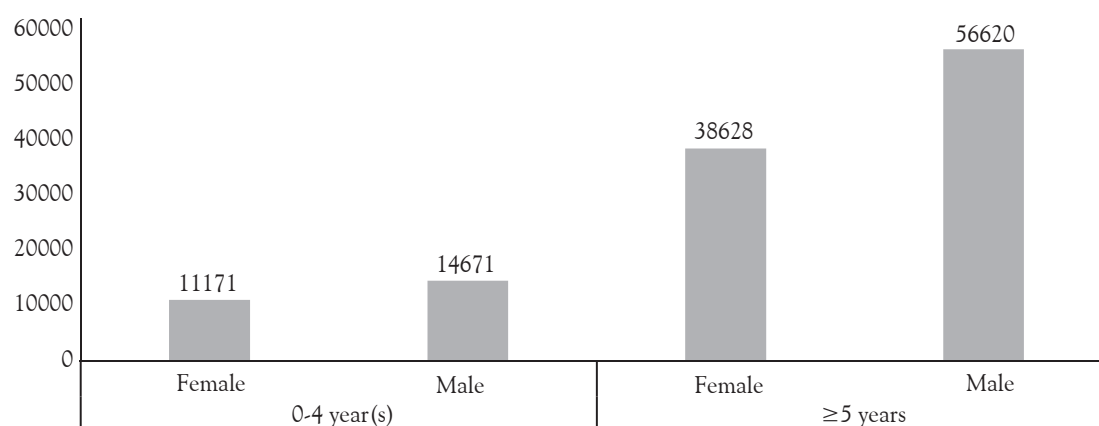


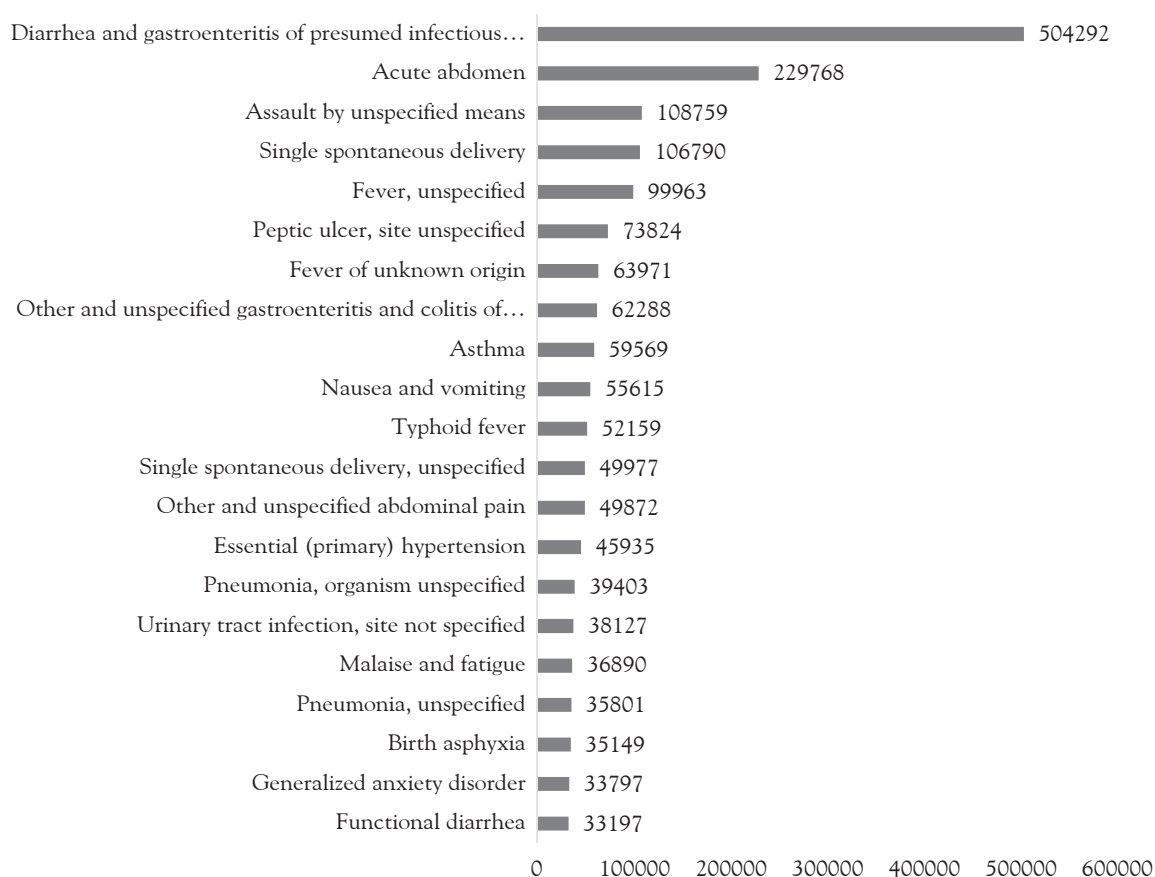
**Figure 5.5.** Total number of outpatient, emergency and inpatient attendance in government facility by sex in 2018

- ◆ At the district/general hospitals, it ranged from 1.6 days at Habiganj District hospital to 10.5 at Shahid Shamsuddin District Hospital, Sylhet (Table 5.9).
- ◆ At the upazila health complexes, the figures ranged from 0.84 days at Dakhin Surma Upazila Health Complex, Sylhet to 7 days at Shajahanpur Upazila Health Complex (Figure 5.9 and 5.10).

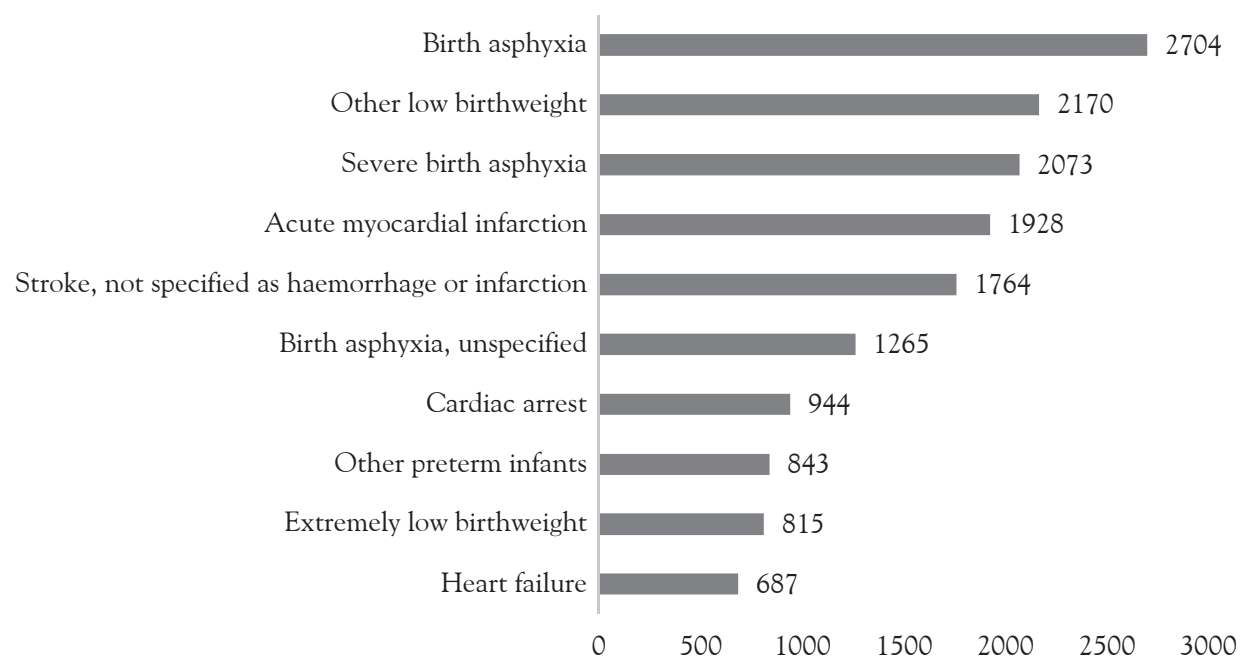
**Table 5.6.** Number of deaths in different health facilities by age and sex, 2018

| Level of care | Type of facilities                             | Number of facilities reporting deaths | Total number of deaths | 0-4 year(s)  |              | ≥ 5 years    |              |
|---------------|------------------------------------------------|---------------------------------------|------------------------|--------------|--------------|--------------|--------------|
|               |                                                |                                       |                        | Female       | Male         | Female       | Male         |
| Primary       | 10-bed hospitals                               | 14                                    | 2                      | 0            | 0            | 2            | 0            |
|               | 20-bed hospitals                               | 26                                    | 27                     | 1            | 4            | 13           | 9            |
|               | 31-bed hospitals                               | 4                                     | 0                      | 0            | 0            | 0            | 0            |
|               | 100-bed hospital, Saidpur, Nilphamari          | 1                                     | 65                     | 2            | 24           | 14           | 25           |
|               | Upazila health complexes                       | 424                                   | 6286                   | 421          | 506          | 2208         | 3151         |
| Secondary     | District hospital/general hospitals            | 62                                    | 25387                  | 3023         | 3878         | 7694         | 10792        |
| Tertiary      | 500-bed hospitals                              | 2                                     | 631                    | 19           | 16           | 231          | 365          |
|               | Medical college hospitals                      | 15                                    | 80290                  | 7564         | 9947         | 25882        | 36897        |
|               | Other tertiary hospitals (national institutes) | 13                                    | 8402                   | 141          | 296          | 2584         | 5381         |
| <b>Total</b>  |                                                | <b>560</b>                            | <b>121090</b>          | <b>11171</b> | <b>14671</b> | <b>38628</b> | <b>56620</b> |

**Figure 5.6.** Number of deaths according to age and sex reported from all government health facilities, 2018



**Figure 5.7.** Top 20 diseases requiring admission in hospital, 2018



**Figure 5.8.** Top 10 causes of death, 2018



**Table 5.7.** Average length of stay (ALS) at government. specialized hospitals, 2018

| Name of organization                                                          | ALS         |
|-------------------------------------------------------------------------------|-------------|
| National Institute of Diseases of the Chest and Hospital, Dhaka               | 13.9        |
| National Institute of ENT, Dhaka                                              | 13.1        |
| National Institute of Mental Health and Research, Dhaka                       | 12.6        |
| National Institute of Neurosciences (NINS) and Hospital, Dhaka                | 11.8        |
| National Institute of Cancer Research and Hospital, Dhaka                     | 10.2        |
| National Institute of Kidney Diseases & Urology, Dhaka                        | 9.3         |
| National Institute of Traumatology & Orthopedic Rehabilitation (NITOR), Dhaka | 9.0         |
| Kurmitola General Hospital ( 500-bed ), Dhaka                                 | 5.9         |
| Mugda 500-bed General Hospital, Dhaka                                         | 4.9         |
| National Institute of Cardiovascular Diseases, Dhaka                          | 4.1         |
| National Institute of Ophthalmology (NIO), Dhaka                              | 3.8         |
| Institute of Child & Mother Health (ICMH), Matuail, Narayanganj               | 3.8         |
| Dhaka Shishu Hospital, Dhaka                                                  | 2.3         |
| <b>Average</b>                                                                | <b>8.05</b> |

**Table 5.8.** Average length of stay (ALS) at government medical college hospitals, 2018

| Name of facility                                         | ALS        |
|----------------------------------------------------------|------------|
| Dhaka Medical College Hospital, Dhaka                    | 6.6        |
| Shaheed Suhrawardy Medical College Hospital, Dhaka       | 6.6        |
| Chittagong Medical College Hospital, Chittagong          | 5.4        |
| Khulna Medical College Hospital, Khulna                  | 5.3        |
| Rangpur Medical College Hospital, Rangpur                | 5.3        |
| Sir Salimullah Medical College Hospital, Mitford, Dhaka  | 5.0        |
| Shahid Ziaur Rahman Medical College Hospital, Bogra      | 4.9        |
| Comilla Medical College Hospital, Comilla                | 4.4        |
| Mymensingh Medical College Hospital, Mymensingh          | 4.4        |
| Sher-E-Bangla Medical College Hospital, Barisal          | 4.2        |
| M Abdur Rahim Medical College Hospital, Dinajpur         | 4.1        |
| Faridpur Medical College Hospital, Faridpur              | 3.8        |
| MAG Osmani Medical College Hospital, Sylhet              | 3.7        |
| Shaheed Tajuddin Ahmad Medical College Hospital, Gazipur | 3.5        |
| Rajshahi Medical College Hospital, Rajshahi              | 3.3        |
| <b>Average</b>                                           | <b>4.7</b> |

**Table 5.9.** Average length of stay (ALS) at government district/general hospital, 2018

| Name of facility                             | ALS  |
|----------------------------------------------|------|
| Shahid Shamsuddin District Hospital, Sylhet  | 10.5 |
| Narayanganj 300-bed Hospital, Narayanganj    | 9.9  |
| Barisal District Hospital, Barisal           | 9.3  |
| B.Baria District Hospital, B.Baria           | 4.9  |
| Khulna District Hospital, Khulna             | 4.8  |
| Dinajpur District Hospital, Dinajpur         | 4.4  |
| Bandarban District Hospital, Bandarban       | 4.2  |
| Mohammad Ali District Hospital, Bogra        | 4.2  |
| Narsingdi (100-bed) Zila Hospital, Narsingdi | 4.1  |
| Patuakhali District Hospital, Patuakhali     | 4.1  |
| Satkhira District Hospital, Satkhira         | 3.5  |
| Kushtia District Hospital, Kushtia           | 3.4  |
| Chittagong General Hospital, Chittagong      | 3.3  |
| Feni District Hospital, Feni                 | 3.3  |
| Cox's Bazar District Hospital, Cox's Bazar   | 3.2  |
| Jhalokati District Hospital, Jhalokati       | 3.1  |
| Joypurhat District Hospital, Joypurhat       | 3.1  |
| Narsingdi District Hospital, Narsingdi       | 3.1  |
| Noakhali District Hospital, Noakhali         | 3.1  |

**Table 5.9 Contd.****Table Continued...**

| Name of facility                             | ALS |
|----------------------------------------------|-----|
| Chandpur District Hospital, Chandpur         | 3.0 |
| Gopalganj District Hospital, Gopalganj       | 2.9 |
| Sirajganj General Hospital, Sirajganj        | 2.9 |
| Bagerhat District Hospital, Bagerhat         | 2.8 |
| Manikganj District Hospital, Manikganj       | 2.8 |
| Jamalpur District Hospital, Jamalpur         | 2.7 |
| Khagrachhari District Hospital, Khagrachhari | 2.7 |
| Comilla District Hospital, Comilla           | 2.7 |
| Lakshmipur District Hospital, Lakshmipur     | 2.7 |
| Lalmonirhat District Hospital, Lalmonirhat   | 2.7 |
| Magura District Hospital, Magura             | 2.7 |
| Nowabganj District Hospital, Nowabganj       | 2.7 |
| Sunamganj District Hospital, Sunamganj       | 2.7 |
| Barguna District Hospital, Barguna           | 2.6 |
| Munshiganj District Hospital, Munshiganj     | 2.6 |
| Narail District Hospital, Narail             | 2.6 |
| Gaibandha District Hospital, Gaibandha       | 2.5 |
| Jhenaidah District Hospital, Jhenaidah       | 2.5 |
| Kurigram District Hospital, Kurigram         | 2.5 |

**Table 5.9 Contd.**

Table Continued...

| Name of facility                                            | ALS |
|-------------------------------------------------------------|-----|
| Meherpur District Hospital, Meherpur                        | 2.5 |
| Thakurgaon District Hospital, Thakurgaon                    | 2.5 |
| Jessore District Hospital, Jessore                          | 2.4 |
| Pabna District Hospital, Pabna                              | 2.4 |
| Pirojpur District Hospital, Pirojpur                        | 2.4 |
| Sherpur District Hospital, Sherpur                          | 2.4 |
| Tangail District Hospital, Tangail                          | 2.4 |
| Panchagarh District Hospital, Panchagarh                    | 2.3 |
| Rangamati District Hospital, Rangamati                      | 2.3 |
| Shariatpur District Hospital, Shariatpur                    | 2.3 |
| Chuadanga District Hospital, Chuadanga                      | 2.2 |
| Faridpur General Hospital, Faridpur                         | 2.2 |
| Kishoreganj District Hospital, Kishoreganj                  | 2.2 |
| Maulvibazar District Hospital, Maulvibazar                  | 2.2 |
| Netrakona District Hospital, Netrakona                      | 2.2 |
| Nilphamari District Hospital, Nilphamari                    | 2.2 |
| Shaheed Ahsan Ullah Master General Hospital, Tongi, Gazipur | 2.1 |
| Bhola District Hospital, Bhola                              | 2.0 |
| Natore District Hospital, Natore                            | 2.0 |
| Rajbari District Hospital, Rajbari                          | 2.0 |

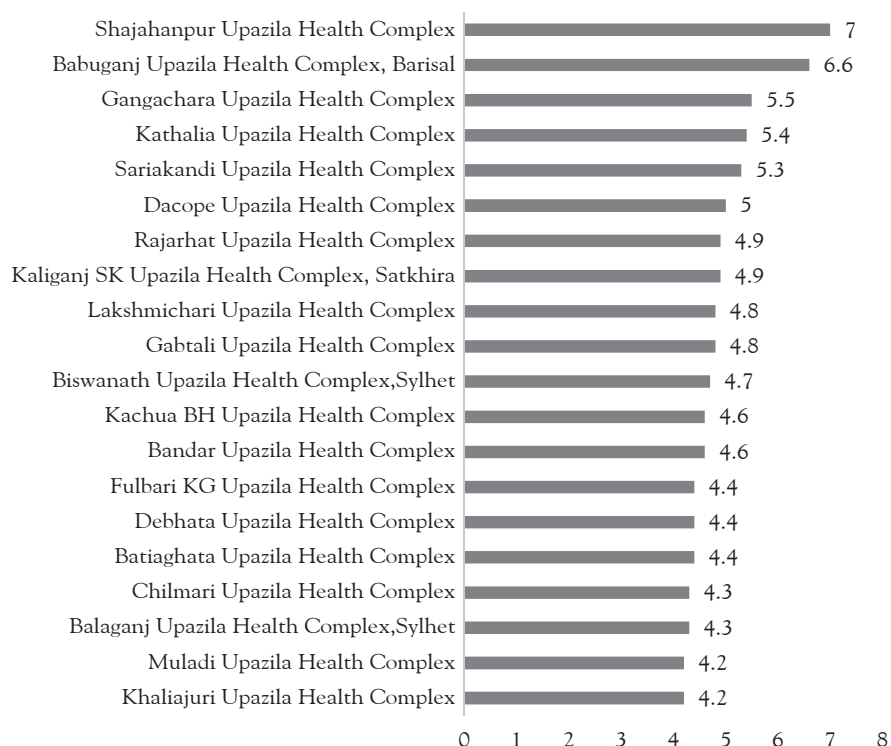
Table 5.9 Contd.

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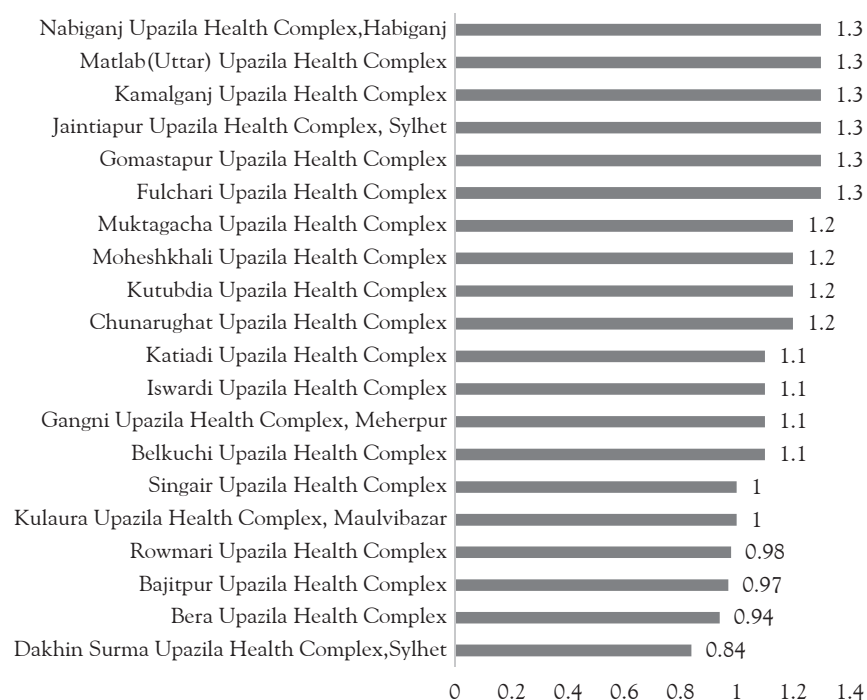
| Name of facility                                  | ALS         |
|---------------------------------------------------|-------------|
| Madaripur District Hospital, Madaripur            | 1.9         |
| Narayanganj General (Victoria) Hospital (100-bed) | 1.9         |
| Naogaon District Hospital, Naogaon                | 1.8         |
| Habiganj District Hospital, Habiganj              | 1.6         |
| <b>Average</b>                                    | <b>3.17</b> |

### Bed-occupancy Rate, 2018

- ◆ At specialized hospitals, the bed-occupancy rates (BOR) ranged from 37.5% in Bangladesh Institute of Tropical and Infectious Diseases, Fouzdarhat to 189.7% in the National Institute of Cardiovascular Diseases, Dhaka (Table 5.10)
- ◆ At medical college hospitals, the BOR ranged from 105.8% in Shaheed Tajuddin Ahmad Medical College Hospital, Gazipur to 259.1 % in Mymensingh Medical College Hospital, with an average bed-occupancy rate of 162.2% (Table 5.11).
- ◆ At the district/general hospitals, the figures ranged from 41.7% at Munshiganj District Hospital, Munshiganj to 334.8% at Thakurgaon District Hospital, with an average bed-occupancy rate of 140.8 (Table 5.12).
- ◆ The upazila health complexes (20 with the highest BOR and 20 with the lowest BOR), with average bed-occupancy rate of 80.69 are shown in Figure 5.11 and 5.12 respectively.
- ◆ All data are from DHIS2. Data of some hospitals could not be gathered from DHIS2 and, thus, not included here.



**Figure 5.9.** Average length of stay by inpatients in 20 upazila health complexes with the highest values, 2018



**Figure 5.10.** Average length of stay by inpatients in 20 upazila health complexes with the lowest values, 2018

**Table 5.10.** Bed-occupancy rate (BOR) in the government specialized hospitals, 2018

| Name of facility                                                              | BOR (%) | No. of sanctioned beds |
|-------------------------------------------------------------------------------|---------|------------------------|
| National Institute of Cardiovascular Diseases, Dhaka                          | 189.7   | 414                    |
| Kurmitola General Hospital (500-bed), Dhaka                                   | 110.8   | 500                    |
| National Institute of Traumatology & Orthopedic Rehabilitation (NITOR), Dhaka | 107.5   | 500                    |
| National Institute of Neurosciences and Hospital (NINS), Dhaka                | 100     | 350                    |
| National Institute of Kidney Diseases & Urology, Dhaka                        | 98.8    | 150                    |
| National Institute of Diseases of the Chest and Hospital, Dhaka               | 91.5    | 670                    |
| National Institute of Cancer Research and Hospital, Dhaka                     | 90.5    | 300                    |
| National Institute of ENT, Dhaka                                              | 88.4    | 100                    |
| Mugda 500-bed General Hospital, Dhaka                                         | 82.8    | 500                    |
| Institute of Child & Mother Health (ICMH), Matuail                            | 81.1    | 200                    |
| National Institute of Mental Health and Research, Dhaka                       | 79.6    | 200                    |
| National Institute of Ophthalmology (NIO), Dhaka                              | 65.5    | 250                    |
| Bangladesh Institute of Tropical and Infectious Diseases, Fouzdarhat          | 37.5    | 100                    |

**Table 5.11.** Bed-occupancy rate (BOR) in the government medical college hospitals, 2018

| Name of facility                                         | BOR (%) | No. of sanctioned beds |
|----------------------------------------------------------|---------|------------------------|
| Mymensingh Medical College Hospital, Mymensingh          | 259.1   | 1000                   |
| Shahid Ziaur Rahman Medical College Hospital, Bogra      | 215.9   | 500                    |
| Chittagong Medical College Hospital, Chittagong          | 208.3   | 1313                   |
| MAG Osmani Medical College Hospital, Sylhet              | 199.2   | 900                    |
| Rangpur Medical College Hospital, Rangpur                | 173.4   | 1000                   |
| Khulna Medical College Hospital, Khulna                  | 164.7   | 500                    |
| Comilla Medical College Hospital, Comilla                | 162.0   | 500                    |
| Sher-E-Bangla Medical College Hospital, Barisal          | 155.8   | 1000                   |
| Rajshahi Medical College Hospital, Rajshahi              | 151.5   | 1200                   |
| Shaheed Suhrawardy Medical College Hospital, Dhaka       | 149.1   | 850                    |
| Dhaka Medical College Hospital, Dhaka                    | 137.7   | 2600                   |
| Sir Salimullah Medical College Hospital, Mitford, Dhaka  | 129.1   | 900                    |
| Faridpur Medical College Hospital, Faridpur              | 124.6   | 500                    |
| M Abdur Rahim Medical College Hospital, Dinajpur         | 121.1   | 500                    |
| Shaheed Tajuddin Ahmad Medical College Hospital, Gazipur | 81.3    | 500                    |

**Table 5.12.** Bed-occupancy rate (BOR) in the government district/general hospitals, 2018

| Name of facility                           | BOR (%) | No. of sanctioned beds |
|--------------------------------------------|---------|------------------------|
| Thakurgaon District Hospital, Thakurgaon   | 334.8   | 100                    |
| Jhenaidah District Hospital, Jhenaidah     | 250.9   | 100                    |
| Chuadanga District Hospital, Chuadanga     | 247.8   | 100                    |
| Kushtia District Hospital, Kushtia         | 237.7   | 250                    |
| Sherpur District Hospital, Sherpur         | 237.2   | 100                    |
| Nowabganj District Hospital, Nowabganj     | 224.5   | 100                    |
| Netrakona District Hospital, Netrakona     | 217.8   | 100                    |
| Cox's Bazar District Hospital, Cox's Bazar | 212.7   | 250                    |
| Nilphamari District Hospital, Nilphamari   | 212.6   | 100                    |
| Tangail District Hospital, Tangail         | 210.1   | 250                    |
| Lakshmipur District Hospital, Lakshmipur   | 209.1   | 100                    |
| Bhola District Hospital, Bhola             | 205.1   | 100                    |
| Noakhali District Hospital, Noakhali       | 204.6   | 250                    |
| Satkhira District Hospital, Satkhira       | 200.4   | 100                    |
| Patuakhali District Hospital, Patuakhali   | 181.8   | 250                    |

**Table 5.12 Contd.****Table Continued...**

| Name of facility                             | BOR (%) | No. of sanctioned beds |
|----------------------------------------------|---------|------------------------|
| Jessore District Hospital, Jessore           | 181.1   | 250                    |
| Kishoreganj District Hospital, Kishoreganj   | 176.0   | 250                    |
| Natore District Hospital, Natore             | 171.6   | 100                    |
| Joypurhat District Hospital, Joypurhat       | 155.2   | 150                    |
| Pabna District Hospital, Pabna               | 154.1   | 250                    |
| Narail District Hospital, Narail             | 152.4   | 100                    |
| Feni District Hospital, Feni                 | 151.6   | 250                    |
| Jamalpur District Hospital, Jamalpur         | 151.4   | 250                    |
| Madaripur District Hospital, Madaripur       | 146.8   | 100                    |
| Naogaon District Hospital, Naogaon           | 143.8   | 100                    |
| Rajbari District Hospital, Rajbari           | 143.5   | 100                    |
| B.Barua District Hospital, B.Barua           | 137.3   | 250                    |
| Narsingdi (100-bed) Zila Hospital, Narsingdi | 128.9   | 100                    |
| Bagerhat District Hospital, Bagerhat         | 128.6   | 100                    |
| Gopalganj District Hospital, Gopalganj       | 123.9   | 250                    |

**Table 5.12 Contd.**

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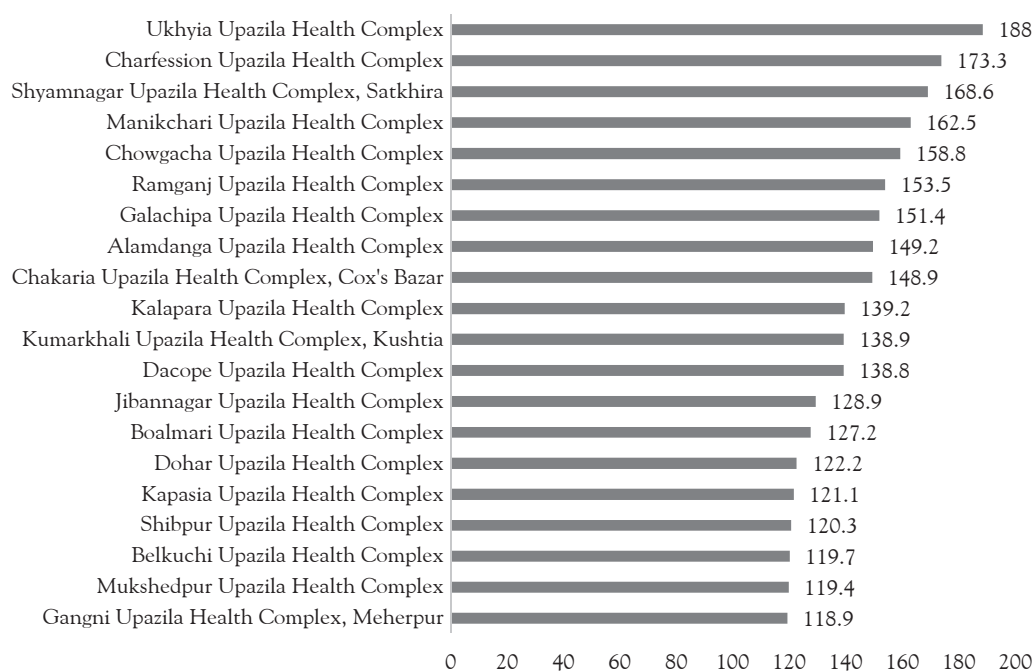
| Name of facility                                     | BOR (%) | No. of sanctioned beds |
|------------------------------------------------------|---------|------------------------|
| Panchagarh District Hospital, Panchagarh             | 119.3   | 100                    |
| Chandpur District Hospital, Chandpur                 | 118.4   | 250                    |
| Manikganj District Hospital, Manikganj               | 117.9   | 250                    |
| Barguna District Hospital, Barguna                   | 117.7   | 100                    |
| Magura District Hospital, Magura                     | 116.5   | 250                    |
| Sirajganj General Hospital, Sirajganj                | 116.4   | 250                    |
| Narayanganj General (Victoria) Hospital, Narayanganj | 115.7   | 100                    |
| Pirojpur District Hospital, Pirojpur                 | 109.8   | 100                    |
| Faridpur General Hospital, Faridpur                  | 108.4   | 100                    |
| Shariatpur District Hospital, Shariatpur             | 107.6   | 100                    |
| Jhalokati District Hospital, Jhalokati               | 106.1   | 100                    |
| Meherpur District Hospital, Meherpur                 | 105.5   | 250                    |
| Comilla District Hospital, Comilla                   | 105.1   | 100                    |
| Lalmonirhat District Hospital, Lalmonirhat           | 104.6   | 100                    |
| Barisal District Hospital, Barisal                   | 102.7   | 100                    |

Table 5.12 Contd.

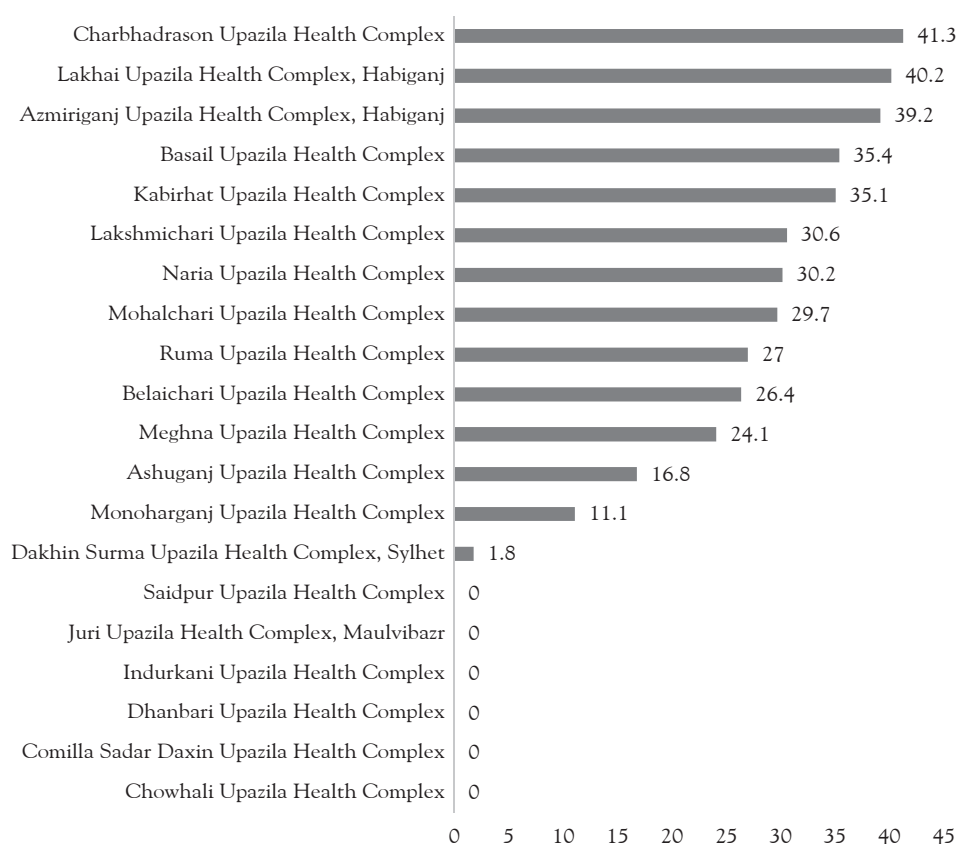
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| Name of facility                             | BOR (%)      | No. of sanctioned beds |
|----------------------------------------------|--------------|------------------------|
| Gaibandha District Hospital, Gaibandha       | 101.0        | 200                    |
| Maulvibazar District Hospital, Maulvibazar   | 99.1         | 250                    |
| Khagrachhari District Hospital, Khagrachhari | 95.8         | 100                    |
| Habiganj District Hospital, Habiganj         | 92.8         | 250                    |
| Rangamati District Hospital, Rangamati       | 92.3         | 100                    |
| Kurigram District Hospital, Kurigram         | 90.7         | 250                    |
| Sunamganj District Hospital, Sunamganj       | 86.5         | 250                    |
| Narayanganj 300-bed Hospital, Narayanganj    | 82.8         | 300                    |
| Shahid Shamsuddin District Hospital, Sylhet  | 81.9         | 100                    |
| Mohammad Ali District Hospital, Bogra        | 80.6         | 250                    |
| Dinajpur District Hospital, Dinajpur         | 80.2         | 250                    |
| Narsingdi District Hospital, Narsingdi       | 74.1         | 100                    |
| Chittagong General Hospital, Chittagong      | 71.4         | 250                    |
| Bandarban District Hospital, Bandarban       | 62.7         | 100                    |
| Khulna District Hospital, Khulna             | 52.9         | 250                    |
| Munshiganj District Hospital, Munshiganj     | 41.7         | 250                    |
| <b>Average</b>                               | <b>140.8</b> |                        |





**Figure 5.11.** Twenty upazila health complexes with the highest BOR, 2018



**Figure 5.12.** Twenty upazila health complexes with the lowest BOR, 2018

## Surgeries

Total number of major surgeries performed during the reporting year was 336,844, and total number of minor surgeries was 1,549,179, including those in all government health facilities.

**Table 5.13.** Number of major and minor surgeries performed at the government specialized hospitals, 2018

| Name of facility                                                              | Major surgery | Minor surgery |
|-------------------------------------------------------------------------------|---------------|---------------|
| National Institute of Neurosciences and Hospital (NINS), Dhaka                | 1,896         | 163           |
| National Institute of ENT, Dhaka                                              | 2,130         | 299           |
| Institute of Child & Mother Health (ICMH) Matuail, Narayanganj                | 3,954         | 825           |
| National Institute of Ophthalmology (NIO), Dhaka                              | 13,970        | 1,476         |
| National Institute of Cancer Research and Hospital, Dhaka                     | 1,021         | 1,536         |
| National Institute of Kidney Diseases & Urology, Dhaka                        | 960           | 1,587         |
| National Institute of Diseases of the Chest and Hospital, Dhaka               | 980           | 2,407         |
| Mugda 500-bed General Hospital, Dhaka                                         | 4,025         | 6,703         |
| Kurmitola General Hospital (500-bed ), Dhaka                                  | 4,127         | 8,339         |
| National Institute of Traumatology & Orthopedic Rehabilitation (NITOR), Dhaka | 15,817        | 14,285        |
| <b>Total</b>                                                                  | <b>48,880</b> | <b>37,620</b> |

**Table 5.14.** Number of major and minor surgeries performed at the government medical college hospitals, 2018

| Name of facility                                         | Major surgery  | Minor surgery  |
|----------------------------------------------------------|----------------|----------------|
| Khulna Medical College Hospital, Khulna                  | 4,638          | 2,940          |
| Shaheed Tajuddin Ahmad Medical College Hospital, Gazipur | 1,641          | 3,248          |
| Faridpur Medical College Hospital, Faridpur              | 6,458          | 3,920          |
| Comilla Medical College Hospital, Comilla                | 10,048         | 9,718          |
| M Abdur Rahim Medical College Hospital, Dinajpur         | 3,498          | 11,026         |
| Sir Salimullah Medical College Hospital, Mitford, Dhaka  | 11,995         | 12,103         |
| Sher-E-Bangla Medical College Hospital, Barisal          | 10,473         | 12,326         |
| Shaheed Suhrawardy Medical College Hospital, Dhaka       | 21,187         | 13,472         |
| Shahid Ziaur Rahman Medical College Hospital, Bogra      | 7,259          | 15,231         |
| Rajshahi Medical College Hospital, Rajshahi              | 17,227         | 22,453         |
| MAG Osmani Medical College Hospital, Sylhet              | 22,798         | 23,170         |
| Chittagong Medical College Hospital, Chittagong          | 15,836         | 23,859         |
| Rangpur Medical College Hospital, Rangpur                | 7,040          | 30,700         |
| Dhaka Medical College Hospital, Dhaka                    | 30,642         | 43,783         |
| Mymensingh Medical College Hospital, Mymensingh          | 17,881         | 81,034         |
| <b>Total</b>                                             | <b>188,621</b> | <b>308,983</b> |

**Table 5.15.** Number of major and minor surgeries performed at district/general hospitals, 2018

| Name of hospital                            | Major surgery | Minor surgery |
|---------------------------------------------|---------------|---------------|
| Rangamati District Hospital, Rangamati      | 494           | 624           |
| Comilla District Hospital, Comilla          | 800           | 951           |
| Chittagong General Hospital, Chittagong     | 2,264         | 1,090         |
| Habiganj District Hospital, Habiganj        | 552           | 1,165         |
| Natore District Hospital, Natore            | 869           | 1,287         |
| Manikganj District Hospital, Manikganj      | 2,896         | 1,497         |
| Bandarban District Hospital, Bandarban      | 61            | 1,609         |
| Nowabganj District Hospital, Nowabganj      | 194           | 1,615         |
| Chandpur District Hospital, Chandpur        | 1,169         | 1,696         |
| Jhalokati District Hospital, Jhalokati      | 642           | 1,905         |
| Shahid Shamsuddin District Hospital, Sylhet | 1,525         | 1,958         |
| Rajbari District Hospital, Rajbari          | 1,161         | 1,977         |
| Dinajpur District Hospital, Dinajpur        | 1,354         | 2,265         |
| Bhola District Hospital, Bhola              | 492           | 2,717         |
| Jamalpur District Hospital, Jamalpur        | 2,302         | 2,828         |

Table 5.15 Contd.

**Table Continued...**

| Name of hospital                                     | Major surgery | Minor surgery |
|------------------------------------------------------|---------------|---------------|
| Munshiganj District Hospital, Munshiganj             | 807           | 2,925         |
| Pirojpur District Hospital, Pirojpur                 | 160           | 3,062         |
| Khagrachhari District Hospital, Khagrachhari         | 80            | 3,086         |
| Cox's Bazar District Hospital, Cox's Bazar           | 1,390         | 3,402         |
| Gaibandha District Hospital, Gaibandha               | 439           | 3,529         |
| Gopalganj District Hospital, Gopalganj               | 2,071         | 3,531         |
| Lalmonirhat District Hospital, Lalmonirhat           | 397           | 4,153         |
| Chuadanga District Hospital, Chuadanga               | 1,719         | 4,208         |
| Narsingdi District Hospital, Narsingdi               | 501           | 4,260         |
| Barguna District Hospital, Barguna                   | 69            | 4,303         |
| Narsingdi (100-bed) Zila Hospital, Narsingdi         | 650           | 4,401         |
| Panchagarh District Hospital, Panchagarh             | 843           | 4,912         |
| Satkhira District Hospital, Satkhira                 | 1,001         | 5,017         |
| Shariatpur District Hospital, Shariatpur             | 872           | 5,308         |
| Mohammad Ali District Hospital, Bogra                | 2,247         | 5,570         |
| Narayanganj General (Victoria) Hospital, Narayanganj | 1,381         | 5,573         |

Table 5.15 Contd.

Table Continued...

| Name of hospital                         | Major surgery | Minor surgery |
|------------------------------------------|---------------|---------------|
| Sirajganj General Hospital, Sirajganj    | 703           | 5,598         |
| Thakurgaon District Hospital, Thakurgaon | 1,078         | 5,751         |
| Sherpur District Hospital, Sherpur       | 1,587         | 6,084         |
| Jhenaidah District Hospital, Jhenaidah   | 1,388         | 6,224         |
| B.Baria District Hospital, B.Baria       | 2,023         | 6,254         |
| Kushtia District Hospital, Kushtia       | 4,559         | 6,472         |
| Barisal District Hospital, Barisal       | 455           | 6,563         |
| Feni District Hospital, Feni             | 1,391         | 6,984         |
| Noakhali District Hospital, Noakhali     | 3,338         | 7,209         |
| Khulna District Hospital, Khulna         | 860           | 7,324         |
| Kurigram District Hospital, Kurigram     | 2,587         | 8,166         |
| Lakshmipur District Hospital, Lakshmipur | 1,390         | 8,360         |
| Joypurhat District Hospital, Joypurhat   | 1,990         | 8,446         |
| Bagerhat District Hospital, Bagerhat     | 143           | 8,542         |
| Meherpur District Hospital, Meherpur     | 518           | 8,560         |

Table 5.15 Contd.

Table Continued...

| Name of hospital                           | Major surgery | Minor surgery  |
|--------------------------------------------|---------------|----------------|
| Maulvibazar District Hospital, Maulvibazar | 1,925         | 9,154          |
| Faridpur General Hospital, Faridpur        | 1,092         | 9,819          |
| Patuakhali District Hospital, Patuakhali   | 3,904         | 9,963          |
| Nilphamari District Hospital, Nilphamari   | 1,040         | 10,744         |
| Magura District Hospital, Magura           | 2,100         | 10,813         |
| Sunamganj District Hospital, Sunamganj     | 322           | 11,306         |
| Narail District Hospital, Narail           | 815           | 11,507         |
| Pabna District Hospital, Pabna             | 3,200         | 13,614         |
| Tangail District Hospital, Tangail         | 5,013         | 14,559         |
| Naogaon District Hospital, Naogaon         | 1,253         | 15,914         |
| Madaripur District Hospital, Madaripur     | 1,069         | 16,098         |
| Jessore District Hospital, Jessore         | 5,085         | 21,033         |
| Narayanganj 300-bed Hospital, Narayanganj  | 2,505         | 23,284         |
| Kishoreganj District Hospital, Kishoreganj | 3,128         | 29,035         |
| Netrakona District Hospital, Netrakona     | 65            | 57,370         |
| <b>Total</b>                               | <b>87,928</b> | <b>463,174</b> |

### Alternative Medical Care

The present Government of Bangladesh pledges modernization of the unani, ayurvedic and homeopathic medical care as an alternative to the widespread allopathic practice. Alternative medicines have been playing a significant role in the healthcare delivery system in Bangladesh.

There are two medical colleges for alternative medicines in Dhaka: Government Unani and Ayurvedic Degree College and the Homeopathic Degree College. Each college has a 100-bed hospital, and people receive medical services from those hospitals. A summarized report is given below:

**Table 5.16.** Total number of patients at the Homoeopathic Medical College Hospital, 2018

| No | Type of care             | Male patients | Female patients | Child patients | Total patients |
|----|--------------------------|---------------|-----------------|----------------|----------------|
| 1  | Outdoor                  | 16996         | 25343           | 13421          | 55760          |
| 2  | Emergency                | 1654          | 1867            | 1139           | 4660           |
| 3  | Indoor/admitted patients | 204           | 131             | 197            | 532            |
| 4  | Pathology                | 540           | 565             | 20             | 1125           |
| 5  | Antenatal care           | 0             | 356             |                | 356            |

**Table 5.17.** Total number of patients at the Government Homoeopathic Medical College Hospital, 2019

| No | Type of care             | Male patients | Female patients | Child patients | Total patients |
|----|--------------------------|---------------|-----------------|----------------|----------------|
| 1  | Outdoor                  | 159987        | 21048           | 14674          | 51709          |
| 2  | Emergency                | 2095          | 2748            | 1287           | 6130           |
| 3  | Indoor/admitted patients | 222           | 228             | 327            | 777            |
| 4  | Pathology                | 756           | 786             | 30             | 1572           |
| 5  | Antenatal care           |               | 495             |                | 495            |

**Table 5.18.** Total number of patients at the Government Unani-Ayurvedic Medical College Hospital for Ayurvedic treatment, 2018

| No. | Type of care             | Male patients | Female patients | Child patients | Total patients |
|-----|--------------------------|---------------|-----------------|----------------|----------------|
| 1   | Outdoor                  | 7215          | 7812            | 1500           | 16527          |
| 2   | Pathology                | 900           | 920             | 300            | 2120           |
| 3   | Emergency                | 1765          | 1850            | 225            | 3840           |
| 4   | Indoor/admitted patients | 3315          | 3914            | 750            | 7979           |
| 5   | Antenatal care           |               | 752             |                | 752            |
| 6   | IMCI & Nutrition         |               |                 | 1250           | 1250           |

**Table 5.19.** Total number of patients at the Government Unani-Ayurvedic Medical College Hospital for Ayurvedic treatment, 2019

| No | Type of care             | Male patients | Female patients | Child patients | Total patients |
|----|--------------------------|---------------|-----------------|----------------|----------------|
| 1  | Outdoor                  | 7073          | 7583            | 2161           | 16817          |
| 2  | Pathology                | 980           | 871             | 336            | 2187           |
| 3  | Emergency                | 1800          | 1785            | 145            | 3730           |
| 4  | Indoor/admitted patients | 4722          | 3636            | 849            | 9207           |
| 5  | Antenatal care           |               | 914             |                | 914            |
| 6  | IMCI & Nutrition         |               |                 | 1614           | 1614           |

**Table 5.20.** Total number of patients at the Government Unani-Ayurvedic Medical College Hospital for Unani treatment, 2018

| No | Type of care             | Male patients | Female patients | Child patients | Total patients |
|----|--------------------------|---------------|-----------------|----------------|----------------|
| 1  | Outdoor                  | 7500          | 7681            | 1800           | 16981          |
| 2  | Pathology                | 998           | 924             | 321            | 2243           |
| 3  | Emergency                | 1775          | 1680            | 130            | 3585           |
| 4  | Indoor/admitted patients | 3122          | 3390            | 765            | 7277           |
| 5  | Antenatal care           |               | 765             |                | 765            |
| 6  | IMCI & Nutrition         |               |                 | 1450           | 1450           |

**Table 5.21.** Total number of patients of at the Government Unani-Ayurvedic Medical College Hospital for Unani treatment, 2019

| No | Type of care             | Male patients | Female patients | Child patients | Total patients |
|----|--------------------------|---------------|-----------------|----------------|----------------|
| 1  | Outdoor                  | 5188          | 6961            | 1909           | 14058          |
| 2  | Pathology                | 800           | 850             | 330            | 1980           |
| 3  | Emergency                | 1874          | 1780            | 137            | 3791           |
| 4  | Indoor/admitted patients | 4816          | 3566            | 1093           | 9475           |
| 5  | Antenatal care           |               | 850             |                | 850            |
| 6  | IMCI & Nutrition         |               |                 | 1620           | 1620           |

**Table 5.22.** Top 10 diseases of the admitted patients under the Government Homoeopathic Medical College Hospital, 2018

| No. | Name of disease          | Number of patients | Percentage |
|-----|--------------------------|--------------------|------------|
| 1   | General weakness         | 105                | 20         |
| 2   | Arthritis and joint pain | 43                 | 8.19       |
| 3   | Peptic ulcer disease     | 28                 | 5.33       |
| 4   | Skin disease             | 25                 | 4.76       |
| 5   | Cancer                   | 23                 | 4.38       |
| 6   | Bronchial asthma         | 22                 | 4.19       |
| 7   | Stroke and paralysis     | 22                 | 4.19       |
| 8   | Leucorrhea               | 12                 | 2.28       |
| 9   | Back pain                | 11                 | 2.09       |
| 10  | Uterine prolapse         | 10                 | 1.90       |

**Table 5.23.** Top 10 diseases of the admitted patients at the Government Homoeopathic Medical College Hospital, 2019

| No. | Name of disease           | Number of patients | Percentage |
|-----|---------------------------|--------------------|------------|
| 1   | Arthritis (HTN/PUD/GW/MC) | 127                | 16.34      |
| 2   | Typhoid/Viral fever       | 108                | 13.90      |
| 3   | Gastritis with PUD        | 88                 | 11.32      |
| 4   | Urinary infection/ MSC    | 45                 | 5.79       |
| 5   | HTN                       | 42                 | 5.40       |
| 6   | Back pain                 | 41                 | 5.27       |
| 7   | Diabetic mellitus         | 39                 | 5.02       |
| 8   | Carcinoma                 | 36                 | 4.63       |
| 9   | Asthma                    | 33                 | 4.25       |
| 10  | Nasal polyps              | 31                 | 3.99       |

**Table 5.24.** Top 10 diseases of the admitted Unani patients at the Government Unani-Ayurvedic Medical College Hospital, 2017-2018

| Sl. No. | Name of disease |
|---------|-----------------|
| 1       | Joint pain      |
| 2       | Skin disease    |
| 3       | Leucorrhea      |
| 4       | Dysentery       |
| 5       | Common cold     |
| 6       | Diabetes        |
| 7       | Piles           |
| 8       | Scabies         |
| 9       | Morrhagia       |
| 10      | Asthma          |

**Table 5.25.** Top 10 diseases of the admitted Ayurvedic patients at the Government Unani-Ayurvedic Medical College Hospital, 2017-2018

| Sl. No. | Name of disease                                         |
|---------|---------------------------------------------------------|
| 1       | Neurological disorders, rheumatic arthritis, joint pain |
| 2       | Hyper acidity                                           |
| 3       | Skin diseases                                           |
| 4       | Jaundice                                                |
| 5       | IBS                                                     |
| 6       | Asthma                                                  |
| 7       | Anemia                                                  |
| 8       | Leucorrhea                                              |
| 9       | Piles                                                   |
| 10      | Retention of urine                                      |



# Bangladesh Scenario of the Health Workforce

## Filling up vacant positions prioritized

Filling up the vacant positions is one of the top priorities of the government health sector in Bangladesh. The health workforce situation, particularly in the DGHS and some of its Operational Plans, are summarized in this chapter. An overview of the number of sanctioned and filled-up posts is presented, along with vacancies. Health workforce deployment and redeployment are ongoing processes; attrition due to death, retirement, resignation, termination, migration, transfer, replacement, and filling-in are constantly occurring. Therefore, the status of health workforce, as shown in this report, may not remain the same by the time this bulletin is published.

An overall health workforce situation of the DGHS is demonstrated in Table 6.1. Key points are highlighted below:

- Out of 105,254 sanctioned posts under the DGHS, physicians (Class I) comprise 25.65% and non-physicians (Class I) 0.50%, Class II 0.94% (excluding posts

Health workforce deployment and redeployment are ongoing processes; attrition due to death, retirement, resignation, termination, migration, transfer, replacement, and filling-in are constantly occurring...

for nursing), Class III 49%, and Class IV employees 23.90%

- For Class II staff, the posts for nursing are not included here as they are the staff of DGNM
- A total of 26,635 sanctioned posts remained vacant which constituted 25.31% of the total sanctioned posts. Vacancy rate was 5.21% (1,407 posts) for doctors and 57.20% for non-doctors (Class I), 30.54% for Class II, 29.59% (1,5261 posts) for Class III staff, and 37.22% (9,363 posts) for Class IV staff.

**Table 6.1.** Number of sanctioned, filled-up and vacant posts under DGHS

| Category  |             | Sanctioned post | Filled up post | Vacant post | Rate of vacancy (%) |
|-----------|-------------|-----------------|----------------|-------------|---------------------|
| Class I   | Doctors     | 27002           | 25594          | 1408        | 5.21                |
|           | Non-Doctors | 528             | 226            | 302         | 57.20               |
| Class II* |             | 989             | 687            | 302         | 30.54               |
| Class III |             | 51580           | 36319          | 15261       | 29.59               |
| Class IV  |             | 25155           | 15792          | 9363        | 37.22               |
| Total     |             | 105254          | 78619          | 26635       | 25.31               |

Source: HRIS, DGHS ; \*Excluding nursing posts

## Administrative, academic and clinical positions

Table 6.2 shows the number of sanctioned, filled-up and vacant posts at administrative, academic and clinical positions under the DGHS.

**Table 6.2** Number of sanctioned, filled-up and vacant posts of physicians at administrative, academic and clinical positions under the DGHS (December 2019)

| Name of the post                                  | Sanctioned<br>(No.) | Filled-up<br>(No.) | Vacant<br>(No.) |
|---------------------------------------------------|---------------------|--------------------|-----------------|
| Director General                                  | 1                   | 1                  | 0               |
| Additional Director General/OSD/ equivalent       | 5                   | 4                  | 1               |
| Director/Principal/Vice Principal/OSD/equivalent* | 128                 | 256                | -128            |
| Deputy Director/OSD/equivalent*                   | 137                 | 611                | -474            |
| Assistant Director/Civil Surgeon/OSD/equivalent*  | 255                 | 642                | -387            |
| Deputy Civil Surgeon/UHFPO/OSD*                   | 994                 | 3027               | -2033           |
| Professor                                         | 691                 | 318                | 373             |
| Associate Professor                               | 1162                | 516                | 646             |
| Assistant Professor                               | 1948                | 1208               | 740             |
| Senior Consultant                                 | 634                 | 230                | 404             |
| Senior Lecturer                                   | 8                   | 7                  | 1               |
| Junior Lecturer                                   | 32                  | 26                 | 6               |
| Junior Consultant                                 | 3940                | 958                | 2982            |
| Assistant Surgeon/OSD/equivalent*                 | 16549               | 17389              | -840            |
| Other posts (Pay scale : 7th + 8th)               | 518                 | 401                | 117             |
| <b>Total</b>                                      | <b>27002</b>        | <b>25594</b>       | <b>1408</b>     |

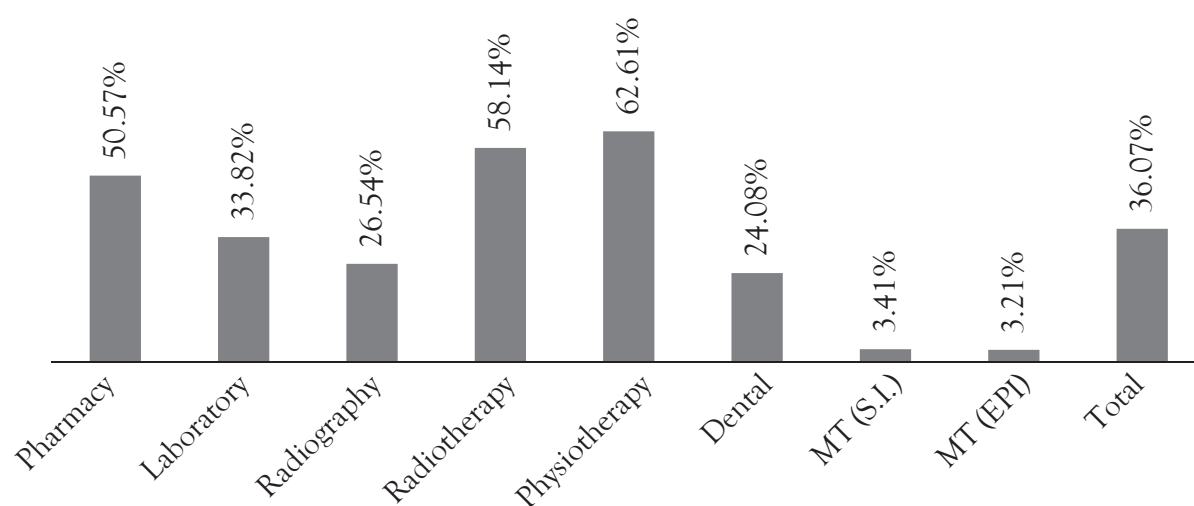
\*Existing manpower is more than sanctioned posts due to personnel placed as OSD

## Availability of Medical Technologists

**Table 6.3** Number of sanctioned, filled-up and vacant posts of medical technologists by discipline (December 2019)

| Category      | Sanctioned  | Filled-up   | Vacant      | % of vacancy |
|---------------|-------------|-------------|-------------|--------------|
| Pharmacy      | 2915        | 1441        | 1474        | 50.57        |
| Laboratory    | 2389        | 1581        | 808         | 33.82        |
| Radiography   | 795         | 584         | 211         | 26.54        |
| Radiotherapy  | 86          | 36          | 50          | 58.14        |
| Physiotherapy | 337         | 126         | 211         | 62.61        |
| Dental        | 627         | 476         | 151         | 24.08        |
| MT (S.I.)     | 498         | 481         | 17          | 3.41         |
| MT (EPI)      | 499         | 483         | 16          | 3.21         |
| <b>Total</b>  | <b>8146</b> | <b>5208</b> | <b>2938</b> | <b>36.07</b> |

Source: HRIS, DGHS



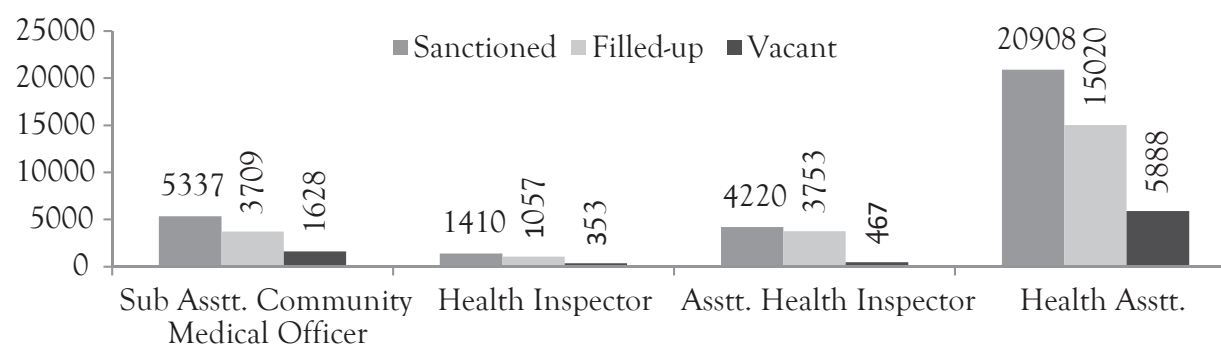
**Figure 6.1** Percentage of vacant posts of medical technologists by discipline

### Availability of Sub-assistant Community Medical Officers and Domiciliary Staff

**Table 6.4** Number of sanctioned, filled-up and vacant posts of Sub-Assistant Community Medical Officer (SACMO) and other domiciliary staff under the DGHS

| Category                                     | Sanctioned   | Filled-up    | Vacant      |
|----------------------------------------------|--------------|--------------|-------------|
| Sub-Asstt. Community Medical Officer (SACMO) | 5337         | 3709         | 1628        |
| Health Inspector                             | 1410         | 1057         | 353         |
| Assistant Health Inspector                   | 4220         | 3753         | 467         |
| Health Assistant                             | 20908        | 15020        | 5888        |
| Sub-total                                    | 26538        | 19830        | 6708        |
| <b>Total</b>                                 | <b>31875</b> | <b>23539</b> | <b>8336</b> |

Source: HRIS, DGHS



**Figure 6.2..** Number of sanctioned, filled-up and vacant posts of SACMOs, Health Inspectors, Assistant Health Inspectors, and Health Assistants

## Availability of Officers and Staff in Alternative Medical Care

**Table 6.5.** Number of sanctioned, filled-up and vacant posts of various officers and staff in alternative medicines under the DGHS as of December 2019

| Name of post                              | Sanctioned post |             | Filled-up post |            | Vacant post |            |
|-------------------------------------------|-----------------|-------------|----------------|------------|-------------|------------|
|                                           | Rev.            | Dev.        | Rev.           | Dev.       | Rev.        | Dev.       |
| Director                                  | 1               | 0           | 1              | 0          | 0           | 0          |
| Principal-cum-Superintendent              | 3               | 0           | 3              | 0          | 0           | 0          |
| Professor                                 | 2               | 0           | 0              | 0          | 2           | 0          |
| Deputy Director                           | 1               | 0           | 1              | 0          | 0           | 0          |
| Assistant Director                        | 1               | 2           | 1              | 0          | 0           | 2          |
| Assistant Professor                       | 16              | 7           | 13             | 2          | 3           | 5          |
| RP(U/A/H)                                 | 0               | 3           | 0              | 2          | 0           | 1          |
| Registrar (U/A/H)                         | 0               | 3           | 0              | 2          | 0           | 1          |
| Asstt. Registrar (U/A/H)                  | 0               | 3           | 0              | 0          | 0           | 3          |
| RMO                                       | 0               | 1           | 0              | 1          | 0           | 0          |
| Lecturer                                  | 21              | 46          | 15             | 31         | 6           | 15         |
| Medical Officer (U/A/H), DGHS             | 0               | 3           | 0              | 3          | 0           | 0          |
| Medical Officer (Unani Medicine)          | 76              | 99          | 25             | 53         | 51          | 46         |
| Medical Officer (Ayurvedic Medicine)      | 75              | 96          | 24             | 49         | 51          | 47         |
| Medical Officer (Homeopathic Medicine)    | 65              | 96          | 14             | 60         | 51          | 36         |
| IMO (U/A/H)                               | 6               | 18          | 0              | 15         | 6           | 3          |
| Drug Superintendent                       | 0               | 3           | 0              | 0          | 0           | 3          |
| Other posts equivalent of medical officer | 14              | 2           | 14             | 2          | 0           | 0          |
| Research Officer (U/A/H)                  | 0               | 4           | 0              | 3          | 0           | 1          |
| Deputy Superintendent                     | 1               | 0           | 1              | 0          | 0           | 0          |
| Nurse/Staff nurse                         | 12              | 0           | 12             | 0          | 0           | 0          |
| Secretary                                 | 1               | 0           | 1              | 0          | 0           | 0          |
| Pharmacist                                | 0               | 3           | 0              | 0          | 0           | 3          |
| Inspector                                 | 0               | 9           | 0              | 0          | 0           | 9          |
| Accountant Officer                        | 0               | 1           | 0              | 0          | 0           | 1          |
| Accountant                                | 0               | 1           | 0              | 0          | 0           | 1          |
| Production Assistant                      | 0               | 4           | 0              | 0          | 0           | 4          |
| Support Personnel (Compounder)            | 220             | 214         | 65             | 0          | 155         | 214        |
| Herbal Assistant (Gardener)               | 0               | 477         | 0              | 430        | 0           | 47         |
| Driver                                    | 2               | 6           | 2              | 0          | 0           | 6          |
| Other Class III personnel                 | 100             | 2           | 82             | 0          | 18          | 2          |
| Other Class IV personnel                  | 184             | 2           | 124            | 2          | 60          | 2          |
| <b>Total</b>                              | <b>801</b>      | <b>1105</b> | <b>398</b>     | <b>655</b> | <b>403</b>  | <b>452</b> |

Source: AMC, DGHS

## Institutions offering degrees and diplomas in Alternative Medical Care, DGHS

**Table 6.6.** Government and private academic institutions offering degrees and diplomas in Alternative Medicines

| Name of institution                                                                  | Number |       |         | Duration of course | Duration of internship | Degree offered                                                                                   | Certifying authority                                                             |
|--------------------------------------------------------------------------------------|--------|-------|---------|--------------------|------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Bachelor Level                                                                       | Total  | Govt. | Private |                    |                        |                                                                                                  |                                                                                  |
| Government Unani and Ayurvedic Medical College/ Non-government Unani Medical College | 4      | 2     | 2       | 5 years            | 1 year                 | BUMS (Bachelor of Unani Medicine and Surgery); BAMS (Bachelor of Ayurvedic Medicine and Surgery) | University of Dhaka.<br>Chattagram Medical University and University of Rajshahi |
| Government/ Bangladesh Homeopathic Medical College                                   | 2      | 1     | 1       | 5 years            | 1 year                 | BHMS (Bachelor of Homeopathic Medicine and Surgery)                                              | University of Dhaka                                                              |
| Diploma Level                                                                        |        |       |         |                    |                        |                                                                                                  |                                                                                  |
| Unani/ Ayurvedic Diploma College                                                     | 23     | 01    | 22      | 4 years            | 6 months               | DUMS (Diploma in Unani Medicine and Surgery); DAMS (Diploma in Ayurvedic Medicine and Surgery)   | Board of Unani and Ayurvedic System                                              |
| Homeopathic Diploma College                                                          | 61     | 0     | 61      | 4 years            | 6 months               | DHMS (Diploma in Homeopathic Medicine and Surgery)                                               | Board of Homeopathic Medicine                                                    |

Source: AMC, DGHS

# Medical Education

## Institutions, courses, and Enrolment capacities

Medical Education and Health Manpower Development (ME&HMD) Unit is an indispensable part of the DGHS since the beginning of its journey; To achieve Vision 2021 and 2041, including Sustainable Development Goals (SDGs), ME&HMD of the DGHS is engaged in the delivery of quality education for the production of a competent and motivated health workforce. The unit also looks after in-service training of the existing health workforce under the DGHS. The ME&HMD leads the implementation of the Medical Education Operational Plan under the 4th health sector program (HPNSP 2017-2022) for institutional and individual capacity development of health workforce.

### Current Status

- Thirty-six public and 70 private medical colleges
- Thirty-nine postgraduate medical teaching institutes, including 4 medical universities: BSMMU, Chittagong Medical University, Rajshahi Medical University, and Sylhet Medical University
- One Armed Forces Medical College and 5 army medical colleges
- Nine public and 200 private medical assistants training schools (MATS)
- Thirteen public and 97 private institutes of health technology (IHT)

The ME&HMD leads the implementation of the Medical Education Operational Plan under the 4th health sector program (HPNSP 2017-2022) for institutional and individual capacity development of health workforces...

### Goal

Provide quality medical education in medical teaching institutes for Universal Health Coverage.

### Objective

Strengthen medical education and health manpower development system for developing medical professionals and health workforce to deliver standard and high-quality services in achieving Universal Health Coverage.

### Major Activities

- Streamlining postgraduate medical education
- Ensuring quality of graduate-level medical education
- Ensuring quality of para-professional education
- Ensuring need-based proper development of health manpower

- Conducting admission tests for students of MBBS, BDS, IHT, MATS, Ayurvedic and Homeopathic Medical College, Diploma and BSc Nursing courses
- Procurement of furniture, machineries, and equipment for all government medical and dental colleges, and other medical institutes
- Strengthening libraries with journals and books and establish a unique e-library.

Thus, the country established the systems for strengthening of medical education and health manpower development.

### Major Achievements

To strengthen quality medical education, the ME&HMD is concerned with streamlining postgraduate medical education, ensuring high-quality graduate-level medical education and that for para-professionals, with need-based proper development of health workforce and supporting infrastructure development in medical institutes, strengthening libraries with journals and books and establishing a unique eLibrary. Training of health workforce is another major activity of the ME&HMD.

A total number of 2,00,295 participants have been trained locally in different clinical, para-clinical and public health subjects, and 1,056 personnel received training from abroad during 2009 to 2018.

### Innovation

- ◆ Medical Education and Health Manpower Development (ME&HMD) Unit of the DGHS conducts national-level admission tests of students for MBBS, BDS, IHT, MATS and alternative medical care courses
- ◆ Digital system is applied from online application submission up to result declaration
- ◆ Global positioning system (GPS) is used during transportation of question papers and dispense of examination materials
- ◆ After evaluation, examination results are informed to the examinees through SMS by teletalk.

### Recommendations

- √ Restructuring of ME&HMD as the Directorate General of Medical Education (DGME)
- √ Procurement and supply of machineries and equipment, furniture and fixture, and provide other logistic support in medical, dental colleges, and other medical institutes, especially newly-declared 5 medical colleges in Netrakona, Chandpur, Naogaon, Nilphamari, and Magura districts
- √ Training on teaching professionally by different programs in home and abroad
- √ A separate medical research institute should be established
- √ A residential training institute should be established.

### Status of MBBS Degree

- Thirty-six government medical colleges with 4,068 seats
- One Armed Forces Medical College and 5 army medical colleges with 375 seats
- 70 private medical colleges with 6,281 seats
- Total number of seats for MBBS students increased by 4% from the last 2 years and by 193% in the last 10 years.



**Table 7.1.** Number of institutions, enrolment capacity of the government and private medical colleges, including dental college/ units during 2018-2019 academic year

| Type of institution                                                     | Number of institutions | Estd. year  | Enrolment capacity (No. of seats) | Remarks |
|-------------------------------------------------------------------------|------------------------|-------------|-----------------------------------|---------|
| Government medical college                                              | 36                     | 1946 - 2018 | 4068                              |         |
| Armed Forces Medical College and Army Medical Colleges                  | 6                      | 1999 - 2014 | 375                               |         |
| Private medical colleges                                                | 70                     | 1985 - 2019 | 6281                              |         |
| Government dental college and dental unit in government medical college | 9                      | 1960 - 2012 | 532                               |         |
| Private dental college and dental units in private medical college      | 26                     | 1995 - 2016 | 1405                              |         |
| Total                                                                   | 147                    |             | 12661                             |         |

Data source: Administrative data, ME&amp;HMD, DGHS

**Table 7.2.** Enrolment capacity of the government medical colleges in 2018-2019 academic year

| SL | Code | Name of institution                                            | Estd. year | Seats |
|----|------|----------------------------------------------------------------|------------|-------|
| 1  | 11   | Dhaka Medical College, Dhaka                                   | 1946       | 220   |
| 2  | 12   | Sir Salimullah Medical College, Mitford, Dhaka                 | 1972       | 220   |
| 3  | 13   | Shaheed Suhrawardy Medical College, Sher-e-Bangla Nagar, Dhaka | 2005       | 175   |
| 4  | 14   | Mymensingh Medical College, Mymensingh                         | 1962       | 220   |
| 5  | 15   | Chittagong Medical College, Chattogram                         | 1962       | 220   |
| 6  | 16   | Rajshahi Medical College, Rajshahi                             | 1962       | 220   |
| 7  | 17   | MAG Osmani Medical College, Sylhet                             | 1966       | 220   |
| 8  | 18   | Sher-E-Bangla Medical College, Barisal                         | 1968       | 220   |
| 9  | 19   | Rangpur Medical College, Rangpur                               | 1972       | 220   |
| 10 | 20   | Cumilla Medical College, Cumilla                               | 1992       | 160   |
| 11 | 21   | Khulna Medical College, Khulna                                 | 1992       | 160   |
| 12 | 22   | Shaheed Ziaur Rahman Medical College, Bogura                   | 1992       | 160   |
| 13 | 23   | Faridpur Medical College, Faridpur                             | 1992       | 160   |
| 14 | 24   | M Abdur Rahim Medical College, Dinajpur                        | 1992       | 160   |
| 15 | 25   | Pabna Medical College, Pabna                                   | 2008       | 70    |
| 16 | 26   | Abdul Malek Ukil Medical College, Noakhali                     | 2008       | 70    |

**Table 7.2 Contd.**

Table Continued...

| SL           | Code | Name of institution                                    | Estd. year | Seats       |
|--------------|------|--------------------------------------------------------|------------|-------------|
| 17           | 27   | Cox's Bazar Medical College, Cox's Bazar               | 2008       | 70          |
| 18           | 28   | Jashore                                                | 2010       | 70          |
| 19           | 29   | Satkhira Medical College, Satkhira                     | 2011       | 65          |
| 20           | 30   | Shaheed Syed Nazrul Islam Medical College, Kishoreganj | 2011       | 65          |
| 21           | 31   | Kushtia Medical College, Kushtia                       | 2011       | 65          |
| 22           | 32   | Sheikh Sayera Khatun Medical College, Gopalganj        | 2011       | 65          |
| 23           | 33   | Shaheed Taj Uddin Ahmad Medical College, Gazipur       | 2013       | 65          |
| 24           | 34   | Sheikh Hasina Medical College, Tangail                 | 2014       | 65          |
| 25           | 35   | Sheikh Hasina Medical College, Jamalpur                | 2014       | 65          |
| 26           | 36   | Colonel Malek Medical College, Manikganj               | 2014       | 65          |
| 27           | 37   | Shaheed M Mansur Ali Medical College, Sirajganj        | 2014       | 65          |
| 28           | 38   | Patuakhali Medical College, Patuakhali                 | 2014       | 51          |
| 29           | 39   | Rangamati Medical College, Rangamati                   | 2014       | 51          |
| 30           | 40   | Mugda Medical College, Dhaka                           | 2015       | 65          |
| 31           | 41   | Sheikh Hasina Medical College, Habiganj                | 2017       | 51          |
| 32           | 42   | Netrakona Medical College, Netrakona                   | 2018       | 50          |
| 33           | 43   | Nilphamari Medical College, Nilphamari                 | 2018       | 50          |
| 34           | 44   | Naogaon Medical College, Naogaon                       | 2018       | 50          |
| 35           | 45   | Magura Medical College, Magura                         | 2018       | 50          |
| 36           | 46   | Chandpur Medical College, Chandpur                     | 2018       | 50          |
| <b>Total</b> |      |                                                        |            | <b>4068</b> |

Data source: Administrative data, ME&amp;HMD, DGHS

**Table 7.3.** Enrolment capacity of Armed Forces Medical College and army medical colleges in 2018-2019 academic year

| Sl No.       | Name of institution                 | Estd. year | Seats      |
|--------------|-------------------------------------|------------|------------|
| 1            | Armed Forces Medical College, Dhaka | 1999       | 125        |
| 2            | Army Medical College, Chittagong    | 2014       | 50         |
| 3            | Army Medical College, Jessore       | 2014       | 50         |
| 4            | Army Medical College, Comilla       | 2014       | 50         |
| 5            | Army Medical College, Rangpur       | 2014       | 50         |
| 6            | Army Medical College, Bogura        | 2014       | 50         |
| <b>Total</b> |                                     |            | <b>375</b> |

Data source: Administrative data, ME&amp;HMD, DGHS

## Status of Degree on Health Technology

- There are 11 government institutes of technology, with an enrolment capacity of 2,685. There are 97 private institutes of technology with an enrolment capacity of 8,940.
- These institutes offer both BSC and diploma courses in health technology and physiotherapy.

## Status of Medical Assistant Training

- ◆ There are 9 government medical assistants training schools with an enrolment capacity of 818.
- ◆ There are 200 private medical assistant training schools with an enrolment capacity of 13,540.

## Postgraduate Education

- There are 39 institutions providing postgraduate degree courses for MBBS and BDS graduates in various disciplines.
- In addition, BCPS provides postgraduate fellowship courses in various disciplines.

## Progress in Medical Education

Number of medical education courses and enrolment capacity both increased manifold in the last decade. The national-level admission tests for MBBS, BDS, IHT, MATS, and alternative medical care courses are being conducted digitally from online application submission up to results declaration. Global Positioning System (GPS) is used during transportation of question papers and dispense

of examination materials. After evaluation, examination results are informed to the examinees through SMS by teletalk telecom network.

Figure 7.1 shows the enrolment capacities of medical institutes in Bangladesh.

## Foreign Student Enrolment in MBBS and BDS Courses

- There are 1,896 foreign students in MBBS and BDS courses in different institutions of Bangladesh.
- Total number of foreign students in MBBS and BDS courses in 2018 across government and private institutions increased by 235% in the last 10 years.
- Foreign students in private medical colleges saw the biggest rise in the number of foreign students.

## In-service Training in 2018-2019

- There were a total of 1,268 batches of local training with 31,700 participants.
- There were a total of 32 batches of foreign training with 492 participants.

## Center for Medical Education

### Curriculum

Need assessment was conducted by the Center for Medical Education (CME) for updating the MBBS Curriculum 2012 in Bangladesh. Out of 112 medical colleges, views of the academic council of 102 medical colleges were collected, and the report was prepared and disseminated among the relevant stakeholders.

**Table 7.4.** Enrolment capacity of government and private institutes of health technology (IHT) in 2018 -2019 academic year

| Sl. No.                                                                                                                     | Type of institution                                                           | Estd. year  | Total seats by discipline |                         |      |      |          |              |     |     |                                               |     |       |
|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------|---------------------------|-------------------------|------|------|----------|--------------|-----|-----|-----------------------------------------------|-----|-------|
|                                                                                                                             |                                                                               |             | LAB                       | RDL&IMG/<br>Radiography | PTY  | DENT | Pharmacy | Radiotherapy | OTA | ICA | Quota for<br>freedom<br>fighter<br>and tribal | SIT | Total |
| 1                                                                                                                           | Enrolment capacity of the government institutes of health technology (IHT)-13 | 1962 - 2018 | 565                       | 505                     | 400  | 405  | 455      | 140          | 25  | 25  | 65                                            | 100 | 2685  |
| 2                                                                                                                           | Enrolment capacity of private institutes of health technology (IHT)-97        | 2001 - 2017 | 4127                      | 935                     | 920  | 2123 | 60       | 235          | 255 | 180 | 5                                             | 100 | 8940  |
| Total                                                                                                                       |                                                                               |             | 4692                      | 1440                    | 1320 | 2528 | 515      | 375          | 280 | 205 | 70                                            | 200 | 11625 |
| LAB= Laboratory; RDL=Radiology; ,PTY=Physiotherapy; SI=Sanitary inspection; DENT=Dentistry; PHAR=Pharmacy; RTY=Radiotherapy |                                                                               |             |                           |                         |      |      |          |              |     |     |                                               |     |       |

Data source: Administrative data, (ME&amp;HMD, DGHS)

Table 7.5. Enrolment capacity of the government institutes of health technology (IHT) in 2018 -2019 academic year by discipline

| Division   | Name of institute with location                         | Estd. year | Discipline |                         |     |      |          |              |     |     |                                                |     |       |
|------------|---------------------------------------------------------|------------|------------|-------------------------|-----|------|----------|--------------|-----|-----|------------------------------------------------|-----|-------|
|            |                                                         |            | LAB        | RDL&IMG/<br>Radiography | PTY | DENT | Pharmacy | Radiotherapy | OTA | ICA | Quota for<br>freedom<br>fighters and<br>tribes | SIT | Total |
| Dhaka      | Institute of Health Technology, Dhaka.                  | 1962       | 50         | 50                      | 50  | 50   | 50       | 20           | 25  | 25  | 7                                              | 50  | 377   |
|            | Institute of Health Technology, Rajshahi.               | 1962       | 50         | 50                      | 50  | 50   | 50       | 20           | 0   | 0   | 7                                              | 50  | 327   |
| Rajshahi   | Institute of Health Technology, Bogra                   | 2007       | 65         | 55                      | 50  | 55   | 55       | 20           | 0   | 0   | 7                                              | 0   | 307   |
| Chattogram | Institute of Health Technology, Sirajganj               | 2017       | 50         | 0                       | 0   | 0    | 50       | 0            | 0   | 0   | 3                                              | 0   | 103   |
|            | Institute of Health Technology, Fouzdarhat, Chattogram. | 2011       | 50         | 50                      | 50  | 50   | 50       | 20           | 0   | 0   | 7                                              | 0   | 277   |
| Barisal    | Institute of Health Technology, Barisal                 | 2011       | 50         | 50                      | 50  | 50   | 50       | 20           | 0   | 0   | 7                                              | 0   | 277   |

Table 7.5 Contd.

Table Continued...

| Division                                                                                                                                                                           | Name of institute with location           | Estd. year | Discipline |                         |     |      |          |              |     |     |                                                |     |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|------------|------------|-------------------------|-----|------|----------|--------------|-----|-----|------------------------------------------------|-----|-------|
|                                                                                                                                                                                    |                                           |            | LAB        | RDL&IMG/<br>Radiography | PTY | DENT | Pharmacy | Radiotherapy | OTA | ICA | Quota for<br>freedom<br>fighters and<br>tribes | SIT | Total |
| Khulna                                                                                                                                                                             | Institute of Health Technology, Jhenaidah | 2011       | 50         | 50                      | 50  | 50   | 50       | 20           | 0   | 0   | 7                                              | 0   | 277   |
|                                                                                                                                                                                    | Institute of Health Technology, Satkhira  | 2018       | 50         | 50                      | 0   | 0    | 0        | 0            | 0   | 0   | 3                                              | 0   | 103   |
| Sylhet                                                                                                                                                                             | Institute of Health Technology, Sylhet    | 2011       | 50         | 50                      | 50  | 50   | 50       | 0            | 0   | 0   | 7                                              | 0   | 257   |
| Mymensingh                                                                                                                                                                         | Institute of Health Technology, Jamalpur  | 2018       | 50         | 50                      | 0   | 0    | 0        | 0            | 0   | 0   | 3                                              | 0   | 103   |
| Total                                                                                                                                                                              |                                           |            | 565        | 505                     | 400 | 405  | 455      | 140          | 25  | 25  | 65                                             | 100 | 2685  |
| Total no. of institutions=11                                                                                                                                                       |                                           |            | 2685       |                         |     |      |          |              |     |     |                                                |     |       |
| LAB= Laboratory; RDL= Radiology; IMG= Imaging, OTA= Operation Theatre Assistant, ICA= Intensive Care Assistant, SIT=Sanitary Inspectorship Training; DENT=Dentistry; PHAR=Pharmacy |                                           |            |            |                         |     |      |          |              |     |     |                                                |     |       |

LAB= Laboratory; RDL= Radiology; IMG= Imaging, OTA= Operation Theatre Assistant, ICA= Intensive Care Assistant, SIT=Sanitary Inspectorship Training; DENT=Dentistry; PHAR=Pharmacy

Data source: Administrative data, (ME&HMD, DGHS

**Table 7.6.** Total enrolment capacity of the government institutes of health technology (IHT) in 2018-2019 academic year

| Sl. No.            | Name of institution                                    | Estd. year | Seats       |
|--------------------|--------------------------------------------------------|------------|-------------|
| 1                  | Institute of Health Technology, Dhaka                  | 1963       | 377         |
| 2                  | Institute of Health Technology, Rajshahi               | 1976       | 327         |
| 3                  | Institute of Health Technology, Bogra                  | 2006       | 307         |
| 4                  | Institute of Health Technology, Fouzdarhat, Chattogram | 2011       | 277         |
| 5                  | Institute of Health Technology, Barisal                | 2011       | 277         |
| 6                  | Institute of Health Technology, Rangpur                | 2011       | 277         |
| 7                  | Institute of Health Technology, Jhenaidah              | 2012       | 277         |
| 8                  | Institute of Health Technology, Sylhet                 | 2012       | 257         |
| 9                  | Institute of Health Technology, Sirajganj              | 2017       | 103         |
| 10                 | Institute of Health Technology, Satkhira               | 2018       | 103         |
| 11                 | Institute of Health Technology, Jamalpur               | 2018       | 103         |
| 12                 | Institute of Health Technology, Tungipara              | 2019       | 103         |
| 13                 | Institute of Health Technology, Gazipur                | 2019       | 103         |
| <b>Total seats</b> |                                                        |            | <b>2891</b> |

Data source: Administrative data, ME&HMD, DGHS.

**Table 7.7.** Enrolment capacity of the government and private medical assistant training schools (MATs) in 2018-2019 academic year

| Sl. No.      | Name of Institution                                                              | Estd. Year       | Enrolment capacity |
|--------------|----------------------------------------------------------------------------------|------------------|--------------------|
| 1            | Enrolment capacity of the government medical assistant training schools (MATs)-9 | 1976–2018        | 818                |
| 2            | Enrolment capacity of the private medical assistant training schools (MATs)-200  | 2008–2015        | 13540              |
| <b>Total</b> |                                                                                  | <b>1976–2018</b> | <b>14358</b>       |

### Quality Assurance Scheme

Assessment and dissemination of the implementation status of National Quality Assurance Scheme (NQAS) at different medical colleges (2nd phase) National Guidelines and Tools for Quality Assurance

Scheme (QAS) for Dental Colleges/Units in Bangladesh.

### Conduction of Courses

- Masters in Medical Education (MMEd) for two-year duration continued.



**Table 7.8.** Enrolment capacity of the government medical assistant training schools (MATS) in 2018-2019 academic year

| Sl. No.            | Name of institution                          | Estd. Year | Enrolment capacity |
|--------------------|----------------------------------------------|------------|--------------------|
| 1                  | Medical Assistant Training School, Comilla   | 1979       | 52                 |
| 2                  | Medical Assistant Training School, Noakhali  | 1976       | 102                |
| 3                  | Medical Assistant Training School, Faridpur  | 1979       | 102                |
| 4                  | Medical Assistant Training School, Tangail   | 1979       | 102                |
| 5                  | Medical Assistant Training School, Bagerhat  | 1979       | 152                |
| 6                  | Medical Assistant Training School, Kushtia   | 1976       | 102                |
| 7                  | Medical Assistant Training School, Jhenaidah | 2011       | 52                 |
| 8                  | Medical Assistant Training School, Sirajganj | 1979       | 102                |
| 9                  | Medical Assistant Training School, Satkhira  | 2018       | 52                 |
| <b>Total seats</b> |                                              |            | <b>818</b>         |

Data source: Administrative data, ME&amp;HMD, DGHS

**Table 7.9.** Postgraduate medical institutions in Bangladesh

| Sl. No.      | Type of Medical Institutions | Number of institutions | Number of seats | Provide degree                                             |
|--------------|------------------------------|------------------------|-----------------|------------------------------------------------------------|
| 1            | Government                   | 22                     | 1048            | Postgraduate Diploma, M.Phil., MD, MS, MPHMMED, FCPS, MCPS |
| 2            | Autonomous                   | 7                      | 1112            |                                                            |
| 3            | Non-government               | 10                     | 112             |                                                            |
| <b>Total</b> |                              | <b>39</b>              | <b>2846</b>     |                                                            |

Data Source: CME Survey 2017

**Table 7.10.** Infrastructural progress during 2009 to 2018

| Type of institution                   | Year |      | Increased in numbers | Increased in percentage (%) |
|---------------------------------------|------|------|----------------------|-----------------------------|
|                                       | 2009 | 2019 |                      |                             |
| Medical university                    | 1    | 4    | 3                    | 0.9                         |
| Government medical college            | 17   | 36   | 19                   | 5.8                         |
| Private medical college               | 40   | 70   | 30                   | 9.2                         |
| Army and Armed Forces Medical College | 1    | 6    | 5                    | 1.5                         |
| Government dental college and unit    | 3    | 9    | 6                    | 1.8                         |
| Private dental college and unit       | 11   | 26   | 15                   | 4.6                         |

**Table 7.10 Contd.**

Table Continued...

| Type of institution                                 | Year       |            | Increased in numbers | Increased in percentage (%) |
|-----------------------------------------------------|------------|------------|----------------------|-----------------------------|
|                                                     | 2009       | 2019       |                      |                             |
| Private institute of health technology (IHT)        | 38         | 97         | 59                   | 18.2                        |
| Government medical assistant training school (MATs) | 7          | 9          | 2                    | 0.6                         |
| Private medical assistant training school (MATs)    | 23         | 200        | 177                  | 54.5                        |
| <b>Total number of institutions</b>                 | <b>144</b> | <b>466</b> | <b>325</b>           | <b>100.0</b>                |

Data source: Administrative data, ME&HMD, DGHS

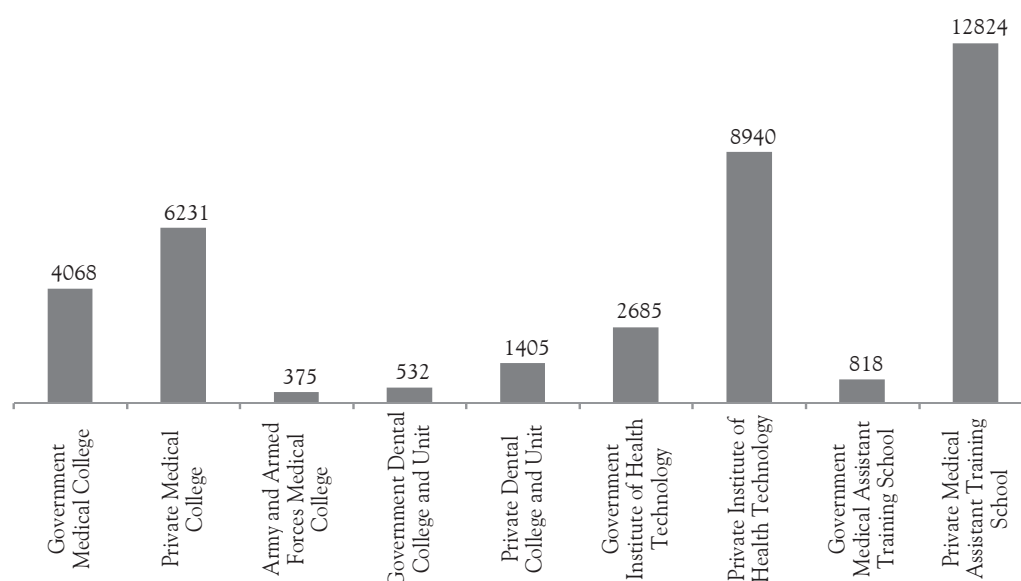


Figure 7.1 Enrolment capacities of medical institutions in Bangladesh

Table 7.11. Foreign student enrolment during 2009 to 2018

| Name of the institute              | 2009<br>(Number of students) | 2018<br>(Number of students) | Increase in<br>the number of<br>foreign students |
|------------------------------------|------------------------------|------------------------------|--------------------------------------------------|
| Government medical college         | 50                           | 81                           | 31                                               |
| Private medical college            | 474                          | 1702                         | 1228                                             |
| Government dental college and unit | 2                            | 15                           | 13                                               |
| Private dental college and unit    | 40                           | 98                           | 58                                               |
| <b>Total</b>                       | <b>566</b>                   | <b>1896</b>                  | <b>1330</b>                                      |

Data source: Administrative data, ME&HMD, DGHS

**Table 7.12.** Training of health workforce from 2009 to 2019 in different clinical subjects, public health and others

| Training         | Training in 2009-2019 |              | Training in 2018-2019 |              |
|------------------|-----------------------|--------------|-----------------------|--------------|
|                  | Batches               | Participants | Batches               | Participants |
| Local training   | 9721                  | 231995       | 1268                  | 31700        |
| Foreign training | 187                   | 1548         | 32                    | 492          |

Data source: Administrative data, ME&HMD, DGHS

- Six students have acquired MMed degree from the 14th batch and 6 students from the 15th batch; they have been enrolled in the Thesis Part, and new 14 students have been enrolled in Part I of two-year MMed course at CME. All the students are teachers of different medical and dental colleges, including the army personnel.
- Till today, a total of 102 have passed the MMed degree.
- Diploma in Health Professional Education (DHPed ) course of one-year duration has been approved and introduced.

Six sessions of the five-day long workshops on Teaching Methodology and Assessment were conducted for the teachers of different government and non-government medical colleges, and postgraduate institutes...

total of 150 teachers of different levels were awarded certificates.

### eLearning Studio

elearning studio for web-based education was developed.

### Training

Six sessions of the five-day long workshops on Teaching Methodology and Assessment were conducted for the teachers of different government and non-government medical colleges, and postgraduate institutes. A

### Planning for international conference

Planning to arrange the South-East Asia Regional Association for Medical Education (SEARAME) Conference has been completed. The conference will be held on 16-17 October 2020 at the Army Golf Garden, Dhaka, Bangladesh.

# Health Information System, eHealth, and MBT

## Ever-shifting ideas for innovations

### 8.1 Prelude

Directorate General of Health Services (DGHS) under the Ministry of Health and Family Welfare (MOHFW) has a department called Management Information System (MIS). The main objective of MIS is to establish Health Information System (HIS) and eHealth in Bangladesh. As a developing country in the South-East Asia, Bangladesh has harnessed impressive rollout of eHealth in its public health sector. This is the first nation in the world to announce its mission to digitalize different functions under the vision “Digital Bangladesh 2021” well ahead in 2009. Bangladesh is among the few countries of the world where the public sector has made tremendous progress in all areas of eHealth and Health Information System (HIS). The health workforce in the entire health sector from national to the sub-national and grassroots community is digitally connected to interact in real-time with robust national databases. The country follows a model of shared eHealth resource hosted by the Ministry of Health and Family Welfare. The model is used by development partners, NGOs, private sector, and other government and non-government organizations. MIS always follows standardization of eHealth data and interoperability. In 2011, the United Nations honored Bangladesh Prime Minister Sheikh Hasina with a “Digital Health for Digital

Bangladesh is among the few countries of the world where the public sector has made tremendous progress in all areas of eHealth and Health Information System (HIS)...

Development” award. In 2014, Bangladesh’s eHealth model was recognized as the World’s best practice by the German Government, exemplified through publication of a book called “A Quiet Revolution: Strengthening the Routine Health Information System in Bangladesh.” Improvement of eHealth in Bangladesh is consistent and rapid, which is now focusing more on newer innovations and improvement of quality. Under the 4th sector-wide program called Health, Population and Nutrition Sector Program (HPNSP 2017-2022), the activities are being continued under HIS and eHealth Operational Plan.

### 8.2 Objectives and Strategies of MIS

**General Objective:** Improve Health Information System, eHealth, and Medical Biotechnology

**Specific objectives:** Improve existing health information system and eHealth through:

- strengthening population-based and institution-based human resources and program-based HIS
- strengthening and improving electronic health records for all citizens
- developing and strengthening logistic tracking and inventory management and procurement system
- expansion of GIS in health services
- improving infrastructure and human resource capacity necessary for HIS and eHealth
- continuation and further development of health call center, mobile phone-based health services (mHealth services)
- strengthening and expansion of video-conferencing and telemedicine service
- introduction of other eHealth services and programs
- sustaining eHealth initiatives and encouraging public-private partnership.

Improve medical biotechnology through:

- achievement of the short- and medium-term deliverables mentioned in the National Guidelines on Medical Biotechnology
- creation of conditions for achieving the long-term deliverables of the National Guidelines on Medical Biotechnology.

## Strategies

- Move towards full digitalization of the entire routine health information system

in Bangladesh by 2021 to build a robust evidence-base for decision-making

- Strengthen monitoring and evaluation (M&E) processes for the sector-wide program through effective utilization of evidences available from all possible sources (routine data, surveillance, CRVS, survey, research, etc.)
- Establish partnerships with relevant institutions for facilitating demand, updating, and utilization of data in the MOHFW and its agencies and departments
- Develop and implement multi-year, comprehensive HIS, eHealth, and M&E training plans for capacity-building of managers and staff
- Increase budgetary allocation for HIS, eHealth, and M&E activities to sustain and further build on the progress.

## 8.3 Organizational Arrangement

The Management Information System Unit has three sections: 1. Health Information System section, 2. eHealth section, and 3. Medical Biotechnology section (MBT). The specific functions of the sections are detailed below:

### 8.3.1 Health Information System

Health Information System (HIS) manages health service information from community to the central level. This system provides the basic information for planning, monitoring, and evaluation of the health system at all levels. The major functions of the HIS are as follows:

- Equipping health facilities with computers, laptops, accessories, gadgets at all levels of health facilities
- Placement, maintenance, and upgradation of software (database, application, customization, etc.) and servers where applicable and, as appropriate, looking into factors, like cost, deployment ability, scalability, integration, interoperability, security, user-friendliness, auto-reporting, dashboard, etc.
- Maintenance and upgradation of Data Center and Disaster Recovery Center(s)
- Strengthening of human resource capacity through training and, in case of shortage, through outsourcing of services
- Liaison, feedback, monitoring, and supportive supervision to ensure data quality in terms of reliability, timeliness, and adequacy, inclusive of non-state providers
- Data analysis, interpretation, report writing, dissemination, and communication to appropriate stakeholders and the public through print, web and electronic media for the sake of evidence-based planning and decision-making as well as ensuring the citizens' rights to information
- Liaison and advocacy with policy-makers to sustain and further boost up the HIS initiatives and public-private-NGO collaboration.

### 8.3.2 eHealth Program

Continue and expand the existing mHealth services, such as SMS-based Complaints & Suggestion System, Shasthya Batayon,

Continue and expand the existing mHealth services, such as SMS-based Complaints & Suggestion System, Shasthya Batayon, bulk SMS, and explore other innovative and effective mHealth services are the main functions of the eHealth Program...

bulk SMS, and explore other innovative and effective mHealth services are the main functions of the eHealth Program. The major pathways are listed below:

- Continue, strengthen, and roll out telemedicine services, especially in remote areas
- Promote the use of video-conferencing for quick and cheaper instant communication, remote monitoring, online meeting, scientific and clinical conference, distance learning, and training on teaching
- Improve infrastructure and human resource capacity necessary for eHealth
- Put in place appropriate database, application, and customized software, where and when applicable, looking into factors, like cost, deployment ability, scalability, interoperability, security, and user-friendliness
- Produce and develop promotional materials to encourage people to utilize the mobile phone-based health services, telemedicine, and other eHealth services

- Sustain the eHealth initiatives, encourage public-private partnership.

### 8.3.3. Medical Biotechnology (MBT) Program

- Establishment of the Center of Excellence on Medical Biotechnology
- Expansion of new MBT laboratories and institutional capacity-building
- Organizing sensitization/orientation training and workshops
- Providing medical biotechnology resources to medical libraries
- Developing and enforcing standards, codes of practice, and regulatory framework for medical biotechnology
- Appropriate communication program with potential entrepreneurs of medical biotechnology.

## 8.4 Major Ongoing Activities of HIS and eHealth Program under MIS-DGHS

The following innovative activities were conducted on a regular or ad-hoc basis in 2018-2019:

### 8.4.1 Online Local Health Bulletins

Automated Local Health Bulletins were updated every month. Almost all health facilities under the DGHS—from community clinics to the specialized hospitals—produced the bulletins. The bulletins contain indicator-wise aggregated information, for example, no. of patient-visits in outpatient and emergency

departments, indoor patients, bed-occupancy and turnover ratios, EmONC, etc. The bulletins are available for downloading and can also be printed and distributed locally (Figure 8.1). [Visit [>>Dashboard >> Local Health Bulletin](http://www.dghs.gov.bd)]

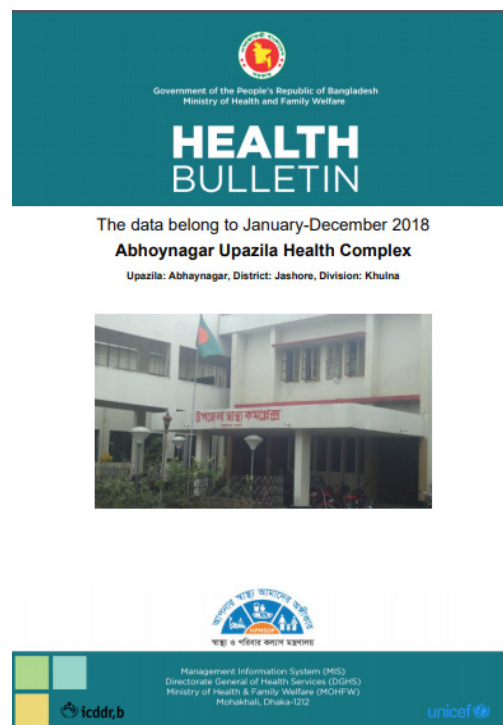


Figure 8.1. Online Local Health Bulletins in dashboard

### 8.4.2 District Health Information Software2

The Ministry of Health and Family Welfare (MOHFW) of Bangladesh has been using District Health Information Software2 (DHIS2) since 2010. Data ranging from community clinics to tertiary-level health facilities and programs, like MNCAH, IMNCI, EPI, TB, NCDs, communicable diseases, HIV/STD, nutrition, cervical cancer and breast cancer screening, obstetric fistula screening and care program, are being flown to the national HMIS through



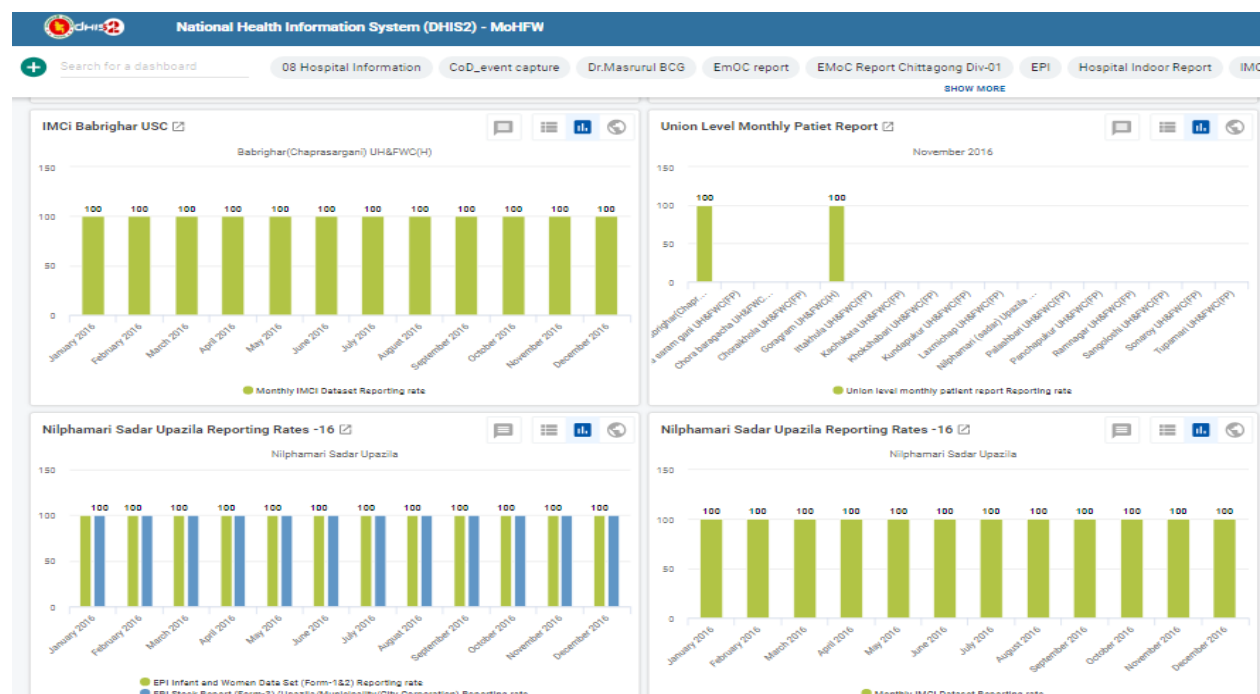


Figure 8.2. DHIS2 (Version 29) screenshot

this software. Main aim of this system is to have electronic repository for national-level health-related data, along with bridging the gap between fragmented systems by bringing together information from various databases. More than 90% of the government health facilities are using DHIS2 for reporting their routine health service data. Figure 8.2 shows the dashboard of DHIS2 (Version 29).

#### 8.4.3 Real-time Online Dashboard on Health Information

An innovative system called Health Dashboard to connect modern digital technology to health services has been developed by Management Information System (MIS) of the DGHS. MIS is using the renowned Business Intelligence (BI) tools, namely Tableau, to visualize this dashboard. Through this dashboard, information on routine health service data and survey

data can be viewed. Also, data on almost all activities can be shown through this dashboard. Along with the various important health indicators, human resource and activities in biometric mode, etc. have been displayed for public view. The features are displayed in the left side and are very rich in information. The dashboard is placed inside the DGHS website which can be accessed through <http://103.247.238.81/webportal/pages/index.php>. Figure 8.3 shows the screenshot of the DGHS dashboard.

#### 8.4.4 Human Resource Information System

Central Human Resource Information System (HRIS) is designed in a way so that it can electronically capture and maintain human resource information for the entire health sector. It is based on 4 registries; Geo-location Registry, Health Facility Registry, Sanctioned

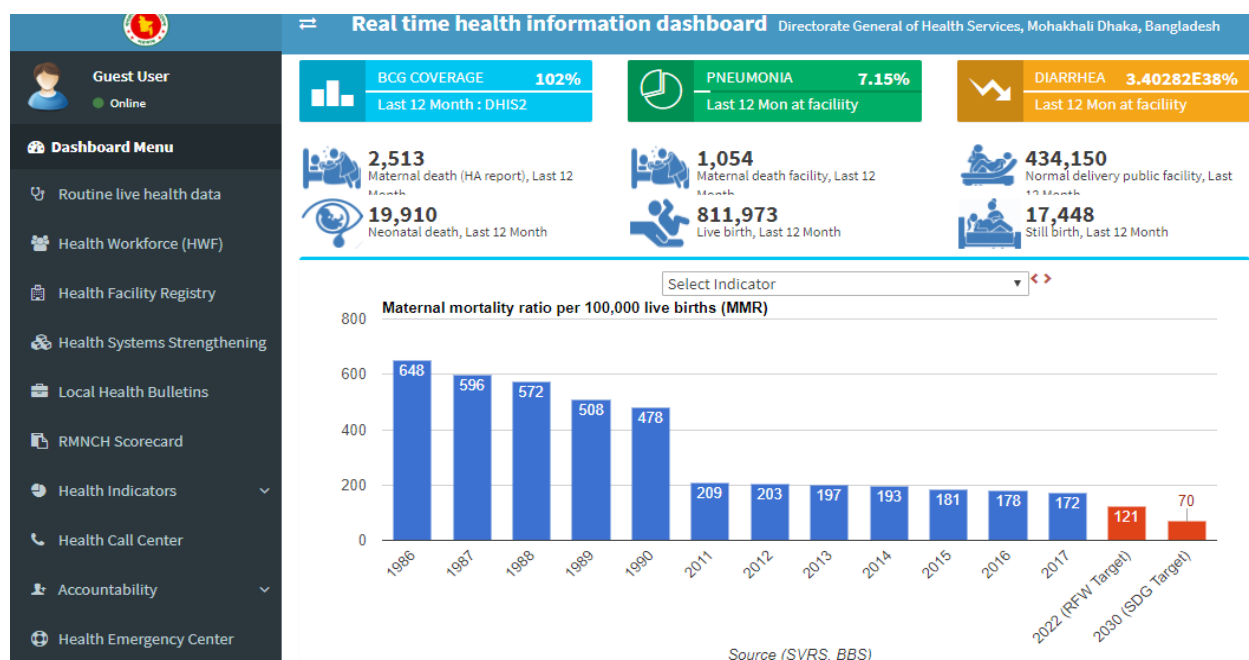


Figure 8.3. Screenshot of DGHS dashboard

Post Registry, and Provider Registry. All human resource-related activities, including transfer, promotion, deputation, leave, annual confidential report (ACR), and others, are automated through this system. The DGHS is the main stakeholder of this system. Initially

named Human Resource Management System (HRMS), this was developed by MIS-DGHS. Currently, the MIS-DGHS plays a vital role in updating the HRIS. This system received three awards for different functions. Figure 8.4 shows the screenshot of HRIS.

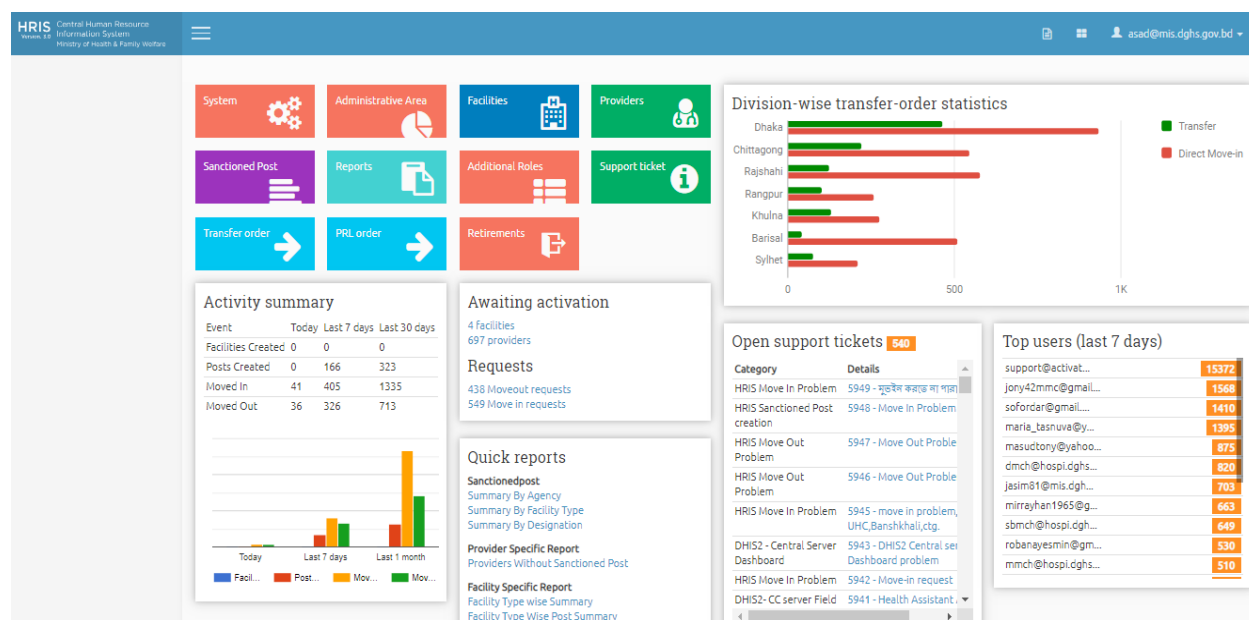


Figure 8.4. Screenshot of the Human Resource Information System (HRIS)

### 8.4.5 Health Facility Registry and GIS

Almost all public health facilities under the DGHS are registered in Health Facility Registry and have unique code numbers, which are very important for interoperability and eHealth strategy. Geo-location Registry is also integrated with the Health Facility Registry. All necessary information on health facilities under the MOHFW can be found, such as name of head of the facility, contact number, address, email ID, etc. Some of the private health facilities are also registered in this system. Health Facility Registry can be accessed publicly through <http://facilityregistry.dghs.gov.bd/>. Figure. 8.5 shows the screenshot of the Health Facility Registry.

### 8.4.6 Health System Strengthening

The Management Information System (MIS) of the Directorate General of

Health Services has launched a revamp performance evaluation platform. WHO provided technical assistance in the conceptualization and development of this revamping process as part of its efforts to support health systems strengthening in Bangladesh. Other partners, such as icddr,b and UNICEF, have also been involved since inception of the initiative. The Health Systems Strengthening (HSS) Project, as it is more commonly known as, aims to promote best practices, transparency, and accountability in health services management through incentivizing better use of the existing resources and generation of better results. Best-performing ones in each tier receive the Health Minister's National Award every year. Performance score shown online can be accessed publicly through [http://103.247.238.81/webportal/pages/hss\\_menu.php](http://103.247.238.81/webportal/pages/hss_menu.php). All score mechanism and performance evaluation indicators are

The screenshot shows the 'Facility Registry' website for the Government of Bangladesh, Ministry of Health and Family Welfare. It features a search bar and several dropdown filters for 'Select Division', 'Select District', 'Select Upazila', 'Select Agency', and 'Select Organization Type'. Below the filters, a green box displays the report details: 'Report displaying form: Division: Dhaka, District: Dhaka, Org Type: All Types' and 'Total 1087 organization found.' with an 'Export Excel' button. At the bottom, a table lists the organizations.

| Organization Name                     | Organization Code | Division | District | Upazila | Paurasava | Union | Ward | Agency | Org Type             | Photo |
|---------------------------------------|-------------------|----------|----------|---------|-----------|-------|------|--------|----------------------|-------|
| Paribaric Health Care Center -Lalbagh | 10020579          | Dhaka    | Dhaka    | Lalbagh |           |       |      | DGHS   | NGO, Hospital/Clinic |       |

Figure 8.5. Screenshot of the Health Facility Registry

defined in the abovementioned link. Figure 8.6 shows the screenshot of Health System Strengthening (HSS) scoreboard.

#### 8.4.7 Shared Health Record and OpenMRS+

Shared Health Record (SHR) is a system that stores the lifetime medical history of citizens in electronic format, and Open Health Information Exchange (OpenHIE) is a gateway to different systems and the data. The electronic medical records (EMR) stored in SHR are shared through OpenHIE across various medical or health facility management systems (like OpenMRS+ combination of OpenMRS, OpenERP, OpenELIS, OpenSRP, etc.), with proper access credentials and privileges where sensitive information is protected with special security. Patients do not need to carry their medical files when they visit any medical facility or a physician.

Initially, SHR was implemented in Kaliganj Upazila Health Complex in Gazipur district and Gazipur General Hospital. Apart from

SHR, the hospital automation system was earlier implemented in 11 hospitals under the MOHFW. This automation system is not interconnected with one another. However, SHR is designed as the central system where every hospital can share information. The real expansion of automation for a reasonable number of hospitals will be seen through scaling-up of the Shared Health Record.

In fiscal year 2018-2019, SHR has expanded in additional 10 district hospitals and 40 upazila health complexes. Figure 8.7 shows the architecture of SHR and OpenHIE.

#### 8.4.8 Video-conferencing

Video-conferencing was started for ease of administration and can convey direct message from the Ministry and DG office to all divisional and civil surgeon's offices as well as medical colleges and medical college hospitals on virtual setting. The system is providing an opportunity for arranging remote meetings, conferences, training, and emergency order (during natural disasters, like flood, landslide, Rohingya influx, etc.), thus minimizing

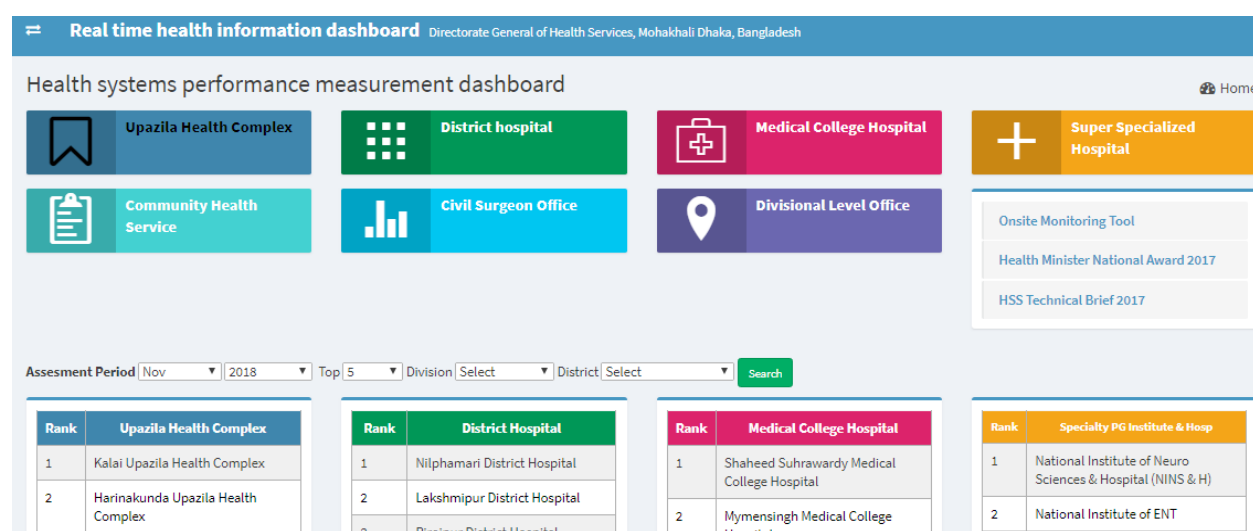


Figure 8.6. Screenshot of the Health System Strengthening (HSS) scoreboard

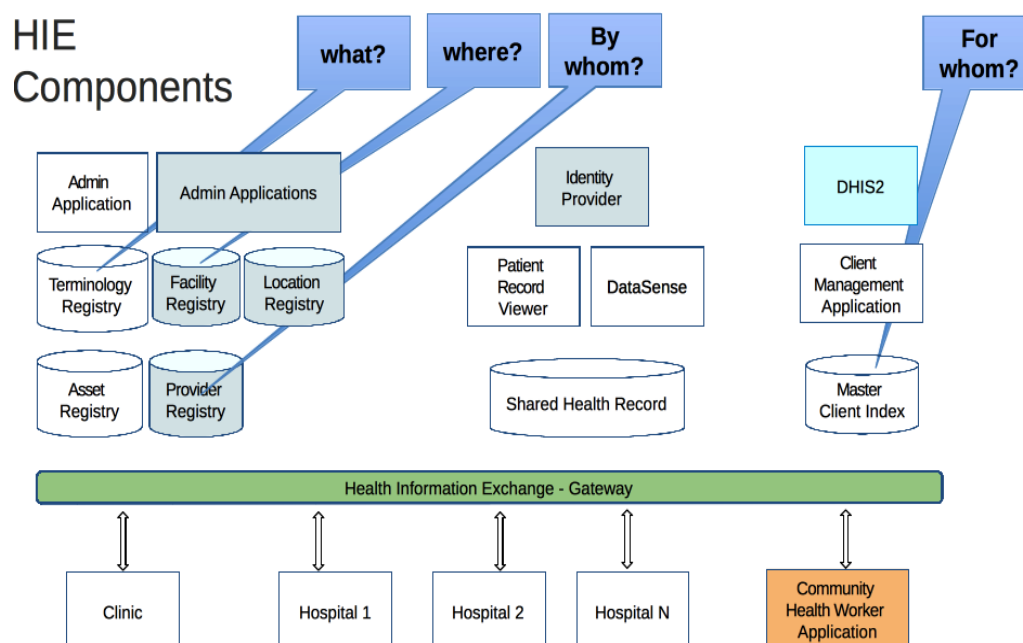


Figure 8.7. Architecture of SHR and OpenHIE

communication gaps among the Ministry, DG office, and peripheral health offices. In the fiscal year 2017-2018, in total 115 video-conferencing systems were installed; an additional 100 video-conferencing setups have been installed in 2018-2019. Now, there are 215 video-conferencing setups installed in different strategic locations. Every week, video-conferencing/meeting takes place from DG office with DHO, CSO, MCH, DH, UHC,

and specialized hospitals. Figure 8.8 shows the video-conferencing with DHO, CSO from DG office.

#### 8.4.9 Telemedicine service

Telemedicine service has started from 8 telemedicine centers in the fiscal year 2009-2010. Till fiscal year 2017-2018, there have been 94 telemedicine centers; among these,



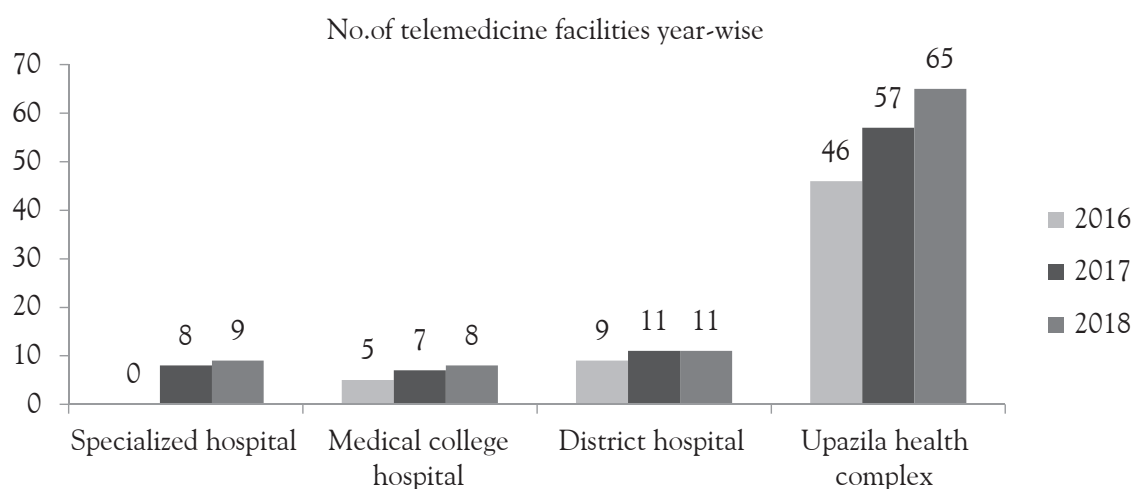
Figure 8.8. Video-conferencing with DHO, CSO from DG office



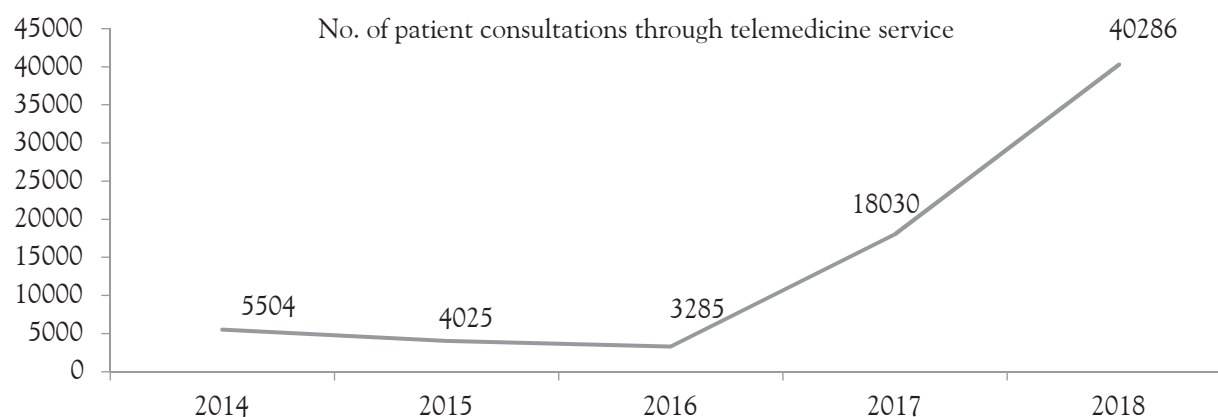
9 are at the specialized hospitals, 8 are at MCHs, 11 are at DHs, 65 are at UHCs, 1 is at MIS-DGHS for monitoring and coordination with other centers which, sometimes, provide consultation service also. Figure 8.9 shows the number of telemedicine centers year-wise (2014-2018).

Telemedicine Monitoring Unit at the MIS-DGHS, especially received complaints and problems faced by other telemedicine centers regarding hardware, software, unavailability of consultants/physicians at the sites. The

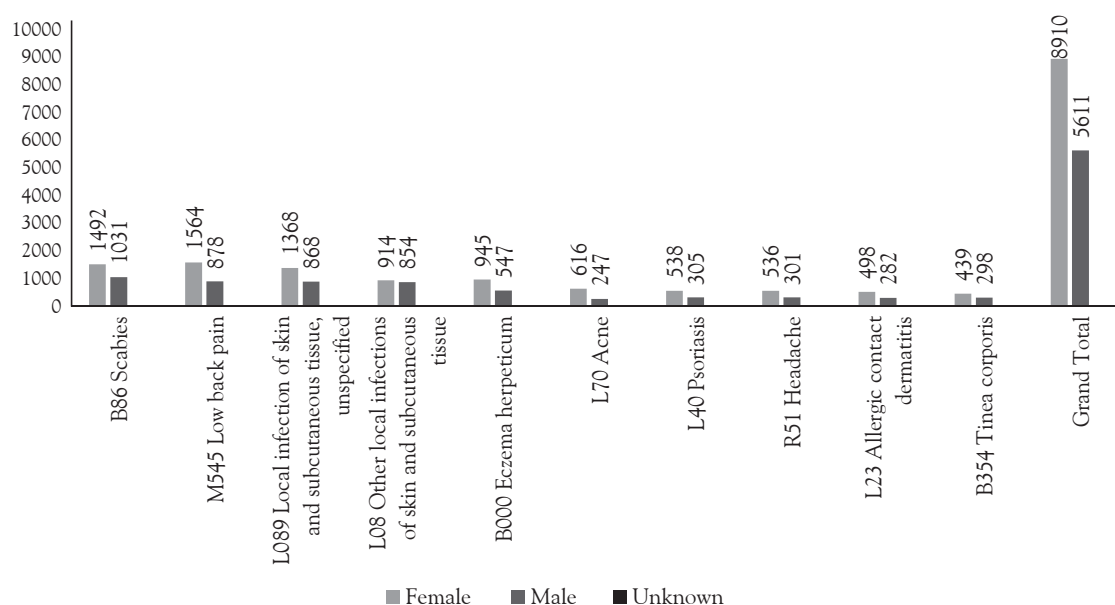
number of patient consultations through this service has shown an increasing trend during 2014-2018 (Figure 8.10). Telemedicine centers are equipped with tele-gadgets, like tele-stethoscope, tele-ECG, tele-microscope, tele-glucometer, etc. and also high-quality camera, big screen monitor, and computer. Each individual consultation has been recorded in DHIS2 from which disease profile could be obtained. Figure 8.11 shows mostly people suffering from skin diseases have been receiving the telemedicine service due to shortage of skin specialists in remote areas.



**Figure 8.9.** Number of telemedicine centers established in various institutions from 2016 to 2018



**Figure 8.10.** Year-wise number of patient consultations through telemedicine service from 2014 to 2018 (Source: DHIS2, MIS, DGHS)



**Figure 8.11.** Disease profile obtained from telemedicine service in 2018 (Source: DHIS2, MIS, DGHS)

#### 8.4.10 Citizen's Grievances Handling System

Since 2012, an exciting and effective innovation—Complaints & Suggestion Box—has been added to the existing service line of the MIS-DGHS. Citizens are now taking part in ensuring accountability of the service providers in health service delivery system. The SMS-based Complaints & Suggestion Box continues to remain as an innovative and effective mechanism to know citizens' feedback on the quality of service in the public hospitals. This system is frequently recommended for further promotion to improve accountability and transparency of public hospitals. It is one of the components to measure DLI (Disbursement Link Indicator). This system is working in 739 public hospitals and health organizations. Presently, on an average, the system receives 15 SMSs per day. The system received 8,166 complaints and suggestions in 2018.). Figure 8.13 shows the types or categories in the Complaints & Suggestion System received

in 2018. Among the categories, 'absence of workforce' was the highest complaint. It shows that absenteeism of health workforce still remains on top of all complaints. The public view of the Complaints & Suggestion Box is available at [www.dghs.gov.bd](http://www.dghs.gov.bd) >> Database Login >> Complaint & Suggestion Box (<http://app.dghs.gov.bd/complaintbox/>).

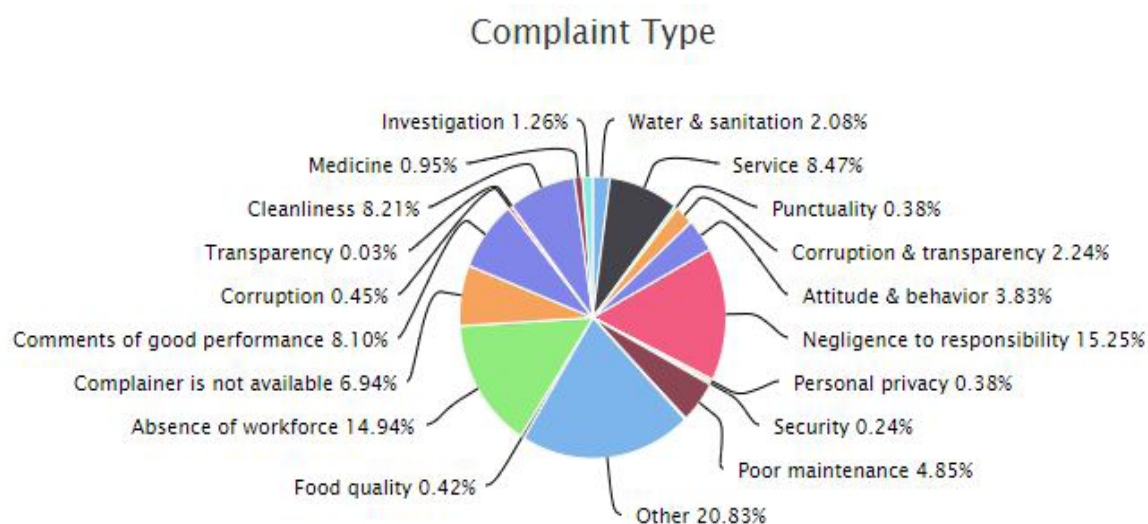
#### 8.4.11 Monitoring Staff Attendance through Biometric System

MIS-DGHS installed remote biometric time-attendance machines in all upazila and district hospitals and in some tertiary hospitals to track the office attendance of government health personnel across the country. These machines have been installed since 2012; gradually, installation has been increased phase by phase; till date, all UHCs and DHs have been covered. Additionally, MIS has planned to cover all government medical colleges and hospitals, and specialized hospitals in the beginning of 2019. Also, decision has been made to merge

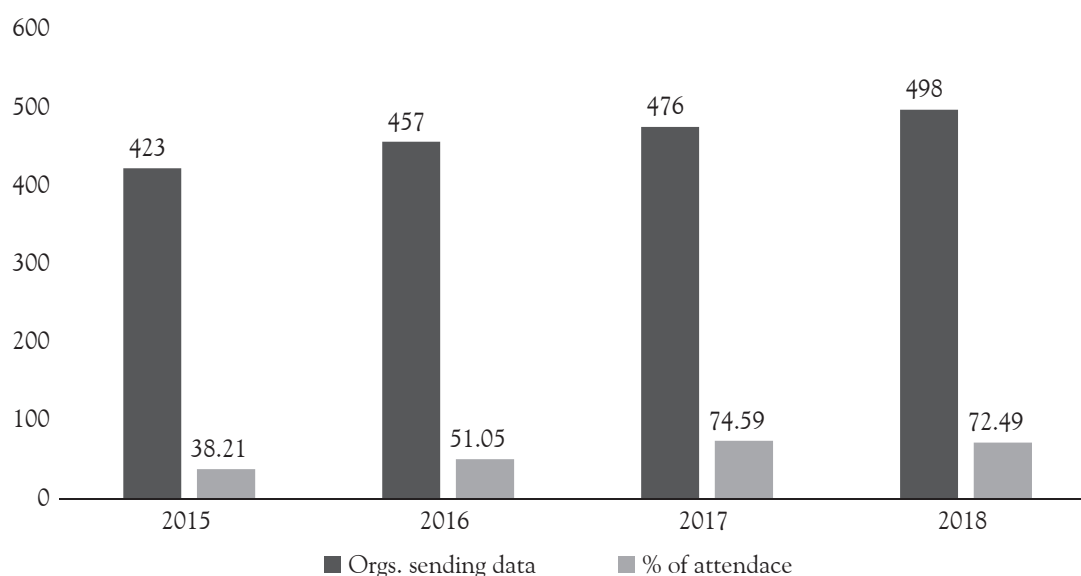


this software with HRIS and to add some more options, like employees' leave, roster and total leave count, individual record, unregistered employees, device monitoring system, etc. The machine can keep in memory 30,000 encounters. Figure 8.14 shows the number of organizations actively sending data and percentage of increase in attendance from 2015 to 2018.

Predefined web-based reports can be generated on the server-side, which can be accessed through web-browser from any place. Furthermore, the GIS map of office attendance also can be viewed from <http://www.dghs.gov.bd> >> Data >> Office Attendance Monitoring >> Office Attendance GIS Map.



**Figure 8.12.** Types or categories in complaints and suggestions received in 2018



**Figure 8.13.** Number of organizations actively sending data, along with % of attendance by Biometric System

#### 8.4.12 Health call center 16263 (Shasthya Batayon)

Shasthya Batayon Call Center 16263 is a government-run national health service center established under the supervision of MIS of DGHS under the Ministry of Health and Family Welfare. This service is called the National Health Call Center. People can get health service from any part of the country round the clock by dialing the short number 16263 from any mobile or land phone number.

Under the supervision of MIS-DGHS, the call center service is being implemented and operated by Synesis IT Limited.

Within three complete shifts in the call center, there are 45 (15 per shift) expert MBBS and master degree-holder doctors providing the service. Besides doctors, there are 15 (5 per shift) specially-trained information service provider or agents working round the clock in the call center of Shasthya Batayon. Currently, Shasthya Batayon has the capacity to take calls from and provide health services to about thirty thousand per day.

#### Key services of Shasthya Batayon

**Doctors' Advice and Treatment:** A team of efficient doctors provides services. People call 16263 and directly communicate with a doctor about their health-related problems and queries any time they want.

**Health Information:** Sometimes, it is hard to find a doctor nearby or to get information about any hospital. Shasthya Batayon provides phone numbers of public or private hospitals, clinics, and doctors with required information.

**Ambulance Information:** Shasthya Batayon has developed a database of ambulance contact numbers with information about locations. Anyone can get information on ambulance

during emergency in specific location by dialing 16263. This service has also been connected with the National Emergency Service 999. Ural EMS has now been integrated with Shasthya Batayon so that people can get ambulance through UBER System—a recent innovation in renting of vehicles.

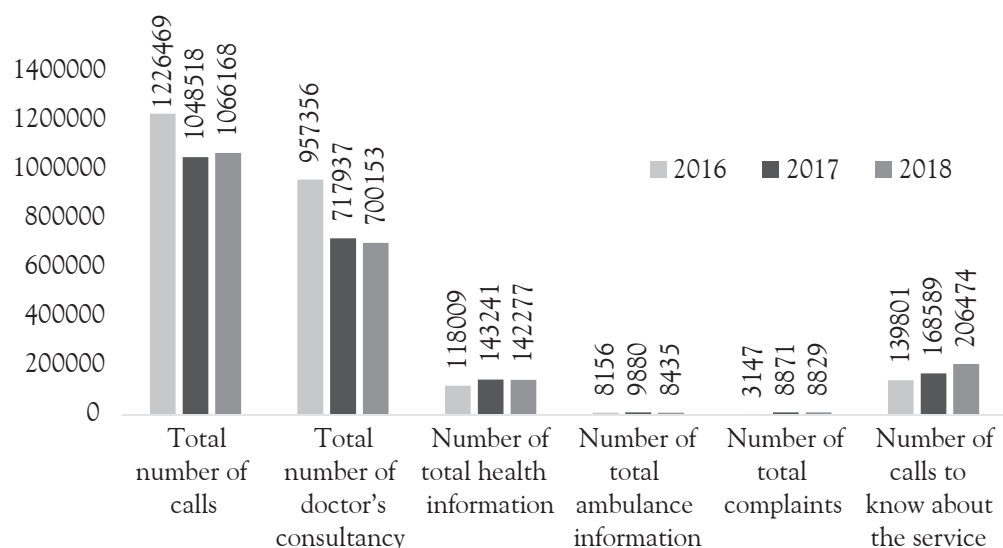
**Health Compliant Management:** To mitigate health service-related problems and complaints and to improve quality of overall health sector of Bangladesh, Shasthya Batayon is taking and solving complaints on any public or private health services. In addition, Shasthya Batayon also records and informs the complainants about measures taken based on the complaints.

**Emergency Accident Information:** Shasthya Batayon 16263 takes emergency accident information and quickly informs the relevant authorities

The total numbers of calls received in the last three years (2016, 2017, and 2018) are shown in Figure 8.14. There is a very slight difference in data of the last two years. People received services mostly for viral fever and diarrheal disease (Figure 8.15). More details of month-wise data can be viewed from public dashboard (<http://16263.dghs.gov.bd/report/report.php>).

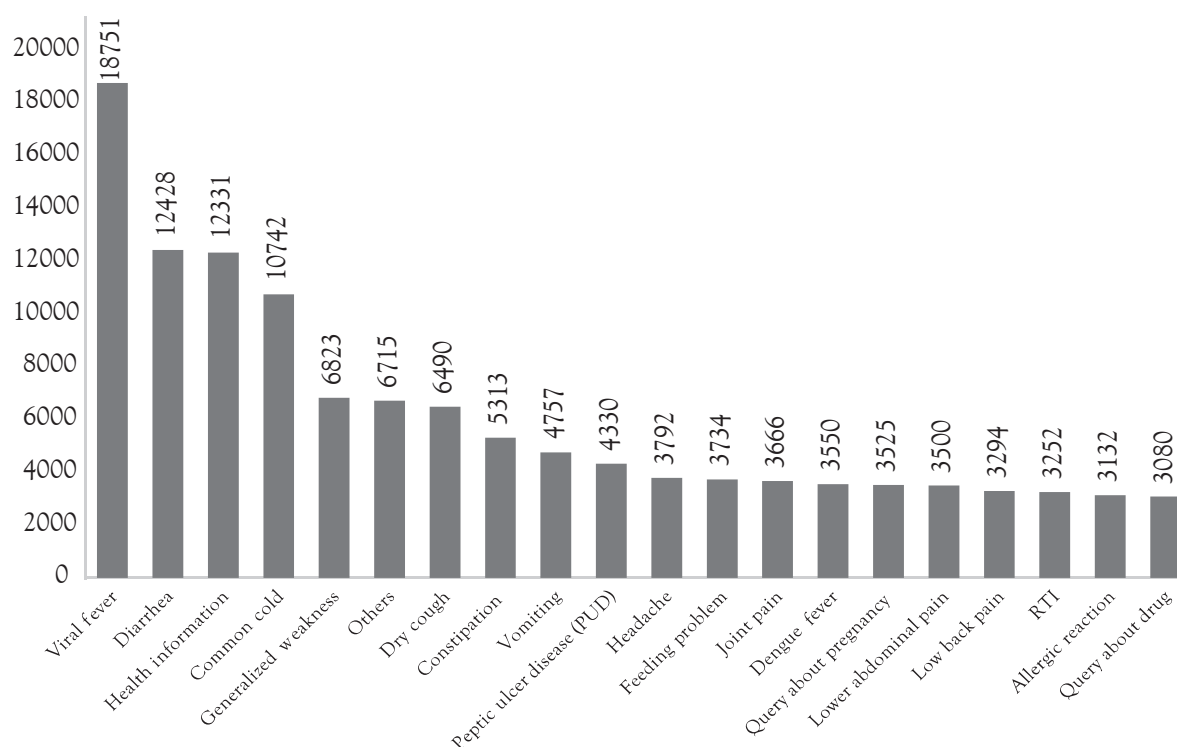
#### 8.4.13 Bulk SMS and Quick email Service and Updating Social Media Portals (Facebook, Twitter, Google+, YouTube)

MIS-DGHS operates and updates social media portals, such as Facebook, Twitter, Google+ YouTube, etc. for sharing information publicly. Bulk SMS is active for sending short messages to respective groups; all UHCs, CSOs, and DHs have official email IDs which are very



**Figure 8.14** Number of calls received through Shasthya Batayon in the last three years

Source- <http://16263.dghs.gov.bd/report/report.php>



**Figure 8.15** No. of persons who gave calls to Shasthya Batayon for the above-mentioned health problems/issues in 2018

Source: Synesis IT Limited

functional. All health offices/facilities have official email IDs (e.g. alluhfpo@dghs.gov.bd, allld@ld.dghs.gov.bd, allcs@cs.dghs.gov.bd,

allhcp@cc.dghs.gov.bd, alldivdir@mis.dghs.gov.bd, allmis@mis.dghs.gov.bd, allhospi@hospi.dghs.gov.bd).

#### **8.4.14 Civil Registration and Vital Statistics**

In 2014, with the assistance from WHO, the MOHFW conducted a comprehensive assessment of the Civil Registration and Vital Statistics (CRVS) system, using the standard assessment tool developed by WHO and the University of Queensland. Following that, Bangladesh developed a CRVS Strategic Action Plan (SAP), including key recommendations, identification of responsible agencies as well as estimated budgets and timelines. Recently, Enterprise Architecture (EA) of the CRVS has been developed, and review of the legal framework has been continuing. Therefore, the SAP is now under the process of being revised.

DGHS and DGFP under the MOHFW support CRVS through notification of birth and death happening in community and in facilities to local registrar office. Hospitals use Form 804 for death declaration as well as the perceived cause of death forms. Birth registration is mandatory for death registration. Bangladesh has recently started practicing the international-standard medical certificate of the cause of death--MCCoD in 105 hospitals. At selected large hospitals where many deaths occur, physicians have been trained to use the ICD certificate for the cause of death, and statistical coding is done according to the Startup Mortality List (ICD-10-SMoL). Smart Verbal Autopsy (VA) also has been introduced in 13 upazilas of 8 districts in 8 divisions of the country for deaths in the community where there is no medical certificate for cause of death. Death registration is mandatory for conducting VA. Since there are no electronic linkages between the ORG's BRIS and DGHS database yet, these deaths, along with causes, are not being notified directly.

#### **Partner-supported activities of CRVS (2018-2019)**

##### ***Nationally focused activities***

- Under the leadership and management of the CRVS Secretariat, WHO and D4H are working with the MOHFW to increase the use of medical certification for the cause of death, using ICD coding and verbal autopsies. So far, 26,000 VAs are done, 90,000 MCCoDs are issued; and 70,000 MCCoD forms have been entered into MCCoD module in DHIS2 of the DGHS. D4H is working with the DGHS to improve the data quality and collection for vital statistics reporting.
- UNHCR is working to end statelessness for migrants.
- Recently, the MOHFW has decided to incorporate detailed cause of death in the undergraduate medical curriculum and create permanent posts for coders.
- UNICEF is helping the DGHS in streamlining service delivery by tracking health information.
- UNFPA is working to strengthen the national statistical system for data collection and analysis so that data for development (D4D) policies can be identified and made available.
- Under the leadership and management of the CRVS Secretariat, the D4H is working with the MOHFW and ORG at 13 upazilas, to facilitate birth and death registration by using frontline health and family planning workers to assist families with the registration process.

- Birth and death registration within 45 days has been enhanced remarkably in Kaliganj upazila. The Kaliganj Model of birth and death notification and registration is scaled up in 12 upazilas using Verbal Autopsy (VA), and MCCoD in 105 hospitals (district-level hospitals, upazila health complexes, medical college hospitals as well as private hospitals).
  - Strengthened and continued the online reporting of routine data through DHIS2 from community clinics to the central-level specialized hospitals/institutions
  - Maintained and updated Data Center (Figure 8.15)
  - Maintained and repaired computer and other electronic hardware in MIS-DGHS (Table 8.1)
  - Procured and distributed different types of hardware and machinery in FY 2017-2018 and FY 2018-2019 (Table 8.2)
  - Conducted several HMIS training sessions and workshops for health managers at all levels, doctors, statisticians, nurses, and field workers throughout the country. Under local training, 24,533 officers and staff members participated in the training courses, workshops, and seminars in FY 2017-2018. (Table 8.3).
- 8.5 Major Activities Done at MIS in Fiscal Year 2018-2019**
- The following are the major activities conducted by MIS-DGHS in 2018-2019:
- Conducted annual divisional HMIS workshop in all 8 divisional health offices
  - Conducted digitally-managed medical and dental admission tests
  - Successfully conducted the yearly National Health Award 2018 as part of health service strengthening with technical support from WHO
  - Prepared, printed, and distributed the National Health Bulletin 2018

### 8.5.1 Technical Partners

Along with the MOHFW, other technical partners assisted the MIS-DGHS to find

**Table 8.1.** Computer and electronic hardware repaired and maintained at MIS-DGHS

| Organization/<br>Facility              | Device category |            |            |          |           |           |          | Grand<br>total |
|----------------------------------------|-----------------|------------|------------|----------|-----------|-----------|----------|----------------|
|                                        | All-in-one      | Desktop    | Laptop     | Monitor  | Printer   | TAB       | UPS      |                |
| CS_Office                              | -               | 1          | 1          | -        | 2         | -         | 1        | 5              |
| DG_Office                              | 22              | 73         | 57         | 2        | 6         | 36        | 1        | 197            |
| District Hospital                      | -               | -          | 1          | -        | -         | -         | -        | 1              |
| Specialized<br>hospital<br>&Institutes | 2               | 6          | 2          | 2        | 4         | -         | -        | 16             |
| UHC with CC                            | -               | 25         | 367        | -        | -         | 63        | -        | 455            |
| <b>Grand total</b>                     | <b>24</b>       | <b>105</b> | <b>428</b> | <b>4</b> | <b>12</b> | <b>99</b> | <b>2</b> | <b>674</b>     |



technology-related solutions, training, and capacity-building; these include a2i Project, World Bank, WHO, UNICEF, DFID, UNFPA, Rockefeller Foundation, JICA, USAID, icddr, Measure Evaluation, CIDA, UNESCAP, JPGSPH-BRAC University, BRAC, JHU, MSH (SIAP), Save the Children, D.Net, CIRPB, CIDA, and others.

### **8.5.2 Maintenance and Repair of Hardware in 2018-2019**

A number of desktop computers, printers, monitors, and tablets were maintained and repaired at MIS-DGHS as received from upazila, districts, and the central level. Table 8.1 shows the number of electronic hardware maintained and repaired in 2018-2019.

### **8.5.3 Supply of ICT Equipment and Computer Stationeries**

MIS-DGHS procured various hardware and ICT equipment and supplied those to other offices at different levels during the reporting period (Table 8.2).

## **8.6 Medical Biotechnology Program**

Compared to the advancements of our neighboring countries in biotechnology, the situation in Bangladesh is just in the sprouting stage. A situation analysis on biotechnology, funded by FAO and UNDP (that ended in 2003), identified that most of the work done in this sector is related to agriculture. Tissue cultures of banana, orchid, jute, wheat, etc. are some examples of regular work. In 2013, the complete genome sequence of a local variety of jute was decoded by famous scientist



**Figure 8.16.** Picture of Data Center, MIS-DGHS

**Table 8.2.** Different types of hardware and machinery procured and distributed in FY 2017-2018 and FY 2018-2019

| Hardware                                           | FY 2017-2018 | FY 2018-2019 | Distribution                                                                                                    |
|----------------------------------------------------|--------------|--------------|-----------------------------------------------------------------------------------------------------------------|
| UPS (offline-600 VA)                               | -            | 4000         | · Accompanied one for each desktop computer                                                                     |
| Tablet device/PDA                                  | 4000         | 6000         | · To community healthcare providers (HA, HI, and AHI)                                                           |
| Video-conferencing tools                           | 60           | 99           |                                                                                                                 |
| Data center equipment/<br>Disaster Recovery Center | 1 Lot        | 1 Lot        | Data Center at the MIS-DGHS established; FY 2013-2014 Disaster Recovery Center established in Khulna district   |
| Medical biotechnology equipment                    | 1 Lot        | 1 Lot        |                                                                                                                 |
| Equipment for Local Area Networking (LAN)          | 1 Lot        | -            | LAN for MCH, tertiary hospital and upazila health complex                                                       |
| Modem                                              | 10000        | 5000         | Community clinic                                                                                                |
| Motor cycle                                        | 200          | 200          | Purchased for statisticians at upazila and district levels                                                      |
| Time attendance machine, finger-print detector     |              | 400          | Distributed to divisional health office, medical college hospital, district hospital and upazila health complex |
| Laptop computer                                    | -            | 6900         |                                                                                                                 |
| Shared Health Record                               | -            | 1 Lot        |                                                                                                                 |

Dr. Maqsoodul Alam. With the recent scientific advancement, medical biotechnology is playing a vital role in patient management, starting from diagnosis and prognosis to treatment. Currently, there are private companies in Bangladesh that use medical biotechnology techniques for diagnostic tests and send samples to neighboring countries for around 1,700 tests costing more than 1.9 million US dollar per year (Source: CMBT Survey 2016). This amount is being transferred

out to neighboring countries only because medical biotechnology is not developed in our country. The Government of Bangladesh has shown interest in the application of medical biotechnology for the development of health and nutrition sectors and has developed the first National Guidelines on Medical Biotechnology in 2006. Later, the guidelines were revised and republished as gazette on 21 December 2010. Six successive meetings of the National Technical Committee on



**Table 8.3.** Local training conducted by MIS-DGHS in fiscal year 2018-2019

| Description of course outline                                                                   | Duration | Central-level (LD office/Directorate and Ministry) | Field-level (division, district upazila, union and ward) | Non-MOHFW personnel | Total participants | Amount spent (Tk in lakh) |
|-------------------------------------------------------------------------------------------------|----------|----------------------------------------------------|----------------------------------------------------------|---------------------|--------------------|---------------------------|
| <b>Local training</b>                                                                           |          |                                                    |                                                          |                     |                    |                           |
| Computer training for doctors, staff, and nurses                                                | 10 days  | 2175                                               | 1134                                                     | 3                   | 3312               | 645.08                    |
| Training on International Form of Medical Certification of Cause of Death (MCCoD)               | 1 day    | 20                                                 | 1140                                                     |                     | 1160               | 91.09                     |
| 5-day training for head office staff                                                            | 5 days   | 201                                                | 0                                                        |                     | 201                | 21.35                     |
| Training on human resource management system (HRIS)                                             | 4 days   | 101                                                | 888                                                      |                     | 989                | 134.10                    |
| Training on DHIS2/Open MRS+                                                                     | 4 days   | 0                                                  | 902                                                      |                     | 902                | 163.04                    |
| Training/workshop for medical teachers                                                          | 2 days   | 0                                                  | 54                                                       |                     | 54                 | 5.80                      |
| 1-day consultative workshop on MBT                                                              | 1 day    | 39                                                 | 0                                                        |                     | 39                 | 3.00                      |
| Training for field workers on Smart Verbal Autopsy (VA)                                         | 2 days   | 0                                                  | 240                                                      |                     | 240                | 16.10                     |
| Sensitization workshop                                                                          | 2 days   | 54                                                 | 16                                                       |                     | 70                 | 16.00                     |
| Hands-on training for medical teachers and scientists                                           | 10 days  | 31                                                 | 13                                                       |                     | 44                 | 15.34                     |
| 1-day consultative workshop on HIS and eHealth                                                  | 1 day    | 49                                                 | 0                                                        |                     | 49                 | 9.00                      |
| 2-day training for health workers on HIS and eHealth, inclusive of the use of mobile device     | 2 days   | 0                                                  | 9438                                                     |                     | 9438               | 257.10                    |
| Annual MIS Conference (Division level) with all upazila health complexes and district hospitals | 1 day    | 0                                                  | 1869                                                     |                     | 1869               | 114.80                    |
| Training on VIA and CBE for physicians, nurses, and health workers                              | 10 days  | 0                                                  | 76                                                       |                     | 76                 | 58.60                     |
| 2-day training for CHCPs on HIS and eHealth                                                     | 2 days   | 0                                                  | 6090                                                     |                     | 6090               | 174.20                    |
| <b>Total</b>                                                                                    |          | <b>2670</b>                                        | <b>21860</b>                                             | <b>3</b>            | <b>24533</b>       | <b>1724.60</b>            |

Medical Biotechnology (NTCMB) were held between 2004 and 2012 on direct recommendations of the National Executive Committee on Biotechnology (NECMB) and supervision of the National Taskforce on Biotechnology of Bangladesh (NTBB), with Hon'ble Prime Minister in the Chair. These ended up with the establishment of the Center for Medical Biotechnology at the Institute of Public Health. National Guidelines on Medical Biotechnology 2010 included guidance to implement provisions in the National Biotechnology Policy and described deliverables for short-, medium- and long-term goals. To implement these goals, Medical Biotechnology Program was included in HIS and eHealth Operational Plan of the HPNSDP 2011-2016, and it is now being continued in the 4th sector-wide program (HPNSP 2017-2022) as a component of HIS and eHealth.

### 8.6.1 Center for Medical Biotechnology

Center for Medical Biotechnology was established in 2012 at the Institute of Public Health (IPH) building to plan, implement, and monitor activities relating to medical biotechnology as per the National Guidelines on Biotechnology 2010. The primary goals and objectives of the center were to create a proper environment for the expansion of medical biotechnology in Bangladesh, preliminarily in the government sector, by providing both technical and resource support in collaboration with other stakeholders.

From the very beginning of its establishment, the Center for Medical Biotechnology (CMBT) began an inclusive effort in cooperation with ideSHi (Institute for Developing Science and Health Initiatives) and relevant stakeholders for capacity



**Figure 8.17.** Hands on training



**Figure 8.18.** Day-long visit to the R&D section of a pharmaceutical company

development. Till today, over 2,000 stakeholders, including medical professionals from various disciplines, journalists, entrepreneurs, scientists, and members of Medical Curriculum Committee have been trained through various training sessions and workshops, including core committee meetings, consultative workshops for updating

medical curriculum, training workshops for journalists, sensitization workshops, and a 10-day hands-on training for medical teachers and scientists. MBT-related books were distributed among the medical university, medical colleges, and other health science institutions, and necessary equipment for setting up biotechnology labs were distributed among different institutions. According to the deliverables for short- and medium-term goals mentioned in National MBT Guidelines, the CMBT is conducting its activities under the Operational Plan of HPNSP 2017-2022 from July 2017.

The following activities have been conducted in the area of medical biotechnology:

1. More than two thousand physicians and researchers have been provided training in medical biotechnology up to 2018 under the 4th sector-wide program (HPNSP 2017-2022).
2. Machines have been provided to 12 institutions. Various studies are underway with the aforementioned instruments.
3. Medical biotechnology-related books have been provided to 4 medical colleges.
4. Grants have been made in 4 research projects.
5. Research work is ongoing as part of the theses in the Masters and PhD courses.
6. PRP therapy is being provided free of charge as part of cell therapy, and the number of recipients is more than 70.

### 8.6.2 Ongoing Functions

1. Detection of HIV virus load, using RT-PCR (Gene Xpert) machine



**Figure 8.19.** Sanger Sequencer for genetic analysis

2. Detection of HBV virus load via RT-PCR
3. Cell therapy (PRP) is provided free of charge
4. Regular training is being provided to physicians to improve the quality of research
5. In collaboration with ideSHI,
  - ◆ antibiotic resistance genes are being collected in the clinical isolates in the spectrum and are being related to Bangladesh in wound infection;
  - ◆ work on Bangladesh is involved in the screening and profiling of plasmid harboring mcr genes in the dissemination of pathogenic bacteria and the underlying mechanism of investigation;

**Table 8.4.** Training and workshops on biotechnological issues conducted in 2018-2019

| Type of training/workshops                             | Venue | Duration (day) | Batch (No.) | Participants (No.) |
|--------------------------------------------------------|-------|----------------|-------------|--------------------|
| Hands-on training for medical teachers and scientists  | Dhaka | 10             | 4           | 40                 |
| Training/workshop for medical teachers                 | Dhaka | 6              | 30          | 180                |
| Consultative workshop for medical teachers             | Dhaka | 4              | 30          | 120                |
| Consultative workshop for review of medical curriculum | Dhaka | 1              | 1           | 10                 |

**Table 8.5.** Diagnostic services provided by CMBT in 2018

| Name of service    | No. of services provided |
|--------------------|--------------------------|
| PRP therapy        | 84                       |
| Hb-electrophoresis | 87                       |
| HIV viral load     | 34                       |

**Table 8.6.** Scientific papers published in international journals from CMBT

| Title of article                                                                                                                                       | Name of Journal                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Activated Autologous Platelet-rich Plasma Therapy in the management of Chronic Non-healing Diabetic Foot Ulcer: One Centre Study from Bangladesh       | Journal of Medical Science and Clinical Research             |
| Activated Platelet-rich Plasma in the Management of Plantar Fasciitis                                                                                  | International Journal of Medical Science and Health Research |
| A Comparative Study on Wound Management by Vacuum-assisted Closure with Low-cost Negative Pressure Wound Therapy and Conventional Moist Wound Dressing | International Journal of Medical Research Professionals      |

- ◆ work on the Foodborne Disease Surveillance (FBIS) Study is underway in collaboration with ideSHI.

in related subjects in medical colleges for research assistance

### 8.6.3 Future Workplans

1. Genetic analysis of congenital diseases with the help of the Sanger Sequencer machine
2. Provision of grants to students studying
3. Contribute to WHO's goal by determining HPV viral load
4. Provision of free delivery of electrophoresis testing to thalassemia patients
5. Accept the procedures for establishing the Stem Cell Laboratory.



# Health Research

## Coordinating role of the DGHS

Health has been accepted as a fundamental right of all people by the Constitution of the World Health Organization (WHO) and in the International Declaration of Human Rights. Several research organizations—both in private and public sectors and departments—are conducting regular health research in their focused area. Most of the research works are carried out in public medical colleges, national institutes, Bangabandhu Sheikh Mujib Medical University (BSMMU), autonomous research organizations/institutes, such as BIRDEM, Heart Foundation, general universities, other academic institutions, and by a variety of private organizations, including the pharmaceutical industries. The Directorate General of Health Services plays a direct role in Medical Biotechnology Research; a coordinating role of the DGHS in research works conducted by other institutions may yield an integrated system of ‘need-based research’ because the DGHS knows the actual health scenario and can set priority research areas.

### Bangladesh Medical Research Council

As the name suggests, the Bangladesh Medical Research Council is the apex body of the government-led health research in the country. The areas of research are as follows:

#### Communicable Diseases

Various studies have been performed in communicable disease, including tuberculosis, malaria, HIV, foodborne diseases, waterborne diseases, and other health problems.

A coordinating role of the DGHS in research works conducted by other institutions may yield an integrated system of ‘need-based research’ because the DGHS knows the actual health scenario and can set priority research areas...

#### Non-communicable Diseases

Researches have also been conducted in non-communicable diseases (NCDs), like dementia, fear of fall among stroke survivors, consequences of arsenic exposure on cardiovascular health, diabetic retinopathy, childhood NCDs, cancer, and others.

#### Health Systems Research

Improving the performance of community health workers, tobacco control policies on smokeless tobacco, menstrual regulation, quality of care, treatment approach, therapy to improve quality of life of cancer patients, emergency healthcare delivery system, health improvement and consequences, and health service delivery were the major areas of health systems research.

#### Maternal and Child Health

Various research activities relating to maternal and child health have been conducted with ethical approval. These include: assessing incidence of hypertension

of pregnant women, assessing the preeclampsia/eclampsia among pregnant women, assessment of the quality of counseling on maternal nutrition and infant feeding, antenatal care, nutrient intake of urban poor pregnant women, determinants of low birthweight, air pollution and its effect on young adolescents' health, sexual reproductive health rights of adolescents.

## Mental Health

National Mental Health Survey, psychometric evaluation of women and girls, integration of mental health in community-based inclusive development were among the major topics of research in this area.

## Nutrition

Nutrition-related research topics included impact of nutrition, quality of adolescents' diet, integrated nutrition intervention for adolescents, evaluation of healthiness of the packaged foods, and non-alcoholic beverages, dietary assessment of macro- and micronutrient intakes by patients with CKD.

## Health Education

Awareness of early child marriage, impact of social protection on birth outcomes, health-related quality of life of auto-rickshaw drivers, awareness regarding body-weight misperception and obesity, drug-abuse among male and female adolescents, sanitation costs and willingness-to-pay among low-income consumers.

## Biological Science

Carrier detection for thalassemia and hemoglobinopathies, genetic diagnosis

of muscular dystrophy, antimicrobial resistance traits in Bangladeshi poultry, role of cholesteryl ester transfer protein gene polymorphisms on high-density lipoprotein cholesterol level, antimicrobial resistance pattern, and virulence factors of enterobacter species, biochemical markers for early diagnosis of neonatal septicemia, use of oral Rifapentine in the preventive treatment of drug susceptible tuberculosis, bioequivalence of two losartan Potassium formulations in healthy volunteers, comparison of serum vitamin D, calcium and phosphate status in children with autism were among the topics of research relating to biological science.

## Public Health

Public health research topics included: food adulteration and its impact on public health, identification of risk factors of diabetes mellitus among secondary school students, epidemiology of post-chikungunya arthralgia following Dhaka chikungunya outbreak 2017, identification of geographical pattern of cancer patients, social protection on birth outcomes, seroprevalence of chikungunya infection, students' perceptions after completion of MPH program.

The following two research projects were conducted by BMRC through tender process as PPR-2008:

- ◆ Situation Analysis on Emergency Healthcare Delivery System of Bangladesh
- ◆ Graduate Medical Education System in Bangladesh: A Situation Analysis

**Table 9.1.** Status of BMRC's research activities, FY 2018-2019

| Area of research               | Number of research protocols |
|--------------------------------|------------------------------|
| Communicable Diseases          | 8                            |
| Non-communicable Diseases      | 7                            |
| Health Systems Research        | 40                           |
| Maternal and Child Health      | 26                           |
| Mental Health-related Research | 6                            |
| Nutrition-related Research     | 11                           |
| Health Education               | 24                           |
| Biological Science             | 26                           |
| Public Health Research         | 26                           |
| <b>Total</b>                   | <b>174</b>                   |

### Research by IEDCR in 2018

The Institute of Epidemiology, Disease Control and Research (IEDCR) was established in 1976 through a bill approved in the Bangladesh Parliament. The Institute was established for epidemiological and communicable disease research as well as functioning of the disease control programs, especially parasitic and entomological containment of vectorborne diseases through application of epidemiological principles. Current major research projects of IEDCR are listed below:

- ◆ Assessment of vaccination coverage and vaccine-preventable disease serology and assessment of exposure to malaria and other parasitic and neglected tropical diseases among Forcibly Displaced Myanmar Nationals (FDMNs) in Cox's Bazar, Bangladesh
- ◆ Oral cholera vaccine coverage evaluation survey among FDMNs
- ◆ The effects of climate variability, seasonal variations and environmental events on drinking-water quality, diarrhea prevalence and WASH behavior in Bangladesh
- ◆ Evaluation of mechanisms to improve mobile phone survey metrics
- ◆ Understanding the ecology of antimicrobial resistance in wildlife
- ◆ OneHealth Economic Analysis of zoonotic diseases in Bangladesh.

### Other researches in 2018

- ◆ Arsenic and Immune Response to Influenza Vaccination in Pregnant Women and Newborn
- ◆ Developing Early Warning System for Climate-sensitive and WASH-associated Diseases through Health Surveillance
- ◆ The Evaluation of mHealth Platform for Diarrheal Disease Decision Support in Hospitals: A Cluster-randomized Control Trial
- ◆ PhD Research: Transmission of Avian Influenza from Animal to Human at the Human-Animal Interface in Bangladesh

### MSc researches in 2018

- ◆ Mental health status in newly-arrived adult Forcibly Displaced Myanmar Nationals (FDMNs) Living in Bangladesh
- ◆ Quality of non-communicable diseases (NCD) in two primary healthcare facilities in Bangladesh



- ◆ Child injury in a rural community of Narsingdi district
- ◆ Needs assessment for reducing childhood overweight and obesity in a selected schools of Dhaka city.
- ◆ Research work on antibiotic resistance genes is going on in collaboration with ideSHi.
- ◆ surveillance (FBIS) study is underway in collaboration with ideSHi.

### Research by Center for Medical Biotechnology (CMBT)

- ◆ Research work on the foodborne disease

### Research by CME in 2018-2019

Research was conducted on Structure and Practices of Institutional Ethics Review Board (IERB) in Medical Institutes of Bangladesh.

**Table 9.2.** Scientific papers published in international journals from CMBT (2018-2019)

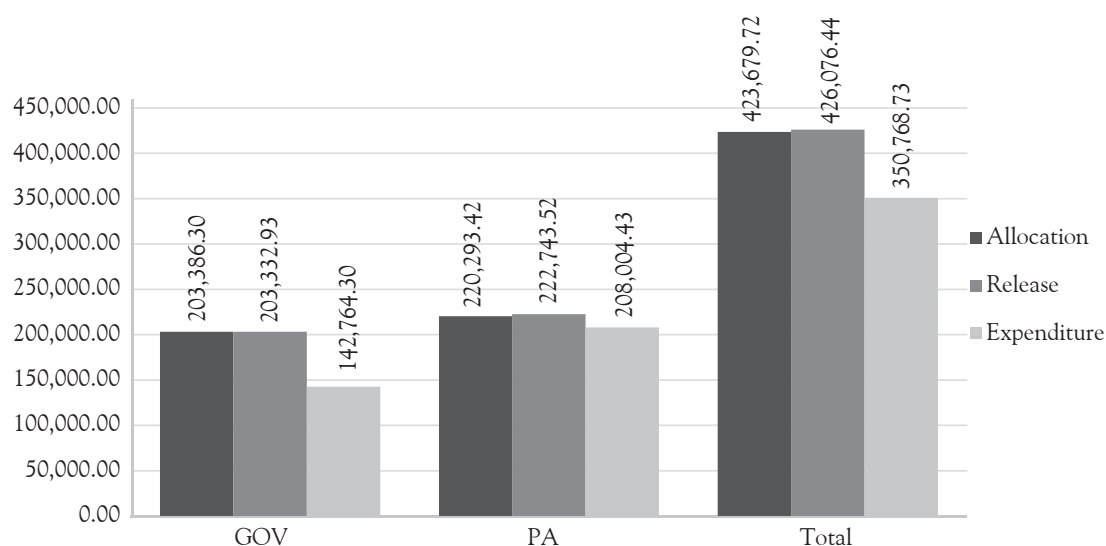
| Title of article                                                                                                                                                                                                          | Name of Journal                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Activated autologous platelet-rich plasma therapy in the management of chronic non-healing diabetic foot ulcer: one centre study from Bangladesh                                                                          | Journal of Medical Science and Clinical Research             |
| Activated platelet-rich plasma in the management of plantar fasciitis                                                                                                                                                     | International Journal of Medical Science and Health Research |
| A Comparative study on wound management by vacuum-assisted closure with low-cost negative pressure wound therapy and conventional moist wound dressing                                                                    | International Journal of Medical Research Professionals      |
| Mutation spectrum in TPO gene of Bangladeshi patients with thyroid dysmorphogenesis and analysis of the effects of different mutations on the structural features and functions of TPO protein through In Silico approach | BioMed Research International                                |

# Financing Healthcare

## Updated status of DGHS

Financial matters of the Ministry of Health and Family Welfare is maintained through the 4th Health, Population, and Nutrition Sector Program (4th HPNSP). Highlights of the 4th HPNSP 2017-2022 are shown below:

- Approved in ECNEC : 21-March 2017
- Overall budget : 11,548,636.03 lakh BDT
  - ◆ Development budget : 11,548,636.03 lakh BDT
  - ◆ Non-development budget : 7,200,000.00 lakh BDT
- Operation Plans in DGHS : 14
- Development projects : 24



**Figure 10.1.** Allocation, release, and expenditure of funds for activities at the DGHS in 2017-2018

### Performance of Operational Plans

Allocation, release, and utilization in FY 2017-2018 (revised ADP) of HPNSP 2017-2022 fund against different Operational Plans of the DGHS are shown in Table 10.1.

### Financial performance in development projects

#### Major Activities of CMSD

1. Procurement of goods through RPA

**Table 10.1.** Allocation, release, and expenditure in FY 2018-2019 of HPNSP 2017-2022 funds against different Operational Plans of the DGHS

| BDT in lakh |                                                   |            |           |           |           |           |           |             |           |           |
|-------------|---------------------------------------------------|------------|-----------|-----------|-----------|-----------|-----------|-------------|-----------|-----------|
| Sl No.      | OP/Project Name                                   | Allocation |           |           | Release   |           |           | Expenditure |           |           |
|             |                                                   | GOB        | PA        | Total     | GOB       | PA        | Total     | GOB         | PA        | Total     |
| 1           | Planning, Monitoring, and Research                | 961.30     | 1,005.42  | 1,966.72  | 908.43    | 735.00    | 1,643.43  | 778.45      | 646.64    | 1,425.09  |
| 2           | Health Information System and eHealth             | 13,490.00  | 3,000.00  | 16,490.00 | 13,490.00 | 3,000.00  | 16,490.00 | 13,330.86   | 3,000.00  | 16,330.86 |
| 3           | Procurement, Storage, and Supplies Management–HS  | 28,300.00  | 1,064.00  | 29,364.00 | 28,300.00 | 475.00    | 28,775.00 | 4,706.01    | 135.56    | 4,841.57  |
| 4           | Medical Education and Health Manpower Development | 36,000.00  | 5,500.00  | 41,500.00 | 35,999.50 | 5,500.00  | 41,499.50 | 8,721.62    | 1,690.20  | 10,411.82 |
| 5           | Maternal, Neonatal, Child and Adolescent Health   | 8,348.00   | 69,017.00 | 77,365.00 | 8,348.00  | 68,280.04 | 76,628.04 | 7,162.55    | 66,650.24 | 73,812.79 |
| 6           | National Nutrition Services                       | 500.00     | 8,400.00  | 8,900.00  | 500.00    | 8,400.00  | 8,900.00  | 461.61      | 8,227.53  | 8,689.14  |

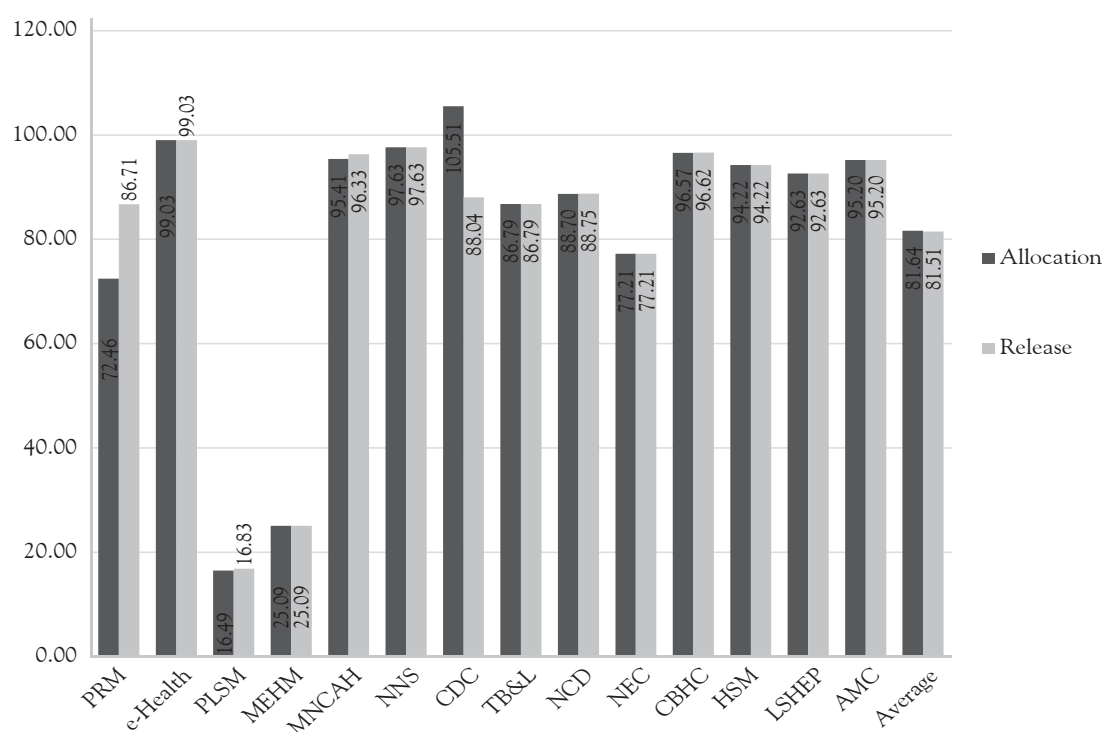
**Table 10.1 Contd.**

Table Continued...

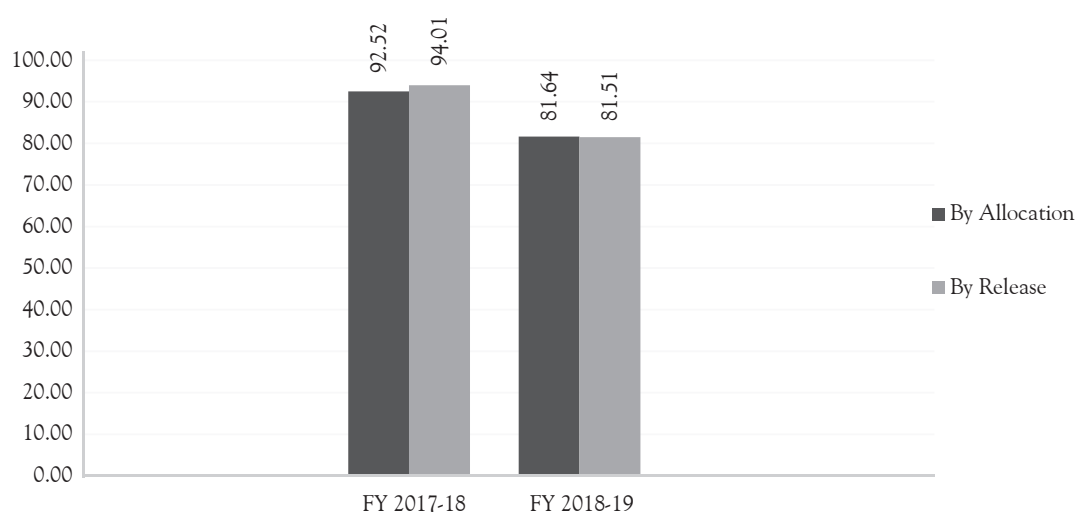
| BDT in lakh |                                              |            |            |            |            |            |            |             |            |            |
|-------------|----------------------------------------------|------------|------------|------------|------------|------------|------------|-------------|------------|------------|
| Sl No       | OP/Project Name                              | Allocation |            |            | Release    |            |            | Expenditure |            |            |
|             |                                              | GOB        | PA         | Total      | GOB        | PA         | Total      | GOB         | PA         | Total      |
| 7           | Communicable Diseases Control                | 8,657.00   | 12,056.00  | 20,713.00  | 8,657.00   | 16,166.14  | 24,823.14  | 7,647.87    | 14,206.72  | 21,854.59  |
| 8           | TB, Leprosy and AIDS/STD Program             | 4,525.00   | 18,798.00  | 23,323.00  | 4,525.00   | 18,798.00  | 23,323.00  | 2,733.16    | 17,508.00  | 20,241.16  |
| 9           | Non-communicable Diseases                    | 9,923.00   | 7,270.00   | 17,193.00  | 9,923.00   | 7,261.32   | 17,184.32  | 8,408.84    | 6,841.79   | 15,250.63  |
| 10          | National Eye Care                            | 440.00     | 2,310.00   | 2,750.00   | 440.00     | 2,310.00   | 2,750.00   | 371.83      | 1,751.50   | 2,123.33   |
| 11          | Community-based Healthcare                   | 59,553.00  | 39,370.00  | 98,923.00  | 59,553.00  | 39,315.02  | 98,868.02  | 58,169.46   | 37,357.03  | 95,526.49  |
| 12          | Hospital Services Management                 | 26,959.00  | 51,113.00  | 78,072.00  | 26,959.00  | 51,113.00  | 78,072.00  | 24,812.75   | 48,743.58  | 73,556.33  |
| 13          | Lifestyle and Health Education and Promotion | 1,909.00   | 940.00     | 2,849.00   | 1,909.00   | 940.00     | 2,849.00   | 1,811.28    | 827.64     | 2,638.92   |
| 14          | Alternative Medical Care                     | 3,821.00   | 450.00     | 4,271.00   | 3,821.00   | 450.00     | 4,271.00   | 3,648.00    | 418.00     | 4,066.00   |
| Summery     |                                              | 203,386.30 | 220,293.42 | 423,679.72 | 203,332.93 | 222,743.52 | 426,076.44 | 142,764.30  | 208,004.43 | 350,768.73 |

Source: ADP fund monitoring system, MOHFW

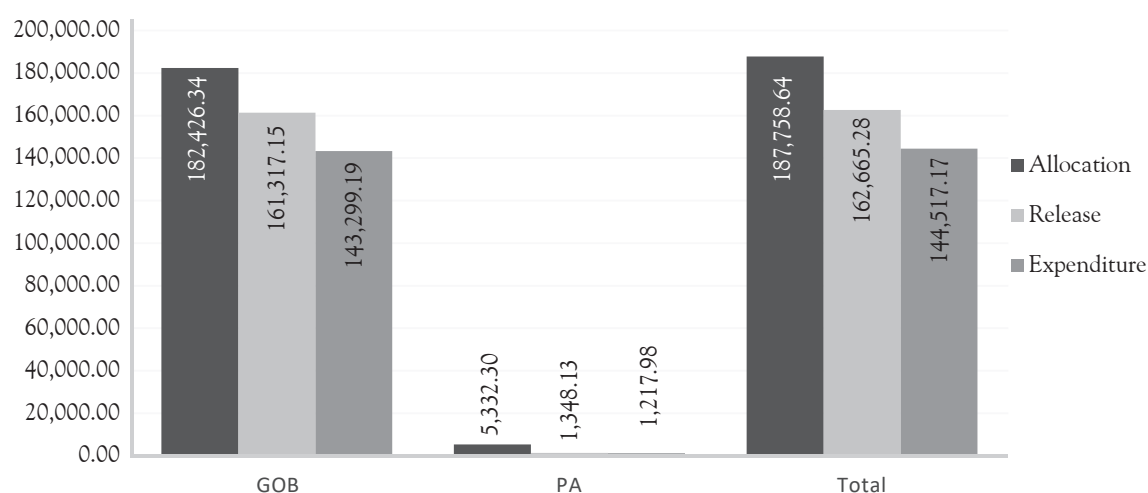
- (GOB), GOB (Development), and GOB (Revenue) funds
2. Clearance of goods from ports providing CD VAT
  3. Storage of goods in CMSD warehouse and distribution of goods to users' ends
  4. Procurement of NCT packages through eGP.



**Figure 10.2.** Financial status (in lakh taka) for Operational Plans of the DGHS in fiscal year 2018-2019 [Source: ADP fund monitoring system, MOHFW]



**Figure 10.3.** Fund utilization rate (%) against allocation (GOB+PA) in 2017-18 and 2018-19 fiscal years [Source: ADP fund monitoring system, MOHFW]



**Figure 10.4.** Fund utilization rate (%) of the DGHS development projects against allocated and released funds in FY 2018-2019

**Table 10. 2.** Allocation, release, and expenditure for the performance of development projects of the DGHS during fiscal year 2018–2019

| Sl No | Investment in projects                                                                      | BDT in lakh |           |             |                      |                                 |
|-------|---------------------------------------------------------------------------------------------|-------------|-----------|-------------|----------------------|---------------------------------|
|       |                                                                                             | Allocation  | Release   | Expenditure | Utilization rate (%) | Utilization rate of Release (%) |
| 1     | Sheikh Hasina National Institute of Burn and Plastic Surgery, Dhaka                         | 46,048.00   | 46,048.00 | 36,144.74   | 78.49                | 78.49                           |
| 2     | Establishment of 250-bedded National Ophthalmology Inst. and Hospital (1st Phase: 250 beds) | 1,831.00    | 15.50     | 8.34        | 0.46                 | 53.81                           |
| 3     | Upgradation of National Institute of Cancer Research and Hospital from 50 beds to 300 beds  | 2,998.00    | 1,454.21  | 638.81      | 21.31                | 43.93                           |
| 4     | Establishment of National Institute of Laboratory Medicine and Referral Center              | 3,392.00    | 3,392.00  | 386.98      | 11.41                | 11.41                           |

**Table 10.2 Contd.**

Table Continued...

| Sl No | Investment in projects                                                                           | BDT in lakh |          |             |                      |                                 |
|-------|--------------------------------------------------------------------------------------------------|-------------|----------|-------------|----------------------|---------------------------------|
|       |                                                                                                  | Allocation  | Release  | Expenditure | Utilization rate (%) | Utilization rate of Release (%) |
| 5     | Establishment Sheikh Sayera Khatun Medical College and Hospital and Nursing Institute, Gopalganj | 12,900.00   | 6,445.03 | 5,000.32    | 38.76                | 77.58                           |
| 6     | Establishment of Shatkhira Medical College & Hospital                                            | 6,000.00    | 6,000.00 | 5,993.28    | 99.89                | 99.89                           |
| 7     | Establishment of Faridpur Medical College & 500-bedded Hospital                                  | 0.00        | 0.00     | 0.00        | 0                    | 0                               |
| 8     | National Institute of Digestive Diseases Research & Hospital                                     | 4,717.00    | 4,717.00 | 4,139.27    | 87.75                | 87.75                           |
| 9     | Establishment of Kushtia Medical College and Hospital Project                                    | 9,942.00    | 9,942.00 | 5,892.90    | 59.27                | 59.27                           |
| 10    | Establishment of Shaheed Syed Nazrul Islam Medical College, Kishoreganj                          | 6,655.00    | 6,655.00 | 4,747.29    | 71.33                | 71.33                           |
| 11    | Extension of Shaheed Sheikh Abu Naser Specialized Hospital, Khulna                               | 0.00        | 0.00     | 0.00        | 0                    | 0                               |
| 12    | Establishment of Trauma Center at Gopalganj                                                      | 69.00       | 69.00    | 68.02       | 98.58                | 98.58                           |
| 13    | Extension of National Institute of Orthopedic Hospital and Rehabilitation Center (NITOR)         | 4,880.00    | 4,880.00 | 1,951.82    | 40                   | 40                              |
| 14    | Establishment of Jamalpur Medical College & Hospital, Jamalpur Nursing College                   | 6,798.00    | 4,999.75 | 4,660.00    | 68.55                | 93.2                            |
| 15    | Establishment of Patuakhali Medical College & Hospital                                           | 4,167.00    | 4,167.00 | 4,014.38    | 96.34                | 96.34                           |

Table 10.2 Contd.



Table Continued...

| Sl No          | Investment in projects                                                                                            | BDT in lakh       |                   |                   |                      |                                 |
|----------------|-------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|-------------------|----------------------|---------------------------------|
|                |                                                                                                                   | Allocation        | Release           | Expenditure       | Utilization rate (%) | Utilization rate of Release (%) |
| 16             | Establishment of Sheikh Lutfar Rahman Dental College                                                              | 1,826.00          | 1,826.00          | 1,816.20          | 99.46                | 99.46                           |
| 17             | MNCH and Health System Improvement Project Comp-2: Strengthening of Diagnostic Imaging System of 8 divisional MCH | 695.00            | 695.00            | 541.01            | 77.84                | 77.84                           |
| 18             | Col. Malek Medical College, Manikganj                                                                             | 20,089.00         | 20,089.00         | 18,821.07         | 93.69                | 93.69                           |
| 19             | Shaheed M. Monsur Ali Medical College and 500-bed Medical College at Sirajganj                                    | 19,987.34         | 19,982.83         | 28,780.61         | 143.99               | 144.03                          |
| 20             | Establishment of Shiekh Hasina Medical College and Hospital, Tangail                                              | 26,279.00         | 13,139.50         | 12,331.34         | 46.92                | 93.85                           |
| 21             | Strengthening Public Health Actions for Emerging Infectious Events in Bangladesh                                  | 1,034.30          | 393.92            | 324.02            | 31.33                | 82.26                           |
| 22             | Expansion of National Institute of Neurosciences & Hospital                                                       | 3,144.00          | 3,484.00          | 4,093.87          | 130.21               | 117.5                           |
| 23             | Shaheed Tajuddin Ahmad Medical College and Hospital, Gazipur                                                      | 4,126.00          | 4,126.00          | 4,096.95          | 99.3                 | 99.3                            |
| 24             | Eye Health Promotion and Prevention of Blindness in Selected Areas of Bangladesh                                  | 0.00              | 0.00              | 0.00              | 0                    | 0                               |
| 25             | Establishment of 500-bedded Hospital and Ancillary Building in Jessore, Cox's Bazar, Pabna, Noakhali              | 181.00            | 144.54            | 65.95             | 36.44                | 45.63                           |
| <b>Summery</b> |                                                                                                                   | <b>187,758.64</b> | <b>162,665.28</b> | <b>144,517.17</b> | <b>61.2528</b>       | <b>70.6056</b>                  |

Source: ADP fund monitoring system, MOHFW

**Table 10.3.** Revenue budget allocation and expenditure summary for the DGHS and organizations under the DGHS in FY 2018-2019

| Name of institution               | Allocation       | Expenditure      | Utilization rate |
|-----------------------------------|------------------|------------------|------------------|
| DGHS (Central Office)             | 61414560         | 58328801         | 94.97553         |
| Divisional Health Offices         | 1715480          | 1493942          | 87.08595         |
| Civil Surgeon Offices             | 62113062         | 56338932         | 90.70384         |
| General Hospitals                 | 49368790         | 39822832         | 80.66398         |
| District Hospitals                | 35813960         | 26140839         | 72.99064         |
| Urban Health Complex & Sub-center | 372880           | 420017           | 112.6413         |
| TB Hospitals                      | 472300           | 385917           | 81.71014         |
| Other TB Hospitals                | 1593450          | 1513359          | 94.97374         |
| Other Hospitals                   | 3917450          | 2079728          | 53.08882         |
| Upazila Health Offices            | 106088310        | 103711675        | 97.75976         |
| Leprosy Hospitals                 | 659730           | 514155           | 77.93415         |
| Trauma Centers                    | 61,170           | 2,413            | 3.944744         |
| School Health Clinics             | 766550           | 467657           | 61.00802         |
| Medical Supply Depots             | 333910           | 387244           | 115.9726         |
| TB Clinics                        | 3192520          | 2479067          | 77.65236         |
| Urban Dispensary                  | 1037750          | 1118031          | 107.7361         |
| Upazila Health Complex            | 123795148        | 97897829         | 79.08051         |
| Communicable Diseases Hospitals   | 1184280          | 233099           | 19.68276         |
| Model Family Planning Clinics     | 744590           | 541188           | 72.68268         |
| <b>Total</b>                      | <b>454645890</b> | <b>393876725</b> | <b>86.63374</b>  |

Source: CMSD

**Table 10.4.** Procurement of packages for CMSD in fiscal year 2018–2019

| Nature of package                | Number of packages | Estimated value (lakh Taka) | Contract value (lakh Tk.) |
|----------------------------------|--------------------|-----------------------------|---------------------------|
| GOB (Revenue—vaccines for Hajjis | 1                  | 762.90                      | 721.82                    |
| GOB (Revenue), MOHFW             | 41                 | 44006.35                    | 30172.19                  |
| GOB (Development)                | 27                 | 12473.55                    | 9382.28                   |
| HSM, DGHS                        | 24                 | 35969.24                    | 17228.97                  |
| RPA (GOB)                        | 49                 | 34663.94                    | 29692.06                  |
| JICA Fund                        | 3                  | 194.10                      | 112.60                    |
| <b>Total</b>                     | <b>145</b>         | <b>128070.08</b>            | <b>87309.92</b>           |

Source CMSD

# Status of Development Activities

## Remarkable achievements in health sector

Bangladesh provides healthcare services to 160 million people with limited resources. Health service network is being expanded, and new areas of services with necessary infrastructural development are being included every year. Recent developments include the following:

### New Frontiers in Health Services and Management

#### *Structural development for health administration*

The following projects have been completed during 2018-2019, including specialized postgraduate medical institutes and hospitals:

1. National Institute of Neurosciences and Hospital, Dhaka (First Phase)
2. National Institute of ENT, Tejgaon, Dhaka
3. Bangladesh Institute of Tropical and Infectious Diseases, Fakirhat, Chottogram
4. National Institute of Digestive Diseases Research and Hospital. DMCH, Dhaka
5. Ahsania Cancer Mission Hospital
6. ENT Head Neck Cancer Hospital
7. Dhaka Shishu Hospital

In 2018-2019, Multipurpose Health Volunteer (MHV) program is being implemented in 19 upazilas and will be scaled up in phases all over the country...

8. Sheikh Hasina National Institute of Burn and Plastic Surgery [World's largest facility of this kind]
9. Sheikh Rasel Gastro Liver Institute and Hospital
10. National Institute of Laboratory Medicine and Referral Center, Dhaka
11. Kurmitola 500-bed General Hospital, Dhaka
12. Khilgaon 500-bed General Hospital
13. 250-bed TB Hospital, Shyamoli, Dhaka
14. 150-bed Sorkari Kormochari Hospital, Fulbaria, Dhaka
15. 100-bed Sheikh Fazilatunnesa Mujib Eye Hospital and Training Institute, Gopalganj

The following projects were ongoing during the reporting period:

1. National Institute of Neurosciences, Dhaka (Second Phase)

2. Sheikh Hasina National Institute of Burn and Plastic Surgery, DMCH, Dhaka
3. 250-bedded National Eye Institute, Sher-e-Bangla Nagar, Dhaka
4. 300-bedded National Cancer Institute, Mohakhali, Dhaka
5. National institute of Laboratory Medicine and Referral Center, Dhaka
6. Sheikh Sayera Khatun Medical College, Gopalganj
7. Satkhira Medical College Hospital
8. 500-bedded Faridpur Medical College Hospital
9. Kushtia Medical College
10. Shaheed Syed Nazrul Islam Medical College, Kishoreganj
11. Trauma Center, Gopalganj
12. Sheikh Lutfar Rahman Dental College, Gopalganj
13. Jamalpur Medical College and Hospital with Nursing College, Jamalpur
14. Patuakhali Medical College Hospital
15. 250-bedded Trauma Center in Manikganj
16. 500-bedded General Hospital in Tangail
17. Shaheed Monsur Ali Medical College and Hospital, Sirajganj
18. Bangabandhu Medical College and Hospital, Sunamganj
19. 100-bedded Cancer Hospital in each divisional medical college hospital

20. Extended National Institute of Trauma and Rehabilitation Center, Sher-e-Bangla Nagar, Dhaka.

### ***Community clinic: a mass-centered approach to healthcare***

Community clinic (CC) is an unprecedented instance of community participation and public-private partnership. Being inspired by community participation, some UN agencies and NGOs have started working for the community clinics. Community-based Healthcare (CBHC) is contributing substantially in achieving the targets of SDGs as did in the MDGs and will continue till achievement of the SDGs by 2030.

In 2018-2019, Multipurpose Health Volunteer (MHV) program is being implemented in 19 upazilas and will be scaled up in phases all over the country. The volunteers will get a small amount of honorarium as per their performance. This initiative will help us in achieving SDG 3, including Universal Health Coverage (UHC) within the specific time limit.

**Table 11.1.** Ongoing CC projects during 2018-2019

| Community clinic            | No.  |
|-----------------------------|------|
| Built newly                 | 404  |
| Ongoing                     | 625  |
| Renovation done             | 791  |
| Renovation ongoing          | 1209 |
| Renovation of delivery room | 200  |

A great number of patients received services at the government healthcare facilities during January–December 2018.

Table 11.2. Number of patients who received care at the government healthcare facilities during January–December 2018

| Type of facilities                      | Reported no. and % of institutions |       | Number of outdoor patients |                  | Number of emergency patients |                 | Number of indoor patients |                |
|-----------------------------------------|------------------------------------|-------|----------------------------|------------------|------------------------------|-----------------|---------------------------|----------------|
|                                         | No.                                | (%)   | Reported                   | Estimated        | Reported                     | Estimated       | Reported                  | Estimated      |
| Postgraduate and specialized facilities | 12                                 | 62.5  | 7291894                    | 11667030         | 1325385                      | 2120616         | 700943                    | 1121509        |
| Medical colleges                        | 17                                 | 92.3  | 9358559                    | 10139284         | 1976668                      | 2141569         | 1696114                   | 1837610        |
| District hospitals                      | 64                                 | 97.9  | 12,002,656                 | 12260118         | 2517062                      | 2571054         | 1899228                   | 1939967        |
| 10, 20, 31, 50 and 100-bedded hospitals | 48                                 | 76.9  | 423409                     | 550597           | 37578                        | 48866           | 33579                     | 43666          |
| Upazila health complexes                | 423                                | 98.99 | 24776798                   | 25029597         | 3800990                      | 3839772         | 2320916                   | 2344596        |
| Chest disease/TB hospitals              | 56                                 | 46.9  | 45703                      | 97448            | 0                            | 0               | 1452                      | 3096           |
| Infectious disease hospitals            | 7                                  | 74    | 40996                      | 55400            | 9375                         | 12669           | 13868                     | 18741          |
| Community clinics                       | 13896                              | 95.17 | 104701823                  | 110015575        | -                            | -               | -                         | -              |
| <b>Total</b>                            |                                    |       | <b>158641838</b>           | <b>169815049</b> | <b>9667058</b>               | <b>10734546</b> | <b>6666100</b>            | <b>7309185</b> |

Modern equipment in healthcare procured and supplied to relevant hospitals and/or institutes during 2018-2019 are shown in Table 11.3.

**Table 11.3.** Modern equipment in healthcare procured and supplied to relevant hospitals and/or institutes during 2014-2018

| Type of modern medical equipment | No.        |
|----------------------------------|------------|
| MRI machine                      | 15         |
| CT scan machine                  | 20         |
| Auto-analyzer machine            | 181        |
| Echo color doppler machine       | 43         |
| Bronchoscope machine             | 17         |
| Brachytherapy machine            | 5          |
| Cobalt machine                   | 4          |
| Ultrasound machine               | 132        |
| X-ray machine                    | 108        |
| <b>Total</b>                     | <b>525</b> |

**Table 11.4.** New healthcare units during 2014-2019

| Unit and/or care         | No.        |
|--------------------------|------------|
| Operation theatre        | 85         |
| ICU                      | 15         |
| CCU                      | 10         |
| HDU                      | 8          |
| NICU                     | 4          |
| SCANU                    | 23         |
| Dialysis                 | 12         |
| Cath lab                 | 5          |
| Burn unit                | 8          |
| Medical waste management | 12         |
| <b>Total</b>             | <b>182</b> |

For expansion of the vehicle pool for healthcare monitoring, 147 cars were delivered for use by the UHFPOs.

**Table 11.5.** Establishment of new teaching institutes during 2014-2019

| Type/Name                                                                                                                                            | No.        |
|------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Construction work in Sir Salimullah Medical College, Noakhali, Cox's Bazar, Jessore, Pabna, Barisal, Chottogram, Comilla and Khulna Medical Colleges | 12         |
| Nursing colleges and/or Institutes (Government)                                                                                                      | 38         |
| Nursing colleges and/or Institutes (Private)                                                                                                         | 132        |
| Medical Assistant Training School (MATs) (Government)                                                                                                | 1          |
| Medical Assistant Training School (MATs) (Private)                                                                                                   | 60         |
| Institute of Health Technology (IHT) (Government)                                                                                                    | 5          |
| Institute of Health Technology (IHT) (Private)                                                                                                       | 12         |
| <b>Total</b>                                                                                                                                         | <b>260</b> |

**Table 11.6.** Present number of total seats in different teaching institutes

| Type/Name                                     | Total number of seats |
|-----------------------------------------------|-----------------------|
| Government medical college                    | 4068                  |
| Private medical college                       | 6281                  |
| Armed forces medical college                  | 375                   |
| Government dental college/unit                | 532                   |
| Private dental college/unit                   | 1405                  |
| Nursing college and/or Institute (Government) | 1125                  |
| Nursing college and/or Institute (Private)    | 8500                  |

**Table 11.6 Contd.**

**Table Continued...**

| Type/Name                                             | Total number of seats |
|-------------------------------------------------------|-----------------------|
| Medical Assistant Training School (MATS) (Government) | 818                   |
| Medical Assistant Training School (MATS) (Private)    | 12824                 |
| Institute of Health Technology (IHT) (Government)     | 2785                  |
| Institute of Health Technology (IHT) (Private)        | 8940                  |
| <b>Total</b>                                          | <b>36113</b>          |

As part of the development of human resources in health sector, 308 and 4,597 new posts for doctors were created during 2018-2019 under the 37th and 39th BCS respectively, totaling 4,905 posts. Two hundred ninety posts for dental surgeons were created during 2018-2019 under the 39th BCS.

During the same period, 550 posts for nurses/midwives were created. All new positions were filled up through recruitment during the said periods.

Remarkable development was made in the key health indicators, such as under-5 mortality rates, neonatal mortality rates, and infant mortality rates as shown in Table 11.7.

**Table 11.7.** Development in key health indicators

| Indicators (per 1000 livebirth) | Period 2019 |
|---------------------------------|-------------|
| Under-5 mortality rate          | 32          |
| Neonatal mortality rate         | 18          |
| Infant mortality rate           | 27          |

## **Golden era in pharmaceutical industries**

On 1 January 2018, the Prime Minister of Bangladesh declared pharmaceutical product as 'the product of the year' for 2018.

During 2018, the country's domestic pharmaceutical market-size stood at Tk 20,511.8 crore with 15.6% compound annual growth rate (CAGR) for the last five years.

On top of that, the sector is expected to grow at 15% year-on-year basis to reach US\$ 5.11 billion by 2023, propelled by high investment by local companies as they seek to grab a bigger share of the global market.

During the reporting period, initiatives were taken to allocate resources in healthcare for the forcibly displaced citizens from Myanmar. Table 11.8, 11.9, and 11.10 show the allocations of resources and other interventions for them.

**Table 11.8.** Resources allocated for the forcibly displaced citizens from Myanmar

| Resources                      | No. |
|--------------------------------|-----|
| Primary healthcare             | 36  |
| Health post                    | 140 |
| Sexual and reproductive health | 09  |
| Other specialized services     | 25  |
| Ambulances                     | 45  |

**Table 11.9.** Healthcare support for the forcibly displaced citizens from Myanmar

| Type/Name | No   |
|-----------|------|
| Doctor    | 293  |
| Nurse     | 186  |
| SACMO     | 224  |
| CHW       | 1259 |



**Table 11.10.** Health interventions for the forcibly displaced citizens from Myanmar

| Type/Name            | No. of doses |
|----------------------|--------------|
| MR Vaccine           | 490501       |
| Oral Polio Vaccine   | 242432       |
| Vitamin A            | 72064        |
| Oral Cholera Vaccine | 2101553      |
| Penta, PCV, boPV     | 492011       |

***Increment in budget allocation in health sector***

Budget allocation for health sector has been increased in the last couple of years. In 2018-2019 financial year, the Government allocated Tk 11,141 crore for development of the Ministry of Health and Family Welfare.

***International recognition during the reporting period***

Hon'ble Prime Minister of Bangladesh Sheikh Hasina received the 2019 Vaccine Hero Award for remarkable development in the area of vaccination.